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(54) **COMPOSITIONS AND METHODS FOR
SPLICING MODULATION OF UNC13A**

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(57) **ABSTRACT**

Antisense oligonucleotides for splicing modulation of UNC13A (e.g., inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA), compositions including the antisense oligonucleotides, and methods of use are described. Also disclosed are pharmaceutical compositions including one or more antisense oligonucleotides and methods of treating an UNC13A-associated disease or a disease associated with TDP-43 dysfunction in a subject by administering the antisense oligonucleotides to the subject.

Specification includes a Sequence Listing.

COMPOSITIONS AND METHODS FOR SPLICING MODULATION OF UNC13A

RELATED APPLICATIONS

[0001] This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application Ser. No. 63/561,593, filed Mar. 5, 2024, entitled “COMPOSITIONS AND METHODS FOR SPLICING MODULATION OF UNC13A”, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The invention relates to compositions (e.g., antisense oligonucleotides) for splicing modulation and/or modulation of gene expression of UNC13A.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

[0003] The contents of the electronic sequence listing (T083370053US01-SEQ-CBD.xml; Size: 3,049,839 bytes; and Date of Creation: Mar. 4, 2025) is herein incorporated by reference in its entirety.

BACKGROUND

[0004] Amyotrophic lateral sclerosis (ALS), also known as Lou Gehrig’s disease, and frontotemporal dementia (FTD) are rare neurodegenerative diseases known as motor neuron diseases (MND). ALS is a terminal MND characterized by upper and lower motor neuron loss, which results in rapid deterioration of muscle function, and eventually death due to respiratory failure. FTD is characterized by progressive neuronal loss predominantly involving the frontal and temporal lobes, and generally presents as a behavioral or language disorder. Some patients with ALS also develop frontotemporal dementia (FTD). There is currently no known cure of ALS or FTD.

[0005] UNC13A is an essential protein for synaptic transmission and transmission at the neuromuscular junction (NMJ). Single nucleotide polymorphisms (SNPs), e.g., intronic SNPs, in UNC13A, have been shown to increase the risk of ALS and FTD. The SNPs have been shown to increase cryptic splicing and alter direct binding of RNA-binding protein TDP-43, which reduces the ability of TDP-43 to repress cryptic exon inclusion in UNC13A during RNA splicing. TDP-43 dysfunction has been shown to result in inclusion of a cryptic exon in an UNC13A mature mRNA. Cryptic exon inclusion in an UNC13A mature mRNA results in reduced UNC13A protein expression due to aberrant RNA degradation or the presence of premature stop codons.

[0006] Additional neurodegenerative diseases involving TDP-43 pathology include Alzheimer’s disease and Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE).

SUMMARY

[0007] ALS and FTD patients could benefit from inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA. There is a need for compositions and methods for treating ALS and FTD, e.g., by inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell in a subject with ALS or FTD.

[0008] The present disclosure, in some aspects, provides antisense oligonucleotides for splicing modulation and/or modulation of gene expression of UNC13A. Dual targeting antisense oligonucleotides comprising a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region allow for effective splicing modulation of an UNC13A cryptic exon while reducing off-target risks. In some embodiments, the UNC13A RNA transcript may be within a cell, e.g., a cell within a subject, such as a human subject. Compositions comprising such antisense oligonucleotides and methods of using such (e.g., for treating an UNC13A-associated disease, such as ALS or FTD) are also provided.

[0009] Some aspects of the present disclosure provide antisense oligonucleotides comprising a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region, wherein the first target region and/or the second target region each comprises an UNC13A sequence, wherein the antisense oligonucleotide modulates splicing of an UNC13A cryptic exon. In some embodiments, the first target region or the second target region comprises a nucleotide sequence as set forth in SEQ ID NO: 313 or SEQ ID NO: 315. In some embodiments, the UNC13A cryptic exon is in intron 20 of an UNC13A pre-mRNA.

[0010] In some embodiments, the first target region and the second target region are in intron 20 of an UNC13A pre-mRNA. In some embodiments, the first target region is adjacent to the second target region in an UNC13A pre-mRNA. In some embodiments, the first target region is downstream of the second target region in an UNC13A pre-mRNA. In some embodiments, the first target region is more than 1 nucleoside downstream of the second target region. In some embodiments, the first target region is downstream of the cryptic exon and the second target region is in the cryptic exon.

[0011] In some embodiments, the first target region comprises a nucleotide sequence as set forth in SEQ ID NO: 315 and/or the second target region comprises a nucleotide sequence as set forth in SEQ ID NO: 313. In some embodiments, the first target region comprises the nucleotide sequence of SEQ ID NO: 315, and the second target region comprises the nucleotide sequence of SEQ ID NO: 313.

[0012] In some embodiments, the first antisense sequence and/or the second antisense sequence is 6-15 nucleosides in length. In some embodiments, the first antisense sequence and the second antisense sequence is 6-15 nucleosides in length. In some embodiments, the first antisense sequence and/or the second antisense sequence is 8-15 nucleosides in length. In some embodiments, the first antisense sequence comprises at least 6 contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC, and/or wherein the second antisense sequence comprises at least 6 contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TGTACTC, and TCACCCA. In some embodiments, the first antisense sequence comprises at least 7 contiguous nucleobases of any

one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC, and/or wherein the second antisense sequence comprises at least 7 contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA. In some embodiments, the first antisense sequence comprises at least 8 contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC, and/or wherein the second antisense sequence comprises at least 8 contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC, and/or wherein the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA.

[0013] In some embodiments, the first antisense sequence and the second antisense sequence are immediately adjacent to each other. In some embodiments, one or more nucleobases are between the first antisense sequence and the second antisense sequence. In some embodiments, one or more (e.g., 1, 2, 3, 4, 5, or more) nucleobases at the 3' end of the first antisense sequence is complementary to one or more (e.g., 1, 2, 3, 4, 5, or more) nucleobases at the 3' end of the second target region, and/or one or more (e.g., 1, 2, 3, 4, 5, or more) nucleobases at the 5' end of the second antisense sequence is complementary to one or more (e.g., 1, 2, 3, 4, 5, or more) nucleobases at the 5' end of the first target region. In some embodiments, a spacer is between the first antisense sequence and the second antisense sequence. In some embodiments, the spacer is a C3 spacer (e.g., a C3 spacer shown in Table 16). In some embodiments, the spacer is a C6 spacer (e.g., a C6 spacer shown in Table 16).

[0014] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 194 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0015] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 194 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0016] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 497 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 249, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0017] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 506 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 259, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0018] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0019] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 198 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0020] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0021] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0022] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 534, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0023] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 534, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0024] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 535, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0025] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 535, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0026] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0027] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0028] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0029] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0030] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0031] In some embodiments, an antisense oligonucleotide provided herein comprises a nucleobase sequence of any one of SEQ ID NOs: 1-190 and 338-494. In some embodiments, an antisense oligonucleotide provided herein comprises a nucleobase sequence of any one of SEQ ID NOs: 1-182, 184-190, 338-434, 442-494.

[0032] In some embodiments, an antisense oligonucleotide provided herein comprises one or more modified nucleosides. In some embodiments, the one or more modified nucleosides comprise a non-bicyclic 2'-modified nucleoside, a 2'-4' bridged nucleoside, or combinations thereof. In some embodiments, each nucleoside of the antisense oligonucleotide is a modified nucleoside.

[0033] In some embodiments, the non-bicyclic 2'-modified nucleoside is a 2'-O-methoxyethyl (2'-MOE) modified nucleoside, a 2'-O—N-methylacetamide (2'-O-NMA) modified nucleoside, a 2'-O-Methyl (2'-O-Me) modified nucleoside, or a 2'-fluoro (2'-F) modified nucleoside, and/or the 2'-4' bridged nucleoside is a locked nucleic acid (LNA, 2'-4' methylene bridge), an ethylene-bridged nucleic acid (ENA, 2'-4' ethylene bridge), a constrained ethyl nucleic acid (cEt, 2'-4' ethylene bridge), a 2'-O, 4'-C-Spirocyclopropylene bridged nucleic acid (scpBNA), an amido-bridged nucleic acid (AmNA), or a 2'-O, 4'-C-aminomethylene bridged nucleic acid (BNA (NC)). In some embodiments, the antisense oligonucleotide comprises one or more 2'-MOE modified nucleosides and one or more LNAs. In some embodiments, each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more modified nucleosides selected from ENA, scpBNA, 2'-O-NMA, AmNA, and BNA (NC). In some embodiments, each nucleoside of the antisense oligonucleotide is a modified nucleoside, wherein the antisense oligonucleotide comprises one or more modified nucleosides selected from ENA, scpBNA, 2'-O-NMA, AmNA, and BNA (NC), and wherein the remaining modified nucleosides are 2'-MOE modified nucleosides, LNAs, or a mix of 2'-MOE modified nucleosides and LNAs.

[0034] In some embodiments, an antisense oligonucleotide provided herein comprises one or more modified internucleoside linkages. In some embodiments, each internucleoside linkage of the oligonucleotide is a modified internucleoside linkage. In some embodiments, the modified internucleoside linkage is a phosphorothioate linkage. In some embodiments, each internucleoside linkage of the oligonucleotide is a phosphorothioate linkage. In some embodiments, the antisense oligonucleotide comprises a mix of phosphodiester linkages and phosphorothioate linkages.

[0035] In some embodiments, the antisense oligonucleotide is a phosphorodiamidate oligomer (PMO).

[0036] In some embodiments, each cytosine of the oligonucleotide is not a 5-methyl-cytosine. In some embodiments, each cytosine of the oligonucleotide is a 5-methyl-cytosine. In some embodiments, one or more of the cytosines of the oligonucleotide is a 5-methyl-cytosine.

[0037] In some embodiments, the antisense oligonucleotide is linked to a lipid. In some embodiments, the lipid is a C16 lipid. In some embodiments, the C16 lipid is linked to the 5' nucleoside of the antisense strand via a DNA linker. In some embodiments, the DNA linker is a phosphodiester d(TCA) linker, wherein each internucleoside linkage is a phosphodiester linkage.

[0038] In some embodiments, an antisense oligonucleotide provided herein comprises a structure as provided in Table 5. In some embodiments, an antisense oligonucleotide provided herein comprises a structure as provided in Table 11. In some embodiments, an antisense oligonucleotide provided herein comprises a structure of any one of ASOs 1-331 or 350-814. In some embodiments, an antisense oligonucleotide provided herein comprises a structure of any one of SEQ ID NOS: 569-925, 927-957, 960-970, 973-1039, 1042-1165, 1170-1350, and 1355-1362. In some embodiments, an antisense oligonucleotide provided herein comprises a structure of any one of ASOs 407, 408, 420, 488, 613, 614, and 796-806. In some embodiments, an antisense oligonucleotide provided herein comprises a structure of any one of SEQ ID NOS: 569-880, 882-925, 927-957, 960-970, 973-1039, 1042-1155, 1165, 1170-1350, 1355-1362 and ASOs 407, 408, 420, 488, 613, 614, and 796-806. In some embodiments, the antisense oligonucleotide inhibits inclusion of the UNC13A cryptic exon into an UNC13A mature mRNA. In some embodiments, the antisense oligonucleotide induces a secondary structure between the first target region and the second target region in the UNC13A pre-mRNA.

[0039] In some embodiments, the antisense oligonucleotide comprises the following structure: [iTs][iGs][iTs][iTs][i⁵Cs][iAs][iAs][iTs][i⁵Cs][iAs][iTs][iTs][i⁵Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iG][iA][iG][iT][iTs][i⁵Cs] (SEQ ID NO: 569), wherein iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0040] In some embodiments, the antisense oligonucleotide comprises the following structure: [iTs][iGs][iTs][iTs][i⁵Cs][iAs][iAs][iTs][i⁵Cs][iAs][iTs][iTs][i⁵Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iG][iA][iG][iT][iTs][i⁵C] (SEQ ID NO: 570), wherein iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0041] In some embodiments, the antisense oligonucleotide comprises the following structure: [i⁵Cs][iGs][iAs][iAs][iTs][i⁵Cs][iTs][iAs][i⁵Cs][i⁵Cs][iAs][i⁵Cs][iAs][iTs][i⁵Cs][iTs][iGs][iTs][i⁵Cs][iAs][iAs][iTs][i⁵Cs][iA] (SEQ ID NO: 571), wherein iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0042] In some embodiments, the antisense oligonucleotide comprises the following structure: [iTs][iG][iT][iTs][i⁵Cs][iAs][iAs][iTs][i⁵Cs][iAs][iTs][iTs][i⁵Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iG][iT][iTs][i⁵C] (SEQ ID NO: 572), wherein iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0043] In some embodiments, the antisense oligonucleotide comprises the following structure: [iTs][iG][iT][iT][i⁵Cs][iAs][iAs][iTs][i⁵Cs][iAs][iTs][iTs][i⁵Cs][iGs][iGs]

[iGs][iAs][iTs][iAs][iAs][iGs][iA][iG][iT][iTs][i⁵Cs] (SEQ ID NO: 573), wherein iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0044] In some embodiments, the antisense oligonucleotide comprises the following structure: [nmaTs][nmaGs][nmaTs][nmaTs][nma⁵Cs][nmaAs][nmaAs][nmaTs][nma⁵Cs][nmaAs][nmaTs][nma⁵Cs][nmaGs][nmaGs][nmaAs][nmaTs][nmaAs][nmaA][nmaG][nmaA][nmaG][nmaT][nmaTs][nma⁵C] (SEQ ID NO: 969), wherein nmaA, nmaG, nma⁵C, and nmaT are 2'-O-NMA modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0045] In some embodiments, the antisense oligonucleotide comprises the following structure: [nma⁵Cs][nmaGs][nmaAs][nmaAs][nmaTs][nma⁵Cs][nmaTs][nmaAs][nma⁵Cs][nma⁵Cs][nma⁵Cs][nmaAs][nma⁵Cs][nmaTs][nma⁵Cs][nmaTs][nmaGs][nmaTs][nmaTs][nma⁵Cs][nmaAs][nmaAs][nmaTs][nma⁵Cs][nmaA] (SEQ ID NO: 1115), wherein nmaA, nmaG, nma⁵C, and nmaT are 2'-O-NMA modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0046] In some embodiments, the antisense oligonucleotide comprises the following structure: [nmaAs][nma⁵Cs][nmaGs][nmaAs][nmaAs][nmaTs][nma⁵Cs][nmaTs][nmaAs][nma⁵Cs][nma⁵Cs][nma⁵Cs][nmaAs][nma⁵Cs][nmaAs][nmaTs][nma⁵Cs][nmaTs][nmaGs][nmaTs][nmaTs][nma⁵Cs][nmaAs][nmaAs][nmaTs][nma⁵C] (SEQ ID NO: 1116), wherein nmaA, nmaG, nma⁵C, and nmaT are 2'-O-NMA modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0047] In some embodiments, the antisense oligonucleotide comprises the following structure: [iGs][nmaAs][nmaAs][nmaTs][nma⁵Cs][nmaTs][nmaAs][nma⁵Cs][nma⁵Cs][nmaAs][nmaTs][nmaGs][nmaTs][nmaAs][nma⁵Cs][iT] (SEQ ID NO: 1355), wherein iA, iG, i⁵C, and iT are 2'-4' methylene bridge (LNA) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; nmaA, nmaG, nma⁵C, and nmaT are 2'-O-NMA modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0048] In some embodiments, the antisense oligonucleotide comprises the following structure: [iGs][nmaAs][nmaAs][nmaTs][nma⁵Cs][nmaTs][nmaAs][nma⁵Cs][nma⁵Cs][nmaAs][nmaTs][nmaGs][nmaTs][nmaAs][nma⁵Cs][iT] (SEQ ID NO: 1356), wherein iA, iG, i⁵C, and iT are 2'-4' methylene bridge (LNA) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; nmaA, nmaG, nma⁵C, and nmaT are 2'-O-NMA modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0049] In some embodiments, the antisense oligonucleotide is in the form of a pharmaceutically acceptable salt.

[0050] Some aspects of the present disclosure provide compositions comprising an antisense oligonucleotide pro-

vided herein. In some embodiments, a composition provided herein further comprises a pharmaceutically acceptable carrier.

[0051] Some aspects of the present disclosure provide methods of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell, comprising contacting the cell with an antisense oligonucleotide or composition provided herein.

[0052] Some aspects of the present disclosure provide methods of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject, comprising administering to the subject an antisense oligonucleotide or composition provided herein. In some embodiments, the method comprises administering to the subject a therapeutically effective amount of an antisense oligonucleotide or composition provided herein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject is human. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0053] Some aspects of the present disclosure provide methods of treating a disease associated with abnormal UNC13A expression in a subject, comprising administering to the subject an antisense oligonucleotide or composition provided herein. In some embodiments, the method comprises administering to the subject a therapeutically effective amount of an antisense oligonucleotide or composition provided herein. In some embodiments, the disease is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is amyotrophic lateral sclerosis (ALS). In some embodiments, the neurodegenerative disease is frontotemporal dementia (FTD). In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE). In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0054] Some aspects of the present disclosure provide methods of treating a disease associated with TDP-43 dysfunction in a subject, comprising administering to the subject an antisense oligonucleotide or composition provided herein. In some embodiments, the method comprises administering to the subject a therapeutically effective amount of an antisense oligonucleotide or composition provided herein. In some embodiments, the disease is a neurodegen-

erative disease. In some embodiments, the neurodegenerative disease is amyotrophic lateral sclerosis (ALS). In some embodiments, the neurodegenerative disease is frontotemporal dementia (FTD). In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE). In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0055] Some aspects of the present disclosure provide pharmaceutical compositions for use in a method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject comprising the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition comprises a therapeutically effective amount of an antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0056] Some aspects of the present disclosure provide pharmaceutical compositions for use in a method of treating a disease associated with abnormal UNC13A expression in a subject comprising the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition comprises a therapeutically effective amount of the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0057] Some aspects of the present disclosure provide pharmaceutical compositions for use in a method of treating

a disease associated with TDP-43 dysfunction in a subject comprising the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition comprises a therapeutically effective amount of the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0058] Some aspects of the present disclosure provide antisense oligonucleotides for use in a method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein. In some embodiments, the method comprises administering to the subject an effective amount of an antisense oligonucleotide provided herein. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0059] Some aspects of the present disclosure provide antisense oligonucleotides for use as a medicament. In some embodiments, the medicament is administered to a subject. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0060] Some aspects of the present disclosure provide antisense oligonucleotides for use in a method of treating a disease associated with abnormal UNC13A expression in a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein. In some embodiments, the method comprises administering to the subject an effective amount of an antisense oligonucleotide provided herein. In some embodiments, the subject is human. In some embodiments, the subject has decreased

expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0061] Some aspects of the present disclosure provide antisense oligonucleotides for use in a method of treating a disease associated with TDP-43 dysfunction in a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein. In some embodiments, the method comprises administering to the subject an effective amount of an antisense oligonucleotide provided herein. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0062] Some aspects of the present disclosure provide a use of an antisense oligonucleotide provided herein for the manufacture of a medicament for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject. In some embodiments, the composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0063] Some aspects of the present disclosure provide a use of an antisense oligonucleotide provided herein for the manufacture of a medicament for treating a disease associated with abnormal UNC13A expression in a subject. In some embodiments, the composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some

embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0064] Some aspects of the present disclosure provide a use of an antisense oligonucleotide provided herein for the manufacture of a medicament for treating a disease associated with TDP-43 dysfunction in a subject. In some embodiments, the composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

DETAILED DESCRIPTION

[0065] Generally, nomenclatures used in connection with cell and tissue culture, molecular biology, immunology, microbiology, genetics and protein and nucleic acid chemistry and hybridization described herein are those well-known and commonly used in the art. Certain methods and techniques provided herein are generally performed according to methods well-known in the art and as described in various general and more specific references that are cited and discussed throughout the present specification unless otherwise indicated. Enzymatic reactions and purification techniques are performed according to manufacturer's specifications, as commonly accomplished in the art or as otherwise described herein. The nomenclatures, laboratory procedures and techniques of analytical chemistry, synthetic organic chemistry, and medicinal and pharmaceutical chemistry described herein are those well-known and commonly used in the art. Standard techniques are used for chemical syntheses, chemical analyses, pharmaceutical preparation, formulation, delivery, and treatment of patients.

[0066] Any of the methods for gene therapy available in the art can be used in the methods provided herein. For general reviews of the methods of gene therapy, see Goldspiel et al. (1993) Clin. Pharmacy 12:488-505; Wu and Wu (1991) Biotherapy 3:87-95; Tolstoshev (1993) Ann. Rev. Pharmacol. Toxicol. 32:573-596; Mulligan (1993) Science 260:926-932; Morgan and Anderson (1993) Ann. Rev. Biochem. 62:191-217; and May (1993) TIBTECH 11 (5): 155-215. Methods commonly known in the art of recombinant DNA technology which can be used are described in Ausubel et al. (eds.), Current Protocols in Molecular Biology, John Wiley & Sons, NY (1993); and Kriegler, Gene Transfer and Expression, A Laboratory Manual, Stockton Press, NY (1990). Detailed description of various methods of gene therapy are disclosed in US Patent Publication No. US20050042664.

Definitions

[0067] In order that the present disclosure may be more readily understood, certain terms are first defined. Unless otherwise defined herein, scientific and technical terms used

in connection with the present disclosure shall have the meanings that are commonly understood by those of ordinary skill in the art. The meaning and scope of the terms should be clear, however, in the event of any latent ambiguity, definitions provided herein take precedent over any dictionary or extrinsic definition. Unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. The use of "or" means "and/or" unless stated otherwise. The use of the term "including", as well as other forms of the term, is not limiting.

[0068] In addition, it should be noted that whenever a value or range of values of a parameter are recited, it is intended that values and ranges intermediate to the recited values are also part of this disclosure.

[0069] As used herein, the singular forms "a", "an" and "the" include plural referents unless the context clearly dictates otherwise. "And" as used herein is interchangeably used with "or" unless expressly stated otherwise. The terms "comprising", "having", "including", and "containing" are to be construed as open-ended terms (i.e., meaning "including, but not limited to") unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value recited or falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited.

[0070] The term "about" or "approximately," as applied to one or more values provided herein, refers to a value that is similar to a stated reference value. In some embodiments, the term "about" or "approximately" refers to a range of values that fall within and include 15%, 14%, 13%, 12%, 11%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, or less in either direction (greater than or less than) of the stated reference value unless otherwise stated or otherwise evident from the context. In some embodiments, "about" or "approximately" can be understood as about 2 standard deviations from the mean. In some embodiments, "about" or "approximately" means up to and including $\pm 10\%$ (e.g., $\pm 10\%$, $\pm 9\%$, $\pm 8\%$, $\pm 7\%$, $\pm 6\%$, $\pm 5\%$, $\pm 4\%$, $\pm 3\%$, $\pm 2\%$, $\pm 1\%$, or less). In some embodiments, "about" or "approximately" means $\pm 5\%$. When "about" or "approximately" is present before a series of numbers or a range, it is understood that it can modify each of the numbers in the series or range.

[0071] The term "administering" or "administration of," as used herein, means to provide an agent, e.g., antisense oligonucleotide to a subject. In some embodiments, "administering" or "administration of" means to provide an antisense oligonucleotide to a subject in a manner that is physiologically and/or pharmacologically useful (e.g., to treat a condition in the subject). Non-limiting examples of routes of administration include intravenous, intramuscular, intraperitoneal, intracerebrospinal, subcutaneous, intra-articular, intrasynovial, intrathecal routes, or intracerebroventricular routes. In some embodiments, the route of administration is intrathecal, or intracerebroventricular routes. In some embodiments, the route of administration is intracerebroventricular. In some embodiments, the route of administration is subcutaneous.

[0072] The term "antisense oligonucleotide (ASO)," as used herein, refers to a single-stranded oligonucleotide which comprises one or more (e.g., 1, 2, or more) regions of complementarity each capable of hybridizing with at least a

part, for example, a target region of a target sequence (e.g., gene sequence, pre-mRNA sequence, or mRNA sequence), and which can modulate the processing of the transcription product of a target gene and/or the expression (e.g., mRNA and/or protein level) of the target gene. In some embodiments, a target region of a target nucleotide sequence can have a length of at least 8 nucleosides, for example, a length of 8, 9, 10, 11, 12, 13, 14, 15, or more nucleosides.

[0073] In some embodiments, an antisense oligonucleotide disclosed herein modulates splicing of an UNC13A pre-mRNA. In some embodiments, an antisense oligonucleotide disclosed herein modulates splicing of an UNC13A cryptic exon present in the UNC13A pre-mRNA (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA). In some embodiments, splicing modulation of UNC13A pre-mRNA and/or an UNC13A cryptic exon present in the UNC13A pre-mRNA results in modulation of UNC13A gene expression (e.g., mRNA and/or protein level).

[0074] In some embodiments, an antisense oligonucleotide disclosed herein targets two target regions of a target sequence (e.g., gene sequence, pre-mRNA sequence, or mRNA sequence). In some embodiments, an antisense oligonucleotide disclosed herein targets two target regions of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence). In some embodiments, the two target regions of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence) have 0-200 nucleobases (e.g., 0-200, 0-150, 0-100, 0-50, 0-20, 0-10, 0-5, 5-200, 5-150, 5-100, 5-50, 5-20, 5-10, 10-200, 10-150, 10-100, 10-50, 10-20, 20-200, 20-150, 20-100, 20-50, 50-200, 50-150, 50-100, 100-200, 100-150, or 150-200 nucleobases) in between.

[0075] In some embodiments, an antisense oligonucleotide comprises a first antisense sequence and a second antisense sequence, each targeting one of the two target regions of a target sequence (e.g., gene sequence, pre-mRNA sequence, or mRNA sequence). In some embodiments, an antisense oligonucleotide comprises a first antisense sequence and a second antisense sequence, each targeting one of the two target regions of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence). In some embodiments, the first antisense sequence is immediately adjacent to the second antisense sequence (i.e., having no nucleobases in between). In some embodiments, one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more) nucleobases are between the first antisense sequence and the second antisense sequence.

[0076] For the purposes of this disclosure, in an antisense oligonucleotide disclosed herein, the first antisense sequence is at the 5' end of the antisense oligonucleotide and the second antisense sequence is at the 3' end of the antisense oligonucleotide. Further, in an antisense oligonucleotide disclosed herein, the first antisense sequence targets the first target region of a target sequence (e.g., UNC13A pre-mRNA) and the second antisense sequence targets the second target region of a target sequence (e.g., UNC13A pre-mRNA). As such, the first target region is 3' to the second target region in the target sequence (e.g., UNC13A pre-mRNA), and the second target region is 5' to the first target region in the target sequence (e.g., UNC13A pre-mRNA).

[0077] The term “at least” prior to a number or series of numbers is understood to include the number adjacent to the term “at least”, and all subsequent numbers or integers that could logically be included, as clear from context. For

example, the number of nucleotides or nucleosides in a nucleic acid molecule must be an integer. For example, “at least 19 nucleosides of a 21-nucleotide nucleic acid molecule” means that 19, 20, or 21 nucleosides have the indicated property. When at least is present before a series of numbers or a range, it is understood that “at least” can modify each of the numbers in the series or range.

[0078] The term “Amyotrophic lateral sclerosis (ALS)” or “ALS,” as used herein, refers to a neurodegenerative disease characterized by progressive upper and lower motor neuron loss.

[0079] Two types of the disease, sporadic ALS (sALS) and familial ALS (fALS) have been described. sALS is the most common form of ALS, accounting for around 90% of all ALS cases, while fALS is rarer and genetic, as it occurs in a series of generations in families. Both forms of ALS show rapid clinical deterioration after onset of the disease, leading to death within a few years often due to respiratory failure. The exact cause of ALS is unknown and there is no known cure.

[0080] The term “frontotemporal dementia” or “FTD,” as used herein, refers to a progressive neurodegenerative disease of humans which has been found to have significant overlap with ALS at clinical, genetic, and pathologic levels. Burrell J R, Halliday G M, Kril J J, et al. The frontotemporal dementia-motor neuron disease continuum. *Lancet*. 2016; 388 (10047): 919-931. FTD is characterized by damage to neurons in the frontal and temporal lobes of the brain. Three types of the disease, behavioral variant frontotemporal dementia (bvFTD), primary progressive aphasia (PPA), and movement disorders, have been described, with bvFTD being the most common and involving changes in personality, behavior, and judgment. Some patients suffer from both Frontotemporal Dementia and ALS. Thus, these diseases are on one molecular and clinical spectrum. The exact cause of FTD is unknown and there is no known cure.

[0081] While causes of ALS and FTD are generally known, single nucleotide polymorphisms (SNPs), e.g., intronic SNPs, in UNC13A have been shown to increase the risk of ALS and FTD. In some instances, the SNPs increase cryptic splicing and alter direct binding of RNA-binding protein TDP-43, which reduces the ability of TDP-43 to repress cryptic exon inclusion in UNC13A during RNA splicing. TDP-43 dysfunction has been shown to result in inclusion of a cryptic exon in an UNC13A mRNA. Cryptic exon inclusion in an UNC13A mature mRNA results in reduced UNC13A protein expression due to aberrant RNA degradation or the presence of premature stop codons. In some embodiments, antisense oligonucleotides disclosed herein inhibit inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restore UNC13A expression in a cell in a subject with ALS or FTD.

[0082] The term “Alzheimer’s disease,” as used herein is a progressive degenerative brain disease and the most common form of dementia. Alzheimer’s disease is associated with the abnormal accumulation of amyloid- β (AB) plaques and neurofibrillary tangles of tau protein aggregates. Alzheimer’s disease is characterized by a progressive pattern of cognitive and functional impairment. TDP-43 pathology also plays a role in the progression of Alzheimer’s disease. Meneses et al., TDP-43 Pathology in Alzheimer’s Disease. *Mol Neurodegener*. 2021 Dec. 20; 16(1):84. There is currently no known cure for Alzheimer’s disease.

[0083] The term “Limbic-Predominant Age-Related TDP-43 Encephalopathy” or “LATE,” as used herein is a late-onset neurodegenerative disease characterized by amnesic dementia resembling Alzheimer’s disease dementia. LATE typically affects persons 80 years of age or older. LATE is characterized by TDP-43 inclusions predominated in the limbic regions of the brain, including the amygdala and hippocampus. Nelson et al., Limbic-predominant age-related TDP-43 encephalopathy (LATE): consensus working group report, *Brain*, Volume 142, Issue 6, June 2019, Pages 1503-1527. There is currently no known cure for LATE.

[0084] The term “biological activity” means any biological property of a molecule, whether present naturally in vivo, or provided or enabled by recombinant means. Biological activities include, but are not limited to, binding to a receptor, inducing cell proliferation, inhibiting cell growth, inducing other cytokines, inducing apoptosis, and enzymatic activity.

[0085] The term “contiguous” in the context of an oligonucleotide refers to nucleosides, nucleobases, sugar moieties, or internucleoside linkages that are immediately adjacent to each other. For example, “contiguous nucleobases” means nucleobases that are immediately adjacent to each other in a sequence.

[0086] The term “complementary,” as used herein, refers to the capacity for base pairing between two nucleobases or two nucleobase sequences. In particular, complementary is a term that characterizes an extent of hydrogen bond pairing that brings about binding between two nucleobases or two nucleobase sequences. For example, if a base at one position of a nucleobase sequence (e.g., of an antisense oligonucleotide described herein) is capable of hydrogen bonding with a base at the corresponding position of another nucleobase sequence (e.g., target gene sequence, pre-mRNA sequence, or mRNA sequence), then the bases are considered to be complementary to each other at that position. The nucleic acid molecules (e.g., antisense oligonucleotide) whose nucleobase sequence is complementary may comprise one or more modified nucleosides and modified internucleoside linkages. The nucleic acid molecules (e.g., antisense oligonucleotide and target sequence) whose nucleobase sequence is complementary may also comprise nucleobase analogues that result in bases at certain positions not being complementary, but the nucleobase sequences of the two molecules must be sufficiently complementary over the entire length to result in a desired biological activity (e.g., modulation of gene expression).

[0087] Complementary sequences include Watson-Crick base pairs or non-Watson-Crick base pairs (e.g., Wobble base pairs (e.g., G-U/T, A-G) and Hoogsteen base pairs) and may include natural or modified nucleosides or nucleoside mimics. For example, in some embodiments, for complementary base pairings, adenosine-type bases (A) are complementary to thymidine-type bases (T) or uracil-type bases (U), that cytosine-type bases (C) are complementary to guanosine-type bases (G), and that universal bases such as 3-nitropyrrole or 5-nitroindole can hybridize to and are considered complementary to any A, C, U, or T. Inosine (I) has also been considered in the art to be a universal base and is considered complementary to any A, C, U or T.

[0088] Complementarity is independent of modifications in the sugar of a nucleoside. For example, 2'-modified A, as

defined herein, are complementary to U (or T) and identical to A for the purposes of determining identity or complementarity.

[0089] The term “perfectly complementary” or “fully complementary” means that all (100%) of the nucleobases, nucleosides, or nucleotides in a contiguous sequence of a first nucleotide sequence (e.g., an antisense oligonucleotide) will hybridize with the same number of nucleobases, nucleosides, or nucleotides in a contiguous sequence of second nucleotide sequence (e.g., a target sequence such as an UNC13A pre-mRNA). The contiguous sequence may comprise all or a part of a first or second nucleotide sequence. The term “partially complementary” means that in a hybridized pair of nucleobase, nucleosides, or nucleotide sequences, at least 70%, but not all, of the bases in a contiguous sequence of a first nucleotide sequence (e.g., an antisense oligonucleotide) will hybridize with the same number of bases in a contiguous sequence of a second nucleotide sequence (e.g., a target sequence such as an UNC13A pre-mRNA). The term “sufficiently complementary” or “substantially complementary” means that in a hybridized pair of nucleobase, nucleosides, or nucleotide sequences, at least 85%, but not all, of the bases in a contiguous sequence of a first nucleotide sequence (e.g., an antisense oligonucleotide) will hybridize with the same number of bases in a contiguous sequence of a second nucleotide sequence (e.g., a target sequence such as an UNC13A pre-mRNA). The terms “complementary,” “fully complementary,” “partially complementary,” and “sufficiently/substantially complementary” herein are used with respect to the nucleobase, nucleosides, or nucleotide matching between an antisense oligonucleotide and a target sequence (e.g., an UNC13A pre-mRNA).

[0090] The term “control” or “reference,” when referring to a substance, means a composition used as a standard or a point of comparison against which other test results are measured. In some embodiments, a “control” or “reference” is a composition known to not contain analyte (“negative control”) or to contain analyte (“positive control”). A positive control can comprise a known concentration of analyte. “Control,” and “positive control,” may be used to refer to a composition comprising a known concentration of analyte. A “positive control” can be used to establish assay performance characteristics and is a useful indicator of the integrity of reagents (e.g., analytes). In some embodiments, an appropriate “control” or “reference” is where only one element is changed in order to determine the effect of the one element. In some embodiments, a control is a level of a target gene (e.g., in a cell or in a subject) before treatment (e.g., with an antisense oligonucleotide described herein).

[0091] The term “control” or “reference” also means a baseline level of a measurement depending upon the context, in which the term is used. A baseline level of a measurement is a standard or a point of comparison against which the measurement is compared. In some embodiments, a “control” or a “reference” refers to a level of a measurement for certain biological activity or substance in a cell, a tissue, an organ, or a subject, e.g., the expression level of a gene, copy number of mRNA for such gene, or level of protein encoded by such gene, without treatment of the cell, the tissue, the organ, or the subject, with an agent, e.g., antisense oligonucleotide. In some embodiments, a “control” or a “reference” refers to a level of an average measurement for a certain biological activity or substance in

a cell, a tissue, an organ, or a subject, e.g., certain enzyme activity of the liver, among a group of healthy subjects, e.g., the general population within in certain geographic or demographic limits or any other limits that may be appropriate for the study of certain disease or disorder, that does not have certain disease or disorder, e.g., liver disease.

[0092] The term “reference” may also be used in “reference sequence.” The term “reference sequence” refers to a sequence, e.g., a nucleic acid sequence or an amino acid sequence, used as a basis for sequence comparison. In certain embodiments, a reference sequence is an RNA sequence, e.g., human UNC13A pre-mRNA sequence, upon which the design of the antisense oligonucleotide is based.

[0093] The term “cross-reactive” means the ability of a binding molecule (e.g., an antisense oligonucleotide) to bind a target molecule (e.g., gene sequence, pre-mRNA sequence, or mRNA sequence) other than that against which it was designed or generated. For example, the binding molecule is capable of specifically binding to more than one target molecule of a similar type or class (e.g., mRNA variants or mRNA homologous from closely related species) with similar affinity. Generally, a binding molecule will bind its target molecule with an appropriately high affinity but can bind to the same target molecule of another species or display a low affinity for non-target molecules. In some embodiments, an antisense oligonucleotide that is cross-reactive against human and non-human primate UNC13A comprises a region of complementarity to human and non-human primate UNC13A gene sequence, pre-mRNA sequence, or mRNA sequence. Individual binding molecules are generally selected to meet two criteria: (1) tissue staining appropriate for the known expression of the target or (2) similar staining pattern between human and toxicology species (mouse and cynomolgus monkey) tissues from the same organ. These and other methods of assessing cross-reactivity are known to one skilled in the art.

[0094] The term “effective amount” or “therapeutically effective amount,” as used herein, refers to that amount of an antisense oligonucleotide to produce a molecular (e.g., reducing inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a subject), biological, pharmacological, therapeutic (e.g., treatment of an UNC13A associated disease or a disease associated with TDP-43 dysfunction in a subject), or preventive result. The amount administered will likely depend on such variables as the overall health status of the patient, the relative biological efficacy of the compound delivered, the formulation of the drug, the presence and types of excipients in the formulation, and the route of administration. Also, it is to be understood that the initial dosage administered can, in some instances, be increased beyond the above upper level to rapidly achieve the desired blood-level or tissue level, or the initial dosage can, in some instances, be smaller than the optimum.

[0095] The terms “hybridize” and “hybridization” refer to the pairing of complementary compounds (e.g., an antisense oligonucleotide and its target nucleic acid). While not limited to a particular mechanism, the most common mechanism of pairing involves hydrogen bonding, which may be Watson-Crick, Wobble, Hoogsteen or reversed Hoogsteen hydrogen bonding, between complementary nucleobases.

[0096] The term “internucleoside linkage,” as used herein, means a covalent linkage between adjacent nucleosides in an oligonucleotide (e.g., antisense oligonucleotide described

herein). An internucleoside linkage may be a natural phosphodiester internucleoside linkage, or may be a modified (non-natural) internucleoside linkage. Modified internucleoside that may be used in an antisense oligonucleotide disclosed herein include, but are not limited to, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates comprising 3' alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates comprising 3'-amino phosphoramidate and aminoalkylphosphoramidates, mesyl phosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and boranophosphates having normal 3'-5' linkages, 2'-5' linked analogs of these, and those having inverted polarity wherein the adjacent pairs of nucleoside units are linked 3'-5' to 5'-3' or 2'-5' to 5'-2'; see U.S. Pat. Nos. 3,687,808; 4,469,863; 4,476,301; 5,023,243; 5,177,196; 5,188,897; 5,264,423; 5,276,019; 5,278,302; 5,286,717; 5,321,131; 5,399,676; 5,405,939; 5,453,496; 5,455,233; 5,466,677; 5,476,925; 5,519,126; 5,536,821; 5,541,306; 5,550,111; 5,563,253; 5,571,799; 5,587,361; and 5,625,050. In some embodiments, in any one of the antisense oligonucleotides disclosed herein, all of the internucleoside linkages are stereorandom.

[0097] The term, “nucleoside,” as used herein, refers to a compound comprising a nucleobase moiety and a sugar moiety. Nucleosides include, but are not limited to, naturally occurring nucleosides (as found in DNA and RNA) and modified nucleosides. Nucleosides may be linked to a phosphate moiety. The term “nucleoside” encompasses a natural nucleoside and chemically modified nucleosides (e.g., with modifications in the base and/or sugar moiety).

[0098] The term “nucleotide,” as used herein, refers to a compound comprising a nucleoside linked to a phosphate group. As used herein, “linked nucleosides” may or may not be linked by phosphate linkages and thus includes, but is not limited to “linked nucleotides.” As used herein, “linked nucleosides” are nucleosides that are connected in a continuous sequence (i.e., no additional nucleosides are present between those that are linked). The term “nucleotide” encompasses a natural nucleotide and chemically modified nucleotides (e.g., with modifications in the base, sugar moiety, and/or phosphate group).

[0099] The term “nucleobase,” as used herein, refers to nitrogen-containing compounds that can be linked to a sugar moiety to create a nucleoside that is capable of incorporation into an oligonucleotide, and wherein the compound is capable of bonding with a complementary naturally occurring nucleobase of another oligonucleotide or nucleic acid. Nucleobases may be naturally occurring or may be modified. As used herein a “naturally occurring nucleobase” is adenine (A), thymine (T), cytosine (C), uracil (U), and guanine (G). The term “nucleobase” encompasses 5'-methylated bases (e.g., 5'-methyl cytosine or 5'-methyl guanine).

[0100] The term “modified internucleoside linkage” refers to a linkage between two nucleosides (e.g., in an oligonucleotide) that is not the natural phosphodiester linkage. Non-limiting examples of modified internucleoside linkages include phosphorothioates, phosphorodiamidates, phosphotriesters, methyl phosphonates, short chain alkyl or cycloalkyl intersugar linkages or short chain heteroatomic or heterocyclic intersugar linkages.

[0101] The term “nucleoside modification” or “modified nucleoside” means a nucleoside that has one or more modi-

fications to the nucleoside, including modifications to the nucleobase moiety and/or the sugar moiety. Any of the modified chemistries or formats of nucleosides described herein can be combined with each other. Non-limiting examples of modified nucleosides includes 2'-fluoro (2'-F), 2'-O-methyl (2'-O-Mc), 2'-O-methoxyethyl (2'-MOE), 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylaminoethyloxyethyl (2'-O-DMAEOE), 2'-O-N-methylacetamide (2'-O-NMA), locked nucleic acid (LNA, methylene-bridged nucleic acid), unlocked nucleic acid (UNA), ethylene-bridged nucleic acid (ENA), 2'-O, 4'-C-Spirocyclopropylene bridged nucleic acid (scpBNA), amido-bridged nucleic acid (AmNA), or 2'-O, 4'-C-aminomethylene bridged nucleic acid (BNA (NC)), and (S)-constrained ethyl-bridged nucleic acid (cEt) modified nucleosides. Further non-limiting examples of modified nucleosides include a conformationally restricted nucleoside, an abasic nucleoside, a 2'-amino-modified nucleoside, a morpholino nucleoside, a phosphoramidate, a non-natural base comprising nucleoside, a tetrahydropyran modified nucleoside, a 1,5-anhydrohexitol modified nucleoside (HNA), a cyclohexenyl modified nucleoside (CeNA), a nucleoside comprising a phosphorothioate group, a nucleoside comprising a methylphosphonate group, a nucleoside comprising a 5'-phosphate, a nucleoside comprising a 5'-phosphate mimic, a thermally destabilizing nucleoside, a glycol modified nucleoside (GNA).

[0102] The term "2'-modified nucleoside" refers to a nucleoside having a sugar moiety modified at the 2' position, meaning the sugar moiety comprises at least one 2'-substituent group other than H or OH. Non-limiting examples of 2'-modified nucleosides include: 2'-O-methoxyethyl (2'-MOE), 2'-O-Methyl (2'-O-Me), 2'-fluoro (2'-F), LNA (2'-4' ethylene bridge), ENA (2'-4' ethylene bridge), or cEt (2'-4' ethylene bridge), 2'-deoxy, 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylaminoethyloxyethyl (2'-O-DMAEOE), 2'-O-N-methylacetamide (2'-O-NMA), 2'-O, 4'-C-Spirocyclopropylene bridged nucleic acid (scpBNA), amido-bridged nucleic acid (AmNA), or 2'-O, 4'-C-aminomethylene bridged nucleic acid (BNA (NC)) modified nucleosides. In some embodiments, any one of the 2'-modified nucleosides described herein are high-affinity modified nucleosides and a modified antisense oligonucleotide has increased affinity to target sequences, relative to an unmodified antisense oligonucleotide. In some embodiments, at least one modified nucleoside is a 2'-modified nucleoside. In some embodiments, the 2'-modified nucleoside is a 2'-MOE modified nucleoside or an LNA or combinations thereof. In some embodiments, the 2'-modified nucleoside is a 2'-MOE modified nucleoside.

[0103] The term "modified oligonucleotide" or "modified antisense oligonucleotide" refers to oligonucleotides or antisense oligonucleotides that comprise one or more modified nucleosides and/or one or more modified internucleoside linkages. In some embodiments, a "modified oligonucleotide" or "modified antisense oligonucleotide" comprises a mix of modified nucleosides and unmodified nucleosides and/or a mix of modified internucleoside linkages and unmodified modified internucleoside linkages. In some embodiments, each nucleoside of a modified oligonucleotide" or "modified antisense oligonucleotide" is a modified nucleoside, and/or each internucleoside linkage of a modi-

fied oligonucleotide" or "modified antisense oligonucleotide" is a modified internucleoside linkage.

[0104] The term "region of complementarity," as used herein, refers to a nucleobase sequence, e.g., of an antisense oligonucleotide, that is sufficiently complementary to a cognate nucleobase sequence, e.g., of a target nucleic acid, such that the two nucleobase sequences are capable of annealing to one another under physiological conditions (e.g., in a cell). In some embodiments, a region of complementarity is fully complementary (e.g., 100% complementarity) to a cognate nucleobase sequence of target nucleic acid. In some embodiments, a region of complementarity is partially complementary to a cognate nucleobase sequence of target nucleic acid (e.g., at least 80%, 90%, 95% or 99% complementarity). In some embodiments, a region of complementarity contains 1, 2, 3, 4, or 5 mismatches compared with a cognate nucleobase sequence of a target nucleic acid.

[0105] The term "sequence identity," as used herein, refers to the extent that sequences are identical (independent of chemical modification) on a nucleobase-by-nucleobase basis or an amino acid-by-amino acid basis over a window of comparison. Thus, a "percentage of sequence identity" may be calculated by comparing two optimally aligned sequences over the window of comparison, determining the number of positions at which the identical nucleic acid base (e.g., A, T, C, G, I) or the identical amino acid residue (e.g., Ala, Pro, Ser, Thr, Gly, Val, Leu, Ile, Phe, Tyr, Trp, Lys, Arg, His, Asp, Glu, Asn, Gln, Cys and Met) occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison (i.e., the window size), and multiplying the result by 100 to yield the percentage of sequence identity. Optimal alignment of sequences for aligning a comparison window may be conducted by computerized implementations of algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package Release 7.0, Genetics Computer Group, 575 Science Drive Madison, Wis., USA) or by inspection and the best alignment (i.e., resulting in the highest percentage homology over the comparison window) generated by any of the various methods selected. Reference also may be made to the BLAST family of programs as for example disclosed by Altschul et al., Nucl. Acids Res. 25:3389, 1997.

[0106] The term "restore" when referring to expression of a given gene (e.g., UNC13A), means that the expression of the gene, as measured by the level of RNA transcribed from the gene or the level of polypeptide, protein or protein subunit translated from the mRNA in a cell, group of cells, tissue, organ, or subject in which the gene is transcribed, is returned to normal levels (e.g., baseline level of gene expression in a subject that does not have an UNC13A-associated disease and/or abnormal UNC13A expression) when the cell, group of cells, tissue, organ, or subject is treated with an antisense oligonucleotide disclosed herein. In some embodiments, when a cell, group of cells, tissue, organ, or subject is treated with an antisense oligonucleotide disclosed herein, expression (e.g., protein and/or mRNA expression) of a target gene (e.g., UNC13A) is restored by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90% or at least 95% relative to a control, e.g., baseline level of gene expression prior to treatment.

[0107] The term “decrease” when referring to expression of TDP-43, means that the expression of the gene, as measured by the level of RNA transcribed from the gene or the level of polypeptide, protein or protein subunit translated from the mRNA in a subject in which the gene is transcribed, is reduced relative to expression of TDP-43 in a healthy subject or a subject that does not have an UNC13A-associated disease (e.g., ALS, FTD, Alzheimer’s disease, or LATE). For example, a subject described herein having decreased expression of TDP-43 gene or protein by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 100%, has decreased expression of TDP-43 relative to expression of TDP-43 in a subject that does not have an UNC13A-associated disease (e.g., ALS, FTD, Alzheimer’s disease, or LATE). Expression of TDP-43 gene or protein may be measured by the level of RNA transcribed from the gene or the level of polypeptide, protein or protein subunit translated from the mRNA in a subject in which the gene is transcribed.

[0108] The term “single nucleotide polymorphism” or “SNP” is a genetic variation caused by changes in a single nucleobase. Without being bound by any particular theory, SNPs in UNC13A (e.g., intronic SNPs) typically increase cryptic splicing and alter direct binding of RNA-binding protein TDP-43, which reduces the ability of TDP-43 to repress cryptic exon inclusion in UNC13A during RNA splicing. SNPs may be associated with one or more diseases and may serve as genetic biomarkers for the one or more diseases. For example, SNPs can be used to predict drug exposure or effectiveness of a treatment in a subject. Non-limiting examples of SNPs associated with ALS include rs12608932, rs12973192, rs56041637, rs116169349, and rs62121687.

[0109] The term “subject,” as used herein, refers to a mammal. In some embodiments, a subject is non-human primate, or rodent. In some embodiments, a subject is a human. In some embodiments, a subject is a patient, e.g., a human patient that has or is suspected of having a disease. In some embodiments, the subject is a human patient who has or is suspected of having an UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) and/or a disease associated with TDP-43 dysfunction in a subject. In some embodiments, the UNC13A-associated disease and/or a disease associated with TDP-43 dysfunction in a subject is a neurodegenerative disease (e.g., amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), Alzheimer’s disease or Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE)). In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. SOD-1 gene mutations have been described, for example, in Miller et al., *N Engl J Med* 2022; 387:1099-110 and Ruffo et al., *Genes* 2022, 13, 537. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0110] The term “specificity” means the ability to inhibit the target RNA without manifest effects on other genes of the cell. The consequences of inhibition can be confirmed by examination of the outward properties of the cell or organism or by biochemical techniques such as RNA solution hybridization, nuclease protection, Northern hybridization,

reverse transcription, gene expression monitoring with a microarray, antibody binding, enzyme linked immunosorbent assay (ELISA), Western blotting, radioimmunoassay (RIA), other immunoassays, and fluorescence activated cell analysis (FACS).

[0111] The term “symptom” as used herein, refers to any manifestation or indication of an underlying disease. A symptom can be any biochemical, cellular, genetic, histological, and/or physiological observation, measurement, and/or test result in a subject that deviate from those of a control or reference. For example, a symptom may be an elevated level of a liver enzyme, such as aminotransferases, as compared to a normal reference range.

[0112] The term “target sequence,” as used herein, refers to a nucleoside sequence whose expression or activity is to be modulated. In some embodiments, the target sequence is a contiguous portion of the nucleoside sequence of a gene, a cDNA, a pre-mRNA, or an mRNA molecule formed during the transcription of a target gene, e.g., UNC13A gene, such as an unprocessed UNC13A pre-mRNA transcript. In some embodiments, a target sequence disclosed herein is in an UNC13A pre-mRNA. In some embodiments, a target sequence disclosed herein is in an intron of an UNC13A pre-mRNA. In some embodiments, a target sequence disclosed herein is in intron 20 of UNC13A pre-mRNA. In some embodiments, a target sequence disclosed herein is in or near (e.g., within 100 nucleobases upstream or downstream of) a cryptic exon in intron 20 of UNC13A pre-mRNA.

[0113] The term “treat,” “treatment,” as used herein, mean the methods or steps taken to provide relief from or alleviation of the number, severity, and/or frequency of one or more symptoms of a disease (e.g., UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) or a disease associated with TDP-43 dysfunction) in a subject. As used herein, “treat” and “treatment” may include the prevention, management, prophylactic treatment, and/or inhibition of the number, severity, and/or frequency of one or more symptoms of a disease (e.g., UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) or a disease associated with TDP-43 dysfunction) in a subject.

[0114] The term “variant” means a molecule (e.g., nucleic acid or polypeptide) that differs from a given molecule (e.g., a reference nucleic acid or polypeptide) in sequence (nucleic acid or amino acid respectively) by the addition (e.g., insertion), deletion, or conservative substitution of nucleic acids or amino acids, respectively, but that retains the biological activity of the given molecule. Changes in the reference nucleic acid sequence of the variant may be silent. That is, they may not alter the amino acid sequence encoded by the nucleic acid. Alternatively, changes in the nucleoside sequence of the variant may alter the amino acid sequence of a polypeptide encoded by the reference polynucleotide. Such nucleoside changes may result in amino acid substitutions, additions, deletions, fusions, and truncations in the polypeptide encoded by the reference sequence. The term “variant” encompasses fragments of a variant unless otherwise defined. A variant may be 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 89%, 88%, 87%, 86%, 85%, 84%, 83%, 82%, 81%, 80%, 79%, 78%, 77%, 76%, or 75% identical to the reference sequence. The degree of homology (percent identity) between a native and a variant sequence can be determined, for example, by comparing the two

sequences using freely available computer programs commonly employed for this purpose on the world wide web (e.g., BLASTn with default settings).

[0115] Definitions of specific functional groups and chemical terms are described in more detail below. The chemical elements are identified in accordance with the Periodic Table of the Elements, CAS version, *Handbook of Chemistry and Physics*, 75th Ed., inside cover, and specific functional groups are generally defined as described therein. Additionally, general principles of organic chemistry, as well as specific functional moieties and reactivity, are described in *Organic Chemistry*, Thomas Sorrell, University Science Books, Sausalito, 1999; Smith and March, *March's Advanced Organic Chemistry*, 5th Edition, John Wiley & Sons, Inc., New York, 2001; Larock, *Comprehensive Organic Transformations*, VCH Publishers, Inc., New York, 1989; and Carruthers, *Some Modern Methods of Organic Synthesis*, 3rd Edition, Cambridge University Press, Cambridge, 1987.

[0116] It is also to be understood that compounds that have the same molecular formula but differ in the nature or sequence of bonding of their atoms or the arrangement of their atoms in space are termed "isomers". Isomers that differ in the arrangement of their atoms in space are termed "stereoisomers".

[0117] Stereoisomers that are not mirror images of one another are termed "diastereomers" and those that are non-superimposable mirror images of each other are termed "enantiomers". When a compound has an asymmetric center, for example, it is bonded to four different groups, a pair of enantiomers is possible. An enantiomer can be characterized by the absolute configuration of its asymmetric center and is described by the R- and S-sequencing rules of Cahn and Prelog, or by the manner in which the molecule rotates the plane of polarized light and designated as dextrorotatory or levorotatory (i.e., as (+) or (−)-isomers respectively). A chiral compound can exist as either individual enantiomer or as a mixture thereof. A mixture containing equal proportions of the enantiomers is called a "racemic mixture".

[0118] The term "TDP-43," as used herein, means an RNA binding protein encoded by the TARDBP gene. TDP-43 has important role in RNA-processing activities and splicing regulatory function as a repressor of cryptic exon inclusion during RNA splicing. A key feature of neurodegenerative diseases, amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), Alzheimer's disease, and Limbic-Pre-

dominant Age-Related TDP-43 Encephalopathy (LATE) is mislocalization of TDP-43 from the nucleus and aggregated in the cytoplasm in human brain, neuronal cell lines motor neurons, and glial cells. It has been shown that TDP-43 dysfunction results in the inclusion of a cryptic exon in UNC13A mRNA and reduction of UNC13A protein expression. Diseases associated with TDP-43 dysfunction in a cell include the neurodegenerative diseases of ALS, FTD, Alzheimer's disease, and LATE.

[0119] The term "UNC-13 homolog A," used interchangeably with the term "UNC13A," refers to the well-known gene and polypeptide, also known in the art as Munc 13-1.

[0120] The term "UNC13A" includes human (*Homo sapiens*) UNC13A, the amino acid and nucleotide sequences of which may be found in, for example, GenBank Accession No. Gene ID: 23025, and NCBI Accession Nos. NG_052872.1: 4991 . . . 92009 (SEQ ID NO: 318), NC_000019.10: 17601336 . . . 17688354 complement (Reference GRCh38.p14 Primary Assembly) (SEQ ID NO: 318), NM_001080421.3 (SEQ ID NO: 307), and NP_001073890.2 (SEQ ID NO: 308). The term "UNC13A" also includes cynomolgus monkey (*Macaca fascicularis*) UNC13A, the amino acid and nucleotide sequences of which may be found in, for example, GenBank Accession No. GI: 102123626 and NCBI Accession Nos. NC_052273.1:17161646 . . . 17254007 complement (Reference MFA1912RKSv2 Primary Assembly) (SEQ ID NO: 319), XM_045380772.1 (SEQ ID NO: 309) and XP_045236707.1 (SEQ ID NO: 310). Additional examples of UNC13A RNA sequences are readily available using, e.g., GenBank, UniProt, and OMIM, and the *Macaca* genome project web site. Exemplary UNC13A nucleotide and amino acid sequences may also be found in Table 1, SEQ ID NOs: 307-310.

[0121] The term "UNC13A," as used herein, also refers to naturally occurring DNA sequence variations of the UNC13A gene. Numerous sequence variations within the UNC13A gene (e.g., SNPs rs12608932, rs12973192, rs56041637, rs116169349, rs62121687) have been identified and may be found at, for example, NCBI dbSNP and UniProt (see, e.g., ncbi.nlm.nih.gov/snp).

[0122] Further information on UNC13A can be found, for example, at ncbi.nlm.nih.gov/gene/23025. The entire contents of each of the foregoing Accession numbers and the Gene database numbers are incorporated herein by reference as of the date of filing this application.

[0123] Table 1 below summarizes exemplary amino acid sequences of UNC13A proteins and DNA sequences of UNC13A genes of human and cynomolgus monkey.

TABLE 1

Exemplary Amino Acid and DNA Sequences of UNC13A		
	Sequence	SEQ ID NO
Human	GCCCCCGGTGCTGAACCAAGATGGCCGGTGGCGGCCGGGCC	307
UNC13A	CCGGCGTGAGCCAAGCGCGGGCTGCAGCCGGGAGATGCCCC	
nucleotide	AGCCAGCGGCCGCTGAGCCCGACCCGACAGAGCCGGCCCG	
(NM_001080421.3)	GCCGCCTCCGGCCACCTGCGAGCTCGGAGACATGTCTCTGC	
	TTTGCGTTGGAGTCAAAAAAGCAAGTTTGTATGGTGCCCAAG	
	AGAAATTAACACGTACGTGACCCTGAAAGTGCAGAATGTC	
	AAGAGCACGACCATCGCGGTGCGGGGCAGCCAGCCAGCTG	
	GGAGCAGGATTTCATGTTTCAGATTAACCGTCTGGATTGGG	
	ACTGACGGTGGAGGTGTGGAATAAGGGTCTCATCTGGGACA	
	CAATGGTGGGCACTGTGTGGATCCCACTGAGGACCATCCGCC	
	AGTCCAATGAGGAGGGCCCTGGAGAGTGGCTGACGCTGGAC	
	TCCAGGTCATCATGGCAGACAGTGAGATCTGTGGCACCAA	
	GGACCCACCTTCCACCGCATCTCTCTGGACACGCGCTTTGA	

TABLE 1-continued

Exemplary Amino Acid and DNA Sequences of UNC13A	
Sequence	SEQ ID NO
GCTACCCCTTAGACATTCTCTGAAGAGGAGGCTCGCTACTGGGC CAAGAAGCTGGAGCAGCTCAATGCTATGCGGGACCAAGATG AATATTCTGTTCCAAGATGAGCAAGACAAGCCTCTGCCTGTCC CCAGCAACCAGTGTGCAACTGGAATTATTTGGCTGGGGTG AGCAGCACAAAGATGACCCCGACAGTGCAGTGGATGATCGT GACAGTGACTACCGCAGTGAACGAGCAACAGCATCCCGCC GCCCTATTATACTACGTCAACCAACCGCTCAGTCCACCA ATATTCTGTTGCCCCACCCCTGGGCTCCCGGAGTCCCTA CAGTGACTCCATGCACAGTTACGAGGAGTTCTCTGAGCCACA AGCCCTCAGCCCCACGGGTAGCAGCCGTATGCCTCTTCCGG GGAGCTGAGCCAGGGAAGCTCTCAGCTGAGCGAGGACTTCG ACCTTGACGAGCACAGCCTGCAGGGCTCCGACATGGAGGAT GAGCGGACCGGGACTCCTACCCTCCTGCCACAGTCCGGTC AGCTACCACAAGACTCGCCTCGCTGGGACCAAGATGAGGA AGAGCTGGAGGAGGACCTGGAGGACTTCTGGAGGAGGAGG AGCTGCCCTGAAGATGAGGAGGAGCTGGAGGAGGAGGAGGA GGAGGTGCTGACGATTGGGCGAGCTATGCCAGCGTGAAG ACGTAGCTGTGGCTGAGCCAAAGACTTCAAACGCATCAGC CTCCCGCCAGCTGCCCCAGGGAAGGAGGACAAGGCCCCAGT GGCACCCACCGAGGCCCGGACATGGCCAAGGTGGCCCCCA AGCCAGCCACGCCGACAAAGGTGCTGCAGCTGAGCAGATC CCTGAGGCTGAGCCACCAAGGACGAGGAGAGTTTCAGGCC GAGAGAGGATGAGGAAGGCCAGGAGGGGAGGACTCCATG TCCAGGGCCAAAGGCCAACTGGCTGCGTGCTTCAACAAAGT GCGGATGCAGCTGCAGGAGGCCCGGGGAGAGGAGAGATGT CTAAATCCCTATGGTTCAAAGGCGGCCAGGGGGCGGTCTCA TCATCATCGACAGCATGCCAGACATCCGCAAGAGGAAACCT ATCCCACTCGTGAGCGACTTGGCCATGTCCCTGGTCCAGTCC AGGAAAGCGGGCATCACCTCGGCCTTGGCCTCCAGCAGCTTG AACAACGAGGAGCTGAAAAACCAAGTTTACAAGAAGACCT GCAAGCCTTAATCTACCCCATCTCGTGACGACGCCACACAA CTTCGAAGTGTGGACGGCCACCAAGGCCCTACTGTACGA GTGCGAGGGGCTGCTGTGGGGCATCGCGAGGCGAGGATGC GCTGCACCGAGTGCCTGTCAAGTGCCACGAGAAGTGCCAG GACCTGCTCAACGCCGACTGCCTGCAGCGGGCTGCGGAGAA GAGCTCCAAGCACGGGGCGGAGGACCGGACACAGAACATCA TCATGGTGCTCAAGGACCGCATGAAGATCCGGGAGCGCAAC AAGCCCGAGATCTTCGAGCTCATCCAGGAGATCTTCGCGGTG ACCAAGACGGCGCACACGACGAGATGAAGGCGGTCAAGCA GAGCGTGCTGACGGCACGCTCAAGTGGTCCGCCAAGATCA GCATCACCGTGGTCTGCGGCCAGGGCTTGCAGGCAAAGGAC AAGACAGGATCCAGTGACCCCTATGTACCGTCCAGGTCCG GAAGACCAAGAAACGGACAAAACCTCTATGGGAACCTCA ACCCGGTGTGGGAGGAGAATTTCACTTTGAATGTCACAATT CCTCCGACCGCATCAAGGTGCGCTCTGGGACGAGGATGAC GACATCAAATCCCGCTGAACAGAGGTTCAAGAGGGAATC TGACGATTTCTGGGGCAGACGATCATTGAGGTGCGGACGCT CAGCGCGAGATGACGTGTGGTACAACTGGACAAGCGAA CTGACAAATCTGCCGTGTGGGTGCCATCCGGCTCCACATCA GTGTGGAGATCAAAGGCGAGGAGAAGGTGGCCCCGTACCAT GTCCAGTACACCTGTCTGCATGAGAACCCTGTTCACCTCTGTG ACCGACGTGCAGAACCAATGGGGTCTGGAAGATCCAGATGC CAAGGGTGACGATGCCTGGAAGGTTTACTACGATGAGACAG CCCAGGAGATTGTGGACGAGTTTGCCATGCGCTACGCGCTCG AGTCCATCTACCAAGCCATGACCACTTGCCTGCCTCTCCTC CAAGTATATGTGCCAGGGGTGCTGCGCTCATGAGCACCCCT GCTCGCCAACATCAATGCCCTACTACGCACACACACCGCCTC CACCAACGTGTCTGCCCTCCGACCGCTTCCGCGCTCCAACCT TGGGAAAGAGCGCTTCTGTAAACTCTGGACCAAGCTGCATA ACTCCCTGCGGATTGACCTCTCCATGTACCGGAATAACTTCC CAGCCAGCAGCCCGGAGAGACTCCAGGACCTCAAATCCACT GTGGACCTTCTACCAAGCATCACCTTCTTCGGATGAAGGTA CAAGAACTCCAGAGCCCGCCGAGCCAGCCAGGTGGTAA GGACTGTGTGAAAGCCTGCTTAATTCTACCTACGATACAT CTTCAATAACTGCCATGAAGTGTACAGCCGGGAGTACCAGAC AGACCCGCCCAGGAAGGGGAAGTTCTCCAGAGGAACAGG GGCCACGATCAAGAACCTCGACTTCTGGTCCAAGCTGATTA CCCTCATAGTGTCCATCATTGAGGAAGACAAGAAATTCCTACA CTCCCTGCCCTCAACAGTTTCCCAGGAGCTGAATGTGGGTA AAATCAGCGCTGAAGTGTGTGGAATCTGTTTGCCTCAAGACA TGAAGTACGCCATGGAGGAGCACGACAAAGCATCGTCTATGC AAGAGTGCCGACTACATGAACCTCACTTCAAGGTGAAATG GCTCTACAATGAGTATGTGACGGAACCTCCCGCTTCAAGGA	

TABLE 1-continued

Exemplary Amino Acid and DNA Sequences of UNC13A	
Sequence	SEQ ID NO
CCGCGTGCCTGAGTACCTGCATGGTTTGAACCCCTTCGTCAT CCAGTGGCTGGATGAGAATGAGGAGGTGTCCCGGGATTTCCT GCACGGTGCCCTGGAGCGAGACAAGAAGGATGGGTTCCAGC AGACCTCAGAGCATGCCCTATTCTCCTGCTCCGTGGTGGATG TTTTCTCCCACTCAACCAGAGCTTTGAAATCATCAAGAAAC TCGAGTGTCCCGACCCCTCAGATCGTGGGGCACTACATGAGGC GCTTTGCCAAGACCATCAGTAATGTGCTCCTCCAGTATGCAG ACATCATCTCCAAGGACTTTGCTCTCTACTGCTCCAAGGAGA AGGAGAAAGTGCCCTGCATTCTCATGAATAACACTCAACAG CTACGAGTTCAGCTGGAGAAGATGTTGAAGCCATGGGAGG AAAGGAGCTGGATGCTGAAGCCAGTGACATCCTGAAGGAGC TTCAGGTGAAACTCAATAACGTCTTGGATGAGCTCAGCCGGG TGTTTGCTACCAGCTTCAGCCGCACATTGAAGAGTGTGTCA AACAGATGGGTGACATCCTTAGCCAGGTTAAGGGCACAGGC AATGTGCCAGCCAGTGCCTGCAGCAGCGTGGCCAGGACGC GGACAATGTGTTCAGCCCATCATGGACCTGCTGGACAGCA ACCTGACCCCTCTTTGCCAAAATCTGTGAGAAGACTGTGTGTA AGCGAGTGTGAAGGAGCTGTGAAGCTGGTTATGAACACC ATGGAGAAAACCATCGTCTGCGCCCTCACTGACCAGACG ATGATCGGGAACCTCTTTGAGAAAACATGGCAAGGGATTAGA AAAGGGCAGGGTGAAATTGCCAAGCCACTCAGACGGAACCC AGATGATCTTCAATGCAGCCAAGGAGCTGGGTGAGCTGTCCA AACTCAAGGATCACATGGTACGAGAAGAAGCCAAGAGCTTG ACCCCAAAGCAGTGCAGCGTTGTTGAGTTGGCCCTGGACACC ATCAAGCAATATTCCACGCGGGTGGCGTGGGCCCTCAAGAA GACCTTCTTGAGAAAGAGCCCGACCTGCAATCCTTGCGCTA TGCCCTGTGCTCTACACGCAAGCCACCGACCTGCTAATCAA GACCTTTGTACAGACGCAATCGGCCAGGGCTTGGGTGTAGA AGACCTGTGGGTGAAGTCTCTGTCCATGTTGAGCTGTTTAC TCATCCAGGAAGTGGGGAACACAAGGTCAAGTGAAGAGTGG TGGCTGCCAATGACCTCAAGTGGCAGACTTCTGGCATCTTCC GGCCGTTTCATCGAGGTCAACATCATTGGGCCCGAGCTCAGCG ACAAGAAACGCAAGTTTGCAGCCAAATCCAAGAACAAATAGC TGGGCTCCCAAGTACAATGAGAGCTTCCAGTTCACGCTGAGC GCCGACGCGGTCCTGAGTGCTATGAGCTGCAGTGTGCGTGC AAGGACTACTGCTTCGCGCGCGAGGACCGCACGGTGGGGCT GGCCGTGCTGCAGCTGCGTGAAGTGGCCAGCGCGGGAGCG CCGCTGCTGGCTGCCGCTCGGCCGCGCATCCACATGGACG ACACGGGCTCACGGTGTGCGAATCCTCTCGCAGCGCAGCA ACGACGAGGTGGCCAAGGAGTTCGTGAAGCTCAAGTCGGAC ACGCGCTCCGCGGAGGAGGCGGTGCGCGCCTGCGCCTTA GCGCGGCGGTCGGCCGAGCGGCACTGCGCTGCGCGGAGG GCGCTGGGCGGGAGGACGGGGCTTGCCTTGGTGGGAC CTCCCCAGGGGCGGGGCTCGGGGGCTCCACGCCAAGGGTG GGCTGCGCCTACGCCCTTGACTCAGCTTCCCTTTTGGGGAA TTAGGAATGGAGGATGCCCCGCCCTCTCGGAGGGCCACGCC CAAGGGCGCGACGAAGGAAGAGCCACATCCCCAACTTGAG GCCACGCCCCCAGCACCTAGGGGGCATTTTGAAGTGGGATG GGGGAAACCTCGTCCCTATGGAGGAGCCACATCCCGGGGC TCTGGTACCGGAGGCACCACTCATGTCCCTTGAAAAGCC ATAAGATGGGACCCAGACCCCTGGGACCCAGACCAATTGC CAAGTATGGAATCTCAGCTCCCTCGAGGGGGGCCCTGGG CAAGGGGTAGGGCTCTCTGAGCGCCCTCTAGTGGCCCTGG GGACTGGAGGACCAAGGATGCTGGTTGGAGGGCCCGGAAT ACCGGAGTCCCTTTAGATATTGTGCAAAAATAAATGGGGG GAGGGGGAGGATGGGATTTCAAAGCACATGCGCCCTTGG GCGCCCCAAACCTGGGGGCGAGGGGACGGCTCTGGTTCC CACGCTGCCCTACTTCCCTTTGGGAGTTTGCCTCTCCCTCTC CCCCAACAAACCCAGTCTCATATCATAGAGTTCAACACACC CATTTGACAGATGGCAAACTGAGGCTTAAGAGCTGCTTG AGACTTGGCCAAGGTTCCAGGTGCCATACCTCTGTGCCCT CCCTTAGGCTGTGTGCCCCATGGAAGGGTGGGTGAGATCG GGATGACCTGACACAGCTCCCTATTGCTGCTAATTCCTCCCTC GGCTCTCTCAAGGGGTGGGAATTCAGGCCAAGACCCCTA CTTCGCTTTCTCTCCGCTGCCAAGCAGGACCTTTGCCCT CAGCCCTTCTCTGGGATCTCCATGGGGATGCCATGAGGG CCTCCACCAAAAAGAGAATTTGGGATCCCTGGTCCCAGG TTTCTCCATCCCTTCTCTCTTTCCAGAATTTCCAAATAGGA AAGAACAGAGGAGACCAAGAACTTAGGGGGAGAAAGA GAATGAGAGAAAGAGAATGAGAGAGAGAGAAACAAACA CAGTGACACAGTGAGAGCTTAGTCTCCAAGAGCCTATTTCATT GATTCAAACACCAAGCCACAGGATACCTCAGATGGCCCTCT TGCCAGCTGGAAGCTCTTCTCCATGAGCAAGTTACAGTG	

TABLE 1-continued

Exemplary Amino Acid and DNA Sequences of UNC13A	
Sequence	SEQ ID NO
ACCTGGCTGGAGTTACCTGGTGCACATAGGACCTAGGGGA AAGTTCAGCGTGGACTACACTTGCTCTGGGATCTGCTTTTCC ACATGTGTGTATGGCAGCCTTTTCTGCTGGATTGGGAAGG ACAAGATTTTGTCTGTCTAGGGAGAAATGAAAACGGGTGA GCTGAGTAGCTGGGTTTCTGGAGGATAGAATCAGATGGG GAGGCTTTCGAGGTGAAGAATGAGAGGGAACCACTACTA GAGAGAAAAGAGCTCCAGGCCTGGGGAACAGCAGCTGCGAA GGCCAGGAGAGAAGAACTGTTGAAACAACGAGAAGGGTGG CACGGCTGGAGCTGAGCCAGCAAGGGGGATCGTGAGGAGCC TTGGGGTTGGGGAGATCTGCAGAAGCATCAGACCAGGCAGG GCCCTGTACGCAGTCTCTGAGGAGTTTACTTTTATTCTAAGA CAGTTGGGAGCTCCAGGAGCTGTTTAAAGTTGGGGAGAGA CTGGATTCCAGCTGCAAAAGCTGTTTGTGAAGACTAAAAAC CAGTGAGGAGAGGTGGAGGTGCTTTGGGGACACTGAAATGG ATTCTTGAAAAGATTCTGAAGGCTGTGTTGAAAAGACACCTA TAGCTGTGGGGACATGACTATAATCCAGCATTTGGGGAGAC CGAGGCTGGCAGATCACTTAAGGTCAGGAGTTTGAGACCAG CCTGGCCAACATGGCGAAAACCCCATCTCTGCTAAAAATACAA AAATTAGCTGGGTGCAGTGGTGCATGCCTGTAGTCCAGCTA CTCAGGAGACTGAGGCGGGAGAAATTGTTGAACCTGGAGG CAGAGGTTGTAGTGAGTCGTGATCACACAACGCACTCCAGC CTGGGCAACAGAACAACTACTCCATCCCTCCCTCTACCCCA CCAAAAAATAAATCTGCCCCCTTAGATGAGCTCTAGG GCTGCTGAGTACAGTTGTCCAGTTGCACAGTGCCCAAGGGT TTGGCATGTCTAAGAAAGCCACGTGCAAAATCCTAGATATTGA GTGTTGTATGTTTGTGACGTGGTTTCCCGACATGTGAATGG CCCAAGTGTCTGGAAGAAGTGGCGCCACTTCTAATTGCTT GGAGATGTTGCATGTCCTTAAATTGAGCAGGTGCAGGTAA CTGGAGGTTCTGAACCAAGGTTAAATGCAAAATCTCATAC AGGGTTGGGAAGTTGTAGCCAGGGATAAGCTTATGTGACTGT TATATGGAAGTGGAGCAGATGTGAATTTGAAACCATGACAT GGCTGAGGCTAGGGGTGGGTGGATGGATGATTGAGGTTG TAACCCATAGAGCCCAAGGGGAAGTGTCTGTGACCTGGG GTGAGGTTGATCTGGAAGATTTTGGATGGCTGGAAAGAAA TGGGGAAGTCGAGCTGCCTGAGAGAGCCAAGTTATTTCCCA AAAGATTCTTAGGAGTCTTCTGTTCAAGACCTCCGTGTGT GTGTGTGTGTGTTAGGGTTCCCGAGCAATGGCCAGGCATG TGAAGGAAACAAGCTTCTCAGGGAATATTGTTGAATGAGT TTTCTGACTCCAGGCTAGAAGTGTGTTTGCATTTCCACCC TCTTTTCTTTCCCGAGAGAACTCTATTCGTCTTCAAACCC CATCACGGAACCCCTCTTGGAGAAAACCTCTCTCTTCCC CTCAGGACTTTCAGCCACCGTCTCTCTCCAGTCCAGCCT GATGCCATGGGACTGGGGTTTCTCTGTCCAGCTCTGTTTCTC CCAGACTGGGTCTGAGGACTCTCAGGACCCCAACTTTTACC TAGCACAGGCTGGGCAAGTGGGTGACAGGAGTCTACGC CTAGTGGAATTATGTATTGGGGCAGGGTCAGTGTGAGAAATAC ACATCCGCATGCATGTCTGTCCATGTCTGCTCCGTACCAACCTT CCCCTTCCACACGACCTGGGCAATAGGAGGTGTCTGAGCC TGACACATGGGACAGAGAGTGGACATGGCTGAGACACGGAC AGAGAAAAGACAAGGAGTCCAGGGGGCTGAAAGCCTTTTGA AATCAGGAAGTTCTGTATTGGCAGAACAAAGCCAGAGAG GAGCAGGGCTTTCTCAACGCCACCCAGCAAGTGGACACAG AGCCCGGCCTTGGATGACACCTCCAGGGTTCTGAACCTTGA CCTCGCTTTATGCAAGGAGCTGGCCCAATTTCCATGAATC GGGGAAACAGCACAAAGAGTTGGCCTGTGGCAGGGCAAGG GTTAAAGGGGTGACATTGAGGATGCTCTCAGAGTCAAAGTC CCCTGACCAAGAGGAATAGAGTAGAAAACACAGAGACAGA GGGTGAGATCACGCCCGATGAGGACGAGAGAGACAGAG ATGGAGAGAGACATAGAGGTGGAATATACAGAGAAAGAT AAATGCAGAGACCAGGCAGGGAGTGTGGGGGAAGTAA GAGGGTGTCTGAAGAAAGAGGATCTGTTCACTCTTACCAG TCTGTCTCGAATGATTGTCATAAAATGAGGAGGTGCTGTCT CACACCCCAATTTCTCTCAGGCCCCAGAGCTGAGACCT CACCATGCCCCATCAGAGATGCAAAAACATAAACCCCAA CTAGAAATCCTTGGGACCTCTCTCGGCTGGGATCTCAGAGCC TTTCTGTCCCTACCCCTACCCATGTGCTGTGATTTTGAG ATGGGGACAACCTGGGGCTTCCCGAACTCTGCCACCCCTGG GGAAGTTGGGGAGGGCCTTAGTCCCGATCACAAACCCGT CTGCTCCCAAGAAATCTTCTAAGAAATCGTTGAGGACCAAA GTTGTCTTTGTCTGACAGTGTGCTTTTCTCTTGCCTTTTAT GTTTCAGAGAAAAATCAAGTTGACTGTGTCAAGTAACACCCC ACCCCTTACCCCGTCCAGCCATAGTGGCTCTCTGGAGACAC AGGTCACAGGCGGAGGTCCCTGATCATCCCCAACACAC	

TABLE 1-continued

Exemplary Amino Acid and DNA Sequences of UNC13A		
	Sequence	SEQ ID NO
	AGCCAGGGGGACTTGACCCCTGTCCACCCCTGTCTCGTGCTC CCTCAGACCCCCACAAACCGGCCAAGCAGTCCGGGGAGGCT TCCCCTCCACACAACTCTAGCATGTGATTGCAGATGTGAAA TCAAAACGTTGTTTGTGTTTGTGTTTGTGATTCTACCCCGT CGGTCCAGTGTCTGCACAGACGCTTCATTCTCTGTAAATA TGTGACTTGGACAAATGTTTAAACACAAACGAGAAGTGGTC ATGAATGCATGGTGTGAGATGTTTGCACATTCTGACTTTT TGGTCTCTGTAAAAATATTTTATTAACAGCAGACATTAATAA AAGAAAACACACACA	
Human UNC13A Protein (NP_001073890.2)	MSLLCVGVKKAKFDGAQEKFNITYVTLKVQNVKSTTIAVRGSQ PSWEQDFMFEINRLDLGLTVEVWNGLIWDTMVGTVWIPLRTI RQSNEEGPGEWLTLDSQVIMADSEICGTDKPTFHRILLDRFELP LDIPEEEARYWAKKLEQLNAMRDQDEYSFQDEQDKPLVPVSNQ CCNWNYPGWGEQHNDPDSAVDDRDSDYRSETSNSIPPPYTT SQPNASVHQYSVRPPPLGSRRESYSDSMHSYEEFSEPQALSPTGSS RYASSGELSQQSSQLSEDFDPDEHSLQGSMDERDRDSYHSC HSSVSYHKDSPRWDQEELEEDLEDLEEEELPEEEEEEEEE EEVPDDLGSYAQREDVAEPAKDFKRISLPPAAPGKEDKAPVA PTEAPDMAKVPKATPDKVPAAEQIPEAEPPKDEESFRPREDE EGQEGQDSMSRAKANWLRAPKVRMLQAEARGEEMSKSLW FKGGPGGLIIIDSMPDIRKRKPIPLVSDLMSLVQSRKAGITSAL ASSTLNNEELKNHVYKKTQLALIYPISTTPHNFVWTATTPTY CYECEGLLWGIARQGMRCETECGVKCEKCDLLNADCLQRAA EKSKHGAEDRTQNIIMVLKDRMKIRERNKPEIFELIQEIFAVTK TAHTQOMKAVKQSVLDGTSKWSAKISITVVCAQGLQAKDKTG SSDPYTVQVGKTKRKTITYGNLNPVWEENFHFECNSSDRI KVRVWDEDDDIKSRVKQRFKRESDDFLGQTIIEVRTLSGEMDV WYNLDKRTDKSAVSGAIRLHISVEIKGEEKVAPYHVQYTCLEH NLFHFVTDVQNNGVVKIPDAKGDDAWKVYDETAQEIIVDEFA MRYGVESIYQAMTHFACLSKYMCPGVPVMSTLLANINAYY AHTTASTNVASDRFAASNFGKERFVKLLDQLHNSLRIDLMSY RNNFPASSPERLQDLKSTVDLLTSITFFRMKVQELQSPPRASQV KDCVKACLNSTYEYIFNNCHELYSREYQTPAKKGEVLPEEQG PSIKNLDWFWSKLITLIVSIIEDKNSYTPCLNQFPQELNVGKISAE VMWNLFQDMKYAMEEHDKHLRCKSADYMNHLFKVKWLYN EYVTELPAPKDRVPEYPAWFEPFVIQWLDENEEVSRDFLHGALE RDKKDGFPQOTSEHALFSCSVVDVFSQLNQSFEEIKKLECPDPQIV GHYMRFAKTISNVLLQYADIIKDFASYCSKEKEKVPICILMNN TQQLRVQLEKMFAMGKELDAEASDILKELQVKLNNVLDLS RVFATSFQPHIEECVKQMGDILSQVKGTVNPASACSSVAQDA DNVLQPIMDLLDNLTLFAKICEKTVLKRVLKELWKLVMNTME KTIIVLPLTDQTMIGNLLRKHGKLEKGRVKLPSHSDGTQMIFN AAKELGQLSKLKDMMVREEAKSLTPKQCAVELALDITIKOYFH AGGVGLKKTFLKSPDLQSLRYALSLYTQATDLLIKTFVQTQSA QGLGVDPVGEVSVHVELFTHPGTGEHKVTVKVVAANDLKWQ TSGIFRPFIEVNIIGPQLSDKKRKFKATSKMNSWAPKYNESFOFT LSADAGPECYELQVCVKDYCFAREDRTVGLAVLQLELAQRGS AACWLPGLRRIHMDDTGLTVLRILSQRSNDEVAKEFVKLKSDT RSAEEGAAPAP	308
Cynomolgus monkey UNC13A nucleotide (XM_045380772.1)	GCCACGTCAGGCGCTGTCCGCGCCCCCGGTGCTGAACCAAG ATGGCCGGTGGCGGCCCGGCCCGGCGTGAGCCAAGCGCGG GCTGCAGCCGGGAGATGCCCCAGCCAGCGGCCGCTGAGCC CGACCCGACAGAGCCGGCCCGGCCGCTCCGGTCCGCCTGC GAGCCCGGAGACATGTCTCTGCTGTGCGTTGGAGAGAAATTC AACACGTACGTGACCTGAAAGTTCAGAAATGTCAAGAGCAC GACCATCGCGGTGCGGGGCAGCCAGCCAGCTGGGAGCAGG ATTTTCATGTTGAGATTAACCGTCTGGATTGGGACTGACAG TGGAGGTGTGGAACAAAGTCTCATCTGGGACACAATGGTG GGCACTGTGTGGATCCCACTGCGGACCATCGCCAGTCCAAT GAGGAGGGCCCTGGAGAGTGGCTGACACTGGACTCCAGGT CATCATGGCAGACAGTGAGATCTGCGGCACCAAGGACCCCA CTTTCCACCGCATCTCTCTGGACACGCGCTTTGAGCTGCCCTT AGACATTCGGAAGAGGAGGCTCGCTACTGGGCCAAGAAGC TGGAGCAGCTCAACGCTATGCGGGACAGGATGAATATTG TTCCAAGATGAGCAAGACAAGCCTCTGCCTGTCCCAGCAAC CAGTGCTGCAACTGGAATTATTTGGCTGGGGTGAGCAGCAC AACGATGACCCGACAGTGAGTGATGATCGGCACAGTGA CTACCGCAGTGAAACGAGCAACAGCATCCCGCCGCCCTATTA CACTACGTCGCAACCAACGCCCTCAGTCCACAGTATTCTGT TCGCCCCACACCCCTGGGCTCCCGGAATCTACAGCGACTC CATGCACAGTTATGAGGAGTTCTCTGAGCCGAGGCCCTCAG	309

TABLE 1-continued

Exemplary Amino Acid and DNA Sequences of UNC13A	
Sequence	SEQ ID NO
CCCACAGGTAGCAGCCGCTACGCCTCCTCAGCGGAGCTGA GCCAGGGAAGCTCTCAGCTGAGTGAGACTTCGACCCCGAC GAGCACAGCCTGCAGGGCTCCGAGATGGAGGACGAGCGGGA CCGGGACTCCTACCACTCCTGCCACAGCTCGGTGAGTATCA CAAAGACTCAGCTCGCTGGGACCAGGATGAGGAGGAGCTGG AGGAGGACCTGGAGGACTTCTGGAGGAGGAGGAGCTGCCT GAAGACGAGGAGGAGCTGGAGGAGGAGGAGGAGGAGCTG ACGATTGGGCGAGCTATGCCAGCGTGAAGAGGTGGCTGTG GCTGAGCCCAAAGACTTCAAACGCATCAGCCTCCCGCCAGCT GCCCCAGGGAAGGAGGACAAGACCCATTGGCACCCACCGA GGCCCCGACACGGCCAAAGTGGCCCCCAAGCCAGCCACAC CCGAGGAGGTGCCTGCAGCTGAGCAGATCCTCGAGGCTGAG CCCCCAAGGCTGAGGAGAGTTTCAAGCCGAGAGAGGATGA GGAAGGCCAGGAGGGGCGAGACTCCATGTCCAGGGCCAAAG CCAACTGGCTGCGTGCTTCAACAAGGTGCGGATGCAGCTGC AGGAGGCCCGGGTGAAGGAGAAATGTCCAATCCTATGG TTCAAAGGCGGCCAGGAGGTGGTCTCATCATCATCGACAGC ATGCCAGACATCCGCAAGAGGAAACCTATCCACTCGTGAG CGACTGGCCATGTCTGTCCAGTCCAGGAAAGCAGGCAT CACCTCGGCCTTGGCCTCCAGCAGCTTGAACAACGAGGAGCT GAAAAACCACTTTACAAGAAGACCTGCAAGCCTTAATCT ACCCATCTCTGTCACGACGCCACACAACCTTCGAAGTGTGA CGGCCACACGCCACCTACTGTACGAGTGCGAGGGGCTG CTGTGGGGCATCGCGAGGCGAGGCATGCGCTGCACGAGTG TGGTGTCAAGTGCCACGAAAGTGCCAGGACCTGCTCAACG CCGATTGCCTGCAGCGGGCTGCGGAGAAGAGCTCCAAGCAC GGGGCGGAGGACCGGACGAGAACATCATATGGTGTCTCAA GGACCGCATGAAGATCCGGGAGCGCAACAAGCCCGAGATCT TCGAGCTCATCCAGGAGATCTTCGCGGTGACCAAGACGGCG CACACGCAACAGATGAAGGCGGTCAAGCAGAGCGTGTGGA CGGCACGTCCAAGTGGTCCGCCAAGATCAGCATCACCGTGGT CTGCGCGCAGGGCTTGCAGGCAAGGACAAGACGGGATCCA GTGACCCCTACGTACCGTCCAGGTGGGAAGACCAAGAAA CGGACAAAACCATCTATGGGAACCTCAACCCGGTGTGGGA GGAGAATTTCCACTTTGAATGTCACAATCCTCCGACCGCAT CAAGGTGCGCGTCTGGGACGAGGATGATGACATCAAATCCC GCGTGAACAGAGGTTCAAGAGGGAATCTGACGATTTCTTG GGGCAGACGATCATTGAGGTGCGGACGCTCAGCGGCAGAT GGACGTGTGGTACAACCTGGACAAGCGAAGTGAACAATCTG CTGTGTGCGGTGCCATCCGGCTCCACATCAGTGTGGAGATCA AAGGCGAGGAGAAGGTGGCCCCATACCATGTCCAGTATACC TGTCTGCATGAGAACCTGTTCCACTTTGTGACCGACGTGCAG AACATGGGGTTTGAAGATCCCGGATGCCAAGGGCGACGA TGCCTGGAAGTTTACTACGATGAGACAGCCAGGAGATTGT GGACGAGTTTGCCTACGCTACGGCGTTGAGTCCATCTACCA AGCCATGACCCACTTTGCCGCTCTCCTCCAAGTATATGTG CCCGGGGTGCCTGCCGTCATGAGCACCCCTACTCGCCAACAT CAATGCCCTACTACGCGCACACCACCGCCTCCACCAAGTGT TGCCTTGACCGCTTCGCCGCTCCAACCTTGGGAAAGAGCG CTTCTGTAAACTCCTGGACAGCTGCATAACTCCCTGCGGAT TGACCTCTCCATGTACCGGAATAACTTCCAGCCAGCAGCCC AGAGAGACTCCAGGACCTCAAATCCACTGTGGACCTTCTCAC CAGCATCACCTTCTTTGCGATGAAGGTACAAGAACTCCAGAG CCCGCCCGAGCCAGCCAGGTGGTAAAGGATTGTGTGAAAG CCTGCCTTAATTCTACCTACGAGTACATCTCAACAACCTGCC ATGAACTCTACAGCCGGGATACAGACAGACCCGGCGAAG AAGGGGAAGTTCCCCAGAGGAACAGGGGCCAGCATCAA GAACCTCGACTTCTGGTCCAAGCTGATTACCTCATAGTGTC CATCATTGAGGAAGACAAGAATCCTTACACTCCTGCTCAA TCAGTTTCCCAGGAGCTGAATGTGGGTAAAATCAGCGCTGA AGTGATGTGGAATCTGTTGCCCAAGACATGAAGTATGCCAT GGAGGAGCAGACAAGCATCGTCTATGCAAGAGTGCCGACT ACATGAACCTCCACTTCAAGGTGAAATGGCTCTACAATGAGT ATGTGACGGAACCTCCGCTTTAAGGACCGCGTGCCTGAGT ACCCTGCATGGTTTGAACCTTCTGTCATCCAGTGGCTGGACG AGAATGAGGAGGTGTCCCGGATTTCTGACCGGTGCCCTGG AGCGAGACAAGAAGGATGGGTTCAGCAGACCTCAGAGCAT GCCCTATTCTCCTGCTCCGTGGTGGACGCTTCTCCCAACTCA ACCAGAGCTTTGAAATCATCAAGAACTTGAGTGTCCCGACC CCCAGATTGTGGGGCACTACATGAGGCGCTTTGCCAAGACCA TCAGTAATGTGCTCCTCCAGTATGCAGACATCATCTCAAAGG ACTTCGCCTCTACTGCTCAAGGAGAAGGAGAAAGTGCCCT GCATCCTCATGAACAACACTCAACAGCTACGAGTTACGCTGG	

TABLE 1-continued

Exemplary Amino Acid and DNA Sequences of UNC13A	
Sequence	SEQ ID NO
AGAAGATGTTTGAAGCCATGGGAGGAAAGGAGCTGGATGCC GAAGCCAGTGACATCTTGAAGGAGCTTCAGGTGAAACTCAA TAACGTCTTGGATGAGCTCAGCCGGGTGTTGCTACCAGCTT CCAGCCGCACATTGAAGAGTGTGTTAAACAGATGGGTGACA TCCTTAGCCAGGTTAAGGGCACAGGCAATGTGCCAGCCAGT GCCTGCAGCAGCGTGGCCAGGACGCTGACAATGTGTTGCA GCCCATCATGACCTGTGGACAGCAACCTGACCCCTCTTTGC CAAAATCTGTGAGAAGACGGTGTGAAGCGAGTGTGAAGG AGCTGTGGAAGCTGGTTATGAACACTATGGAGAAAACCATC GTCTTGCACCCCTCACCGACCAGACGATGATCGGGAACCTC TTGAGAAAACATGGCAAGGGATTAGAAAAGGGCAGGGTGAA ATTGCCAAGCCACTCAGACGGAACCCAGATGATCTTCAATGC AGCCAAGGAGCTGGGTGAGTGTCCAACTCAAGGACCAACA TGGTACGAGAAGAAGCCAGAGCTTGACCCCAAGCAGTGC GCGGTGTTGAGTTGGCCCTGGACACAATCAAGCAATATTTC CATGCGGGTGGCTGGGCCCTCAAGAACCTTCTCGAGAA GAGCCCGGACCTGCAATCCCTGCGCTATGCCCTGTGCTCTA CACACAGGCCACCGACCTGCTCATTAAGACCTTTGTACAGAC GCAATCGGCCCAGGTCCATGGTGAAAAGGTACTAGGTTTA CCCTTAGTGAAGACATTTATCCTGAGAAGGGCTTGGGTGTAG AAGACCTGTGGGTGAAGTCTCCGTCCACGTTGAGCTGTTTA CTCATCCAGGAACCTGGGGAACACAGGTCAAGTGAAAGTG GTGGCTGCCAATGACCTCAAATGGCAGACTTCTGGCATCTTC CGGCCGTTTCATCGAGGTCAACATATTGGGCCCCAGCTCAGC GACAAGAAACGCAAGTTTTCGACCAAATCCAAGAACATAG CTGGGCTCCCAAGTACAATGAGAGCTTCCAGTTTACGCTGAG CGCCGACGCGGGCCCCGAGTGTACGAGCTGCAGGTGTGCG TCAAGGACTACTGCTTCGCGCGCGAGGACCGCACGGTGGGG CTGGCCGTGCTGCAGCTGCGCGAGCTGGCCAGCGCGGGAG TGCCGCTGTGCTGGCTGCCGCTAGGCGCTGCATCCACATGGA CGACACCGGCTCAGGCTCTGCGAATCCTCTCGCAGCGCAG CAACGACGAGGTGGCCAAAGAGTTCGTGAAACTCAAGTCGG ACACGCGCTCAGCCGAGGAGGGTGTGCGCGCTGCGCCT TAGCGCGGGGGGGCGGCCGAGCGGCACTGCGCTGCGCGG AGGGCGCTGGGCGGGGAGGGCTTGCCTCTCGGTGGGACCT CCCGAGGGCGGGGCTCGGGAGACTACACGCCGAGGGTGGG CTGCGCTACGCCCTTGACTGAGCTCTCCCTTTTGGAGAATT AGGAATGGAGGATGCCCGCCCTCTCGGAGGGCCACGCCCA AGGGCGGATGAAGGAAGAAGCCACATCCCAACTTGAGGC CACGCCCCAGCATCTAGGGGGCATTGAGCTGGGATGGG GGAAACCTCGTCCCATGGAGGAGGCCACATCCCGGCGCTC CGGTACTGGGAGGCACAGCCTCCTGTCCCTGGAAAAGCCAT AAGACGGGACCCAGACCCCTGGGACCCAGACCAATTGCCA AGTATGAAATCTCAGCTCCCTCAAGGGGGGCTGGGCA AGGGATAGAGCTTCTGGAGGGCCCTCTAGGTGATCTGGG GACTGGAGGGACAGGATTCTGGTTGGAGGGCCCGGAATA CCGGAGTCCCTTTAGATGTTTGTCAAAAAATAAATGGGGGG AGGGGGGAGGATGGGATTTCAAAAGCACATGCGCTTGGGC GCCCAAACCTTGGGGGCGAGGGGACGCTCTGGGGCACCC ACGCTGCCCTACTTCCCTTTGGGAGTTTGCCTCTCCCTCTCC CCCAACAAACCCAGTCTCATATCATAGATTTCAACACACCC ATTTGACAGATGGCAAACTGAGGCTCAAAGAGCTGCTTGA GACTTGGCCGAGGTTCCAGGTGCCATACCTCTATTCTCTC CCTTAGGCTGTGTGCCCCATGAAAGGGTGGGCTGAGATCGG GATGACCTGACACAGCTCCCTATTGCTGCTAATCCCCCTCG GGGGCTCTCCAAGGGGTGGGAGTTCCAGGCCAAGACCCC TACTTTGCCCTTCTTCTCGGCTGCCAAGCAAGACCCCTGCC CTCAGCCCTTCTCTGCGATTCATGGGGATGCCATGAGG ACCTCCCAACCAAAAGAGAATTTGGGATCCCTGGTCCAG GTTTCTCACCCCTTCTTCTTCCAGAAATTTCCAAATAGG AAAGAACAGAAAGAAACAGAACTCTAGGGGGAGAGAG AGAATGAGAGAGAGAGAAAGAGAGAGAGAGAGAAACCAA ACACAGTGAGAGCTTAGTCTCCAAGATCCTATTATTGATT AAACACCAAGCCACAGGATACCTCAGATGGCCCTCTAGCCT GCTGGAAGCCCTTATCCAGTAAGCAAAGTTGCCGTGACTTG GCTGCAGTTAACCTTGTGCACACAGGACCTTAGGGGAAAGTT CAGTGTGGACCATATTGCTCTGGGATCTGCTTTCTCTGTC TCGTATGGCACGCCCTTTCTGCTGGATTGAGAAAGACAAGA TTTGTGTGTAGGGAGAAATGAAAACGGGGTGAGATGAG TAGCTGGGTTCTGGAGGATGGAACATCAGATGGGGAGGCT TCTGAGGTGAAGAATGAGAGGGAACCACTTACTAGAGAGA AAAGAGCTCCAGGCTGGGAACAGCATGTGCGAAGACAGG AGAGGAGAACTGTTGAAACAAGGAGAAAGTGGCACAGCTG	

TABLE 1-continued

Exemplary Amino Acid and DNA Sequences of UNC13A	
Sequence	SEQ ID NO
GAGCTGAGCCAGCAAGGGGGATCATGAGGAGCCTTGGGGTT GGGGAGATCTGCAGAAGCATCAGACCAGGCAGGACCTTGTA CGCTGTCTTGAGGAGTTTCACTTTTATTTCTAAGACAGTCGGG GAGCTCCAGGAGATGTTTAAGCTGGGAGAGATTGGATTCC AGCCTGCAAAAGCTGTTTTGTGAAGACTAAAACCACTGAGG AGGGGTGGGGTGCTTTGGAGACACTGAAATGGATTATTGG AAAGATTCTGAAGCTGTATTGAAAAGACCTTATACTGTG GACATGGCTATAATCCCAGCATTTTGGGGAGGCTGAGGCTGG TGGATCATTTAAGGTCAGGAGTTTGAAGCAGCCTGGCCAAC GTGGTGAACCTGTCTCTGCTAAAAATACAAAATTAGCTG GGTGTGGTGGTGCATGCCTGTAATCCCAGCTACACAGGAGAC TGAGGTGGGAGAATTGCTTGAACCTGGAGGCAGAGGTGCT AGTGAGCCAGATCACACCACTGCACTCCAGCCTGGGTAAAC AGACCAATACTCCGTTCCCTCCCTCTACCCCCCCCCAAAAA AAAAAAATCCTGCGCTTAGATGAGCTCCATGGCTGCTGAGT ACAGTTGTCCAGTTGCACACTGCCAAGGGTTTGGCATCGC TAAGAAGGCCACGTGCAAACTTAGACATTGAGTGTGTATG TTTATGACCTTGGTGTCCCAACATGTGAATGGCCCAAGTGT TGGAAGAAGTGGCGCACTTCTAATTGCTTGAAGTGTG CGTGTCCCTAAATTCAGACAGATGCAGGTAACCTGGAGGTTT TGAACCAAAGGCTAAAATGCAAAATCTCATAAAGGATTGG GAAGCTGTAGCCAGGGATAAGCTTATGTGACTGTAATATGG GCTGAGGAGCAGATGTGAATTCGGACCATGCCATGCCTGA GGGTGGGGTGGGTGGATGGATGATTCAAGGTTGTAACCC ATAGAGCCCAAAGGGGAAGTTATCTGTGACCTGTGGTGAGG GTGACCTGGAACATTTTGGATGGCTGGAAGAAATGGGGA AATTGAGCTGCCAGATAGTCAGGATATTTCCAAAAGATT CCTTAGGAGTCTTCTGGTCAAGACGTGTGTGTGTGTGTG TGTGTGTTTAGGGTTCCCAAGAAATGGCTCAGGCATGTGAAG GAAACAAGCTTCAGGGAATATTTGTGAATGAGTTTCTCTGC CTCCAGGCTAGAACTGTTTTGCAATTTCCACCCTCTTTCT TTCCCTCCAGAGAACTCCTATTATCCTTTGAAACCCATCAT GAAAATCCCTCTTGGAGAAAACCTCCTTCTTCCCTCAGG ACTTTCAGCCTCCATCCCTCCTCCAGTCCAGCCTGATGCCA TGGGACTGGGGTTCTCTCTTCACTCTGTTTCTCCAGACTG GGGTCTGAGGCTCTCAGGACCCCACTTTACCTAGCACAG GTTGGGCACAGGTGGATGGCAGGAGTCTGTGCTAGTGGA ATTATGTGTTGGGGCAGTGTGAGTGCAGAAATACACATCTGC ATGTGTATCTGTCCATGTCTGTCCATACCAACCTTCCCTTCC ACACGGACCTGGGCACATAGGAGGTGTCTGAGCCCAACACA TGGGACAGAGAGTGGACACGGCTGAGACACGGACAGAGAC AAGACAAGGAGTCCAGGCGGCTGAAAGCCTTTTGAATCAG GAAGTTCCTGGATTGGCAGAACAAAGCCAGAGAGGAGCAG GGCTTTCCCATCGCCACCCAGCAAGTGGCCACAGAGCCAG CCTTGGATGACACCTCCAGGATTCTGAACCTGGACCTCACT TTATGCAAGGAGCTGGCCGAGATTTCATGAATCAGGGAA ACAGCACAGAGGGTTGGCCTGTGGCAGGCGAGGATTAAA GGGGTAACATTGAGGGATGCCCTCAGAGTCAAAGTCCCTGA CCAAGAGGAATAAAGAGTAGAAGGCACAGAGAGGGTGAGA TGACGCCCTGATGAGAACGGAGAGAGACATAGAGGTGAAA TATACAGAGAAAGAGAAATGCAGAGACCAGGGCAGGGGT GTCGGGGAAAGTAAAGAGGGTGTCCCGAGGAAAGAGGCTC TGTCCATTGTCAACAGTCTGTCTCGAATGATTTGCATAAAA TGAGGAGGTGCTGTCCATGCCCAATTCTCTCTCAGGTG CCAGAGCCTGAGACCTCACCATGCCCCATCAGAGATGCAA AAAATAAACACCCAACTAGAAATCCTTGGGACCTCTCTCGG CTGGGATCTTAGAGCCTTTCTGTCCCGACCCCTACCCCATGT GCTGTGATTTTGCAGATGGGGACAACCTGGGGCTCCCGGA ACTCTGCCACCTGGGGAAGTTGGGGGAGGGCGTTGGTCC CGGATCACACCCCGTCTGCTCCCAAGAACTTTCTCTAAGA ATCGTTGAGGACCAAAGTCGTCTTGTGACACGTGTGCTT TTCCTTTGCTTTTATTGTTTTAGAGAAAAATCAAGTTGACT GTGTCAAGTAACACCCACCCCTTACCCCGTCCAGCCATA GTGGCTCTCTGGAGACACAGGTCACAGGCGAGGGTCCCT GATCATCCCTGACCACACAGCCAGGGGTGTTTGACCCCTGTC CACCCCTGTCTCGTGTCTCCCTCAGACCCCAACACGGCCA AGCAGTCCGGGATGCTTCCCTCCACACAACCTCTCTTAGC ATGTGATTGCAGATGTGAAATCAAAGCGTTGTTGTTTTTGT TTTTTTTTGATTTCCACCCCGTGGTCTGGTGTCTGCACAGACG CCTTCATTTCTCTGTAATATGTGACTTGAACAAATGTTTAA CACAAACGAGAAGTGGTCATGAATGCATGGTGTGAGATGT TTTGCACTATCTGACTTTTGGTCTCTGTAATAATATTTTAT TAACAGCAGACATTAATAAAGAAAAACCA	

TABLE 1-continued

Exemplary Amino Acid and DNA Sequences of UNC13A		
	Sequence	SEQ ID NO
Cynomolgus	MSLLCVGKEFNTYVTLKVQNVKSTTIIVRGSQPSWEQDFMFEI	310
monkey	NRLDLGLTVEVWNKGLIWDTMVGTVWIPLRTIQQSNEEGPGE	
UNC13A	WLTLDQVIMADSEICGTDPTFHRILLDTRFELPLDIPEEEARY	
Protein	WAKKLEQLNAMRDQDEYSFQDEQDKPLVPVSNQCCNWNYPFG	
(XP_045236707.1)	WGEQHNDPDSAVDDRDSDYRSETSNSIPPPYTTSQPNASVH	
	QYSVRPPPLGSRSEYSDSMHSYEEFSEFQALSPTGSSRYASSAEL	
	SQSSSLSSEDFDPDEHSLQGSSEMEDERDRDSYHSHSSVSYHK	
	DSPRWQDEELEEEDLEDFLEEEELPEDEEELEEEVEVPDDLGS	
	YAQREEVAVAEPKDFKRISLPPAAPGKEDKTPLAPTEAPDTAKV	
	APKPATPEEVPAEQIPEAEPPKAEESFRPREDEEGQEGQDSMSR	
	AKANWLRAPFNKVRMQLQEQARGEGEMSKSLWFKGGPGGLII	
	DSMPDIRKRKPIPLVSDLAMSLVQSRKAGITSALASSTLNNEELK	
	NHVKKTLQALIYPISCTTPHNFEVWTATPTCYCEGELLWGI	
	ARQGMRCTECGVKCHEKCDLLNADCLQRAAEKSKHGAEDR	
	TQNIIMVLKDRMKIRERNKPEIFELIQEIFAVTKTAHTQQMKAV	
	KQSVLDGTSKWSAKISITVCAQGLQAKDKTGSSDPYVTVQVG	
	KTKKRTKTIYGNLNPVWEENFHFHECHNSSDRIKVRVWDEDDDI	
	KSRVKQRKFRESDDFLGQTIIEVRTLSGEMDVWYNLDKRTDKS	
	AVSGAIRLHISVEIKGEEKVAPYHVQYTCLENLHFHFTVDVQNN	
	GVVKIPDAKGDDAWKVYYDETAQEIIDEFAMRYGVESIYQAM	
	THFACLSKVMCPGVPVAVMSTLLANINAYAHHTASTNVSASD	
	RFAASNFGKERFVKLLDQLHNSLRIDLMSYRNPNFPASSPERLQD	
	LKSTVDLLTSITFFRMKVQELQSPPRASQVVKDCVKACLNSTYE	
	YIFNNCHELYSREYQTPAKKGEVPPPEEQGPSIKNLDWFWSKLIITL	
	IVSIIIEEDKNSYTPCLNQFPQELNVGKISAEVMNLFQAQDMKYA	
	MEEHDKHRLCKSADYMNLFKVKWLYNEYVTELPFKDRVPE	
	YPAWFEPFVIQWLDENEVSRDLFHGALERDKKDGFGQQTSEHA	
	LFSCSVVDVFSQLNQSFELIKKLECPDPQIVGHYMRRAKTISNV	
	LLQYADII SKDFASYCSKEKEKVPCLIMNNTQQLRVQLEKMFPA	
	MGGKELDAEASDILKELQVKLNNVLDLSRVFATSFQPHIEECV	
	KQMGDILSQVKGTVGNVPASACSSVAQDADNVLPIMDLLDSNL	
	TLFAKICEKTVLKRVLKELWKLVMNTMEKTIIVLPPLTDQTMIG	
	NLLRKHKGLKLEKGRVKLPSHSDGTQMI FNAAKELGQLSKLDH	
	MVREEAKSLTPKQCAVVELALDTIKQYFHAGGVGLKKTFLKS	
	PDLQSLRYALSLYTQATDLLIKTFVQTQSAQVHGGKGRFTLSE	
	DIYPEKGLGVEDPVGEVSVHVELFTHPGTGEHKVTVKVVAAND	
	LKWQTSIGIFRPFIEVNIIGPQLSDKKRKFKATSKNNSWAPKYNES	
	FQFTLSADAGPECYELQVCVKDYCFAREDRTVGLAVLQQLRELA	
	QRGSAACWLPLGRRIRHMDTGLTVLRLLSQRSNDEVAKEFVKL	
	KSDTRSAEEGGAAPAP	

[0124] As used herein, “intron 20” of UNC13A refers to the intron between exon 20 and exon 21 of UNC13A pre-mRNA (e.g., as described in Brown et al., Nature. 2022 March; 603(7899): 131-137). Intron 20 of UNC13A is also sometimes known in the art as intron 20-21 (see, for example, Ma et al. TDP-43 represses cryptic exon inclusion in the FTD-ALS gene UNC13A. Nature 603, 124-130 (2022)). In some embodiments, intron 20 of UNC13A corresponds to positions 45511-46798 of an UNC13A gene sequence as set forth in accession numbers: NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009 (human UNC13A). In some embodiments, intron 20 of UNC13A comprises the nucleotide sequence set forth in SEQ ID NO: 311. One skilled in the art would understand that in an UNC13A pre-mRNA, the intron 20 sequence is SEQ ID NO: 311 with each thymine replaced with uracil (see SEQ ID NO: 325).

(SEQ ID NO: 311)
gtgagggtcattgctgcggccctcccatgccacttccactcaccattcct
gcctgccagctcttctctcttctctggccacaccatccactctctctggc

-continued
cctctgagactgccccccatgccattccctttacctggaaaactcctccc
tatccatcaaagtcagattcagggtcacctcctctgggaagcccacctt
ggcctccagggtgactctcactactcatcatcagggtctctctctctatc
cagccctaaccactcaggattgggcccgtttgtgtctgggtatgtctcttc
cagctGCCTGGGTTTCTTGAAAGAACTCTTATCCCCAGGAAGTAGTTTG
TTGAATAAATGCTGGTGAAATGAATGAATGATTGAACAGATGAATGAGTGA
TGAGTAGATAAAAGGATGGATGGAGAGATGGgtgagtagatgtagatgata
gatggatgagttggtgggtagattcgtggctagatggatgtagatggat
ggacagatggatggatatatgattgaactattgaaagtatagatgtagtg
atgggtgaatttgggggtaattgttagatgtagatgagtagatgtagat
gatggatggataacttgatgagtgtagatagatgtagtctgtagatgtag
tgactgggtggatagatgaaatgttgtagtagcagattgaattgtattgg
atgggtaggtaggaagtgtggttagttagttagaaggaagattgtagtaga
taggtgaattgttagtagtagcagatgggttagataggttagatggatggatg

-continued

gatggatggatgtataggcagatggacaaatggatgaatgggtgggtgga
tgaatggaaggatgtgtggtgaactattgcaagtattgataattgggtt
cataatttctgaatatttagatggatggtgtgagtggtggtggacaga
cgaaaaatggatggttgataaattgatgggtggatggatggttggtgt
atgaagaatgaatgattgggtaggtgattaagttgcggatcaatgat
gggatggatgaatggatggatggatggatggtgtggtgaattactgaaag
gttgaagagtggtgggtgaaatttgggttagttagatgggtgggtgtg
tggatggataaagagtagatgaatgaattaatgaataaacaggcagatg
gatgatgaagctgccccagaccctgggacctctgacccccggcgacccc
ttgcactctccatgacactttctctccatggtggcag

(SEQ ID NO: 325)

gugagggucuuugcucggccccucccaugccacuuccacucacacauccu
gccugcccagcucuuuccuuucugggcacaccauccacacucuccuggc
ccucugagacugcccgcgaugccauuccuuuaccuggaaaacuccuccc
uauccaucaaguccagauucaggguaccuccucugggagcccaccuu
ggccuccagguagacucacacacucaucaucagguucuuuccuuauuc
cagcccucaaccacucaggauugggcgguugugucuggguaugucucuc
cagCUGCCUGGGUUUCCUGGAAAGAACUCUUAUCCCGAGGAACUAGUUUG
UGAAUAAAUGCUGUGAAUGAAUGAAUGAUGAACAGAUAAUGAGUGA
UGAGUAGAUAAGGAUGGAUGGAGAGAGUGGugaguacauuggauggau
gauggaugaguuggugguagauucgugguagauggaugauggaugau
ggacagauggauggaauaauagaauaagaaugauagaugauugg
augggugaauuugggguaauuguuagauggauggaugauagaugaau
gauggauggaauaacuugaugaguggaugauagaauugcuggaugau
ugacuggguggaugagaauaauuggaugagcagauaauaguuuug
augggauuggauggaugugguagauuuaagaaggaugauagauaga
uaggugaauuugauagucagaugguagauagguagauggauggaug
gauggauggaugauaaggcagauggacaaauggaugaauugguggggga
ugauggaaggaugugguuagacuaauugcaugauuagaauuuggguu
cauaauuucugaauuuuagauggaugguugaguggcugguaggacaga
cgaaaaauggaugguugauaaaauagauggguggauggaugguugguu
augaagaagaugaauugguuagguagauuagauugcggaucagaau
gggauggaugaauuggauggauggaugaugugguuagaauucgaaag
guuggaagaguggauggguaauuuggguuagauagugggugguugu
ggauggaauaaagaguagaugaauaauaagaauaaacaggcagau
augaugaagcugccccagaccugggaccucugacccccggcgacccu
ugcacuccaagacacuuucucccaugguaggcag

[0125] The term “cryptic exon,” as used herein, refers to intronic sequences that are erroneously included in mature mRNA as a result of a splicing defect. Cryptic exons may introduce frameshifts or stop codons, among other changes

in the resulting mRNA and can arise through single nucleotide polymorphisms (SNPs) in acceptor/donor splice sites or the activity of spliceosome or RNA-binding proteins such as TDP-43. Cryptic exons can be found in patient tissues from people affected by ALS, FTD, Alzheimer’s disease, and LATE. In some embodiments, cryptic exons can be found in patient tissues from people affected by frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuropathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington’s disease, multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen’s encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression. In some embodiments, a cryptic exon disclosed herein is 128 nucleosides in length. In some embodiments, a cryptic exon disclosed herein is 178 nucleosides in length. In some embodiments, a cryptic exon disclosed herein is 431 nucleosides in length.

[0126] In some embodiments, a cryptic exon disclosed herein comprises the sequence set forth in SEQ ID NO: 312 and SEQ ID NO: 326 (see also, nucleosides 304-431 in SEQ ID NO: 311 and SEQ ID NO: 325; capitalized sequence):

(SEQ ID NO: 312)

CTGCCTGGGTTTCCTGGAAGAACTCTTATCCCCAGGAAGTGTGTTG
AATAAATGCTGGTGAATGAATGAATGATTGAACAGATGAATGAGTGATGA
GTAGATAAAAGGATGGATGGAGAGATGG

(SEQ ID NO: 326)

CUGCCUGGGUUUCCUGGAAAGAACUCUUAUCCCGAGGAACUAGUUUGUUG
AAUAAAUGCUGGUGAAUGAAUGAAUGAUGAACAGAUAAUGAGUGAUGA
GUAGAUAAAAGGAUGGAUGGAGAGAUUG

[0127] In some embodiments, a cryptic exon disclosed herein comprises the sequence set forth in SEQ ID NO: 334 and SEQ ID NO: 335 (see also, nucleosides 254-431 in SEQ ID NO: 311 and SEQ ID NO: 325; italicized sequence):

(SEQ ID NO: 334)

CCCTAACCACTCAGGATTGGGCCGTTTGTGTCTGGGTATGTCTCTCCAG
CTGCCTGGGTTTCCTGGAAGAACTCTTATCCCCAGGAAGTGTGTTG
AATAAATGCTGGTGAATGAATGAATGATTGAACAGATGAATGAGTGATGA
GTAGATAAAAGGATGGATGGAGAGATGG

(SEQ ID NO: 335)

CCCUAACCACTCAGGAUUGGGCCGUUGUGUCUGGGUAGUCUCUCCAG
CUGCCUGGGUUUCCUGGAAAGAACUCUUAUCCCGAGGAACUAGUUUGUUG
AAUAAAUGCUGGUGAAUGAAUGAAUGAUGAACAGAUAAUGAGUGAUGA
GUAGAUAAAAGGAUGGAUGGAGAGAUUG

[0128] In some embodiments, a cryptic exon disclosed herein comprises the sequence set forth in SEQ ID NO: 336

and SEQ ID NO: 337 (see also, nucleosides 1-431 in SEQ ID NO: 311 and SEQ ID NO: 325; underlined sequence):

(SEQ ID NO: 336)
 GTGAGGGTCATTGCTCGGCCCTCCCATGCCACTTCCACTCACCATTCTT
 GCCTGCCAGCTCTTCTCTTTCTGGCCACACCATCCACACTCTCTCTGGC
 CCTCTGAGACTGCCCGCCATGCCATTCCCTTTACCTGGAAACTCCTCCC
 TATCCATCAAAGTCCAGATTTCAGGGTCACTTCTCTGGGAAGCCACCTT
 GGCTCCAGGTTGACTCTCACTACTCATCATCAGTTTCTTCTCTATTCT
 CAGCCCTAACCACTCAGGATTGGGCGTTTGTGTCTGGGTATGTCTCTTC
 CAGCTGCCTGGGTTTCTCGAAAGAACTCTTATCCCCAGGAAGTGTG
 TTGAATAAATGCTGGTGAATGAATGAATGATTGAACAGATGAATGAGTGA
 TGAGTAGATAAAAGGATGGATGGAGAGATGG
 (SEQ ID NO: 337)
 GUGAGGGUCAUUGCUCGGCCCCUCCCAUGCCACUCCACCAUCCU
 GCCUGCCCAGCUCUCCUUCUUCUGGCCACACCAUCCACACUCCUGGC
 CCUCUGAGACUGCCCGCCAUCCAUUCCUUAUCCUGGAAACUCCUCC
 UAUCCAUCAAAGUCCAGAUUCAGGGUCCACCUCUCUGGAAGCCACCU
 GGCCUCCAGGUUGACUCUACUACUACUACUAGGUUCUUCUUAUUC
 CAGCCCUAACACUCAGGAUUGGGCGUUGUGUCUGGGUAUGUCUUC
 CAGCUGCCUGGUUUCUGGAAAGAACUUAUCCCAAGGAACUAGUUUG
 UUGAAUAAAUGCUGGUGAAUGAAUUAUUAUUAACAGAUAAUGAGUGA
 UGAGUAGAUAAGGAUGGAUGGAGAGAUGG

[0129] As used herein, the term “UNC13A-associated disease,” is a disease or disorder that is caused by, or associated with, abnormal UNC13A expression and/or activity. The term “UNC13A-associated disease” includes a disease, disorder or condition that would benefit from inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell. In some embodiments, a subject having an UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) may benefit from inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell in the subject. In some embodiments, an UNC13A-associated disease is a motor neuron disease that is characterized by neuronal loss. An UNC13A-associated disease may be neurodegenerative disease. In some embodiments, an UNC13A-associated disease is ALS. In some embodiments, an UNC13A-associated disease is FTD. In some embodiments, an UNC13A-associated disease is Alzheimer’s disease. In some embodiments, an UNC13A-associated disease is LATE. In some embodiments, an UNC13A-associated disease is frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington’s disease,

multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen’s encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression.

[0130] Further details regarding signs and symptoms of the various diseases or conditions are provided herein and are well known in the art.

Compositions

UNC13A Antisense Oligonucleotides

[0131] Some aspects of the present disclosure provide antisense oligonucleotides that target UNC13A and/or modulates UNC13A splicing and/or expression. In some embodiments, an antisense oligonucleotide disclosed herein comprises sequences complementary to an UNC13A sequence (e.g., UNC13A gene sequence or pre-mRNA sequence). In some embodiments, an antisense oligonucleotide disclosed herein targets more than one (e.g., two or more) target regions of UNC13A (e.g., UNC13A gene or pre-mRNA). In some embodiments, an antisense oligonucleotide disclosed herein is designed for splicing modulation of an UNC13A pre-mRNA (e.g., to inhibit the inclusion of a cryptic exon present in UNC13A pre-mRNA). In some embodiments, an antisense oligonucleotide disclosed herein is designed for modulation (e.g., restoring) of UNC13A gene expression (e.g., RNA expression and/or protein expression) and/or function.

[0132] In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more regions of complementarity to one or more (e.g., one, two, or more) target regions in a human UNC13A sequence (e.g., as set forth in SEQ ID NO: 318). In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more regions of complementarity to one or more (e.g., one, two, or more) target regions in an intron (e.g., intron 20 such as the intron 20 as set forth in SEQ ID NO: 311) of a human UNC13A sequence (e.g., as set forth in SEQ ID NO: 318). In some embodiments, an antisense oligonucleotide disclosed herein comprises regions of complementarity to one or more (e.g., one, two, or more) target regions in intron 20 (e.g., the intron as set forth in SEQ ID NO: 311) of an UNC13A sequence (e.g., as set forth in SEQ ID NO: 318), wherein each target region each comprises an UNC13A sequence that modulates splicing of an UNC13A cryptic exon (e.g., an UNC13A cryptic exon present in intron 20 such as the cryptic exon as set forth in SEQ ID NO: 312, SEQ ID NO: 334, or SEQ ID NO: 336). In some embodiments, an antisense oligonucleotide described herein inhibits inclusion of an UNC13A cryptic exon (e.g., an UNC13A cryptic exon present in intron 20) into an UNC13A mature mRNA.

[0133] In some embodiments, an antisense oligonucleotide disclosed herein targets one or more (e.g., one, two, or more) target regions in intron 20 (e.g., the intron 20 as set forth in SEQ ID NO: 311) of an UNC13A sequence in or near a cryptic exon (e.g., the cryptic exon as set forth in SEQ ID NO: 312, SEQ ID NO: 334, or SEQ ID NO: 336). “Near” a position or a sequence, as used herein, means within 125 (e.g., 125, 120, 115, 110, 105, 100, 95, 90, 85, 80, 75, 70, 65, 60, 55, 50, 45, 40, 35, 30, 25, 20, 15, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1) nucleosides upstream (5’ end) or downstream (3’ end) of the position or sequence. “Upstream” of a position

or a sequence, as used herein, means at the 5' side of the position or sequence. "Downstream" of a position or sequence, as used herein, means at the 3' side of the position or sequence. For example, a target region "near" a cryptic exon in intron 20 of UNC13A sequence means the target region is within 125 (e.g., 125, 120, 115, 110, 105, 100, 95, 90, 85, 80, 75, 70, 65, 60, 55, 50, 45, 40, 35, 30, 25, 20, 15, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1) nucleosides upstream (e.g., counting towards the 5' end from the 5' terminal nucleotide of the cryptic exon) or downstream (e.g., counting towards the 3' end from the 3' terminal nucleotide of the cryptic exon) of the cryptic exon in intron 20 of UNC13A sequence.

[0134] As used herein, a "target region" is represented by a nucleotide sequence in a reference sequence. For example, for target regions 1-4 provided in Table 2, the reference sequence used is human UNC13A sequence (SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009). For the purposes of the present disclosure, when referring to antisense oligonucleotides that target a target region in a reference sequence (e.g., in human UNC13A sequence as set forth in NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009; SEQ ID NO: 318), such antisense oligonucleotides encompass antisense oligonucleotides that target the target region in the reference sequence (SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009), and antisense oligonucleotides that target a corresponding target region in another UNC13A sequence (e.g., gene sequence, pre-mRNA sequence, or a variant or a homologue of human UNC13A, e.g., from a closely related species, such as a cynomolgus monkey).

[0135] To identify a target region in another UNC13A sequence that corresponds to a certain target region in the reference sequence (e.g., SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009), such other UNC13A sequences can be aligned to the reference sequence and the corresponding target region can be determined. For example, in some embodiments, a target region in human UNC13A comprises a nucleotide sequence as set forth in SEQ ID NO: 327 (CTGGGTATGTCTCTTCCAGCTGCCTGGGTTTCCTGGAAAGAACTCTTATCCCCAGGA ACTAGTTTGTGAATAAATGCTGGTGAATGAATGA). Target region 1 corresponds to nucleosides at positions 45795-45886 of SEQ ID NO: 318 (human UNC13A; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009). Target region 1 of SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009; upper sequence) is aligned to the corresponding target region of the cynomolgus UNC13A sequence (nucleosides at positions 50846-50936 of cynomolgus UNC13A SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement; lower sequence)) as follows:

```
CTGGGTATGTCTCTTCCAGCTGCCTGGGTTTCCTGGAAAGAACTCTTATCCCCAGGAAGTCTAGTT
|||||
CTGGGTATGTCTCTTCCAGCTGCCTGGGTTTCCTGGAAAGAACTG-TATCCCCAGGAAGTCTAGTT
```

TGTTGAATAAATGCTGGTGAATGAATGA (SEQ ID NO: 327)

|||||

TGTTGAATGAATGCTGGTGAATGAATGA (SEQ ID NO: 330)

[0136] As such, positions 50846-50936 of cynomolgus UNC13A, SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement) is the corresponding target region 1. When referring to antisense oligonucleotides targeting target region 1 of an UNC13A sequence, the present disclosure encompasses antisense oligonucleotides targeting target region 1 of NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009 (SEQ ID NO: 318) and antisense oligonucleotides that target the corresponding target region 1 of NC_052273.1: 17161646 . . . 17254007 complement (SEQ ID NO: 319).

[0137] In determining a corresponding target region of another sequence in relation to a reference sequence, the other sequence and the reference sequence are optimally aligned over the window of comparison, which comprises sufficient number of nucleosides for the alignment. Optimal alignment of sequences for aligning a comparison window may be conducted by computerized implementations of algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package Release 7.0, Genetics Computer Group, 575 Science Drive Madison, Wis., USA) or by inspection and the best alignment (i.e., resulting in the highest percentage homology over the comparison window) generated by any of the various methods selected. Reference also may be made to the BLAST family of programs as for example disclosed by Altschul et al., Nucl. Acids Res. 25:3389, 1997.

[0138] One or more insertions in another UNC13A sequence from a closely related species, such as a cynomolgus monkey, relative to the human reference sequence (SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009) are represented as position "N-M," wherein N represents the corresponding position immediately before the insertions in the reference sequence, and M represents the position of the nucleoside within the one or more insertions. For example, in the following alignment between an area of the sequence set forth in SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009; upper sequence) and the sequence set forth in SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement; lower sequence),

```
TGA----TGGGTG
|||      |||||
TGATGGGTGGGTG (SEQ ID NO: 331)
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[0139] The insertions TGGG in SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement) follows the position 46487 (bolded A in the alignment above) in SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009). Accordingly, the positions of insertions TGGG are designated as 46487-1, 46487-2, 46487-3, and 46487-4, respectively.

[0140] One or more deletions or mismatches in another UNC13A sequence from a closely related species, such as a cynomolgus monkey, relative to the human reference sequence (SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009) may result in the lack of a “corresponding position.” In these instances, as long as the deletions/mismatches (e.g., no more than 5, 4, 3, 2, 1 mismatches) are within a stretch of sequences of sufficient length (e.g., at least 20, 25, 30, 35, or 40 nucleosides) that aligned between the two sequences, the mismatched positions are represented as the position in the reference sequence that directly aligned even if the nucleoside in the reference is different. The position for the deletion in the other sequence is “skipped.”

[0141] For example, the nucleosides at positions 46024-46036 of the human UNC13A sequence SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009; upper sequence) are aligned to the nucleoside 51070-51082 of the cynomolgus UNC13A sequence SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement; lower sequence) as follows:

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GATATATGATTGA (SEQ ID NO: 332)
||||| ||| |||
GATATGTGACTGA (SEQ ID NO: 333)
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[0142] The bolded A at position 46029 of SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009) aligns with the bolded G at position 51075 of SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement). When referring to antisense oligonucleotides that target an UNC13A sequence at a target region comprising the nucleoside at position 46029, the present disclosure encompasses antisense oligonucleotides that target a target region comprising the nucleoside at position 46029 of NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009 (SEQ ID NO: 318) and antisense oligonucleotides that target a target region comprising the nucleoside at position 51075 of NC_052273.1: 17161646 . . . 17254007 complement (SEQ ID NO: 319).

[0143] In some embodiments, an antisense oligonucleotide provided herein comprising a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region, wherein the first target region and/or the second target region each comprises an UNC13A sequence. In some embodiments, each of the first antisense sequence and the second antisense sequence comprises a region of complementary to an UNC13A sequence as set forth in SEQ ID NO: 318. In some embodiments, the region of complementary consists of at least 6 (e.g., at least 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length. In some embodiments, the region of complementary consists of at least 7 (e.g., at least 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length. In some embodiments, the region of complementary consists of at least 8 (e.g., at least 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length.

[0144] In some embodiments, each of the first target region and the second target region is in intron 20 of an

UNC13A sequence (e.g., gene sequence or pre-mRNA sequence). In some embodiments, each of the first antisense sequence and the second antisense sequence comprises a region of complementary to intron 20 of an UNC13A sequence, wherein the intron 20 is as set forth in SEQ ID NO: 311 or SEQ ID NO: 325. In some embodiments, the region of complementary consists of at least 6 (e.g., at least 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length. In some embodiments, the region of complementary consists of at least 7 (e.g., at least 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length. In some embodiments, the region of complementary consists of at least 8 (e.g., at least 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length.

[0145] In some embodiments, each of the first target region and the second target region is near a cryptic exon in intron 20 of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence). In some embodiments, each of the first target region and the second target region comprises an UNC13A sequence, wherein targeting such sequences modulates splicing of an UNC13A cryptic exon (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA). In some embodiments, targeting the UNC13A sequences using an antisense oligonucleotide described herein that modulate splicing of an UNC13A cryptic exon inhibits inclusion of the cryptic exon (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA) in a mature UNC13A mRNA.

[0146] In some embodiments, the first target region and second target region are adjacent (e.g., within 300 nucleobases) to each other. In some embodiments, the first target region is downstream of the second target region. In some embodiments, the first target region is more than one nucleotide (2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, 50, etc.) downstream of the second target region. In some embodiments, the first target region and the second target region are both in the cryptic exon (e.g., the cryptic exon in intron 20 of an UNC13A pre-mRNA). In some embodiments, the first target region is downstream of the cryptic exon (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA) and the second target region is in the cryptic exon (e.g., a cryptic exon in intron 20). In some embodiments, the first target region and the second target region are both downstream of the cryptic exon (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA). In some embodiments, the 5' end of the first target region and 3' end of the second target region have 1-300 nucleobases (e.g., 1-300, 3-300, 5-300, 7-300, 10-300, 1-250, 3-250, 5-250, 7-250, 10-250, 1-200, 3-200, 5-200, 7-200, 10-200, 1-150, 3-150, 5-150, 7-150, or 10-150 nucleobases) in between each other.

[0147] Non-limiting examples of target regions in an UNC13A pre-mRNA that may be targeted by an antisense oligonucleotide described herein are provided in Table 2. Such target regions are near the cryptic exon in intron 20 of an UNC13A sequence. One of ordinary skill in the art would appreciate that the target regions overlap and are not discrete regions. In some embodiments, antisense oligonucleotides disclosed herein that comprise antisense sequences targeting target regions 1 and 3 are particularly effective in skipping the cryptic exon, inhibiting inclusion of the UNC13A cryptic exon into an UNC13A mature mRNA.

TABLE 2

Exemplary target regions in intron 20 of UNC13A pre-mRNA		
Target regions	Sequence	SEQ ID NO:
Target region 1	CUGGGU AUGUCU UCCAGCUGCCUGGG UUUCCUGGAAAGAACUCUUAUCCCCAGG AACUAGUUUGUUGAAUAAUUGCUGGUG AAUGAAUGA	313
Target region 2	GAAUAAUUGCUGGUGAAUGAAUGAAUG AUUGAACAGAUAGAAUGAGUGAUGAGUA GAUAAAGGAUGGAUGGAGAGAUGGGU GAGUACAUGGAUGGA	314
Target region 3	GAGAUGGGUGAGUACAUGGAUGGAUAG AUGGAUGAGUUGGUGGUGAGAUUCGUG GCUGAUGGAUGGAUGGAUGGAUGGACA GAUGGAUGGAU	315
Target region 4	AUGGAUGGAUGGACAGAUUGGAUGGAUA UAUGAUUGAACUAUUGAAAGUAUA	316

†It is to be understood that if the target sequence is a gene sequence (DNA), each uracil base (U) in any one of the target region sequences provided in Table 2 may be replaced with a thymine base (T).

[0148] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of any one of SEQ ID NO: 313-316. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of in SEQ ID NO: 313 or SEQ ID NO: 315. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of any one of SEQ ID NO: 313-316. In some embodiments, the second antisense sequence targets the second target region comprising the nucleotide sequence of SEQ ID NO: 313 or SEQ ID NO: 315.

[0149] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence targets the first target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the first antisense sequence targets a first target region compris-

ing the nucleotide sequence of SEQ ID NO: 316, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 316, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 316, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 316, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 315, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 315, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 315, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 314, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 314, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 313, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313.

[0150] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide

sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 316.

[0151] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target

[illegible]

prises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 313.

[0160] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 313 and the second antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 313.

[0161] Non-limiting examples of the nucleobase sequences of an antisense oligonucleotide described herein, the first antisense sequence and/or the second antisense sequence of an antisense oligonucleotide described herein, are provided in Table 3. Non-limiting examples of the nucleobase sequences of an antisense oligonucleotide described herein, the first antisense sequence and/or the second antisense sequence of an antisense oligonucleotide described herein, are provided in Table 9.

[0162] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 191-245. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 246, 247, 274, 288, and 495-515. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 246, 247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14)

contiguous nucleobases of any one of SEQ ID NOS: 191-245. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 246, 247, 274, 288, and 495-515. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOS: 191-245. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOS: 246, 247, 274, 288, and 495-515. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC. In some embodiments, the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 516-535. In some embodiments, the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TGTACTC, and TCACCCA. In some embodiments, the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 516-535. In some embodiments, the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOS: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOS: 516-535. In some embodiments, the second antisense sequence comprises the nucleobase sequence of any one of CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TGTACTC, and TCACCCA.

[0163] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least

9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-245 and the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-245 and the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 191-245 and the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306.

[0164] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, and 495-515 and the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, and 516-535. In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC and the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TGTACTC, and TCACCCA. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous

nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, and 495-515 and the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, and 516-535. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC and the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTAT, TGTACTC, and TCACCCA.

[0165] In some embodiments, in any one of the antisense oligonucleotides described herein, the first antisense sequence and second antisense sequence are immediately adjacent (e.g., having no nucleobases between the first antisense sequence and second antisense sequence) to each other. In some embodiments, in any one of the antisense oligonucleotides described herein, one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) are between the first antisense sequence and second antisense sequence. In some embodiments, the present disclosure provides antisense oligonucleotides in which one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) at the 3' end of the first antisense sequence is complementary to one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) at the 3' end of the second target region, and/or one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) at the 5' end of the second antisense sequence is complementary to one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) at the 5' end of the first target region.

[0166] In some embodiments, any one of the antisense oligonucleotides described herein comprises a spacer between the first antisense sequence and second antisense sequence. In some embodiments, a spacer comprises one or more (1, 2, 3, 4, 5, 6, or more) abasic moieties. In some embodiments, the spacer is a C3 spacer (e.g., a C3 spacer as shown in Table 16). In some embodiments, the spacer is a C6 spacer (e.g., a C5 spacer as shown in Table 16).

[0167] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 194 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0168] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 194 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3' end of the first antisense sequence is linked to the 5' end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

35) and comprises at least 16 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) contiguous nucleobases of any one of SEQ ID NOs: 338-494. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises at least 16 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) contiguous nucleobases of any one of SEQ ID NOs: 1-190. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises at least 16 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) contiguous nucleobases of any one of SEQ ID NOs: 338-494. In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, or 35) and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 1-190. In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, or 35) and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 338-494. In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, or 35) and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 1-182, 184-190, 338-434, 442-494. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 338-494. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 1-182, 184-190, 338-434, 442-494.

[0185] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence, wherein the first antisense sequence comprises a nucleobase sequence selected from (5'→3'):

(i) (SEQ ID NO: 191)
CACGAATCTACCC

(ii) (SEQ ID NO: 192)
CGAATCTACCCAC

-continued

(iii) (SEQ ID NO: 198)
CCATCCATCATCC

(iv) (SEQ ID NO: 199)
GCATTTATTCAAC

(v) (SEQ ID NO: 193)
CCACGAATCTACC

(vi) (SEQ ID NO: 194)
TGTTCATCATTC

(vii) (SEQ ID NO: 204)
CGAATCTACCCA

(viii) (SEQ ID NO: 212)
CGAATCTACC

(ix) (SEQ ID NO: 196)
ACGAATCTACCCA

(x) (SEQ ID NO: 205)
ACGAATCTACCC

(xi) (SEQ ID NO: 222)
TCCATCCATCATC
and

(xii) (SEQ ID NO: 233)
CATCCATCTATCC.

[0186] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence, wherein the second antisense sequence that comprises a nucleobase sequence selected from (5'→3'):

(i) (SEQ ID NO: 246)
CCATCCATGTACT

(ii) (SEQ ID NO: 247)
ATCTGTTCAATCA

(iii) (SEQ ID NO: 253)
GGGATAAGAGTTC

(iv) (SEQ ID NO: 254)
GGGATAAGAGTT

(v) (SEQ ID NO: 252)
GGATAAGAGTTCT

(vi) (SEQ ID NO: 261)
ATCTGTTCAATC

(vii) (SEQ ID NO: 271)
ATCCATGTAC

-continued
 (viii) (SEQ ID NO: 276)
 CATCTGTTCAATC
 and
 (ix) (SEQ ID NO: 262)
 CCATCCATGTAC.

[0187] In some embodiments, an antisense oligonucleotide disclosed herein comprising a first antisense sequence at the 5' end of the antisense oligonucleotide and a second antisense sequence at the 3' end of the antisense oligonucleotide, wherein the first antisense sequence comprises a nucleobase sequence selected from (5'->3'):

(i) (SEQ ID NO: 191)
 CACGAATCTACCC
 (ii) (SEQ ID NO: 192)
 CGAATCTACCCAC
 (iii) (SEQ ID NO: 198)
 CCATCCATCATCC
 (iv) (SEQ ID NO: 199)
 GCATTATTCAAC
 (v) (SEQ ID NO: 193)
 CCACGAATCTACC
 (vi) (SEQ ID NO: 194)
 TGTTCATCATTC
 (vii) (SEQ ID NO: 204)
 CGAATCTACCCA
 (viii) (SEQ ID NO: 212)
 CGAATCTACC
 (ix) (SEQ ID NO: 196)
 ACGAATCTACCCA
 (x) (SEQ ID NO: 205)
 ACGAATCTACCC
 (xi) (SEQ ID NO: 222)
 TCCATCCATCATC
 and
 (xii) (SEQ ID NO: 233)
 CATCCATCTATCC;

wherein the second antisense sequence comprises a nucleobase sequence selected from (5'→3'):

(i) (SEQ ID NO: 245)
 CCATCCATGTACT

-continued
 (ii) (SEQ ID NO: 247)
 ATCTGTTCAATCA
 (iii) (SEQ ID NO: 253)
 GGGATAAGAGTTC
 (iv) (SEQ ID NO: 254)
 GGGGATAAGAGTT
 (v) (SEQ ID NO: 252)
 GGATAAGAGTTCT
 (vi) (SEQ ID NO: 261)
 ATCTGTTCAATC
 (vii) (SEQ ID NO: 271)
 ATCCATGTAC
 (viii) (SEQ ID NO: 276)
 CATCTGTTCAATC
 and
 (ix) (SEQ ID NO: 262)
 CCATCCATGTAC.

[0188] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence selected from (5'→3'):

(i) (SEQ ID NO: 1)
 CACGAATCTACCCCATCCATGTACT
 (ii) (SEQ ID NO: 2)
 CGAATCTACCCACATCTGTTCAATCA
 (iii) (SEQ ID NO: 16)
 CCATCCATCATCCGGGATAAGAGTTC
 (iv) (SEQ ID NO: 17)
 GCATTATTCAACGGGATAAGAGTT
 (v) (SEQ ID NO: 8)
 CCACGAATCTACCGGATAAGAGTTCT
 (vi) (SEQ ID NO: 9)
 TGTTCATCATTCGGGATAAGAGTTC
 (vii) (SEQ ID NO: 14)
 CCACGAATCTACCGGATAAGAGTTC
 (viii) (SEQ ID NO: 36)
 CGAATCTACCCAATCTGTTCAATC
 (ix) (SEQ ID NO: 82)
 CGAATCTACCATCCATGTAC
 (x) (SEQ ID NO: 93)
 ACGAATCTACCCACATCTGTTCAATC

-continued

(xi) (SEQ ID NO: 37)
ACGAATCTACCCCATCCATGTAC

(xii) (SEQ ID NO: 121)
TCCATCCATCATCGGGATAAGAGTTC
and

(xiii) (SEQ ID NO: 159)
CATCCATCTATCCGGGATAAGAGTTC

[0189] It is to be understood that, for the purposes of the present disclosure, an antisense oligonucleotide comprising nucleobase sequences (e.g., nucleobase sequences of the antisense oligonucleotide, first antisense sequence, and second antisense sequence) of the antisense oligonucleotides listed in Table 3 encompasses antisense oligonucleotides comprising such nucleobase sequences and comprising no chemical modifications (e.g., modified nucleosides and/or modified internucleoside linkages), and an antisense oligonucleotides comprising such nucleobase sequences and

comprising chemical modifications (e.g., one or more modified nucleosides and/or one or more modified internucleoside linkages; e.g., those provided in Table 5), and encompasses such modified or unmodified antisense oligonucleotides unconjugated or conjugated to a targeting moiety.

[0190] It is to be understood that, for the purposes of the present disclosure, an antisense oligonucleotide comprising nucleobase sequences (e.g., nucleobase sequences of the antisense oligonucleotide, first antisense sequence, and second antisense sequence) of the antisense oligonucleotides listed in Table 9 encompasses antisense oligonucleotides comprising such nucleobase sequences and comprising no chemical modifications (e.g., modified nucleosides and/or modified internucleoside linkages), and an antisense oligonucleotides comprising such nucleobase sequences and comprising chemical modifications (e.g., one or more modified nucleosides and/or one or more modified internucleoside linkages; e.g., those provided in Table 11), and encompasses such modified or unmodified antisense oligonucleotides unconjugated or conjugated to a targeting moiety.

TABLE 3

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2	SEQ ID NO:
1	CACGAATCTACCCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT	246
2	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
3	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC	248
4	CGAATCTACCCACGAGTTCTTTCCAG	4	CGAATCTACCCAC	192	GAGTTCTTTCCAG	249
5	CGAATCTACCCACTAAGAGTTCTTTC	5	CGAATCTACCCAC	192	TAAGAGTTCTTTC	250
6	CCACGAATCTACCGATAAGAGTTCTT	6	CCACGAATCTACC	193	GATAAGAGTTCTT	251
7	CACGAATCTACCCGATAAGAGTTCT	7	CACGAATCTACCC	191	GGATAAGAGTTCT	252
8	CCACGAATCTACCGGATAAGAGTTCT	8	CCACGAATCTACC	193	GGATAAGAGTTCT	252
9	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
10	GAATCTACCCACCGGATAAGAGTTC	10	GAATCTACCCACC	195	GGGATAAGAGTTC	253
11	CGAATCTACCCACGGGATAAGAGTTC	11	CGAATCTACCCAC	192	GGGATAAGAGTTC	253
12	ACGAATCTACCCAGGGATAAGAGTTC	12	ACGAATCTACCCA	196	GGGATAAGAGTTC	253
13	CACGAATCTACCCGGGATAAGAGTTC	13	CACGAATCTACCC	191	GGGATAAGAGTTC	253
14	CCACGAATCTACCGGATAAGAGTTC	14	CCACGAATCTACC	193	GGGATAAGAGTTC	253
15	CTAGCCACGAATCGGGATAAGAGTTC	15	CTAGCCACGAATC	197	GGGATAAGAGTTC	253
16	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC	253
17	GCATTTATTCAACGGGGATAAGAGTT	17	GCATTTATTCAAC	199	GGGGATAAGAGTT	254
18	CCACCAACTCATCGGGATAAGAGTT	18	CCACCAACTCATC	200	GGGGATAAGAGTT	254
19	GAATCTACCCACCGGGATAAGAGTT	19	GAATCTACCCACC	195	GGGGATAAGAGTT	254
20	CGAATCTACCCACGGGGATAAGAGTT	20	CGAATCTACCCAC	192	GGGGATAAGAGTT	254
21	CACGAATCTACCCGGGGATAAGAGTT	21	CACGAATCTACCC	191	GGGGATAAGAGTT	254
22	CCACGAATCTACCGGGATAAGAGTT	22	CCACGAATCTACC	193	GGGGATAAGAGTT	254

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein					
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2
					SEQ ID NO:
23	GCCACGAATCTACGGGATAAGAGTT	23	GCCACGAATCTAC	201	GGGATAAGAGTT
24	CGAATCTACCCACCCTGGGGATAAGA	24	CGAATCTACCCAC	192	CCTGGGGATAAGA
25	ACGAATCTACCCATCTCGGGATAAG	25	ACGAATCTACCCA	196	TCCTGGGGATAAG
26	CGAATCTACCCACGTTCTCGGGATA	26	CGAATCTACCCAC	192	GTTCTCGGGATA
27	CACGAATCTACCCGTTCTCGGGATA	27	CACGAATCTACCC	191	GTTCTCGGGATA
28	CTAGCCACGAATCGTTCTCGGGATA	28	CTAGCCACGAATC	197	GTTCTCGGGATA
29	CGAATCTACCCACTAGTTCCTGGGGA	29	CGAATCTACCCAC	192	TAGTTCCTGGGGA
30	CTAGCCACGAATCTAGTTCCTGGGGA	30	CTAGCCACGAATC	197	TAGTTCCTGGGGA
31	ACGAATCTACCCACTAGTTCCTGGGG	31	ACGAATCTACCCA	196	CTAGTTCCTGGGG
32	ATCTAGCCACGAATCTAGTTCCTGGGG	32	ATCTAGCCACGAA	202	CTAGTTCCTGGGG
33	GAATCTACCCACTCTGTTCAATCA	33	GAATCTACCCAC	203	TCTGTTCAATCA
34	CGAATCTACCCATCTGTTCAATCA	34	CGAATCTACCCA	204	TCTGTTCAATCA
35	GAATCTACCCACATCTGTTCAATC	35	GAATCTACCCAC	203	ATCTGTTCAATC
36	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
37	GAATCTACCCACTCTGTTCAATCA	33	GAATCTACCCAC	203	TCTGTTCAATCA
38	CGAATCTACCCATCTGTTCAATCA	34	CGAATCTACCCA	204	TCTGTTCAATCA
39	GAATCTACCCACATCTGTTCAATC	35	GAATCTACCCAC	203	ATCTGTTCAATC
40	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
41	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
42	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC
43	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
44	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
45	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
46	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC
47	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
48	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
49	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
50	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC
51	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
52	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
53	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
54	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC
55	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
56	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
57	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
58	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein					
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2
					SEQ ID NO:
59	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
60	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
61	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
62	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC
63	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
64	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
65	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
66	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC
67	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
68	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
69	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
70	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC
71	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
72	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
73	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
74	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC
75	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
76	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC
77	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
78	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
79	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
80	ACGAATCTACCCCATCCATGTAC	37	ACGAATCTACCC	205	CCATCCATGTAC
81	CACGAATCTACCCCATCCATGTACT	38	CACGAATCTACC	206	CCATCCATGTACT
82	ACGAATCTACCCCATCCATGTACT	39	ACGAATCTACCC	205	CCATCCATGTACT
83	CACGAATCTACCCCATCCATGTAC	40	CACGAATCTACCC	191	CCATCCATGTAC
84	CACGAATCTACCCCTCCATCCATGTACT	41	CACGAATCTACCC	191	TCCATCCATGTACT
85	CACGAATCTACCCACCATCCATGTACT	42	CACGAATCTACCCA	207	CCATCCATGTACT
86	ACGAATCTACCCCATCCATGTACT	43	ACGAATCTACCC	205	CATCCATGTACT
87	CACGAATCTACCCATCCATGTACTC	44	CACGAATCTACC	206	CATCCATGTACTC
88	ACGAATCTACCCCATCCATGTACTC	45	ACGAATCTACCC	205	CATCCATGTACTC
89	CACGAATCTACCCCATCCATGTACTCA	46	CACGAATCTACCC	191	CATCCATGTACTCA
90	CCATCCATCATCCGGGATAAGAGTT	47	CCATCCATCATC	208	GGGATAAGAGTT
91	CCATCCATCATCCGGGATAAGAGTTC	48	CCATCCATCATC	208	GGATAAGAGTTC
92	CATCCATCATCCGGGATAAGAGTT	49	CATCCATCATCC	209	GGGATAAGAGTT
93	CATCCATCATCCGGGATAAGAGTTC	50	CATCCATCATCC	209	GGATAAGAGTTC
94	CCATCCATCATCCGGGATAAGAGTTC	51	CCATCCATCATC	208	GGGATAAGAGTTC

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein					
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2
					SEQ ID NO:
95	CATCCATCATCCGGGATAAGAGTTC	52	CATCCATCATCC	209	GGGATAAGAGTTC
96	CCATCCATCATCCGGGATAAGAGTT	53	CCATCCATCATCC	198	GGGATAAGAGTT
97	CCATCCATCATCCGGGATAAGAGTTC	54	CCATCCATCATCC	198	GGATAAGAGTTC
98	CCATCCATCATCCGGGGATAAGAGTTC	55	CCATCCATCATCC	198	GGGGATAAGAGTTC
99	CCATCCATCATCCGGGATAAGAGTTCT	56	CCATCCATCATCC	198	GGGATAAGAGTTCT
100	TCCATCCATCATCCGGGATAAGAGTTC	57	TCCATCCATCATCC	210	GGGATAAGAGTTC
101	GTTCAATCATTCGGGATAAGAGTT	58	GTTCAATCATTC	211	GGGATAAGAGTT
102	GTTCAATCATTCGGGATAAGAGTTC	59	GTTCAATCATTC	211	GGATAAGAGTTC
103	GTTCAATCATTCGGGATAAGAGTTC	60	GTTCAATCATTC	211	GGGATAAGAGTTC
104	TGTTCAATCATTCGGGATAAGAGTTC	61	TGTTCAATCATTC	194	GGATAAGAGTTC
105	TGTTCAATCATTCGGGATAAGAGTTCT	62	TGTTCAATCATTC	194	GGGATAAGAGTTCT
106	GAATCTACCCACATCTGTTCAATCA	63	GAATCTACCCAC	203	ATCTGTTCAATCA
107	CGAATCTACCCACATCTGTTCAATC	64	CGAATCTACCCAC	192	ATCTGTTCAATC
108	CGAATCTACCCACCATCTGTTCAATCA	65	CGAATCTACCCAC	192	CATCTGTTCAATCA
109	CACGAATCTACCCCCCATCCATGTACT	66	CACGAATCTACCC	191	CCATCCATGTACT
110	CACGAATCTACCCGCCATCCATGTACT	67	CACGAATCTACCC	191	CCATCCATGTACT
111	CACGAATCTACCCCTCATCCATGTACTC	68	CACGAATCTACCC	191	CATCCATGTACTC
112	CACGAATCTACCCGCATCCATGTACTC	69	CACGAATCTACCC	191	CATCCATGTACTC
113	TGTTCAATCATTCGGGATAAGAGTTC	70	TGTTCAATCATTC	194	GGGATAAGAGTTC
114	CCATCCATCATCCGGGATAAGAGTTC	71	CCATCCATCATCC	198	GGGATAAGAGTTC
115	CACGAATCTACCCACCATCCATGTACT	72	CACGAATCTACCC	191	CCATCCATGTACT
116	CACGAATCTACCCCTCCATCCATGTACT	73	CACGAATCTACCC	191	CCATCCATGTACT
117	CACGAATCTACCCCCCATCCATGTACT	74	CACGAATCTACCC	191	CCATCCATGTACT
118	CACGAATCTACCCCTCATCCATGTACTC	75	CACGAATCTACCC	191	CATCCATGTACTC
119	CACGAATCTACCCCTTCATCCATGTACTC	76	CACGAATCTACCC	191	CATCCATGTACTC
120	TGTTCAATCATTCAGGGATAAGAGTTC	77	TGTTCAATCATTC	194	GGGATAAGAGTTC
121	TGTTCAATCATTCCTGGGATAAGAGTTC	78	TGTTCAATCATTC	194	GGGATAAGAGTTC
122	CCATCCATCATCCAGGGATAAGAGTTC	79	CCATCCATCATCC	198	GGGATAAGAGTTC
123	CCATCCATCATCCTCGGGATAAGAGTTC	80	CCATCCATCATCC	198	GGGATAAGAGTTC
124	CCATCCATCATCCTTGGGATAAGAGTTC	81	CCATCCATCATCC	198	GGGATAAGAGTTC
125	ACGAATCTACCCCATCCATGTAC	37	ACGAATCTACCC	205	CCATCCATGTAC
126	ACGAATCTACCCCATCCATGTACT	43	ACGAATCTACCC	205	CATCCATGTACT
127	CCATCCATCATCCGGGATAAGAGTT	47	CCATCCATCATC	208	GGGATAAGAGTT
128	CCATCCATCATCGGATAAGAGTTC	48	CCATCCATCATC	208	GGATAAGAGTTC
129	CATCCATCATCCGGGATAAGAGTT	49	CATCCATCATCC	209	GGGATAAGAGTT
130	CATCCATCATCCGGGATAAGAGTTC	50	CATCCATCATCC	209	GGATAAGAGTTC

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein					
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2
					SEQ ID NO:
131	GTTCAATCATTCGGGATAAGAGTT	58	GTTCAATCATTC	211	GGGATAAGAGTT
132	GTTCAATCATTCGGGATAAGAGTTC	59	GTTCAATCATTC	211	GGATAAGAGTTC
133	CGAATCTACCCATCTGTTCAATCA	34	CGAATCTACCCA	204	TCTGTTCAATCA
134	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
135	ACGAATCTACCCCATCCATGTAC	37	ACGAATCTACCC	205	CCATCCATGTAC
136	ACGAATCTACCCCATCCATGTACT	43	ACGAATCTACCC	205	CATCCATGTACT
137	CCATCCATCATCGGGATAAGAGTT	47	CCATCCATCATC	208	GGGATAAGAGTT
138	CCATCCATCATCGGATAAGAGTTC	48	CCATCCATCATC	208	GGATAAGAGTTC
139	CATCCATCATCCGGGATAAGAGTT	49	CATCCATCATCC	209	GGGATAAGAGTT
140	CATCCATCATCCGGATAAGAGTTC	50	CATCCATCATCC	209	GGATAAGAGTTC
141	GTTCAATCATTCGGGATAAGAGTT	58	GTTCAATCATTC	211	GGGATAAGAGTT
142	GTTCAATCATTCGGGATAAGAGTTC	59	GTTCAATCATTC	211	GGATAAGAGTTC
143	CGAATCTACCCATCTGTTCAATCA	34	CGAATCTACCCA	204	TCTGTTCAATCA
144	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
145	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC
146	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
147	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
148	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
149	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
150	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
151	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
152	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
153	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
154	AATCTACCCACCTGTTCAATCA	83	AATCTACCCAC	213	CTGTTCAATCA
155	AATCTACCCACATCTGTTCAAT	84	AATCTACCCAC	213	ATCTGTTCAAT
156	CGAATCTACCCCTGTTCAATCA	85	CGAATCTACCC	214	CTGTTCAATCA
157	CGAATCTACCCATCTGTTCAAT	86	CGAATCTACCC	214	ATCTGTTCAAT
158	ACGAATCTACCCAATCTGTTCAATCA	87	ACGAATCTACCCA	196	ATCTGTTCAATCA
159	CACGAATCTACCCATCTGTTCAATCA	88	CACGAATCTACCC	191	ATCTGTTCAATCA
160	GAATCTACCCACCATCTGTTCAATCA	89	GAATCTACCCACC	195	ATCTGTTCAATCA
161	AATCTACCCACCAATCTGTTCAATCA	90	AATCTACCCACCA	215	ATCTGTTCAATCA
162	CGAATCTACCCACTCTGTTCAATCAT	91	CGAATCTACCCAC	192	TCTGTTCAATCAT
163	CGAATCTACCCACTCATCTGTTCAAT	92	CGAATCTACCCAC	192	TCATCTGTTCAAT
164	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC
165	GAATCTACCCACCATCTGTTCAATC	94	GAATCTACCCACC	195	CATCTGTTCAATC
166	ACGAATCTACCCATCTGTTCAATCAT	95	ACGAATCTACCCA	196	TCTGTTCAATCAT

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein					
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2
					SEQ ID NO:
167	GAATCTACCCACCTCTGTTCAATCAT	96	GAATCTACCCACC	195	TCTGTTCAATCAT
168	CCACGAATCTACCTCAATCATTCATT	97	CCACGAATCTACC	193	TCAATCATTCATT
169	CGAATCTACCCACCTCTGTTCAATCA	98	CGAATCTACCCACC	216	TCTGTTCAATCA
170	ACGAATCTACCCACTCTGTTCAATCA	99	ACGAATCTACCCAC	217	TCTGTTCAATCA
171	CGAATCTACCCAATCTGTTCAATCAT	100	CGAATCTACCCA	204	ATCTGTTCAATCAT
172	GAATCTACCCACATCTGTTCAATCAT	101	GAATCTACCCAC	203	ATCTGTTCAATCAT
173	AATCTACCCACCTGTTCAATCA	83	AATCTACCCAC	213	CTGTTCAATCA
174	AATCTACCCACATCTGTTCAAT	84	AATCTACCCAC	213	ATCTGTTCAAT
175	AATCTACCCACTCTGTTCAATC	102	AATCTACCCAC	213	TCTGTTCAATC
176	CGAATCTACCCCTGTTCAATCA	85	CGAATCTACCC	214	CTGTTCAATCA
177	CGAATCTACCCATCTGTTCAAT	86	CGAATCTACCC	214	ATCTGTTCAAT
178	CGAATCTACCCCTCTGTTCAATC	103	CGAATCTACCC	214	TCTGTTCAATC
179	GAATCTACCCACTGTTCAATCA	104	GAATCTACCCA	218	CTGTTCAATCA
180	GAATCTACCCAATCTGTTCAAT	105	GAATCTACCCA	218	ATCTGTTCAAT
181	GAATCTACCCATCTGTTCAATC	106	GAATCTACCCA	218	TCTGTTCAATC
182	AATCTACCCACTCTGTTCAATC	102	AATCTACCCAC	213	TCTGTTCAATC
183	CGAATCTACCCCTCTGTTCAATC	103	CGAATCTACCC	214	TCTGTTCAATC
184	GAATCTACCCACTGTTCAATCA	104	GAATCTACCCA	218	CTGTTCAATCA
185	GAATCTACCCAATCTGTTCAAT	105	GAATCTACCCA	218	ATCTGTTCAAT
186	GAATCTACCCATCTGTTCAATC	106	GAATCTACCCA	218	TCTGTTCAATC
187	GAATCTACCCATCTGTTCAATC	106	GAATCTACCCA	218	TCTGTTCAATC
188	GAATCTACCCATCTGTTCAATC	106	GAATCTACCCA	218	TCTGTTCAATC
189	GAATCTACCCATCTGTTCAATC	106	GAATCTACCCA	218	TCTGTTCAATC
190	GAATCTACCCATCTGTTCAATC	106	GAATCTACCCA	218	TCTGTTCAATC
191	GAATCTACCCATCTGTTCAATC	106	GAATCTACCCA	218	TCTGTTCAATC
192	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
193	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
194	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
195	CGAATCTACCCACTCATCTGTTCAAT	92	CGAATCTACCCAC	192	TCATCTGTTCAAT
196	CGAATCTACCCACTCATCTGTTCAAT	92	CGAATCTACCCAC	192	TCATCTGTTCAAT
197	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC
198	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC
199	CAAATCTACCCACATCTGTTCAATCA	107	CAAATCTACCCAC	219	ATCTGTTCAATCA
200	CGAATCTACCCACATCTATTCAATCA	108	CGAATCTACCCAC	192	ATCTATTCAATCA
201	CAAATCTACCCACATCTATTCAATCA	109	CAAATCTACCCAC	219	ATCTATTCAATCA
202	CGAATCTACCCAATCTATTCAATC	110	CGAATCTACCCA	204	ATCTATTCAATC

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein					
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2
					SEQ ID NO:
203	CGAATCTACCCACATCTGTTCAAT	111	CGAATCTACCCA	204	CATCTGTTCAAT
204	CGAATCTACCCATCATCTGTTCAA	112	CGAATCTACCCA	204	TCATCTGTTCAA
205	ACGAATCTACCCATCTGTTCAATC	113	ACGAATCTACCC	205	ATCTGTTCAATC
206	ACGAATCTACCCCATCTGTTCAAT	114	ACGAATCTACCC	205	CATCTGTTCAAT
207	CACGAATCTACCCATCCATGTACT	115	CACGAATCTACC	206	CATCCATGTACT
208	CACGAATCTACCCATCCATGTAC	116	CACGAATCTACC	206	CCATCCATGTAC
209	CATTTATTCAACGGGATAAGAGTT	117	CATTTATTCAAC	220	GGGATAAGAGTT
210	CATTTATTCAACGGGGATAAGAGT	118	CATTTATTCAAC	220	GGGGATAAGAGT
211	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
212	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
213	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
214	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA
215	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
216	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
217	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
218	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
219	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
220	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
221	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
222	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC
223	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
224	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
225	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
226	CACGAATCTACCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT
227	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
228	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
229	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
230	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC
231	GCATTTATTCAACGGGATAAGAGTT	17	GCATTTATTCAAC	199	GGGGATAAGAGTT
232	GCATTTATTCAACGGGATAAGAGTT	17	GCATTTATTCAAC	199	GGGGATAAGAGTT
233	GCATTTATTCAACGGGATAAGAGTT	17	GCATTTATTCAAC	199	GGGGATAAGAGTT
234	GCATTTATTCAACGGGATAAGAGTT	17	GCATTTATTCAAC	199	GGGGATAAGAGTT
235	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC
236	ATCCATCCATCATTGGGGATAAGAGT	119	ATCCATCCATCAT	221	TGGGGATAAGAGT
237	TCCATCCATCATCGGGATAAGAGTT	120	TCCATCCATCATC	222	GGGGATAAGAGTT
238	TCCATCCATCATCGGGATAAGAGTTC	121	TCCATCCATCATC	222	GGGATAAGAGTTC

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein					
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2
					SEQ ID NO:
239	TCCATCCATCATCGGATAAGAGTTCT	122	TCCATCCATCATC	222	GGATAAGAGTTCT
240	CCATCCATCATCCGATAAGAGTTCTT	123	CCATCCATCATCC	198	GATAAGAGTTCTT
241	GCATTTATTCAACGGGATAAGAGTTTC	124	GCATTTATTCAAC	199	GGGATAAGAGTTTC
242	GCATTTATTCAACGGGATAAGAGTTCT	125	GCATTTATTCAAC	199	GGATAAGAGTTCT
243	CGAATCTACCCACCCTTTTATCTACT	126	CGAATCTACCCAC	192	CCTTTTATCTACT
244	CGAATCTACCCACCAGTACTCACCC	127	CGAATCTACCCAC	192	CATGTACTCACCC
245	CACGAATCTACCCCATCCTTTTATC	128	CACGAATCTACCC	191	CCATCCTTTTATC
246	CACGAATCTACCCCTGTTCAATCATT	129	CACGAATCTACCC	191	CTGTTCAATCATT
247	CACGAATCTACCCGTTCAATCATTCA	130	CACGAATCTACCC	191	GTTCAATCATTCA
248	CGAATCTACCCACTCCTTTTATCTAC	131	CGAATCTACCCAC	192	TCCTTTTATCTAC
249	CGAATCTACCCACCTGTTCAATCATT	132	CGAATCTACCCAC	192	CTGTTCAATCATT
250	CGAATCTACCCACCTTTTATCTACTC	133	CGAATCTACCCAC	192	CTTTTATCTACTC
251	CGAATCTACCCACGTTCAATCATTCA	134	CGAATCTACCCAC	192	GTTCAATCATTCA
252	ACGAATCTACCCAGTACTCACCACATC	135	ACGAATCTACCCA	196	GTACTCACCACATC
253	CGAATCTACCCACTGTTCAATCATTTC	136	CGAATCTACCCAC	192	TGTTCAATCATTTC
254	CACGAATCTACCCCATCTGTTCAATC	137	CACGAATCTACCC	191	CATCTGTTCAATC
255	CACGAATCTACCCCTTTTATCTACT	138	CACGAATCTACCC	191	CCTTTTATCTACT
256	CACGAATCTACCCCATGTACTCACC	139	CACGAATCTACCC	191	CCATGTACTCACC
257	CGAATCTACCCACATTACCAGCATT	140	CGAATCTACCCAC	192	ATTACCAGCATT
258	CGAATCTACCCACCATCTGTTCAATC	141	CGAATCTACCCAC	192	CATCTGTTCAATC
259	ACGAATCTACCCACCATCCTTTTATC	142	ACGAATCTACCCA	196	CCATCCTTTTATC
260	CGAATCTACCCACGTACTCACCACATC	143	CGAATCTACCCAC	192	GTACTCACCACATC
261	CACGAATCTACCCGTACTCACCACATC	144	CACGAATCTACCC	191	GTACTCACCACATC
262	ACGAATCTACCCAATTACCAGCATT	145	ACGAATCTACCCA	196	ATTACCAGCATT
263	CGAATCTACCCACGAAACCCAGGCAG	146	CGAATCTACCCAC	192	GAAACCCAGGCAG
264	CCACGAATCTACCGGAAACCCAGGCA	147	CCACGAATCTACC	193	GGAAACCCAGGCA
265	CAATCATATATCCGGAACCCAGGCA	148	CAATCATATATCC	223	GGAACCCAGGCA
266	CAATAGTTCAATCGGAAACCCAGGCA	149	CAATAGTTCAATC	224	GGAACCCAGGCA
267	TCTAGCCACGAATTTCCAGGAAACCC	150	TCTAGCCACGAAT	225	TTCCAGGAAACCC
268	GTACTCACCACATCGATAAGAGTTCTT	151	GTACTCACCACATC	226	GATAAGAGTTCTT
269	CCATGTACTCACCAGGATAAGAGTTCT	152	CCATGTACTCACC	227	GGATAAGAGTTCT
270	TGTCCATCCATCCGGATAAGAGTTCT	153	TGTCCATCCATCC	228	GGATAAGAGTTCT
271	TTCATTCAACAGCGGGATAAGAGTTTC	154	TTCATTCAACAGC	229	GGGATAAGAGTTTC
272	CTACTCATCACTCGGGATAAGAGTTTC	155	CTACTCATCACTC	230	GGGATAAGAGTTTC
273	TCTCCATCCATCCGGGATAAGAGTTTC	156	TCTCCATCCATCC	231	GGGATAAGAGTTTC
274	GTACTCACCACATCGGGATAAGAGTTTC	157	GTACTCACCACATC	226	GGGATAAGAGTTTC

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein					
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2
					SEQ ID NO:
275	CATGTACTCACCCGGGATAAGAGTTC	158	CATGTACTCACCC	232	GGGATAAGAGTTC
276	CATCCATCTATCCGGGATAAGAGTTC	159	CATCCATCTATCC	233	GGGATAAGAGTTC
277	CCATCCATCTGTCCGGGATAAGAGTTC	160	CCATCCATCTGTC	234	GGGATAAGAGTTC
278	CCAGCATTATTTCGGGATAAGAGTT	161	CCAGCATTATTTC	235	GGGGATAAGAGTT
279	TTCATTACCAGCGGGATAAGAGTT	162	TTCATTACCAGC	229	GGGGATAAGAGTT
280	TGTTCAATCATTCGGGATAAGAGTT	163	TGTTCAATCATTC	194	GGGGATAAGAGTT
281	TTATCTACTCATCGGGATAAGAGTT	164	TTATCTACTCATC	236	GGGGATAAGAGTT
282	CACCCATCTCTCCGGGATAAGAGTT	165	CACCCATCTCTCC	237	GGGGATAAGAGTT
283	GTACTCACCCATCGGGATAAGAGTT	166	GTACTCACCCATC	226	GGGGATAAGAGTT
284	CATGTACTCACCCGGGATAAGAGTT	167	CATGTACTCACCC	232	GGGGATAAGAGTT
285	CCATCTATCCATCGGGATAAGAGTT	168	CCATCTATCCATC	238	GGGGATAAGAGTT
286	CATCCATCTAGCCGGGATAAGAGTT	169	CATCCATCTAGCC	239	GGGGATAAGAGTT
287	CTGTCCATCCATCGGGATAAGAGTT	170	CTGTCCATCCATC	240	GGGGATAAGAGTT
288	CAATCATATATCCGGGATAAGAGTT	171	CAATCATATATCC	223	GGGGATAAGAGTT
289	CCATCTAGCCAGTTCCTGGGGATAA	172	CCATCTAGCCACG	241	TTCCTGGGGATAA
290	CATGTACTCACCCGTTCTGGGGATA	173	CATGTACTCACCC	232	GTTCTGGGGATA
291	CTAGCCACGAATCACTAGTTCCTGGG	174	CTAGCCACGAATC	197	ACTAGTTCCTGGG
292	CCACGAATCTACCGCATTATTCAAC	175	CCACGAATCTACC	193	GCATTTATTCAAC
293	CTAGCCACGAATCGTACTCACCCATC	176	CTAGCCACGAATC	197	GTACTCACCCATC
294	GAATCTACCCACTTCACCAGCATT	177	GAATCTACCCAC	203	TTCACCAGCATT
295	CGAATCTACCCATTTCACCAGCATT	178	CGAATCTACCCA	204	TTCACCAGCATT
296	GAATCTACCCACATTTCACCAGCAT	179	GAATCTACCCAC	203	ATTACCAGCAT
297	CGAATCTACCCAATTTCACCAGCAT	180	CGAATCTACCCA	204	ATTACCAGCAT
298	GAATCTACCCACTTCACCAGCATT	177	GAATCTACCCAC	203	TTCACCAGCATT
299	CGAATCTACCCATTTCACCAGCATT	178	CGAATCTACCCA	204	TTCACCAGCATT
300	GAATCTACCCACATTTCACCAGCAT	179	GAATCTACCCAC	203	ATTACCAGCAT
301	CGAATCTACCCAATTTCACCAGCAT	180	CGAATCTACCCA	204	ATTACCAGCAT
302	CGAATCTACCCACATTTCACCAGCATT	140	CGAATCTACCCAC	192	ATTACCAGCATT
303	CGAATCTACCCACATTTCACCAGCATT	140	CGAATCTACCCAC	192	ATTACCAGCATT
304	CGAATCTACCCACATTTCACCAGCATT	140	CGAATCTACCCAC	192	ATTACCAGCATT
305	GAATCTACCCACATTTCACCAGCATT	181	GAATCTACCCAC	203	ATTACCAGCATT
306	CGAATCTACCCACATTTCACCAGCAT	182	CGAATCTACCCAC	192	ATTACCAGCAT
307	CGAATCTACCCATTTCACCAGCATT	178	CGAATCTACCCA	204	TTCACCAGCATT
308	CGAATCTACCCATTTCACCAGCATT	178	CGAATCTACCCA	204	TTCACCAGCATT
309	CATATATCCATCTAGCCACG	183	CATATATCCA	242	TCTAGCCACG
310	ACGAATCTACCCATCCATGT	184	ACGAATCTAC	243	CCATCCATGT

TABLE 3-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO: (5'→3')	Antisense Sequence 1	SEQ ID NO: (5'→3')	Antisense Sequence 2	SEQ ID NO:
311	TGTCCATCCACGAATCTACC	185	TGTCCATCCA	244	CGAATCTACC	212
312	CGAATCTACCAACTCATCCA	186	CGAATCTACC	212	AACTCATCCA	302
313	ACGAATCTACCATCCATGTA	187	ACGAATCTAC	243	CATCCATGTA	303
314	CGAATCTACCCATCTCTCCA	188	CGAATCTACC	212	CATCTCTCCA	304
315	CACGAATCTACCCATCTCTC	189	CACGAATCTA	245	CCCATCTCTC	305
316	ACGAATCTACCCATCTCTCC	190	ACGAATCTAC	243	CCATCTCTCC	306
317	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
318	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
319	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
320	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
321	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
322	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
323	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
324	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
325	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
326	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
327	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
328	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
329	CACGAATCTACCCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT	246
330	CCATCCATCATCCGGGATAAGAGTTC	16	CCATCCATCATCC	198	GGGATAAGAGTTC	253
331	GCATTTATTCAACGGGATAAGAGTT	17	GCATTTATTCAAC	199	GGGGATAAGAGTT	254

*When C is provided in a nucleobase sequence disclosed herein, without modification, such as in Table 3, it encompasses cytidine or 5'-methyl cytidine

*Each thymine base (T) in any one of the oligonucleotide sequences provided in Table 3 may independently and optionally be replaced with a uracil base (U).

TABLE 9

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
350	GCATTTATTCAACGGGATAAGAGTT	17	GCATTTATTCAAC	199	GGGATAAGAGTT	254
351	GCATTTATTCAACGGGATAAGAGTT	17	GCATTTATTCAAC	199	GGGATAAGAGTT	254
352	GCATTTATTCAACGGGATAAGAGTTC	124	GCATTTATTCAAC	199	GGGATAAGAGTTC	253
353	GCATTTATTCAACGGGATAAGAGTTC	124	GCATTTATTCAAC	199	GGGATAAGAGTTC	253
354	GCATTTATTCAACGGGATAAGAGTTC	124	GCATTTATTCAAC	199	GGGATAAGAGTTC	253
355	GCATTTATTCAACGGGATAAGAGTTC	124	GCATTTATTCAAC	199	GGGATAAGAGTTC	253
356	GCATTTATTCAACGGGATAAGAGTTC	124	GCATTTATTCAAC	199	GGGATAAGAGTTC	253
357	GCATTTATTCAACGGATAAGAGTTCT	125	GCATTTATTCAAC	199	GGATAAGAGTTCT	252

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
358	CATTATTCAACGGATAAGAGTTC	338	CATTATTCAAC	220	GGATAAGAGTTC	267
359	CATTATTCAACGGATAAGAGTTC	338	CATTATTCAAC	220	GGATAAGAGTTC	267
360	CATTATTCAACGGGATAAGAGTT	117	CATTATTCAAC	220	GGGATAAGAGTT	266
361	CATTATTCAACGGGGATAAGAGTT	339	CATTATTCAAC	220	GGGGATAAGAGTT	254
362	GCATTATTCAAGGGGATAAGAGTT	340	GCATTATTCAA	495	GGGGATAAGAGTT	254
363	GCATTATTCAACGGGATAAGAGTT	341	GCATTATTCAAC	199	GGGATAAGAGTT	266
364	GCATTATTCAACGGGGATAAGAGT	342	GCATTATTCAAC	199	GGGGATAAGAGT	284
365	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
366	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
367	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
368	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
369	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
370	TGTTCAATCATTCGGGATAAGAGTTC	343	TGTTCAATCATT	496	GGGATAAGAGTTC	253
371	TGTTCAATCATTCGGGATAAGAGTT	344	TGTTCAATCATTC	194	GGGATAAGAGTT	266
372	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
373	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
374	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
375	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
376	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
377	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
378	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
379	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
380	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
381	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
382	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
383	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
384	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
385	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
386	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
387	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
388	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
389	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
390	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
391	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
392	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
393	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
394	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
395	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
396	TGTTCAATCATTCGGGATAAGAGTTC	345	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
397	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
398	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
399	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
400	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
401	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
402	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
403	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
404	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
405	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
406	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
407	TGTTCAATCATTC[n3]GGGATAAGAGTTC	194/ 253	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
408	TGTTCAATCATTC[n6]GGGATAAGAGTTC	194/ 253	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
409	TCTGTTCAATCATTTGGGGATAAGAGT	347	TCTGTTCAATCAT	274	TGGGGATAAGAGT	285
410	TCTGTTCAATCATGGGGATAAGAGTT	348	TCTGTTCAATCAT	274	GGGGATAAGAGTT	254
411	TCTGTTCAATCATGGGATAAGAGTTC	349	TCTGTTCAATCAT	274	GGGATAAGAGTTC	253
412	CTGTTCAATCATTTGGGGATAAGAGT	350	CTGTTCAATCAT	288	TGGGGATAAGAGT	285
413	CTGTTCAATCATTTGGGGATAAGAGTT	351	CTGTTCAATCAT	288	GGGGATAAGAGTT	254
414	CTGTTCAATCATTTGGGATAAGAGTTC	352	CTGTTCAATCAT	288	GGGATAAGAGTTC	253
415	TGTTCAATCATTC TGGGGATAAGAGT	353	TGTTCAATCATTC	194	TGGGGATAAGAGT	285
416	TGTTCAATCATTCGGGGATAAGAGTT	163	TGTTCAATCATTC	194	GGGGATAAGAGTT	254
417	TGTTCAATCATTCGGGATAAGAGTTCT	354	TGTTCAATCATTC	194	GGATAAGAGTTCT	252
418	TGTTCAATCATTCGGGATAAGAGTTC	9	TGTTCAATCATTC	194	GGGATAAGAGTTC	253
419	TCCATCCATCCTTGAGTTCTTTCCAG	355	TCCATCCATCCTT	497	GAGTTCTTTCCAG	249
420	TCCATCCATCCTT[n6]GAGTTCTTTCCAG	497/ 249	TCCATCCATCCTT	497	GAGTTCTTTCCAG	249
421	CTCTCCATCCATCTTTTCCAGGAA	357	CTCTCCATCCATC	498	TTCTTTCCAGGAA	516
422	TCCATCCATCCTTGAGTTCTTTCCAG	355	TCCATCCATCCTT	497	GAGTTCTTTCCAG	249
423	CATCCATCTATCCGGGATAAGAGTTC	159	CATCCATCTATCC	233	GGGATAAGAGTTC	253
424	CATCCATCTATCCGGGATAAGAGTTC	159	CATCCATCTATCC	233	GGGATAAGAGTTC	253
425	CATCCATCTATCCGGGATAAGAGTTC	159	CATCCATCTATCC	233	GGGATAAGAGTTC	253
426	CATCCATCTATCCGGGATAAGAGTTC	159	CATCCATCTATCC	233	GGGATAAGAGTTC	253
427	CATCCATCTATCCGGGATAAGAGTTC	159	CATCCATCTATCC	233	GGGATAAGAGTTC	253

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
428	ATCCATCTATCCGGATAAGAGTTC	358	ATCCATCTATCC	499	GGATAAGAGTTC	267
429	ATCCATCTATCCGGATAAGAGTT	359	ATCCATCTATCC	499	GGGATAAGAGTT	266
430	ATCCATCTATCCGGATAAGAGTTC	358	ATCCATCTATCC	499	GGATAAGAGTTC	267
431	ATCCATCTATCCGGATAAGAGTT	359	ATCCATCTATCC	499	GGGATAAGAGTT	266
432	CAACTCATCCATCAGAGTTCTTTCCA	360	CAACTCATCCATC	500	AGAGTTCTTTCCA	517
433	GCCACGAATCTACGGATAAGAGTTCT	361	GCCACGAATCTAC	201	GGATAAGAGTTCT	252
434	CCACGAATCTACCGGATAAGAGTTC	14	CCACGAATCTACC	193	GGGATAAGAGTTC	253
435	CCACGAATCTACCGGATAAGAGTTC	14	CCACGAATCTACC	193	GGGATAAGAGTTC	253
436	CCACGAATCTACCGGATAAGAGTTC	14	CCACGAATCTACC	193	GGGATAAGAGTTC	253
437	CCACGAATCTACCGGATAAGAGTTC	14	CCACGAATCTACC	193	GGGATAAGAGTTC	253
438	CCACGAATCTACCGGATAAGAGTTC	14	CCACGAATCTACC	193	GGGATAAGAGTTC	253
439	CCACGAATCTACCGGATAAGAGTTCT	8	CCACGAATCTACC	193	GGATAAGAGTTCT	252
440	CCACGAATCTACCGGATAAGAGTTCT	8	CCACGAATCTACC	193	GGATAAGAGTTCT	252
441	CCACGAATCTACCGGATAAGAGTTCT	8	CCACGAATCTACC	193	GGATAAGAGTTCT	252
442	CCACGAATCTACCGGATAAGAGTTCT	8	CCACGAATCTACC	193	GGATAAGAGTTCT	252
443	CCACGAATCTACCGGATAAGAGTTCT	8	CCACGAATCTACC	193	GGATAAGAGTTCT	252
444	CACGAATCTACCGATAAGAGTTCT	362	CACGAATCTACC	206	GATAAGAGTTCT	518
445	CCACGAATCTACGATAAGAGTTCT	363	CCACGAATCTAC	501	GATAAGAGTTCT	518
446	CACGAATCTACCGGGATAAGAGT	364	CCACGAATCTAC	501	GGGGATAAGAGT	284
447	CCACGAATCTACCGGGATAAGAGT	365	GCCACGAATCTA	502	GGGGATAAGAGT	284
448	CACGAATCTACCGGATAAGAGTTC	366	CACGAATCTACC	206	GGATAAGAGTTC	267
449	CACGAATCTACCGGGATAAGAGTT	367	CACGAATCTACC	206	GGGATAAGAGTT	266
450	CCACGAATCTACGGATAAGAGTTC	368	CCACGAATCTAC	501	GGATAAGAGTTC	267
451	CCACGAATCTACGGGATAAGAGTT	369	CCACGAATCTAC	501	GGGATAAGAGTT	266
452	CACGAATCTACCGATAAGAGTTCT	362	CACGAATCTACC	206	GATAAGAGTTCT	518
453	CCACGAATCTACGGGGATAAGAGT	365	CCACGAATCTAC	501	GGGGATAAGAGT	284
454	CCACGAATCTACCGGGATAAGAGTT	22	CCACGAATCTACC	193	GGGGATAAGAGTT	254
455	CCACGAATCTACCGGGATAAGAGTT	22	CCACGAATCTACC	193	GGGGATAAGAGTT	254
456	CCACGAATCTACCGGGATAAGAGTT	22	CCACGAATCTACC	193	GGGGATAAGAGTT	254
457	CCACGAATCTACCGGGATAAGAGTT	22	CCACGAATCTACC	193	GGGGATAAGAGTT	254
458	CCACGAATCTACCGGGATAAGAGTT	22	CCACGAATCTACC	193	GGGGATAAGAGTT	254
459	ACGAATCTACCCAGGGATAAGAGTT	370	ACGAATCTACCCA	196	GGGGATAAGAGTT	254
460	CTAGCCACGAATCTAAGAGTTCTTTC	371	CTAGCCACGAATC	197	TAAGAGTTCTTTC	250
461	CCACGAATCTACCATAAGAGTTCTTT	372	CCACGAATCTACC	193	ATAAGAGTTCTTT	519
462	CCATCCATCATCCGGGATAAGAGTCC	373	CCATCCATCATCC	198	GGGATAAGAGTTC	253
463	TCCATCCATCATCGGGATAAGAGTT	120	TCCATCCATCATC	222	GGGGATAAGAGTT	254

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
464	TCCATCCATCATCGGGATAAGAGTT	120	TCCATCCATCATC	222	GGGGATAAGAGTT	254
465	TCCATCCATCATCGGGATAAGAGTT	120	TCCATCCATCATC	222	GGGGATAAGAGTT	254
466	TCCATCCATCATCGGGATAAGAGTT	120	TCCATCCATCATC	222	GGGGATAAGAGTT	254
467	TCCATCCATCATCGGGATAAGAGTT	120	TCCATCCATCATC	222	GGGGATAAGAGTT	254
468	TCCATCCATCATCGGGATAAGAGTTC	121	TCCATCCATCATC	222	GGGATAAGAGTTC	253
469	TCCATCCATCATCGGGATAAGAGTTC	121	TCCATCCATCATC	222	GGGATAAGAGTTC	253
470	TCCATCCATCATCGGGATAAGAGTTC	121	TCCATCCATCATC	222	GGGATAAGAGTTC	253
471	TCCATCCATCATCGGGATAAGAGTTC	121	TCCATCCATCATC	222	GGGATAAGAGTTC	253
472	TCCATCCATCATCGGGATAAGAGTTC	121	TCCATCCATCATC	222	GGGATAAGAGTTC	253
473	TCCATCCATCATCGGATAAGAGTTCT	122	TCCATCCATCATC	222	GGATAAGAGTTCT	252
474	CCATCCATCATCCGATAAGAGTTCTT	123	CCATCCATCATCC	198	GATAAGAGTTCTT	251
475	CCATCCATCATCCGATAAGAGTTCTT	123	CCATCCATCATCC	198	GATAAGAGTTCTT	251
476	CCATCCATCATCCGATAAGAGTTCTT	123	CCATCCATCATCC	198	GATAAGAGTTCTT	251
477	CCATCCATCATCCGATAAGAGTTCTT	123	CCATCCATCATCC	198	GATAAGAGTTCTT	251
478	CCATCCATCATCGGGATAAGAGT	374	CCATCCATCATC	208	GGGGATAAGAGT	284
479	CCATCCATCATCGGGATAAGAGT	374	CCATCCATCATC	208	GGGGATAAGAGT	284
480	CCATCCATCATCGGATAAGAGTTC	48	CCATCCATCATC	208	GGATAAGAGTTC	267
481	TCATCCATCTAGCGGGATAAGAGTTC	375	TCATCCATCTAGC	503	GGGATAAGAGTTC	253
482	CATCCATCTAGCCGGGATAAGAGTTC	376	CATCCATCTAGCC	239	GGGATAAGAGTTC	253
483	TCCATCTAGCCACGGGATAAGAGTT	377	TCCATCTAGCCAC	504	GGGGATAAGAGTT	254
484	ATCCATCTAGCCGGGATAAGAGTT	378	ATCCATCTAGCC	505	GGGATAAGAGTT	266
485	ATCCATCTAGCCGGGATAAGAGT	379	ATCCATCTAGCC	505	GGGGATAAGAGT	284
486	CCATCCATCATCCGGGATAAGAGTT	380	CCATCCATCATCC	198	GGGGATAAGAGTT	254
487	ATATATCCATCCACTAGTTCCTGGGG	381	ATATATCCATCCA	506	CTAGTTCCTGGGG	259
488	ATATATCCATCCA[n6]CTAGTTCCTGGGG	506/ 259	ATATATCCATCCA	506	CTAGTTCCTGGGG	259
489	ATATATCCATCCACTAGTTCCTGGGG	381	ATATATCCATCCA	506	CTAGTTCCTGGGG	259
490	TAGTTCAATCATAAGAGTTCTTTCCA	383	TAGTTCAATCATA	507	AGAGTTCTTTCCA	517
491	TCATCTGTTCAACAACTAG	384	TCATCTGTTT	508	AACAACTAG	520
492	ATCTGTTCAATCAGCATTTATTCAAC	385	ATCTGTTCAATCA	247	GCATTTATTCAAC	199
493	ATCTACCCACCAGCATTTAT	386	ATCTACCCAC	509	CAGCATTTAT	521
494	CCACGAATCTACCGTTCAATCATTCA	387	CCACGAATCTACC	193	GTTCAATCATTCA	289
495	CACGAATCTACCCATTACCAGCATT	388	CACGAATCTACCC	191	ATTACCAGCATT	292
496	ACGAATCTACCCAGTTCAATCATTCA	389	ACGAATCTACCCA	196	GTTCAATCATTCA	289
497	CCACGAATCTACCGTTCAATCATTCA	387	CCACGAATCTACC	193	GTTCAATCATTCA	289
498	CCACGAATCTACCGTTCAATCATTCA	387	CCACGAATCTACC	193	GTTCAATCATTCA	289
499	CACGAATCTACCCATTACCAGCATT	388	CACGAATCTACCC	191	ATTACCAGCATT	292

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
500	CACGAATCTACCCATTACCAGCATT	388	CACGAATCTACCC	191	ATTCACCAGCATT	292
501	CCATCCATGTACTCAATCAT	390	CCATCCATGTACT	246	CAATCAT	
502	GAATCTACCCATCTGTTCAATC	106	GAATCTACCCA	218	TCTGTTCAATC	279
503	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
504	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
505	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
506	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC	276
507	CGAATCTACCCACATCTGTTCAAT	111	CGAATCTACCCAC	192	ATCTGTTCAAT	273
508	ACGAATCTACCCATCTGTTCAATC	113	ACGAATCTACCC	205	ATCTGTTCAATC	261
509	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC	276
510	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC	276
511	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC	276
512	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC	276
513	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
514	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
515	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
516	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
517	CGAATCTACCCACCTGTTCAATCA	391	CGAATCTACCCAC	192	CTGTTCAATCA	272
518	CGAATCTACCCACATCTGTTCAAT	111	CGAATCTACCCAC	192	ATCTGTTCAAT	273
519	CGAATCTACCCACTCTGTTCAATC	392	CGAATCTACCCAC	192	TCTGTTCAATC	279
520	CGAATCTACCCAATCTGTTCAATCA	393	CGAATCTACCCA	204	ATCTGTTCAATCA	247
521	CGAATCTACCCACTCTGTTCAATCA	394	CGAATCTACCCAC	192	TCTGTTCAATCA	260
522	ACGAATCTACCCCATCTGTTCAATC	395	ACGAATCTACCC	205	CATCTGTTCAATC	276
523	ACGAATCTACCCAATCTGTTCAATC	396	ACGAATCTACCCA	196	ATCTGTTCAATC	261
524	CACGAATCTACCCTGTTCAATCATTC	397	CACGAATCTACCC	191	TGTTCAATCATTC	194
525	ACGAATCTACCCATGTTCAATCATTC	398	ACGAATCTACCCA	196	TGTTCAATCATTC	194
526	CCACGAATCTACCCTGTTCAATCATT	399	CCACGAATCTACC	193	CTGTTCAATCATT	288
527	ACGAATCTACCCACTGTTCAATCATT	400	ACGAATCTACCCA	196	CTGTTCAATCATT	288
528	CCACGAATCTACCCATCTGTTCAATC	401	CCACGAATCTACC	193	CATCTGTTCAATC	276
529	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
530	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
531	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
532	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
533	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
534	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
535	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
536	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
537	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
538	CGAATCTACCCAATCTGTTCAATC	36	CGAATCTACCCA	204	ATCTGTTCAATC	261
539	AGCCACGAATCTGTTCAATC	402	AGCCACGA		ATCTGTTCAATC	261
540	CCACCAACTCATTCACTGT	403	CCACCAACTC	510	ATTCACTGT	522
541	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
542	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
543	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
544	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
545	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
546	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
547	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
548	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
549	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
550	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
551	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
552	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
553	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
554	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
555	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
556	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
557	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
558	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
559	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
560	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
561	CGAATCTACCCACATCTGTTCAATCA	404	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
562	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
563	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC	276
564	CGAATCTACCCACATCTGTTCAATCA	2	CGAATCTACCCAC	192	ATCTGTTCAATCA	247
565	ACGAATCTACCCACATCTGTTCAATC	93	ACGAATCTACCCA	196	CATCTGTTCAATC	276
566	CCATCCATCATCTGTTCAAT	405	CCATCCATCA	511	TCTGTTCAAT	523
567	CCATCCATGTACTGTTCAAT	406	CCATCCATGTACT	246	GTTCAAT	
568	CCATCCATGTACTTTCAATC	407	CCATCCATGTACT	246	TTCAATC	
569	CCATCCATGTACTTCAATCA	408	CCATCCATGTACT	246	TCAATCA	
570	ACGAATCTACCCAATCCTTTTATCTA	409	ACGAATCTACCCA	196	ATCCTTTTATCTA	524
571	CGAATCTACCCACATCCTTTTATCTA	410	CGAATCTACCCAC	192	ATCCTTTTATCTA	524

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
572	CACGAATCTACCCCTTTTATCTACTC	411	CACGAATCTACCC	191	CTTTTATCTACTC	291
573	ACGAATCTACCCATCCTTTTATCTAC	412	ACGAATCTACCCA	196	TCCTTTTATCTAC	290
574	ACGAATCTACCCACTTTTATCTACTC	413	ACGAATCTACCCA	196	CTTTTATCTACTC	291
575	CACGAATCTACCCATCCTTTTATCTA	414	CACGAATCTACCC	191	ATCCTTTTATCTA	524
576	CCATCCATGTACTTTTTATC	415	CCATCCATGTACT	246	TTTTATC	
577	CCATCCATGTACTCTTTTAT	416	CCATCCATGTACT	246	CTTTTAT	
578	CCATCCATGTACTTGACTC	417	CCATCCATGTACT	246	TGTACTC	
579	CGAATCTACCCACCATCCATGTACTC	418	CGAATCTACCCAC	192	CATCCATGTACTC	248
580	CGAATCTACCCACATCCATGTACTCA	419	CGAATCTACCCAC	192	ATCCATGTACTCA	525
581	GCCACGAATCTACTCCATGTACTCAC	420	GCCACGAATCTAC	201	TCCATGTACTCAC	526
582	ACGAATCTACCCACATCCATGTACTC	421	ACGAATCTACCCA	196	CATCCATGTACTC	248
583	ACGAATCTACCCAATCCATGTACTCA	422	ACGAATCTACCCA	196	ATCCATGTACTCA	525
584	CGAATCTACCCACTCCATGTACTCAC	423	CGAATCTACCCAC	192	TCCATGTACTCAC	526
585	CACGAATCTACCCCATCCATGTACTC	3	CACGAATCTACCC	191	CATCCATGTACTC	248
586	ACGAATCTACCCATCCATGTACTC	424	ACGAATCTACCC	205	ATCCATGTACTC	527
587	ACGAATCCATCCATGTACTC	425	ACGAATC		CATCCATGTACTC	248
588	ACGAATCATCCATGTACTCA	426	ACGAATC		ATCCATGTACTCA	525
589	CGAATCTCATCCATGTACTC	427	CGAATCT		CATCCATGTACTC	248
590	CGAATCTATCCATGTACTCA	428	CGAATCT		ATCCATGTACTCA	525
591	GAATCTACATCCATGTACTC	429	GAATCTA		CATCCATGTACTC	248
592	AATCTACCATCCATGTACTC	430	AATCTAC		CATCCATGTACTC	248
593	ACGAATCCATCCATGTACTC	425	ACGAATC		CATCCATGTACTC	248
594	ACGAATCCATCCATGTACTC	425	ACGAATC		CATCCATGTACTC	248
595	GAATCTACCCATCTCTCCAT	431	GAATCTA		CCCATCTCTCCAT	528
596	TCCATCATCCATGTACTCAC	432	TCCATCATCC	512	ATGTACTCAC	529
597	GAATCTACCCATCCATGTAC	433	GAATCTACCC	513	ATCCATGTAC	271
598	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
599	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
600	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
601	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
602	CCATCCATGTACTTCACCCA	434	CCATCCATGTACT	246	TCACCCA	
603	CAATCATCCATCCATGTACT	435	CAATCAT		CCATCCATGTACT	246
604	CAATCATCCATCCATGTACT	435	CAATCAT		CCATCCATGTACT	246
605	CAATCATCCATCCATGTACT	435	CAATCAT		CCATCCATGTACT	246
606	GTTCAATCCATCCATGTACT	436	GTTCAAT		CCATCCATGTACT	246
607	TTCAATCCCATCCATGTACT	437	TTCAATC		CCATCCATGTACT	246

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
608	TCAATCACCATCCATGTACT	438	TCAATCA		CCATCCATGTACT	246
609	TTTTATCCCATCCATGTACT	439	TTTTATC		CCATCCATGTACT	246
610	CTTTTATCCATCCATGTACT	440	CTTTTAT		CCATCCATGTACT	246
611	TCACCCACCATCCATGTACT	441	TCACCCA		CCATCCATGTACT	246
612	CCATCCATGTACTCCATCCATGTACT	442	CCATCCATGTACT	246	CCATCCATGTACT	246
613	CCATCCATGTACT[n6]CCATCCATGTACT	246/ 246	CCATCCATGTACT	246	CCATCCATGTACT	246
614	CCATCCATCATCC[n6]GGGATAAGAGTTC	198/ 253	CCATCCATCATCC	198	GGGATAAGAGTTC	253
615	GCCACGAATCTACTCCATCCATGTAC	445	GCCACGAATCTAC	201	TCCATCCATGTAC	530
616	GCCACGAATCTACCCATCCATGTACT	446	GCCACGAATCTAC	201	CCATCCATGTACT	246
617	CACGAATCTACCCCTCCATCCATGTAC	447	CACGAATCTACCC	191	TCCATCCATGTAC	530
618	ACGAATCTACCCACCATCCATGTACT	448	ACGAATCTACCCA	196	CCATCCATGTACT	246
619	CCACGAATCTACCCCATCCATGTACT	449	CCACGAATCTACC	193	CCATCCATGTACT	246
620	CACGAATCTACCCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT	246
621	CACGAATCTACCCCCATCCATGTACT	1	CACGAATCTACCC	191	CCATCCATGTACT	246
622	ACGAATCTACCCCCATCCATGTAC	37	ACGAATCTACCC	205	CCATCCATGTAC	262
623	ACGAATCTACCCCCATCCATGTAC	37	ACGAATCTACCC	205	CCATCCATGTAC	262
624	ACGAATCTACCCCCATCCATGTAC	37	ACGAATCTACCC	205	CCATCCATGTAC	262
625	ACGAATCTACCCCCATCCATGTAC	37	ACGAATCTACCC	205	CCATCCATGTAC	262
626	ACGAATCTACCCCCATCCATGTAC	37	ACGAATCTACCC	205	CCATCCATGTAC	262
627	ACGAATCTATCCATCCATGTAC	450	ACGAATCTA		TCCATCCATGTAC	530
628	CGAATCTACCCACCCATCCATGTACT	451	CGAATCTACCCAC	192	CCATCCATGTACT	246
629	ACGAATCCCATCCATGTACT	452	ACGAATC		CCATCCATGTACT	246
630	GCCACGACCATCCATGTACT	453	GCCACGA		CCATCCATGTACT	246
631	CGAATCTCCATCCATGTACT	454	CGAATCT		CCATCCATGTACT	246
632	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
633	AATCTACCCATCCATGTACT	456	AATCTAC		CCATCCATGTACT	246
634	ACGAATCTCCATCCATGTAC	457	ACGAATC		TCCATCCATGTAC	530
635	CGAATCTTCCATCCATGTAC	458	CGAATCT		TCCATCCATGTAC	530
636	GAATCTATCCATCCATGTAC	459	GAATCTA		TCCATCCATGTAC	530
637	GAATCTACCATCCATGTACT	455	ACGAATCTAC	243	CCATCCATGTACT	246
638	GAATCTACCATCCATGTACT	455	ACGAATCTAC	243	CCATCCATGTACT	246
639	GAATCTACCATCCATGTACT	455	ACGAATCTAC	243	CCATCCATGTACT	246
640	GAATCTACCATCCATGTACT	455	ACGAATCTAC	243	CCATCCATGTACT	246
641	GAATCTACCATCCATGTACT	455	ACGAATCTAC	243	CCATCCATGTACT	246
642	GAATCTACCATCCATGTACT	455	ACGAATCTAC	243	CCATCCATGTACT	246

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
643	GAATCTACCATCCATGTACT	455	ACGAATCTAC	243	CCATCCATGTACT	246
644	GAATCTACCATCCATGTACT	455	ACGAATCTAC	243	CCATCCATGTACT	246
645	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
646	CGAATCTACCATCCATGTAC	82	CGAATCTACC	212	ATCCATGTAC	271
647	ACGAATCTCCATCCATGTAC	457	ACGAATC		TCCATCCATGTAC	530
648	ACGAATCTCCATCCATGTAC	457	ACGAATC		TCCATCCATGTAC	530
649	ACGAATCCCATCCATGTACT	452	ACGAATC		CCATCCATGTACT	246
650	CGAATCTCCATCCATGTACT	454	CGAATCT		CCATCCATGTACT	246
651	CGAATCTCCATCCATGTACT	454	CGAATCT		CCATCCATGTACT	246
652	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
653	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
654	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
655	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
656	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
657	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
658	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
659	GAATCTACCATCCATGTAC	460	GAATCTA		CCATCCATGTAC	262
660	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
661	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
662	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
663	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
664	CGAATCTACCATCCATGTACT	462	CGAATCTA		CCATCCATGTACT	246
665	ACGAATCTACCATCCATGTACT	463	ACGAATCTA		CCATCCATGTACT	246
666	CACGAATCTACCATCCATGTACT	464	CACGAATCTA	245	CCATCCATGTACT	246
667	CCACGAATCTACCATCCATGTACT	465	CCACGAATCTA	514	CCATCCATGTACT	246
668	CCACGAATCTACCATCCATGTACT	465	CCACGAATCTA	514	CCATCCATGTACT	246
669	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
670	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
671	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
672	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
673	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
674	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
675	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
676	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
677	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
678	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
679	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
680	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
681	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
682	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
683	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
684	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
685	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
686	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
687	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
688	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
689	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
690	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
691	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
692	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
693	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
694	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
695	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
696	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
697	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
698	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
699	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
700	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
701	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
702	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
703	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
704	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
705	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
706	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
707	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
708	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
709	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
710	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
711	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
712	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
713	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
714	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
715	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
716	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
717	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
718	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
719	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
720	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
721	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
722	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
723	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
724	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
725	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
726	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
727	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
728	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
729	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
730	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
731	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
732	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
733	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
734	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
735	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
736	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
737	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
738	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
739	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
740	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
741	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
742	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
743	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
744	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
745	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
746	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
747	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
748	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
749	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
750	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
751	GAATCTACCATCCATGTACT	466	GAATCTA		CCATCCATGTACT	246
752	GAATCTACCATCCATGTACT	467	GAATCTAC		CCATCCATGTACT	246
753	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
754	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
755	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
756	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
757	CGAATCTACCATCCATGTACT	468	CGAATCTAC		CCATCCATGTACT	246
758	GAATCTACCATCCATGTACT	469	GAATCTACC		CCATCCATGTACT	246
759	ACGAATCTACCATCCATGTACT	470	ACGAATCTAC	243	CCATCCATGTACT	246
760	CGAATCTACCATCCATGTACT	471	CGAATCTACC	212	CCATCCATGTACT	246
761	GAATCTACCATCCATGTACT	472	GAATCTACCC	513	CCATCCATGTACT	246
762	CACGAATCTACCATCCATGTACT	115	CACGAATCTAC	515	CCATCCATGTACT	246
763	CGAATCTACCATCCATGTACT	473	CGAATCTACCC	214	CCATCCATGTACT	246
764	GAATCTACCATCCATGTACT	474	GAATCTACCCA	218	CCATCCATGTACT	246
765	CGAATCTACATCCATGTACT	475	CGAATCTA		CATCCATGTACT	264
766	ACGAATCTACATCCATGTACT	476	ACGAATCTA		CATCCATGTACT	531
767	GAATCTACCATCCATGTACT	477	GAATCTACC		CCATCCATGTACT	532
768	CACGAATCTATCCATGTACT	478	CACGAATCTA	245	TCCATGTACT	533
769	CACGAATCTATCCATGTACT	479	CACGAATCTA	245	ATCCATGTACT	271
770	CACGAATCTATCCATGTACT	480	CACGAATCTA	245	CATCCATGTACT	303
771	GAATCTACCATCCATGTACT	481	GAATCTACCC	513	CCATCCATGTACT	301
772	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
773	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
774	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
775	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
776	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
777	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
778	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
779	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
780	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
781	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
782	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
783	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
784	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
785	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
786	GAATCTACCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246

TABLE 9-continued

Nucleobase sequences of antisense oligonucleotides provided herein						
ASO #	Nucleobase Sequence (5'→3')	SEQ ID NO:	Antisense Sequence 1 (5'→3')	SEQ ID NO:	Antisense Sequence 2 (5'→3')	SEQ ID NO:
787	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
788	GAATCTACACATCCATGTACT	482	GAATCTAC		CCATCCATGTACT	246
789	GAATCTACACATCCATGTACT	482	GAATCTAC		CCATCCATGTACT	246
790	GAATCTACACATCCATGTACT	482	GAATCTAC		CCATCCATGTACT	246
791	GAATCTACACATCCATGTACT	482	GAATCTAC		CCATCCATGTACT	246
792	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
793	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
794	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
795	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
796	GAATCTA[n3]CCATCCATGTACT	246	GAATCTA		CCATCCATGTACT	246
797	GAATCTA[n6]CCATCCATGTACT	246	GAATCTA		CCATCCATGTACT	246
798	GAATCTA[n3]CCCATCCATGTACT	534	GAATCTA		CCCATCCATGTACT	534
799	GAATCTA[n6]CCCATCCATGTACT	534	GAATCTA		CCCATCCATGTACT	534
800	GAATCTACC[n3]ATCCATGTACT	535	GAATCTACC		ATCCATGTACT	535
801	GAATCTACC[n6]ATCCATGTACT	535	GAATCTACC		ATCCATGTACT	535
802	GAATCTACC[n3]CATCCATGTACT	264	GAATCTACC		CATCCATGTACT	264
803	GAATCTACC[n6]CATCCATGTACT	264	GAATCTACC		CATCCATGTACT	264
804	GAATCTAC[n3]CATCCATGTACT	264	GAATCTAC		CATCCATGTACT	264
805	GAATCTAC[n3]CCATCCATGTACT	246	GAATCTAC		CCATCCATGTACT	246
806	GAATCTAC[n6]CCATCCATGTACT	246	GAATCTAC		CCATCCATGTACT	246
807	GAATCTACCATCCATGTACT	455	GAATCTA		CCATCCATGTACT	246
808	GAATCTACCCATCCATGTACT	461	GAATCTAC		CCATCCATGTACT	246
809	TCTACCCCCATCCATGTACT	489	TCTACCC		CCATCCATGTACT	246
810	ATCTACCCCCATCCATGTACT	490	ATCTACC		CCATCCATGTACT	246
811	GTCGCCGCCATCCATGTACT	491	GTCGCCG		CCATCCATGTACT	246
812	GGGGTCGCCATCCATGTACT	492	GGGGTCG		CCATCCATGTACT	246
813	GGGTCGCCCATCCATGTACT	493	GGGTCGC		CCATCCATGTACT	246
814	GGTCGCCCATCCATGTACT	494	GGTCGCC		CCATCCATGTACT	246

*When C is provided in a nucleobase sequence disclosed herein, without modification, such as in Table 9, it encompasses cytidine or 5'-methyl cytidine

*Each thymine base (T) in any one of the oligonucleotide sequences provided in Table 9 may independently and optionally be replaced with a uracil base (U).

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[n3] is a C3 spacer as shown in Table 16.

[n6] is a C6 spacer as shown in Table 16.

Modified Antisense Oligonucleotides

[0191] Some aspects of the present disclosure provide modified (e.g., chemically modified) antisense oligonucleotides. For example, any one of the antisense oligonucleotides provided herein (e.g., in Table 3 or Table 9) may

comprise one or more chemical modifications including, e.g., one or more modified nucleosides and/or one or more modified nucleosides and/or modified internucleoside linkages.

[0192] In some embodiments, an antisense oligonucleotide disclosed herein, comprises a first antisense sequence

and a second antisense sequence, and comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, or more) modified nucleosides and/or one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) modified internucleoside linkages. In some embodiments, a modified nucleoside comprises a modification at the 2' position of the sugar (referred to herein as a "2'-modified nucleoside"). In some embodiments, a 2'-modified nucleoside is a not bicyclic (referred to herein as "non-bicyclic 2'-modified nucleoside"). In some embodiments, a 2'-modified nucleoside comprises a modification at the 2' position and 4' position of the sugar creating a bridged structure (referred to herein as a "2'-4' bridged nucleoside"). In some embodiments, the one or more modified nucleosides comprise a non-bicyclic 2'-modified nucleoside, a 2'-4' bridged nucleoside, or combinations thereof. In some embodiments, all modified nucleosides in an antisense oligonucleotide are 2'-modified nucleosides. In some embodiments, the non-bicyclic 2'-modified nucleoside is a 2'-O-methoxyethyl (2'-MOE) modified nucleoside, a 2'-O-Methyl (2'-O-Me) nucleoside, or a 2'-fluoro (2'-F) modified nucleoside, and/or the 2'-4' bridged nucleoside is a locked nucleic acid (LNA, 2'-4' methylene bridge), an ethylene-bridged nucleic acid (ENA, 2'-4' ethylene bridge) or an constrained ethyl nucleic acid (cEt, 2'-4' ethylene bridge). In some embodiments, the 2'-modified nucleoside is a 2'-O, 4'-C-Spirocyclopropylene bridged nucleic acid (scpBNA). In some embodiments, the 2'-modified nucleoside is a 2'-O—N-methylacetamide (2'-O-NMA) modified nucleoside. In some embodiments, the 2'-modified nucleoside is an amide-bridged nucleic acid (AmNA). In some embodiments, the 2'-modified nucleoside is a 2'-O, 4'-C-aminomethylene bridged nucleic acid (BNA (NC)). In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more 2'-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more LNAs. In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more 2'-MOE modified nucleosides and one or more LNAs. In some embodiments, each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside.

[0193] In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the following 2'-modified nucleosides: 2'-O-Me, 2'-F, 2'-MOE, LNA, ENA, cEt, scpBNA, 2'-O-NMA, AmNA, and BNA (NC). In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the following 2'-modified nucleosides: ENA, cEt, scpBNA, 2'-O-NMA, AmNA, and BNA (NC). In some embodiments, each nucleoside in an antisense oligonucleotide disclosed herein is a modified nucleoside (e.g., 2'-modified nucleoside). In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an ENA. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are cEt. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are a scpBNA. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are 2'-O-NMA modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an

AmNA. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are a BNA (NC). In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are selected from the following 2'-modified nucleosides: 2'-O-Me, 2'-F, ENA, cEt, scpBNA, 2'-O-NMA, AmNA, and BNA (NC) and the remaining modified nucleosides are 2'-MOE modified nucleosides, LNAs, or a mix of 2'-MOE modified nucleosides and LNAs. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are selected from the following 2'-modified nucleosides: ENA, scpBNA, 2'-O-NMA, AmNA, and BNA (NC) and the remaining modified nucleosides are 2'-MOE modified nucleosides, LNAs, or a mix of 2'-MOE modified nucleosides and LNAs. In some embodiments, the remaining modified nucleosides are 2'-MOE modified nucleosides. In some embodiments, the remaining modified nucleosides are LNAs. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an AmNA and the remaining modified nucleosides are a mix of 2'-MOE modified nucleosides and LNAs. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an ENA and the remaining modified nucleosides are 2'-MOE modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are a scpBNA and the remaining modified nucleosides are 2'-MOE modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an AmNA and the remaining modified nucleosides are 2'-MOE modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are a BNA (NC) and the remaining modified nucleosides are 2'-MOE modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are 2'-O-NMA modified nucleosides and the remaining modified nucleosides are LNAs.

[0194] In some embodiments, an antisense oligonucleotide disclosed herein, comprises a first antisense sequence and a second antisense sequence, and comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) modified internucleoside linkages. In some embodiments, each internucleoside linkage of the antisense oligonucleotide is a modified internucleoside linkage. Non limiting examples of internucleoside linkages include phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates comprising 3' alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates comprising 3'-amino phosphoramidate and aminoalkylphosphoramidates, mesyl phosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and boranophosphates. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. It is to be understood that, in any one of the antisense oligonucleotides disclosed herein, the rest of internucleoside linkages, unless otherwise specified, are all phosphodiester internucleoside linkages. In some embodiments, each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage. In some embodiments, the antisense oligonucleotide comprises a mix of phosphodiester

linkages and phosphorothioate linkages. In some embodiments, the antisense oligonucleotide comprises one or more phosphorodiamidate morpholinos. In some embodiments, the antisense oligonucleotide is a phosphorodiamidate morpholino oligomer (PMO).

[0195] In some embodiments, an antisense oligonucleotide disclosed herein, comprises a first antisense sequence and a second antisense sequence, wherein each cytosine of the oligonucleotide is a 5-methyl-cytosine. In some embodiments, each cytosine of the oligonucleotide is not a 5-methyl-cytosine. In some embodiments, one or more of the cytosines of the oligonucleotide is a 5-methyl-cytosine.

[0196] Table 4 lists various exemplary nucleosides with 3'-phosphate or 3'-phosphorothioate and the structures thereof.

TABLE 4		
Exemplary nucleosides with 3'-phosphate or 3'-phosphorothioate and structures		
Abbreviation	Nucleosides or linkage	Structure
1A	2'-O,4'-C-methylene bridge adenosine-3'-phosphate	
1As	2'-O,4'-C-methylene bridge adenosine-3'-phosphorothioate	
1C	2'-O,4'-C-methylene bridge cytidine-3'-phosphate	

TABLE 4-continued		
Exemplary nucleosides with 3'-phosphate or 3'-phosphorothioate and structures		
Abbreviation	Nucleosides or linkage	Structure
1Cs	2'-O,4'-C-methylene bridge cytidine-3'-phosphorothioate	
1 ² C	2'-O,4'-C-methylene bridge-5'-methyl cytidine-3'-phosphate	
1 ⁵ Cs	2'-O,4'-C-methylene bridge-5'-methylcytidine-3'-phosphorothioate	

TABLE 4-continued

Exemplary nucleosides with 3'-phosphate or 3'-phosphorothioate and structures		
Abbreviation	Nucleosides or linkage	Structure
IG	2'-O,4'-C-methylene bridge guanosine-3'-phosphate	
IGs	2'-O,4'-C-methylene bridge guanosine-3'-phosphorothioate	
IU	2'-O,4'-C-methylene bridge uridine-3'-phosphate	
IUs	2'-O,4'-C-methylene bridge uridine-3'-phosphorothioate	

TABLE 4-continued

Exemplary nucleosides with 3'-phosphate or 3'-phosphorothioate and structures		
Abbreviation	Nucleosides or linkage	Structure
IT	2'-O,4'-C-methylene bridge thymidine-3'-phosphate	
ITs	2'-O,4'-C-methylene bridge thymidine-3'-phosphorothioate	
iA	2'-O-methoxyethyl-adenosine-3'-phosphate	
iAs	2'-O-methoxyethyl-adenosine-3'-phosphorothioate	

TABLE 4-continued

Exemplary nucleosides with 3'-phosphate or 3'-phosphorothioate and structures		
Abbreviation	Nucleosides or linkage	Structure
iC	2'-O-methoxyethyl-cytidine-3'-phosphate	
iCs	2'-O-methoxyethyl-cytidine-3'-phosphorothioate	
i ⁵ C	2'-O-methoxyethyl-5-methylcytidine-3'-phosphate	
i ⁵ Cs	2'-O-methoxyethyl-5-methylcytidine-3'-phosphorothioate	

TABLE 4-continued

Exemplary nucleosides with 3'-phosphate or 3'-phosphorothioate and structures		
Abbreviation	Nucleosides or linkage	Structure
iG	2'-O-methoxyethyl-guanosine-3'-phosphate	
iGs	2'-O-methoxyethyl-guanosine-3'-phosphorothioate	
iU	2'-O-methoxyethyl-uridine-3'-phosphate	
iUs	2'-O-methoxyethyl-uridine-3'-phosphorothioate	

TABLE 4-continued

Exemplary nucleosides with 3'-phosphate or 3'-phosphorothioate and structures		
Abbreviation	Nucleosides or linkage	Structure
iT	2'-O-methoxyethyl-thymidine-3'-phosphate	
iTs	2'-O-methoxyethyl-thymidine-3'-phosphorothioate	
s	Phosphorothioate linkage	

TABLE 10-continued

Additional exemplary modified nucleosides		
Abbreviation	Name	Structure
b	BNA(NC)	
ena	ENA	
scp	scpBNA	
nma	2'-O-NMA	

TABLE 10

Additional exemplary modified nucleosides		
Abbreviation	Name	Structure
z	AmNA	

**“base” represents any nucleobase or derivative thereof

TABLE 16

Additional exemplary abbreviations		
Abbreviation	Name	Structure
x	C16 lipid (6-[(1-oxohexadecyl)amino]hexyl]phosphoryl)	RO is 5' terminal of oligonucleotides
n3	C3 spacer	
n6	C6 spacer	

[0197] In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more modified nucleosides (e.g., one or more 2'-modified nucleosides). In some embodiments, the antisense oligonucleotide comprises a non-bicyclic 2' modified nucleoside, a 2'-4' bridged nucleoside, or combinations thereof (e.g., any combination of 2'-O-methoxyethyl (2'-MOE) modified nucleoside, 2'-O-Methyl (2'-O-Me) modified nucleoside, 2'-fluoro (2'-F) modified nucleosides, LNA, ENA, and cEt). In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more modified nucleosides (e.g., one or more 2'-modified nucleosides). In some embodiments, the antisense oligonucleotide comprises a non-bicyclic 2'-modified nucleoside, a 2'-4' bridged nucleoside, or combinations thereof (e.g., any combination of 2'-MOE modified nucleoside, 2'-O-Me modified nucleoside, 2'-F modified nucleosides, LNA, ENA, cEt, AmNA, BNA (NC), scpBNA, and 2'-O-NMA). In some embodiments, the antisense oligonucleotide comprises a combination of 2'-MOE modified nucleosides and LNAs. In some embodiments, the antisense oligonucleotide comprises a combination of 2'-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) ENA. In some embodiments, the antisense oligonucleotide comprises a combination of 2'-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) AmNA. In some embodiments, the antisense oligonucleotide comprises a combination of 2'-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) BNA (NC). In some embodiments, the antisense oligonucleotide comprises a combination of 2'-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) scpBNA. In some embodiments, the antisense oligonucleotide comprises a combination of 2'-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) 2'-O-NMA modified nucleosides. In some embodiments, the antisense oligonucleotide comprises a combination of LNAs and one or more (e.g., 1, 2, 3, or 4) 2'-O-NMA modified nucleosides. In some embodiments, the antisense oligonucleotide comprises a combination of 2'-MOE modified nucleosides, LNAs, and one or more (e.g., 1, 2, 3, or 4) AmNA. In some embodiments, each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside. In some embodiments, the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) positions of the antisense

oligonucleotide are nucleosides with the same 2' chemistry in the sugar moiety selected from, e.g., 2'-MOE modified nucleoside, 2'-O-Methyl (2'-O-Mc) modified nucleoside, 2'-fluoro (2'-F) modified nucleoside, LNA, ENA, and cEt, and the remaining nucleosides are nucleosides with a different 2' chemistry in the sugar moiety, selected from e.g., 2'-MOE modified nucleoside, 2'-O-Methyl (2'-O-Me) modified nucleoside, 2'-fluoro (2'-F) modified nucleoside, LNA, ENA, and cEt. In some embodiments, the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and the remaining nucleosides are nucleosides with different 2' chemistry in the sugar moiety, e.g., 2'-MOE, 2'-O-Methyl (2'-O-Mc), 2'-fluoro (2'-F), ENA, or cEt. In some embodiments, the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides.

[0198] In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2'-4' bridged nucleoside (e.g., LNA, ENA, or cEt). In some embodiments, the remaining nucleosides are non-bicyclic 2' modified nucleoside (e.g., 2'-MOE modified nucleoside, 2'-O-Methyl (2'-O-Me) modified nucleoside, 2'-fluoro (2'-F) modified nucleoside). In some embodiments, at least one of the one or more motifs is located at the 5' end of the antisense oligonucleotide. In some embodiments, at least one of the one or more motifs is located at the 3' end of the antisense oligonucleotide. In some embodiments, at least one of the one or more motifs is located within any of positions 9-16 (counting 5'→3') of the antisense oligonucleotide.

[0199] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length and comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive LNAs at the 5' end of the antisense oligonucleotide, the 3' end of the antisense oligonucleotide, and/or within any of positions 9-16 (counting 5'→3') (e.g., 9-10, 9-12, 10-11, 10-12, 10-13, 10-15, 10-16, 11-12, 11-14, 11-16, 12-13, 12-15, and 13-15), and all of the remaining nucleosides are 2'-MOE modified nucleosides.

[0200] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside.

[0201] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length, wherein the nucleosides at one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8 or more) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-O-NMA modified nucleosides.

[0202] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside.

[0203] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length, wherein one or more nucleobases at the 3' end of the first antisense sequence is complementary to one or more nucleobases at the 3' end of the second target region, and/or one or more nucleobases at the 5' end of the second antisense sequence is complementary to one or more nucleobases at the 5' end of the first target region.

[0204] In some embodiments, an antisense oligonucleotide disclosed herein is 28 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0205] In some embodiments, an antisense oligonucleotide disclosed herein is 28 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0206] In some embodiments, an antisense oligonucleotide disclosed herein is 27 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0207] In some embodiments, an antisense oligonucleotide disclosed herein is 27 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0208] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length,

wherein the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length and comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2'-4' bridged nucleoside (e.g., LNA, ENA, or cEt) within positions 1-6, 10-17, and/or 22-26 (counting 5'→3'). In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0209] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 11-16 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0210] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 1-4 and 23-26 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0211] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 1-4, 12-15, and 23-26 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0212] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 1-3, 11-16, and 24-26 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0213] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 1-4 and 11-16 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0214] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 11-16 and 23-26 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0215] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside link-

ages. In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0216] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages.

[0217] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises two phosphodiester internucleoside linkages and all of the remaining internucleoside linkages are phosphorothioate internucleoside linkages.

[0218] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises five phosphodiester internucleoside linkages and all of the remaining internucleoside linkages are phosphorothioate internucleoside linkages.

[0219] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0220] In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0221] In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0222] In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and the antisense oligonucleotide

comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages.

[0223] In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0224] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2'-4' bridged nucleoside (e.g., LNA, ENA, or cEt) within positions 1-5, 10-16, and/or 20-24. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0225] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 4, 8, 12, 16, 20, and 24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0226] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 10-15 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0227] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1-4, 11-14, and 21-24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0228] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1-4, 11-14, and 21-24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein the antisense oligonucleotide comprises two phosphodiester internucleoside linkages and all of the remaining internucleoside linkages are phosphorothioate internucleoside linkages.

[0229] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 4, 11-14, 21, and 24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0230] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 4, 12, 13, 21, and 24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0231] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 2, 12, 13, 23, and 24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0232] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 2, 7, 8, 17, 18, 23, and 24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0233] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 13-15 and 22-24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0234] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1-3 and 10-12 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0235] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 6, 12, 13, 19, and 24 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0236] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0237] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and the antisense oligonucleotide

comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21 or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 or more) phosphodiester internucleoside linkages.

[0238] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0239] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length and comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2'-4' bridged nucleoside (e.g., LNA, ENA, or cEt) within positions 1-5, 9-14, and/or 20-22. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0240] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at positions 1, 2, 10-13, 21, and 22 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0241] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at positions 1, 2, 10-13, 21, and 22 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and five or more (e.g., 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages.

[0242] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at positions 1, 2, 10-13, 21, and 22 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and twelve or less (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12) phosphodiester internucleoside linkages.

[0243] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein each

8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length and comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2'-4' bridged nucleoside (e.g., LNA, ENA, or cEt) within positions 1-5, 9-13, and/or 16-20. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0246] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 19, and 20 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0247] In some embodiments, an antisense oligonucleotide disclosed herein is 21 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) phosphodiester internucleoside linkages.

[0248] In some embodiments, an antisense oligonucleotide disclosed herein is 21 nucleosides in length, wherein the nucleosides at positions 1 and 21 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-O-NMA modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0249] In some embodiments, an antisense oligonucleotide disclosed herein is 21 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0250] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 19, and 20 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein the antisense oligonucleotide comprises two phosphodiester internucleoside linkages and all of the remaining internucleoside linkages are phosphorothioate internucleoside linkages.

[0251] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 9, and 10 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0252] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 9-12 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining

9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7,

nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0253] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 11, 12, 19, and 20 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0254] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 10, 11, 19, and 20 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0255] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 4, and 5 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0256] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 16, 17, 19, and 20 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0257] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17 or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9 or more) phosphodiester internucleoside linkages.

[0258] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1 and 20 (counting 5'→3') of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2'-O-NMA modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0259] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0260] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage. In some embodiments, an antisense oligonucleotide disclosed

herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage, wherein one or more nucleobases at the 3' end of the first antisense sequence is complementary to one or more nucleobases at the 3' end of the second target region, and/or one or more nucleobases at the 5' end of the second antisense sequence is complementary to one or more nucleobases at the 5' end of the first target region.

[0261] In some embodiments, an antisense oligonucleotide disclosed herein comprises any one of the nucleobase sequences described herein (e.g., listed in Table 3 and Table 9) and further comprises modified nucleosides selected from one of the following patterns (5'→3'):

MMMMMMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMLLLLLMMMMMMMMMM,
MMMMMMMMMMLLLLLMMMMMMMMMM,
LLLLMMMMMMLLLLLMMMMMMMMMM,
LLLLMMMMMMLLLLLMMMMMMLLL,
LLLLMMMMMMLLLLLMMMMMMLLL,
LLLLMMMMMMLLLLLMMMMMMLLL,
LLLLMMMMMMLLLLLMMMMMMLLL,
MMMMMMLMMLMMLMMLMMLMMLMML,
LMMLMMMMMMLLLLLMMMMMMLMML,
LMMLMMMMMMLLLLLMMMMMMLMML,
LLMMMMMMMMMMLMMMMMMMMMMLL,
LLMMMMMMLMMMMMMMMMMLMMMMMLL,
MMMMMMMMMMMMMLLLMMMMMMLLL,
LLLLMMMMMMLLLLLMMMMMMMMMM,
LMMMMLMMMMMMLMMMMMMLMMMMML,
LLMMMMMMMLLLLLMMMMMMMLL,
LLMMMMMMMMMMMMMMMMMMLL,
LLMMMMMMLMMMMMMMMMM,
MMMMMMMMMLLLLLMMMMMM,
MMMMMMMMMMLLLLLMMMMMLL,

-continued
LLMMMMMMMLLLMMMMMMMLL,

LLMLLLMMMMMMMMMMMMMM,
and

MMMMMMMMMMMMMMMLLMLL,

wherein M is a 2'-O-methoxyethyl (2'-MOE)
modified nucleoside and L is an LNA.

[0262] In some embodiments, an antisense oligonucleotide disclosed herein comprises any one of the nucleobase sequences described herein (e.g., listed in Table 3 and Table 9) and further comprises modified nucleosides selected from one of the following patterns (5'→3'):

MLMMMMMMMMMMMMMMMMMMMLL,

MMLMMMMMMMLLMMMMMMMMMMMMMM,

MMMMMLLMMMMMMMMMMMMMMMMMM,

MMMMMLLMMMMMMMMMMMLLMMMMMM,

MMMMMMMMMMLLLMMMMMMMMMMMMMM,

MLMMMMMMMMMMMMMMMMMMMMMMMMMM,

LLMMMMMMMMMMLLMMMMMMMLL,

LLMMMMMMMMMMLLLMMMMMMMLLLL,

LLLLMMMMMMMLLLMMMMMMMLL,

LLLLMMMMMMMLLMMMMMMMLLLL,

LLLMMMMMMMLLLMMMMMMMLLL,

LLLMMMMMMMMLLMMMMMMMLLLL,

LLMLLMMMMMMLLLMMMMMMMLLLL,

LLMMMMMMMMMMMMMMMMMMMMMMMMMM,

MMLMMMMMMMMMMMMMMMMMMMMMMMMMM,

MMMMLLMMMMMMMMMMMMMMMMMMMMMM,

MMMMMMLLMMMMMMMMMMMMMMMMMMMMMM,

MMMMMMMMMLLMMMMMMMMMMMMMMMMMM,

MMMMMMMMMMLLMMMMMMMMMMMMMMMMMM,

MMMMMMMMMMMLLMMMMMMMMMMMMMMMMMM,

MMMMMMMMMMMMMMLLMMMMMMMMMMMMMM,

MMMMMMMMMMMMMMMLLMMMMMMMMMM,

MMMMMMMMMMMMMMMLLMMMMMM,

MMMMMMMMMMMMMMMLLMMMMMM,

MMMMMMMMMMMMMMMLLMMMMMM,

MMMMMMMMMMMMMMMLLMMMMMM,

MMMMMMMMMMMMMMMLLMMMMMM,

MMMMMMMMMMMMMMMLLMMMMMM,

MMMMMMMMMLLMMMMMMLLMMMMMMMMMM,

-continued
LMMMMMMMMMMMMMMMMMMMLL,

LLMMMMMMMMMMMMMMMMMMMMMMMLL,

MMMMMMMMMLLMMMMMMMMMMMMMM,

LLMMMLLMMMMMMMMMMMLL,

LLMMMMMLLMMMMMMMMMMMLL,

LLMMMMMLLMMMMMMMMMMMLL,

LLMMMMMMMMMMLLMMMMMMMLL,

LMMMMMMMMMMMMMMMMMMMLL,

LMMMMMLLMMMMMMMMMMMLL,

LLMMMMMMMMMMMMMMMLL,

LMMMMMLLMMMMMMMMMMMLL,

LMMMMMLLMLLMMMMMMMMMMMLL,

MMLMMMMLMMMMMMMLLMMMLMM,

LMMMMMLLMMMMMMMLLMMMMMLL,

MMLLMMMMMMMMMMMLLMMMLLMM,

MMMMMMMLLMMMLLMMMMMMMLL,

LLMMMMMMMMMMMMMMMMMMMMMM,

MMLLMMMMMMMMMMMMMMMMMM,

MMMMMLLMMMMMMMMMMMMMM,

MMMMMMMLLMMMMMMMMMMMMMM,

MMMMMMMMMLLMMMMMMMMMMMMMM,

MMMMMMMMMMMLLMMMMMMMMMM,

MMMMMMMMMMMLLMMMMMM,

MMMMMMMMMMMMMMMLLMMMLL,

MMMMMMMMMMMMMMMLLMMMLL,

MMMMMMMMMMMMMMMLLMMMLL,

LLMMMMMMMMMMMMMMMMMMMMMM,

MMLLMMMMMMMMMMMMMMMMMM,

MMMMMLLMMMMMMMMMMMMMM,

MMMMMMMLLMMMMMMMMMMMMMM,

MMMMMMMMMMMLLMMMMMMMMMM,

MMMMMMMMMMMLLMMMMMM,

MMMMMMMMMMMMMMMLLMMMLL,

MMMMMMMMMMMMMMMLLMMMLL,

MMMMMMMMMMMMMMMLLMMMLL,

MMMMMMMMMMMMMMMLLMMMLL,

MLMMMMMMMMMMMMMMMLLMMMLL,

MMMLLMMMMMMMMMMMLLMMMLL,

MMMMMLLMMMMMMMMMMMLLMMMLL,

L - nma - nma - nma - nma - nma - nma - nma - nma -
 nma - nma - nma - nma - nma - nma - nma - nma -
 nma - nma - L ,
 L - nma - nma - nma - nma - nma - nma - nma - nma -
 nma - nma - nma - nma - nma - nma - nma - nma -
 nma - nma - nma - L .

nma-nma-nma-nma-nma-nma-nma-nma-nma-nma-
nma-nma-nma-nma-nma-nma-nma-nma-
nma-nma-nma-nma-nma-nma-nma-nma-nma,
xdddMMMMMMMMMMMMMMMMMMMMMMMMMM,
zzMMMMMMzZZZMMMMMMzL,
zzzzMMMMMzzzzMMMMMMzzL,
enaMMMMMMMMMMMMMMMMMMMMMena,
bMMMMMMMMMMMMMMMMMMMMMb,
enaMMMMMMMMMMMMMMMMMMMMMena,
scpMMMMMMMMMMMMMMMMMMMMMscp,
bMMMMMMMMMMMMMMMMMMMMMb,
LMMMMMMMMMMMMMMMMMLM,
MzMMMMMMMMMMMMMMMMMzM,
enaMMMMMMMMMMMMMMMMMMMMMenaM,
MenaMMMMMMMMMMMMMMMMMMMMMenaM,
MenaMMMMMMMMMMMMMMMMMMMMMena,
MzMMMMMMMMMMMMMMMMMMMMMzM,
MspMMMMMMMMMMMMMMMMMMMMMsp,
enaMMMMMMMMMMMMMMMMMMMMMenaM,
MenaMMMMMMMMMMMMMMMMMMMMMenaM,
MenaMMMMMMMMMMMMMMMMMMMMMena,
xdddLMMMMMMMMMMMMMMMMMMMML,
xdddLMMMMMMMMMMMMMMMMMMMML,
MMMMMMMMMMMMMM[n]MMMMMMMMMMMMMM,
LMMMMMM[n]MMMMMMMMMMMMMML,
LMMMMMM[n]MMMMMMMMMMMMMMML,
LMMMMMMMM[n]MMMMMMMMMMMMML,
LMMMMMMMM[n]MMMMMMMMMMMMML,
and

LMMMMMMMM[n]MMMMMMMMMMMMML,
wherein M is a 2'-O-methoxyethyl (2'-MOE) modified nucleoside, L is an LNA, x is a C16 lipid, 6-[(1-oxohexadecyl)amino]hexyl phosphoryl, [n] is a C3 or C6 spacer (e.g., a C3 or C6 spacer as shown in Table 16), dis DNA, ena is ENA, b is BNA(NC), z is AmNA, nma is 2'-NMA modified nucleoside, and scp is scpBNA.

S-
S-S-S-S.

-continued

[illegible]

[0272] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications.

[0279] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2'-O-NMA modified nucleoside and the antisense oligonucleotide comprises modified internucleoside linkages

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
7	CACGAATCTACCCGGATAAGAGTTCT	7	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iT]	579	
8	CCACGAATCTACCCGGATAAGAGTTCT	8	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iT]	580	
9	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	581	
10	GAATCTACCCACCGGATAAGAGTTC	10	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	582	
11	CGAATCTACCCACCGGATAAGAGTTC	11	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	583	
12	ACGAATCTACCCAGGATAAGAGTTC	12	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	584	
13	CACGAATCTACCCGGGATAAGAGTTC	13	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	585	
14	CCACGAATCTACCCGGGATAAGAGTTC	14	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	586	
15	CTAGCCACGAATCGGGATAAGAGTTC	15	[i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	587	
16	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	588	
17	GCATTTATTCAACGGGGATAAGAGTT	17	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	589	
18	CCACCAACTCATCGGGGATAAGAGTT	18	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	590	
19	GAATCTACCCACCGGGGATAAGAGTT	19	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	591	
20	CGAATCTACCCACGGGGATAAGAGTT	20	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	592	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
21	CACGAATCTACCCGGGATAAGAGTT	21	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT]	593	
22	CCACGAATCTACCCGGGATAAGAGTT	22	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT]	594	
23	GCCACGAATCTACGGGGATAAGAGTT	23	[iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT]	595	
24	CGAATCTACCCACCTGGGGATAAGA	24	[i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iT s][iGs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iA]	596	
25	ACGAATCTACCCATCCTGGGGATAAG	25	[iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT s][i ⁵ Cs][i ⁵ Cs][iT s][iGs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iG]	597	
26	CGAATCTACCCACGTTCTGGGGATA	26	[i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iT s][iT s][i ⁵ Cs][i ⁵ Cs][iT s][iGs][iGs][iGs][iGs][iAs][iT s][iA]	598	
27	CACGAATCTACCCGTTCTGGGGATA	27	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iGs][iT s][iT s][i ⁵ Cs][i ⁵ Cs][iT s][iGs][iGs][iGs][iGs][iAs][iT s][iA]	599	
28	CTAGCCACGAATCGTTCTGGGGATA	28	[i ⁵ Cs][iT s][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iGs][iT s][iT s][i ⁵ Cs][i ⁵ Cs][iT s][iGs][iGs][iGs][iAs][iT s][iA]	600	
29	CGAATCTACCCACTAGTTCTGGGGA	29	[i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iT s][iAs][iGs][iT s][iT s][i ⁵ Cs][i ⁵ Cs][iT s][iGs][iGs][iGs][iGs][iA]	601	
30	CTAGCCACGAATCTAGTTCTGGGGA	30	[i ⁵ Cs][iT s][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][iGs][iT s][iT s][i ⁵ Cs][i ⁵ Cs][iT s][iT s][iGs][iGs][iGs][iGs][iA]	602	
31	ACGAATCTACCCACTAGTTCTGGGG	31	[iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT s][i ⁵ Cs][i ⁵ Cs][iT s][iGs][iGs][iGs][iGs][iG]	603	
32	ATCTAGCCACGAAGTAGTTCTGGGG	32	[iAs][iT s][i ⁵ Cs][iT s][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][i ⁵ Cs][iT s][iAs][iGs][iT s][iT s][i ⁵ Cs][i ⁵ Cs][iT s][iT s][iGs][iGs][iGs][iG]	604	
33	GAATCTACCCACTCTGTTCAATCA	33	[iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iT s][i ⁵ Cs][iT s][iGs][iT s][iT s][i ⁵ Cs][iA s][iAs][iT s][i ⁵ Cs][iA]	605	
34	CGAATCTACCCATCTGTTCAATCA	34	[i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT s][i ⁵ Cs][iT s][iGs][iT s][iT s][i ⁵ Cs][iA s][iAs][iT s][i ⁵ Cs][iA]	606	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
35	GAATCTACCCACATCTGTTCAATC	35	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	607	
36	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	608	
37	GAATCTACCCACTCTGTTCAATCA	33	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs]	609	
38	CGAATCTACCCATCTGTTCAATCA	34	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs]	610	
39	GAATCTACCCACATCTGTTCAATC	35	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	611	
40	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	612	
41	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ Cs][iA][i ⁵ C][iG][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iA][iT][iG][iA][iTs][i ⁵ C]	613	
42	CACGAATCTACCCCATCCATGTACTC	3	[i ⁵ Cs][iA][i ⁵ C][iG][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iT][iG][iT][iA][i ⁵ C][iTs][i ⁵ C]	614	
43	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ Cs][i ⁵ C][iA][iT][i ⁵ C][i ⁵ C][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iG][iGs][iGs][iAs][iTs][iAs][iA][iG][iA][iG][iT][iTs][i ⁵ C]	615	
44	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iG][iT][iT][i ⁵ C][iA][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iA][iG][iA][iG][iT][iTs][i ⁵ C]	616	
45	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ Cs][iA][i ⁵ C][iG][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C][i ⁵ C][i ⁵ C][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iA][iT][iG][iT][iA][i ⁵ Cs][iT]	617	
46	CACGAATCTACCCCATCCATGTACTC	3	[i ⁵ Cs][iA][i ⁵ C][iG][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C][i ⁵ C][i ⁵ C][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iT][iG][iT][iA][i ⁵ C][iTs][i ⁵ C]	618	
47	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ Cs][i ⁵ C][iA][iT][i ⁵ C][i ⁵ C][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ C][i ⁵ C][iG][iGs][iGs][iAs][iTs][iAs][iA][iG][iA][iG][iT][iTs][i ⁵ C]	619	
48	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iG][iT][iT][i ⁵ C][iA][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ C][iG][iGs][iGs][iAs][iTs][iAs][iA][iG][iA][iG][iT][iTs][i ⁵ C]	620	
49	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iA][i ⁵ C][i ⁵ C][i ⁵ C][i ⁵ C]	621	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			C[[i ⁵ C][iA][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iT]		
50	CACGAATCTACCCCATCCATGTACTC	3	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA][i ⁵ C][i ⁵ C][i ⁵ C][iA][iT][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][i ⁵ C]	622	
51	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA][iT][i ⁵ C][i ⁵ C][iG][iG][iG][iA <i>s</i>][iT <i>s</i>][iA <i>s</i>][iA <i>s</i>][iG <i>s</i>][iA <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	623	
52	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA][iT][iT][i ⁵ C][iG][iG][iG][iA <i>s</i>][iT <i>s</i>][iA <i>s</i>][iA <i>s</i>][iG <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	624	
53	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iT]	625	
54	CACGAATCTACCCCATCCATGTACTC	3	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][i ⁵ C]	626	
55	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	627	
56	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	628	
57	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	629	
58	CACGAATCTACCCCATCCATGTACTC	3	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	630	
59	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	631	
60	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	632	
61	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	633	
62	CACGAATCTACCCCATCCATGTACTC	3	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	634	
63	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	635	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			s][lGs][lGs][iGs][iAs][iT s][iAs][iA s][iGs][iAs][lGs][lTs][lTs][1 ³ C]		
64	TGTTCAATCATTCGGGATAAGAGTTC	9	[lTs][lGs][lTs][lTs][i ⁵ Cs][iAs][iAs][iT s][i ³ Cs][iAs][iT s][lTs][1 ³ Cs][lGs][lGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][lGs][lTs][lTs][1 ³ C]	636	
65	CACGAATCTACCCCATCCATGTACT	1	[1 ⁵ Cs][lAs][1 ⁵ Cs][iGs][iAs][iAs][iT s][i ³ Cs][iT s][iAs][1 ³ Cs][1 ³ Cs][1 ³ Cs][1 ³ Cs][lAs][iT s][i ³ Cs][i ³ Cs][iAs][iT s][iGs][iT s][lAs][1 ³ Cs][lT]	637	
66	CACGAATCTACCCCATCCATGTACTC	3	[1 ⁵ Cs][lAs][1 ⁵ Cs][iGs][iAs][iAs][iT s][i ³ Cs][iT s][iAs][1 ³ Cs][1 ³ Cs][1 ³ Cs][lAs][lTs][i ³ Cs][i ³ Cs][iAs][iT s][iGs][iT s][iAs][1 ⁵ Cs][lTs][1 ³ C]	638	
67	CCATCCATCATCCGGGATAAGAGTTC	16	[1 ⁵ Cs][1 ³ Cs][lAs][iT s][i ⁵ Cs][i ⁵ Cs][iAs][iT s][i ³ Cs][iAs][lTs][1 ³ Cs][1 ³ Cs][lGs][lGs][lGs][iAs][iT s][iAs][iA s][iGs][iAs][iGs][lTs][lTs][1 ³ C]	639	
68	TGTTCAATCATTCGGGATAAGAGTTC	9	[lTs][lGs][lTs][iT s][i ³ Cs][iAs][iAs][iT s][i ³ Cs][iAs][lTs][lTs][1 ³ Cs][lGs][lGs][lGs][iAs][iT s][iAs][iAs][iGs][iAs][lTs][lTs][1 ³ C]	640	
69	CACGAATCTACCCCATCCATGTACT	1	[1 ⁵ Cs][lAs][1 ⁵ Cs][lGs][iAs][iAs][iT s][i ³ Cs][iT s][iAs][1 ³ Cs][1 ³ Cs][1 ³ Cs][1 ³ Cs][1 ³ Cs][lAs][iT s][i ³ Cs][i ³ Cs][iAs][iT s][iGs][iT s][iAs][i ³ Cs][lT]	641	
70	CACGAATCTACCCCATCCATGTACTC	3	[1 ⁵ Cs][lAs][1 ⁵ Cs][lGs][iAs][iAs][iT s][i ³ Cs][iT s][iAs][1 ³ Cs][1 ³ Cs][1 ³ Cs][1 ³ Cs][lAs][lTs][i ³ Cs][i ³ Cs][iAs][iT s][iGs][iT s][iAs][i ³ Cs][iT s][i ³ C]	642	
71	CCATCCATCATCCGGGATAAGAGTTC	16	[1 ⁵ Cs][1 ³ Cs][lAs][lTs][i ⁵ Cs][i ⁵ Cs][iAs][iT s][i ³ Cs][iAs][lTs][1 ³ Cs][1 ³ Cs][lGs][lGs][lGs][iAs][iT s][iAs][iA s][iGs][iAs][iGs][iT s][iT s][i ³ C]	643	
72	TGTTCAATCATTCGGGATAAGAGTTC	9	[lTs][lGs][lTs][lTs][i ³ Cs][iAs][iAs][iT s][i ³ Cs][iAs][lTs][lTs][1 ³ Cs][lGs][lGs][lGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT s][i ³ C]	644	
73	CACGAATCTACCCCATCCATGTACT	1	[1 ⁵ Cs][iAs][i ³ Cs][iGs][iAs][iAs][iT s][i ³ Cs][iT s][iAs][1 ³ Cs][1 ³ Cs][1 ³ Cs][1 ³ Cs][1 ³ Cs][lAs][iT s][i ³ Cs][i ³ Cs][iAs][iT s][iGs][lTs][lAs][1 ³ Cs][lT]	645	
74	CACGAATCTACCCCATCCATGTACTC	3	[1 ⁵ Cs][iAs][i ³ Cs][iGs][iAs][iAs][iT s][i ³ Cs][iT s][iAs][1 ³ Cs][1 ³ Cs][1 ³ Cs][1 ³ Cs][lAs][lTs][i ³ Cs][i ³ Cs][iAs][iT s][iGs][iT s][lAs][1 ⁵ Cs][lTs][1 ³ C]	646	
75	CCATCCATCATCCGGGATAAGAGTTC	16	[1 ⁵ Cs][i ³ Cs][iAs][iT s][i ³ Cs][i ³ Cs][iAs][iT s][i ³ Cs][1 ³ Cs][1 ³ Cs][lGs][lGs][lGs][iAs][iT s][iAs][iA s][iGs][iAs][lGs][lTs][lTs][1 ³ C]	647	
76	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT s][iGs][iT s][iT s][i ⁵ Cs][iAs][iAs][iT s][i ³ Cs][iAs][lTs][lTs][1 ³ Cs][lGs][lGs][lGs][iAs][iT s][iAs][iAs][iGs][iAs][lGs][lTs][lTs][1 ³ C]	648	
77	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iT s][iAs][i ³ Cs][i ³ Cs][1 ³ Cs][1 ³ Cs][1 ³ Cs][lAs][lTs][1 ³ Cs][iT s][iGs][iT s][iT s][i ⁵ Cs][iAs][iAs][iT s][i ³ Cs][iA]	649	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
78	CGAATCTACCCACATCTGTTCAATCA	2	[1 ⁵ Cs][lGs][lAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][lAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][lAs][lTs][l ⁵ Cs][lA]	650	
79	CGAATCTACCCACATCTGTTCAATCA	2	[1 ⁵ Cs][lGs][lAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][lAs][l ⁵ Cs][lAs][lTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][lAs][lTs][l ⁵ Cs][lA]	651	
80	ACGAATCTACCCCATCCATGTAC	37	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	652	
81	CACGAATCTACCCCATCCATGTACT	38	[1 ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][l ⁵ Cs][lT]	653	
82	ACGAATCTACCCCATCCATGTACT	39	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	654	
83	CACGAATCTACCCCATCCATGTAC	40	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	655	
84	CACGAATCTACCTCCATCCATGTACT	41	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	656	
85	CACGAATCTACCCACCATCCATGTACT	42	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	657	
86	ACGAATCTACCCCATCCATGTACT	43	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	658	
87	CACGAATCTACCCCATCCATGTACTC	44	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	659	
88	ACGAATCTACCCCATCCATGTACTC	45	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	660	
89	CACGAATCTACCCCATCCATGTACTCA	46	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	661	
90	CCATCCATCATCGGATAAGAGTT	47	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	662	
91	CCATCCATCATCGGATAAGAGTTC	48	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	663	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			[[iGs][iAs][iTs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]		
92	CATCCATCATCCGGGATAAGAGTT	49	[i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	664	
93	CATCCATCATCCGGGATAAGAGTTC	50	[i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	665	
94	CCATCCATCATCCGGGATAAGAGTTC	51	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	666	
95	CATCCATCATCCGGGATAAGAGTTC	52	[i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	667	
96	CCATCCATCATCCGGGATAAGAGTT	53	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][iT]	668	
97	CCATCCATCATCCGGGATAAGAGTTC	54	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	669	
98	CCATCCATCATCCGGGATAAGAGTTC	55	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	670	
99	CCATCCATCATCCGGGATAAGAGTTCT	56	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iT]	671	
100	TCCATCCATCATCCGGGATAAGAGTTC	57	[iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	672	
101	GTTCAATCATTCGGGATAAGAGTT	58	[iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	673	
102	GTTCAATCATTCGGGATAAGAGTTC	59	[iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	674	
103	GTTCAATCATTCGGGATAAGAGTTC	60	[iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	675	
104	TGTTCAATCATTCGGGATAAGAGTTC	61	[iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	676	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
105	TGTTCAATCATTCGGGATAAGAGTTCT	62	[iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iT]	677	
106	GAATCTACCCACATCTGTTCAATCA	63	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][iCs][iA]	678	
107	CGAATCTACCCACATCTGTTCAATC	64	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	679	
108	CGAATCTACCCACCATCTGTTCAATCA	65	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	680	
109	CACGAATCTACCCCCATCCATGTACT	66	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	681	
110	CACGAATCTACCCGCCATCCATGTACT	67	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	682	
111	CACGAATCTACCCCTCATCCATGTACTC	68	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	683	
112	CACGAATCTACCCGCATCCATGTACTC	69	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	684	
113	TGTTCAATCATTCGGGATAAGAGTTC	70	[iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	685	
114	CCATCCATCATCCCGGGATAAGAGTTC	71	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	686	
115	CACGAATCTACCCACCATCCATGTACT	72	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	687	
116	CACGAATCTACCCCTCCATCCATGTACT	73	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	688	
117	CACGAATCTACCCCCCATCCATGTACT	74	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs]	689	

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[¹³ Cs][¹⁵ Cs][iAs][iT s][iGs][iT s][iA s][¹³ Cs][iT]	
118	CACGAATCTACCCTCATCCATGTACTC	75	[¹³ Cs][iAs][¹⁵ Cs][iGs][iAs][iAs][iT s][¹³ Cs][iT s][iAs][¹³ Cs][¹³ Cs][¹³ Cs]][¹³ Cs][iT s][¹³ Cs][iAs][iT s][¹³ Cs][¹³ Cs][iAs][iT s][iGs][iT s][iAs][¹³ Cs][iT s][¹³ C]	690
119	CACGAATCTACCCTTCATCCATGTACTC	76	[¹³ Cs][iAs][¹³ Cs][iGs][iAs][iAs][iT s][¹⁵ Cs][iT s][iAs][¹⁵ Cs][¹³ Cs][¹³ Cs]][iT s][iT s][¹³ Cs][iAs][iT s][¹³ Cs][¹³ Cs][iAs][iT s][iGs][iT s][iAs][¹⁵ Cs] [iT s][¹³ C]	691
120	TGTTCAATCATTCAGGGATAAGAGTTC	77	[iT s][iGs][iT s][iT s][¹⁵ Cs][iAs][iAs][iT s][¹³ Cs][iAs][iT s][iT s][¹³ Cs][¹³ Cs][iAs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT s][¹³ C]	692
121	TGTTCAATCATTCCTGGGATAAGAGTTC	78	[iT s][iGs][iT s][iT s][¹⁵ Cs][iAs][iAs][iT s][¹³ Cs][iAs][iT s][iT s][¹³ Cs][¹³ Cs][iT s][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT s][¹³ C]	693
122	CCATCCATCATCCAGGGATAAGAGTTC	79	[¹³ Cs][¹³ Cs][iAs][iT s][¹⁵ Cs][¹³ Cs][iAs][iT s][¹³ Cs][iAs][iT s][¹³ Cs][¹³ C s][¹³ Cs][iAs][iGs][iGs][iGs][iAs][i Ts][iAs][iAs][iGs][iAs][iGs][iT s][i Ts][¹³ C]	694
123	CCATCCATCATCCTCGGGATAAGAGTTC	80	[¹³ Cs][¹³ Cs][iAs][iT s][¹⁵ Cs][¹³ Cs][iAs][iT s][¹³ Cs][iAs][iT s][¹³ Cs][¹³ C s][iT s][¹³ Cs][iGs][iGs][iGs][iAs][i Ts][iAs][iAs][iGs][iAs][iGs][iT s][i Ts][¹³ C]	695
124	CCATCCATCATCCTTGGGATAAGAGTTC	81	[¹⁵ Cs][¹⁵ Cs][iAs][iT s][¹⁵ Cs][¹⁵ Cs][iAs][iT s][¹³ Cs][iAs][iT s][¹⁵ Cs][¹³ C s][iT s][iT s][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT s][¹³ C]	696
125	ACGAATCTACCCCCATCCATGTAC	37	[iAs][¹³ Cs][iGs][iAs][iAs][iT s][¹³ C s][iT s][iAs][¹³ Cs][¹³ Cs][¹³ Cs][¹³ Cs]][¹³ Cs][iAs][iT s][¹³ Cs][¹³ Cs][iAs][iT s][iGs][iT s][iAs][¹³ C]	697
126	ACGAATCTACCCCATCCATGTACT	43	[iAs][¹⁵ Cs][iGs][iAs][iAs][iT s][¹⁵ C s][iT s][iAs][¹³ Cs][¹³ Cs][¹³ Cs][¹³ Cs]][iAs][iT s][¹³ Cs][¹³ Cs][iAs][iT s][i Gs][iT s][iAs][¹³ Cs][iT]	698
127	CCATCCATCATCGGGATAAGAGTT	47	[¹⁵ Cs][¹⁵ Cs][iAs][iT s][¹⁵ Cs][¹⁵ Cs][iAs][iT s][¹³ Cs][iAs][iT s][¹³ Cs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT]	699
128	CCATCCATCATCGGATAAGAGTTC	48	[¹⁵ Cs][¹⁵ Cs][iAs][iT s][¹⁵ Cs][¹⁵ Cs][iAs][iT s][¹³ Cs][iAs][iT s][¹³ Cs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT s][¹³ C]	700
129	CATCCATCATCCGGGATAAGAGTT	49	[¹⁵ Cs][iAs][iT s][¹⁵ Cs][¹⁵ Cs][iAs][i Ts][¹³ Cs][iAs][iT s][¹³ Cs][¹³ Cs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iT s][iT]	701

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
130	CATCCATCATCCGGATAAGAGTTC	50	[1 ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][iAs][1T _s][1 ⁵ Cs][1 ⁵ Cs][1G _s][1G _s][1As][iT _s][iAs][iAs][iG _s][iAs][iG _s][iT _s][iT _s][i ⁵ C]	702	
131	GTTCAATCATTCGGGATAAGAGTT	58	[iG _s][iT _s][iT _s][i ⁵ Cs][iAs][iAs][iT _s][i ⁵ Cs][iAs][iT _s][1T _s][1 ⁵ Cs][1G _s][1G _s][1G _s][iAs][iT _s][iAs][iAs][iG _s][iAs][iG _s][iT _s][1T]	703	
132	GTTCAATCATTCGGATAAGAGTTC	59	[iG _s][iT _s][iT _s][i ⁵ Cs][iAs][iAs][iT _s][i ⁵ Cs][iAs][iT _s][1T _s][1 ⁵ Cs][1G _s][1G _s][1As][iT _s][iAs][iAs][iG _s][iAs][iG _s][iT _s][i ⁵ C]	704	
133	CGAATCTACCCATCTGTTCAATCA	34	[i ⁵ Cs][iG _s][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][1 ⁵ Cs][1As][1T _s][1 ⁵ Cs][1T _s][iG _s][iT _s][iT _s][i ⁵ Cs][iAs][iAs][iT _s][i ⁵ Cs][iA]	705	
134	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iG _s][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][1 ⁵ Cs][1 ⁵ Cs][1As][1As][1T _s][1 ⁵ Cs][iT _s][iG _s][iT _s][iT _s][i ⁵ Cs][iAs][iAs][iT _s][i ⁵ C]	706	
135	ACGAATCTACCCCATCCATGTAC	37	[1As][1 ⁵ Cs][1G _s][1As][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][1 ⁵ Cs][1 ⁵ Cs][1 ⁵ Cs][1 ⁵ Cs][iAs][iT _s][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][1G _s][1T _s][1As][1 ⁵ C]	707	
136	ACGAATCTACCCCATCCATGTACT	43	[1As][1 ⁵ Cs][1G _s][1As][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][1 ⁵ Cs][1 ⁵ Cs][1 ⁵ Cs][1As][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iG _s][1T _s][1As][1T]	708	
137	CCATCCATCATCCGGATAAGAGTT	47	[1 ⁵ Cs][1 ⁵ Cs][1As][1T _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][iAs][1T _s][1 ⁵ Cs][1G _s][1G _s][iG _s][iAs][iT _s][iAs][iAs][iG _s][1As][1G _s][1T _s][1T]	709	
138	CCATCCATCATCCGGATAAGAGTTC	48	[1 ⁵ Cs][1 ⁵ Cs][1As][1T _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][iAs][1T _s][1 ⁵ Cs][1G _s][1G _s][iAs][iT _s][iAs][iAs][iG _s][iAs][1G _s][1T _s][1T][1 ⁵ C]	710	
139	CATCCATCATCCGGATAAGAGTT	49	[1 ⁵ Cs][1As][1T _s][1 ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][iAs][iT _s][1 ⁵ Cs][1 ⁵ Cs][1G _s][1G _s][iG _s][iAs][iT _s][iAs][iAs][iG _s][1As][1G _s][1T _s][1T]	711	
140	CATCCATCATCCGGATAAGAGTTC	50	[1 ⁵ Cs][1As][1T _s][1 ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][iAs][iT _s][1 ⁵ Cs][1 ⁵ Cs][1G _s][1G _s][iAs][iT _s][iAs][iAs][iG _s][iAs][1G _s][1T _s][1T][1 ⁵ C]	712	
141	GTTCAATCATTCGGGATAAGAGTT	58	[1G _s][1T _s][1T _s][1 ⁵ Cs][iAs][iAs][iT _s][i ⁵ Cs][iAs][iT _s][1T _s][1 ⁵ Cs][1G _s][1G _s][iG _s][iAs][iT _s][iAs][iAs][iG _s][1As][1G _s][1T _s][1T]	713	
142	GTTCAATCATTCGGATAAGAGTTC	59	[1G _s][1T _s][1T _s][1 ⁵ Cs][iAs][iAs][iT _s][i ⁵ Cs][iAs][iT _s][1T _s][1 ⁵ Cs][1G _s][1G _s][iAs][iT _s][iAs][iAs][iG _s][iAs][1G _s][1T _s][1T][1 ⁵ C]	714	
143	CGAATCTACCCATCTGTTCAATCA	34	[1 ⁵ Cs][1G _s][1As][1As][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][1 ⁵ Cs][1As][1T _s][1 ⁵ Cs][iT _s][iG _s][iT _s][iT _s][i ⁵ Cs][iAs][1As][1T _s][1 ⁵ Cs][1A]	715	
144	CGAATCTACCCAATCTGTTCAATC	36	[1 ⁵ Cs][1G _s][1As][1As][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][1 ⁵ Cs][1As][1As]	716	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			[lTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][lAs][lAs][lTs][l ⁵ C]		
145	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	717	
146	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iA][iT][i ⁵ C][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iT][i ⁵ Cs][iT][iGs][iT][iT][i ⁵ C][iA][iA][iT][i ⁵ Cs][iA]	718	
147	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iA][iT][i ⁵ C][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ C][iA][iT][i ⁵ Cs][iT][iGs][iT][iT][i ⁵ C][iA][iA][iT][i ⁵ Cs][iA]	719	
148	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C][i ⁵ C][iA][i ⁵ C][iA][iT][i ⁵ C][iT][iGs][iT][iT][i ⁵ Cs][iAs][iAs][iT][i ⁵ Cs][iA]	720	
149	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iGs][iT][iT][i ⁵ C][iA][iA][iT][i ⁵ Cs][iA]	721	
150	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iA][iT][i ⁵ C][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iT][i ⁵ Cs][iT][iGs][iT][iT][i ⁵ Cs][iAs][iAs][iT][i ⁵ Cs][iA]	722	
151	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iT][i ⁵ Cs][iT][iGs][iT][iT][i ⁵ C][iA][iA][iT][i ⁵ Cs][iA]	723	
152	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iT][i ⁵ Cs][iT][iGs][iT][iT][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	724	
153	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iAs][iA][iT][i ⁵ C][iTs][iA][i ⁵ Cs][i ⁵ C][i ⁵ Cs][iA][i ⁵ Cs][iA][iT][i ⁵ C][iT][iG][iT][iT][i ⁵ Cs][iAs][iAs][iT][i ⁵ Cs][iA]	725	
154	AATCTACCCACCTGTTCAATCA	83	[lAs][lAs][iTs][l ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][lAs][l ⁵ Cs][l ⁵ Cs][lTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][l ⁵ Cs][lA]	726	
155	AATCTACCCACATCTGTTCAAT	84	[lAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][lAs][l ⁵ Cs][lAs][lTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][lAs][lT]	727	
156	CGAATCTACCCCTGTTCAATCA	85	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][l ⁵ Cs][l ⁵ Cs][lTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][l ⁵ Cs][lA]	728	
157	CGAATCTACCCATCTGTTCAAT	86	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][l ⁵ Cs][l ⁵ Cs][lAs][lTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][lAs][lT]	729	
158	ACGAATCTACCCAATCTGTTCAATCA	87	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs]	730	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			[iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]		
159	CACGAATCTACCCATCTGTTCAATCA	88	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	731	
160	GAATCTACCCACCATCTGTTCAATCA	89	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	732	
161	AATCTACCCACCAATCTGTTCAATCA	90	[iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	733	
162	CGAATCTACCCACTCTGTTCAATCAT	91	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	734	
163	CGAATCTACCCACTCATCTGTTCAAT	92	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	735	
164	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	736	
165	GAATCTACCCACCCATCTGTTCAATC	94	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	737	
166	ACGAATCTACCCATCTGTTCAATCAT	95	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	738	
167	GAATCTACCCACCTCTGTTCAATCAT	96	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	739	
168	CCACGAATCTACCTCAATCATTCATT	97	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	740	
169	CGAATCTACCCACCTCTGTTCAATCA	98	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	741	
170	ACGAATCTACCCACTCTGTTCAATCA	99	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	742	
171	CGAATCTACCCAATCTGTTCAATCAT	100	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	743	
172	GAATCTACCCACATCTGTTCAATCAT	101	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	744	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
173	AATCTACCCACCTGTTCAATCA	83	[iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s][i ⁵ Cs][iA]	745	
174	AATCTACCCACATCTGTTCAAT	84	[iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s]	746	
175	AATCTACCCACTCTGTTCAATC	102	[iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s][i ⁵ C]	747	
176	CGAATCTACCCCTGTTCAATCA	85	[i ⁵ Cs][iGs][iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s][i ⁵ Cs][iA]	748	
177	CGAATCTACCCATCTGTTCAAT	86	[i ⁵ Cs][iGs][iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s]	749	
178	CGAATCTACCCCTCTGTTCAATC	103	[i ⁵ Cs][iGs][iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s][i ⁵ C]	750	
179	GAATCTACCCACTGTTCAATCA	104	[iGs][iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s][i ⁵ Cs][iA]	751	
180	GAATCTACCCAATCTGTTCAAT	105	[iGs][iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s]	752	
181	GAATCTACCCATCTGTTCAATC	106	[iGs][iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ^s][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s][i ⁵ C]	753	
182	AATCTACCCACTCTGTTCAATC	102	[lAs][lAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][lAs][l ⁵ Cs][lTs][l ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][lTs][l ⁵ C]	754	
183	CGAATCTACCCCTCTGTTCAATC	103	[l ⁵ Cs][lGs][iAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][l ⁵ Cs][lTs][l ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][lTs][l ⁵ C]	755	
184	GAATCTACCCACTGTTCAATCA	104	[lGs][lAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][lAs][l ⁵ Cs][lTs][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][iT ^s][l ⁵ Cs][lA]	756	
185	GAATCTACCCAATCTGTTCAAT	105	[lGs][lAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][lAs][lAs][lTs][i ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][lAs][lT]	757	
186	GAATCTACCCATCTGTTCAATC	106	[lGs][lAs][iAs][iT ^s][i ⁵ Cs][iT ^s][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][lAs][lTs][l ⁵ Cs][iT ^s][iGs][iT ^s][iT ^s][i ⁵ Cs][iAs][iAs][lTs][l ⁵ C]	758	

TABLE 5-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
187	GAATCTACCCATCTGTTCAATC	106	[iGs][A][iA][lT][i ⁵ C][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iTs][lT][i ³ C][iA][iA][iTs][i ⁵ C]	759
188	GAATCTACCCATCTGTTCAATC	106	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iT][iT][i ³ C][iA][iA][iTs][i ³ C]	760
189	GAATCTACCCATCTGTTCAATC	106	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][iA][iT][i ³ Cs][iTs][iGs][iTs][iTs][i ³ C][iA][iA][iTs][i ³ C]	761
190	GAATCTACCCATCTGTTCAATC	106	[iGs][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ C][iA][iT][i ³ C][iTs][iGs][iTs][iTs][i ³ Cs][iA][iA][iTs][i ⁵ C]	762
191	GAATCTACCCATCTGTTCAATC	106	[lGs][A][iA][lT][i ⁵ C][iTs][iAs][i ⁵ Cs][i ³ Cs][l ³ Cs][lAs][iTs][l ³ Cs][iTs][iGs][iTs][iT][i ³ C][iA][iA][lTs][i ⁵ C]	763
192	CGAATCTACCCAATCTGTTCAATC	36	[i ³ Cs][iGs][iAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][l ³ Cs][lAs][lAs][lTs][i ³ Cs][iTs][iGs][iTs][iTs][iTs][i ³ Cs][lAs][iAs][iTs][l ³ C]	764
193	CGAATCTACCCAATCTGTTCAATC	36	[i ³ Cs][iGs][iAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][lAs][lAs][iTs][i ³ Cs][iTs][iGs][iTs][iTs][iTs][i ³ Cs][lAs][iAs][iTs][i ⁵ C]	765
194	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][i ³ Cs][lAs][lAs][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][lTs][l ³ C]	766
195	CGAATCTACCCACTCATCTGTTCAAT	92	[i ³ Cs][iG][iA][iA][iT][i ³ Cs][iTs][iAs][i ⁵ Cs][i ³ Cs][i ³ Cs][iAs][i ³ Cs][iTs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iT][iT][i ³ C][iA][iAs][iT]	767
196	CGAATCTACCCACTCATCTGTTCAAT	92	[i ³ Cs][iG][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ³ Cs][i ³ Cs][iAs][i ⁵ Cs][iTs][i ³ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iT][i ³ C][iA][iAs][iT]	768
197	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ³ C][iG][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][i ³ Cs][iAs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iTs][iT][i ³ C][iA][iTs][i ³ C]	769
198	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ³ C][iG][iA][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ³ C][iA][iA][iTs][i ³ C]	770
199	CAAATCTACCCACATCTGTTCAATCA	107	[i ⁵ Cs][iAs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][i ³ Cs][iAs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ³ Cs][iAs][iAs][iTs][i ³ Cs][iA]	771
200	CGAATCTACCCACATCTATTCAATCA	108	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][i ³ Cs][iAs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iAs][iTs][iTs][iTs][i ³ Cs][iAs][iAs][iTs][i ³ Cs][iA]	772

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
201	CAAATCTACCCACATCTATTCAATCA	109	[i ⁵ Cs][iAs][iAs][iAs][iT ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][iAs][iAs][iT ⁵ Cs][iA]	773	
202	CGAATCTACCCAATCTATTCAATC	110	[l ⁵ Cs][lGs][lAs][lAs][iT ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][lAs][lAs][lTs][i ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][lAs][lAs][lTs][l ⁵ C]	774	
203	CGAATCTACCCACATCTGTTCAAT	111	[l ⁵ Cs][lGs][lAs][lAs][iT ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][l ⁵ Cs][lAs][l ⁵ Cs][lAs][iT ⁵ Cs][i ⁵ Cs][iT ⁵ Cs][iGs][iT ⁵ Cs][iT ⁵ Cs][lAs][lAs][lT]	775	
204	CGAATCTACCCATCATCTGTTCAA	112	[l ⁵ Cs][lGs][lAs][lAs][iT ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][l ⁵ Cs][lAs][lTs][l ⁵ Cs][iAs][iT ⁵ Cs][iT ⁵ Cs][iGs][iT ⁵ Cs][lTs][l ⁵ Cs][lAs][lA]	776	
205	ACGAATCTACCCATCTGTTCAATC	113	[lAs][l ⁵ Cs][lGs][lAs][iAs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][l ⁵ Cs][l ⁵ Cs][lAs][lTs][i ⁵ Cs][iT ⁵ Cs][iGs][iT ⁵ Cs][iT ⁵ Cs][i ⁵ Cs][lAs][lAs][lTs][l ⁵ C]	777	
206	ACGAATCTACCCATCTGTTCAAT	114	[lAs][l ⁵ Cs][lGs][lAs][iAs][iT ⁵ Cs][iT ⁵ Cs][iAs][l ⁵ Cs][l ⁵ Cs][l ⁵ Cs][l ⁵ Cs][lAs][iT ⁵ Cs][i ⁵ Cs][iT ⁵ Cs][iGs][iT ⁵ Cs][lTs][l ⁵ Cs][lAs][lAs][iT]	778	
207	CACGAATCTACCCATCCATGTACT	115	[l ⁵ Cs][lAs][l ⁵ Cs][lGs][iAs][iAs][iT ⁵ Cs][i ⁵ Cs][iT ⁵ Cs][iAs][l ⁵ Cs][l ⁵ Cs][l ⁵ Cs][lAs][iT ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ Cs][iGs][lTs][lAs][l ⁵ Cs][lT]	779	
208	CACGAATCTACCCATCCATGTAC	116	[l ⁵ Cs][lAs][l ⁵ Cs][lGs][iAs][iAs][iT ⁵ Cs][i ⁵ Cs][iT ⁵ Cs][iAs][l ⁵ Cs][l ⁵ Cs][l ⁵ Cs][l ⁵ Cs][iAs][iT ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ Cs][lGs][lTs][lAs][l ⁵ C]	780	
209	CATTTATTCAACGGGATAAGAGTT	117	[l ⁵ Cs][lAs][lTs][lTs][iT ⁵ Cs][iAs][iT ⁵ Cs][iT ⁵ Cs][iAs][lAs][l ⁵ Cs][lGs][lGs][iGs][iAs][iT ⁵ Cs][iAs][iAs][iGs][lAs][lGs][lTs][lT]	781	
210	CATTTATTCAACGGGATAAGAGT	118	[l ⁵ Cs][lAs][lTs][lTs][iT ⁵ Cs][iAs][iT ⁵ Cs][iT ⁵ Cs][iAs][lAs][l ⁵ Cs][lGs][lGs][iGs][iGs][iAs][iT ⁵ Cs][iAs][iAs][lGs][lAs][lGs][lT]	782	
211	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iAs][iT ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ Cs][iGs][iT ⁵ Cs][iT ⁵ Cs][i ⁵ Cs][iAs][iAs][iT ⁵ Cs][iA]	783	
212	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ Cs][iGs][iT ⁵ Cs][iT ⁵ Cs][i ⁵ Cs][iAs][iAs][iT ⁵ Cs][iA]	784	
213	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ Cs][iGs][iT ⁵ Cs][iT ⁵ Cs][i ⁵ Cs][iAs][iAs][iT ⁵ Cs][iA]	785	
214	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT ⁵ Cs][iT ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][iT ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ Cs][iGs][iT ⁵ Cs][iT ⁵ Cs][i ⁵ Cs][iAs][iA][iT ⁵ Cs][iA]	786	
215	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ Cs][lGs][iAs][iAs][iT ⁵ Cs][lTs][lAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iAs]	787	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			[iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][lT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][lT <i>s</i>][l ⁵ C]		
216	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][l ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][lT <i>s</i>][l ⁵ C]	788	
217	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ C <i>s</i>][lG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C]	789	
218	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][l ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][lT <i>s</i>][l ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C]	790	
219	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ C <i>s</i>][lG][lA][lA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][l ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][l ⁵ C]	791	
220	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ C <i>s</i>][lG <i>s</i>][lA <i>s</i>][lA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C][l ⁵ C][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][l ⁵ C]	792	
221	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ C <i>s</i>][lG <i>s</i>][lA <i>s</i>][lA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][l ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT][i ⁵ C][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][l ⁵ C]	793	
222	CGAATCTACCCAATCTGTTCAATC	36	[l ⁵ C <i>s</i>][lG <i>s</i>][lA <i>s</i>][lA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][l ⁵ C <i>s</i>][lA <i>s</i>][lA <i>s</i>][lT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][lA][lA][lT <i>s</i>][l ⁵ C]	794	
223	CACGAATCTACCCCCATCCATGTACT	1	[i ⁵ C <i>s</i>][iA][i ⁵ C][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iT]	795	
224	CACGAATCTACCCCCATCCATGTACT	1	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C][i ⁵ C][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iT]	796	
225	CACGAATCTACCCCCATCCATGTACT	1	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C][iA][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iT]	797	
226	CACGAATCTACCCCCATCCATGTACT	1	[i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT][iA][i ⁵ C <i>s</i>][iT]	798	
227	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ C <i>s</i>][i ⁵ C][iA][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA <i>s</i>][iT <i>s</i>][iA <i>s</i>][iA <i>s</i>][iG <i>s</i>][iA <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	799	
228	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C][i ⁵ C <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA <i>s</i>][iT <i>s</i>][iA <i>s</i>][iA <i>s</i>][iG <i>s</i>][iA <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	800	
229	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][i ⁵ C]	801	

TABLE 5-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			s][iGs][iG][iG][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][i ³ C]	
230	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ³ Cs][iAs][iTs][i ³ Cs][i ³ C s][iGs][iGs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iG][iT][iTs][i ³ C]	802
231	GCATTTATTCAACGGGATAAGAGTT	17	[iGs][i ⁵ C][iA][iTs][iTs][iTs][iAs][iTs][iTs][i ³ Cs][iAs][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	803
232	GCATTTATTCAACGGGATAAGAGTT	17	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][i ³ Cs][iA][iA][i ³ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	804
233	GCATTTATTCAACGGGATAAGAGTT	17	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][i ³ Cs][iAs][iAs][i ³ Cs][i Gs][iG][iG][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	805
234	GCATTTATTCAACGGGATAAGAGTT	17	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][i ³ Cs][iAs][iAs][i ³ Cs][i Gs][iGs][iGs][iGs][iAs][iTs][iAs][i As][iGs][iA][iG][iTs][iT]	806
235	CGAATCTACCATCCATGTAC	82	[i ³ Cs][iGs][iAs][iAs][iTs][i ³ Cs][iT s][iAs][i ³ Cs][i ³ Cs][iAs][iTs][i ³ Cs] [i ³ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	807
236	ATCCATCCATCATTTGGGATAAGAGT	119	[iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ³ Cs][iAs][iTs][i ³ Cs][iAs][iTs] [iTs][iGs][iGs][iGs][iGs][iAs][iTs] [iAs][iAs][iGs][iAs][iGs][iT]	808
237	TCCATCCATCATCGGGATAAGAGTT	120	[iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ³ Cs][iAs][iTs][i ³ Cs][iAs][iTs][i ³ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	809
238	TCCATCCATCATCGGGATAAGAGTTC	121	[iTs][i ⁵ Cs][i ³ Cs][iAs][iTs][i ³ Cs][i ³ Cs][iAs][iTs][i ³ Cs][iAs][iTs][i ³ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	810
239	TCCATCCATCATCGGATAAGAGTTCT	122	[iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ³ Cs][iAs][iTs][i ³ Cs][iAs][iTs][i ³ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iT]	811
240	CCATCCATCATCCGATAAGAGTTCTT	123	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ³ Cs][iAs][iTs][i ³ Cs][i ³ C s][iGs][iAs][iTs][iAs][iAs][iGs][iA s][iGs][iTs][iTs][i ⁵ Cs][iTs][iT]	812
241	GCATTTATTCAACGGGATAAGAGTTC	124	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][i ³ Cs][iAs][iAs][i ³ Cs][i Gs][iGs][iGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][iTs][iTs][i ³ C]	813
242	GCATTTATTCAACGGGATAAGAGTTCT	125	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][i ³ Cs][iAs][iAs][i ³ Cs][i Gs][iGs][iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][iTs][i ⁵ Cs][iT]	814
243	CGAATCTACCCACCCCTTTTATCTACT	126	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ³ Cs][iT s][iAs][i ³ Cs][i ³ Cs][i ³ Cs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][iTs][iTs][iTs][iTs][i As][iTs][i ³ Cs][iTs][iAs][i ³ Cs][iT]	815

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
244	CGAATCTACCCACCATGTACTACCC	127	[i ⁵ Cs][iGs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ C]	816	
245	CACGAATCTACCCCATCCTTTTATC	128	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs] [iTs][iTs][iTs][iTs][iAs][iTs][i ⁵ C]	817	
246	CACGAATCTACCCCTGTTCAATCATT	129	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][i As][iAs][iTs][i ⁵ Cs][iAs][iTs][iT]	818	
247	CACGAATCTACCCGTTCAATCATTCA	130	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iT s][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iA]	819	
248	CGAATCTACCCACTCCTTTTATCTAC	131	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][iTs][iTs][i Ts][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ C]	820	
249	CGAATCTACCCACCTGTTCAATCATT	132	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][i As][iAs][iTs][i ⁵ Cs][iAs][iTs][iT]	821	
250	CGAATCTACCCACCTTTTATCTACTC	133	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][iTs][iTs][iTs][iAs][iT s][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ C]	822	
251	CGAATCTACCCACGTTCAATCATTCA	134	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iT s][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iA]	823	
252	ACGAATCTACCCAGTACTACCCATC	135	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ C s][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs] [iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iA s][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ C]	824	
253	CGAATCTACCCACTGTTCAATCATTC	136	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iA s][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ C]	825	
254	CACGAATCTACCCCATCTGTTCAATC	137	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][i Ts][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	826	
255	CACGAATCTACCCCTTTTATCTACT	138	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iTs][iTs][iTs][iTs][i As][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iT]	827	
256	CACGAATCTACCCCATGTACTACCC	139	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][i As][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ C]	828	
257	CGAATCTACCCACATTACACGACATT	140	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs]	829	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			[[iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iTs][iT]		
258	CGAATCTACCCACCATCTGTTCAATC	141	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	830	
259	ACGAATCTACCCACCATCCTTTTATC	142	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][iTs][iTs][iAs][iTs][i ⁵ C]	831	
260	CGAATCTACCCACGTACTACCCATC	143	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ C]	832	
261	CACGAATCTACCCGTACTACCCATC	144	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ C]	833	
262	ACGAATCTACCCAATTACCCAGCATT	145	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iTs][iT]	834	
263	CGAATCTACCCACGAAACCCAGGCAG	146	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iAs][iGs][i ⁵ Cs][iAs][iG]	835	
264	CCACGAATCTACCGAAACCCAGGCA	147	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iAs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][iGs][i ⁵ Cs][iA]	836	
265	CAATCATATATCCGGAAACCCAGGCA	148	[i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iAs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iAs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][iGs][i ⁵ Cs][iA]	837	
266	CAATAGTTCAATCGGAAACCCAGGCA	149	[i ⁵ Cs][iAs][iAs][iTs][iAs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iGs][iGs][iAs][iAs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][iGs][i ⁵ Cs][iA]	838	
267	TCTAGCCACGAATTTCCAGGAAACCC	150	[iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][iTs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][iGs][iAs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C]	839	
268	GTACTACCCATCGATAAGAGTTCTT	151	[iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iTs][iT]	840	
269	CCATGTACTACCCGATAAGAGTTCT	152	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iT]	841	
270	TGTCCATCCATCCGGATAAGAGTTCT	153	[iTs][iGs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iT]	842	
271	TTCATTACCCAGCGGATAAGAGTTC	154	[iTs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs]	843	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			[iGs][iGs][iGs][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][i ⁵ C]		
272	CTACTCATCACTCGGGATAAGAGTTC	155	[i ⁵ Cs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	844	
273	TCTCCATCCATCCGGGATAAGAGTTC	156	[iTs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	845	
274	GTACTCACCATCGGGATAAGAGTTC	157	[iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	846	
275	CATGTACTCACCCGGGATAAGAGTTC	158	[i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	847	
276	CATCCATCTATCCGGGATAAGAGTTC	159	[i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	848	
277	CCATCCATCTGTCCGGGATAAGAGTTC	160	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	849	
278	CCAGCATTTATTCGGGGATAAGAGTT	161	[i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	850	
279	TTCATTACACGCGGGGATAAGAGTT	162	[iTs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	851	
280	TGTTCAATCATTCGGGGATAAGAGTT	163	[iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	852	
281	TTATCTACTCATCGGGGATAAGAGTT	164	[iTs][iTs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	853	
282	CACCCATCTCTCCGGGATAAGAGTT	165	[i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	854	
283	GTACTCACCATCGGGGATAAGAGTT	166	[iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	855	
284	CATGTACTCACCCGGGGATAAGAGTT	167	[i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	856	
285	CCATCTATCCATCGGGGATAAGAGTT	168	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs]	857	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
			[[iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iTs][iT]		
286	CATCCATCTAGCCGGGGATAAGAGTT	169	[i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	858	
287	CTGTCCATCCATCGGGGATAAGAGTT	170	[i ⁵ Cs][iTs][iGs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	859	
288	CAATCATATATCCGGGGATAAGAGTT	171	[i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iAs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	860	
289	CCATCTAGCCACGTTCTGGGGATAA	172	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iTs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iGs][iGs][iAs][iTs][iAs][iA]	861	
290	CATGTACTACCCGTTCTCTGGGGATA	173	[i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][iGs][iTs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iGs][iGs][iGs][iAs][iTs][iA]	862	
291	CTAGCCACGAATCACTAGTTCTCTGGG	174	[i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iAs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iAs][iGs][iTs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iGs][iG]	863	
292	CCACGAATCTACCGCATTATTCAAC	175	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iA][iTs][iTs][i ⁵ Cs][iAs][iAs][i ⁵ C]	864	
293	CTAGCCACGAATCGTACTACCCATC	176	[i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iGs][iAs][iAs][iTs][i ⁵ Cs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iA][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ C]	865	
294	GAATCTACCCACTTCACCAGCATT	177	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iTs][iT]	866	
295	CGAATCTACCCATTACACCAGCATT	178	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iTs][iT]	867	
296	GAATCTACCCACATTACACCAGCAT	179	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iT]	868	
297	CGAATCTACCCAATTACACCAGCAT	180	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iT]	869	
298	GAATCTACCCACTTCACCAGCATT	177	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iTs][iT]	870	
299	CGAATCTACCCATTACACCAGCATT	178	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iT]	871	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
300	GAATCTACCCACATTACCAGCAT	179	[iGs][iAs][iAs][lTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][iTs][l ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][lAs][iGs][i ⁵ Cs][iAs][lT]	872	
301	CGAATCTACCCAATTACCAGCAT	180	[i ⁵ Cs][iGs][iAs][lAs][iTs][i ⁵ Cs][iTs][lAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iAs][iTs][iTs][l ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][lT]	873	
302	CGAATCTACCCACATTACCAGCATT	140	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][lAs][l ⁵ Cs][lAs][lTs][lTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iTs][lT]	874	
303	CGAATCTACCCACATTACCAGCATT	140	[i ⁵ Cs][lGs][lAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][l ⁵ Cs][lAs][lTs][lT]	875	
304	CGAATCTACCCACATTACCAGCATT	140	[i ⁵ Cs][lGs][lAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][lAs][l ⁵ Cs][lAs][lTs][lTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][l ⁵ Cs][lAs][lTs][lT]	876	
305	GAATCTACCCACATTACCAGCATT	181	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iTs][iT]	877	
306	CGAATCTACCCACATTACCAGCAT	182	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][i ⁵ Cs][iAs][iT]	878	
307	CGAATCTACCCATTACCAGCATT	178	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][lAs][lTs][lTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][l ⁵ Cs][lAs][lTs][lT]	879	
308	CGAATCTACCCATTACCAGCATT	178	[i ⁵ Cs][lGs][lAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][lAs][lTs][lTs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iGs][l ⁵ Cs][lAs][lTs][lT]	880	
309	CATATATCCATCTAGCCAG	183	[i ⁵ Cs][iAs][iTs][iAs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG]	881	
310	ACGAATCTACCCATCCATGT	184	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iT]	882	
311	TGTCCATCCACGAATCTACC	185	[iTs][iGs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iTs][iAs][i ⁵ Cs][i ⁵ C]	883	
312	CGAATCTACCAACTCATCCA	186	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iA]	884	
313	ACGAATCTACCATCCATGTA	187	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iA]	885	

TABLE 5-continued

Exemplary antisense oligonucleotides					
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:	
314	CGAATCTACCCATCTCTCCA	188	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iA]	886	
315	CACGAATCTACCCATCTCTC	189	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][i ⁵ C]	887	
316	ACGAATCTACCCATCTCTCC	190	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ C]	888	
317	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][lGs][iA][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][lAs][l ⁵ C]	889	
318	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iA][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][lAs][l ⁵ C]	890	
319	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][lAs][l ⁵ C]	891	
320	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	892	
321	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][lAs][l ⁵ C]	893	
322	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	894	
323	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][lAs][l ⁵ C]	895	
324	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][lAs][l ⁵ C]	896	
325	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	897	
326	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	898	
327	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][iTs][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	899	
328	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][iTs][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	900	

TABLE 5-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
329	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ C][iA][iT][iG][iT][iA][i ⁵ Cs][iT]	901
330	CCATCCATCATCCGGGATAAGAGTTC	16	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iA][iG][iA][iG][iT][iTs][i ⁵ C]	902
331	GCATTATTCAACGGGGATAAGAGTT	17	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iA][iA][iG][iA][iG][iTs][iT]	903

iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; iA, iG, i⁵C, and iT are 2'-4' methylene bridge (LNA) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage, and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.
 †Each 5'-methyl cytidine (5C) in any of the sequences provided in Table 5 may independently and optionally be replaced with cytidine.

TABLE 11

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
350	GCATTATTCAACGGGGATAAGAGTT	17	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iAs][iTs][iTs][iTs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	904
351	GCATTATTCAACGGGGATAAGAGTT	17	[iGs][i ⁵ C][iA][iTs][iTs][iTs][iTs][iAs][iTs][iTs][iTs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iA][iGs][iAs][iGs][iTs][iT]	905
352	GCATTATTCAACGGGGATAAGAGTTC	124	[iGs][i ⁵ C][iA][iTs][iTs][iTs][iTs][iAs][iTs][iTs][iTs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iG][iAs][iGs][iTs][iTs][i ⁵ C]	906
353	GCATTATTCAACGGGGATAAGAGTTC	124	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iTs][iAs][iTs][iTs][iTs][i ⁵ Cs][iA][iA][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iG][iAs][iGs][iTs][iTs][i ⁵ C]	907
354	GCATTATTCAACGGGGATAAGAGTTC	124	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iTs][iAs][iTs][iTs][iTs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iGs][iG][iG][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	908
355	GCATTATTCAACGGGGATAAGAGTTC	124	[iGs][i ⁵ Cs][iAs][iTs][iTs][iTs][iTs][iAs][iTs][iTs][iTs][i ⁵ Cs][iAs][iAs][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iG][iAs][iG][iT][iTs][i ⁵ C]	909

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
356	GCATTATTCAACGGGATAAGAGTTC	124	[iGs][i ⁵ Cs][iAs][iTs][iTs] [iTs][iAs][iTs][iTs][i ³ Cs] [iAs][iAs][i ³ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i ³ C]	910
357	GCATTATTCAACGGATAAGAGTTCT	125	[iGs][i ⁵ Cs][iAs][iTs][iTs] [iTs][iAs][iTs][iTs][i ³ Cs] [iAs][iAs][i ³ Cs][iGs][iGs] [iAs][iTs][iAs][iAs][iG][i A][iG][iT][iT][i ³ Cs][iT]	911
358	CATTATTCAACGGATAAGAGTTC	338	[l ⁵ Cs][lAs][lTs][lTs][iTs] [iAs][iTs][iTs][i ³ Cs][iAs] [lAs][l ⁵ Cs][lGs][lGs][iAs] [iTs][iAs][iAs][iGs][iAs][lGs][lTs][lTs][l ⁵ C]	912
359	CATTATTCAACGGATAAGAGTTC	338	[i ⁵ Cs][iAs][iTs][iTs][iTs] [iAs][iTs][iTs][i ³ Cs][iAs] [iAs][i ³ Cs][iGs][iGs][iAs] [iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	913
360	CATTATTCAACGGGATAAGAGTT	117	[i ⁵ Cs][iAs][iTs][iTs][iTs] [iAs][iTs][iTs][i ³ Cs][iAs] [iAs][i ³ Cs][iGs][iGs][iGs] [iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	914
361	CATTATTCAACGGGATAAGAGTT	339	[i ⁵ Cs][iAs][iTs][iTs][iTs] [iAs][iTs][iTs][i ³ Cs][iAs] [iAs][i ³ Cs][iGs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	915
362	GCATTATTCAAGGGGATAAGAGTT	340	[iGs][i ⁵ Cs][iAs][iTs][iTs] [iTs][iAs][iTs][iTs][i ³ Cs] [iAs][iAs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][iTs][iT]	916
363	GCATTATTCAACGGGATAAGAGTT	341	[iGs][i ⁵ Cs][iAs][iTs][iTs] [iTs][iAs][iTs][iTs][i ³ Cs] [iAs][iAs][i ³ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	917
364	GCATTATTCAACGGGATAAGAGT	342	[iGs][i ⁵ Cs][iAs][iTs][iTs] [iTs][iAs][iTs][iTs][i ³ Cs] [iAs][iAs][i ³ Cs][iGs][iGs] [iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iT]	918
365	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iG][iT][iTs][i ⁵ Cs][i As][iAs][iTs][i ³ Cs][iAs][i Ts][iTs][i ³ Cs][iGs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i ³ C]	919
366	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ³ Cs][iAs] [iT][iT][i ³ Cs][iGs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i ³ C]	920
367	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ³ Cs][iAs] [iTs][iTs][i ³ Cs][iGs][iG][iG][iAs][iTs][iAs][iAs][iG]	921

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			s][iAs][iGs][iTs][iTs][i ⁵ C]	
368	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][iGs][iAs][iG][iT][iTs][i ⁵ C]	922
369	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i ⁵ C]	923
370	TGTTCAATCATTCTGGGATAAGAGTTC	343	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][iTs][i ⁵ C]	924
371	TGTTCAATCATTCTGGGATAAGAGTT	344	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	925
372	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][iG][iA][iG][iT][iTs][i ⁵ C]	569
373	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][iGs][iA][iG][iT][iTs][i ⁵ C]	570
374	TGTTCAATCATTCTGGGATAAGAGTTC	9	[lTs][lGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i ⁵ C]	927
375	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][lTs][lTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i ⁵ C]	928
376	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [lAs][iAs][iTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i ⁵ C]	929
377	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][lAs][lTs][i ⁵ Cs][iAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][lT][iTs][i ⁵ C]	930
378	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][lAs] [iTs][iTs][i ⁵ Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i ⁵ C]	931
379	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ Cs][iAs] [lTs][lTs][i ⁵ Cs][iGs][iGs]	932

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	
380	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	933
381	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	934
382	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	935
383	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	936
384	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	937
385	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	938
386	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	939
387	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	940
388	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	941
389	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	942
390	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	943
391	TGTTCAATCATTCGGGATAAGAGTTC	9	[iT ^s][iGs][iT ^s][iT ^s][i ³ Cs] [iAs][iAs][iT ^s][i ³ Cs][iAs] [iT ^s][iT ^s][i ³ Cs][iGs][iGs] [iGs][iAs][iT ^s][iAs][iA][iG][iA][iG][iT ^s][iT ^s][i ³ C]	944

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[[iAs][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i ³ C]	
392	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iG][iT][iTs][i ⁵ Cs][iAs][iAs][iTs][i ³ Cs][iAs][iT][iT][i ³ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	945
393	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iG][iT][iTs][i ⁵ Cs][iAs][iAs][iTs][i ³ Cs][iAs][iTs][iTs][i ³ Cs][iGs][iG][iG][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	946
394	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iG][iT][iTs][i ⁵ Cs][iAs][iAs][iTs][i ³ Cs][iAs][iTs][iTs][i ³ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iG][iAs][iG][iT][iTs][i ³ C]	572
395	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iG][iT][iT][i ⁵ Cs][iAs][iAs][iTs][i ³ Cs][iAs][iT][iTs][i ³ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iG][iT][iTs][i ³ C]	573
396	TGTTCAATCATTCGGGATAAGAGTTC	345	[x][dt][dc][da][iTs][iGs][iTs][iTs][i ³ Cs][iAs][iAs][iTs][i ³ Cs][iAs][iTs][iT][i ³ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iA][iG][iA][iG][iT][iTs][i ³ C]	947
397	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iG][iT][iT][i ⁵ Cs][iAs][iAs][iTs][i ³ Cs][iAs][iT][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	948
398	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iT][i ⁵ C][iA][iAs][iTs][i ³ Cs][iAs][iT][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	949
399	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ³ Cs][iA][iA][iT][i ³ Cs][iAs][iT][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	950
400	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ³ Cs][iAs][iAs][iT][i ³ C][iA][iT][iTs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	951
401	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ³ Cs][iAs][iAs][iTs][i ³ Cs][iA][iT][iT][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	952
402	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ³ Cs][iAs][iAs][iTs][i ³ Cs][iAs][iT][iT][i ³ C][iG][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ³ C]	953
403	TGTTCAATCATTCGGGATAAGAGTTC	9	[iTs][iGs][iTs][iTs][i ³ Cs][iAs][iAs][iTs][i ³ Cs][iAs]	954

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iG][iG][iG][iA <i>s</i>][iT <i>s</i>][iA <i>s</i>][iA <i>s</i>][iG <i>s</i>][iA <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	
404	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG][iA][iT][iA <i>s</i>][iA <i>s</i>][iG <i>s</i>][iA <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	955
405	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA <i>s</i>][iA <i>s</i>][iG <i>s</i>][iA <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	956
406	TGTTCAATCATTCTGGGATAAGAGTTC	9	[iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA <i>s</i>][iA <i>s</i>][iG <i>s</i>][iA <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C]	957
407	TGTTCAATCATT[n3]GGGATAAGAGTTC	194/253	[iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][n3 <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA <i>s</i>][iA <i>s</i>][iG][iT][iT <i>s</i>][i ⁵ C]	958/959
408	TGTTCAATCATT[n6]GGGATAAGAGTTC	194/253	[iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][n6 <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA <i>s</i>][iA <i>s</i>][iG][iT][iT <i>s</i>][i ⁵ C]	958/959
409	TCTGTTCAATCATTTGGGATAAGAGT	347	[iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA <i>s</i>][iT][iA][iA][iG][iA][iG <i>s</i>][iT]	960
410	TCTGTTCAATCATTTGGGATAAGAGTT	348	[iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA][iA][iG][iA][iG <i>s</i>][iT]	961
411	TCTGTTCAATCATTTGGGATAAGAGTTC	349	[iT <i>s</i>][i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA][iA][iG][iA][iG <i>s</i>][iT]	962
412	CTGTTCAATCATTGGGATAAGAGT	350	[i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA][iA][iG][iA][iG <i>s</i>][iT]	963
413	CTGTTCAATCATTGGGATAAGAGTT	351	[i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA][iA][iG][iA][iG <i>s</i>][iT]	964
414	CTGTTCAATCATTGGGATAAGAGTTC	352	[i ⁵ C <i>s</i>][iT <i>s</i>][iG <i>s</i>][iT <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iA <i>s</i>][iT <i>s</i>][i ⁵ C <i>s</i>][iA <i>s</i>][iT <i>s</i>][iT <i>s</i>][iG <i>s</i>][iG <i>s</i>][iG <i>s</i>][iA][iT][iA][iA][iG][iA][iG <i>s</i>][iT]	965

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
415	TGTTCAATCATTCTGGGGATAAGAGT	353	[iT _s][iG _s][iT _s][iT _s][i ⁵ C _s][iA _s][iA _s][iT _s][i ⁵ C _s][iA _s][iT _s][iT _s][i ⁵ C _s][iT _s][iG _s][iG _s][iG _s][iA _s][iT][iA][iA][iG][iA][iG _s][iT]	966
416	TGTTCAATCATTCTGGGGATAAGAGTT	163	[iT _s][iG _s][iT _s][iT _s][i ⁵ C _s][iA _s][iA _s][iT _s][i ⁵ C _s][iA _s][iT _s][iT _s][i ⁵ C _s][iG _s][iG _s][iG _s][iA _s][iT][iA][iG][iA][iG][iT _s][iT]	967
417	TGTTCAATCATTCTGGGATAAGAGTTCT	354	[iT _s][iG _s][iT _s][iT _s][i ⁵ C _s][iA _s][iA _s][iT _s][i ⁵ C _s][iA _s][iT _s][iT _s][i ⁵ C _s][iG _s][iG _s][iA _s][iT _s][iA _s][iA _s][iG][iA][iG][iT][iT][i ⁵ C _s][iT]	968
418	TGTTCAATCATTCTGGGATAAGAGTTC	9	[nmaT _s][nmaG _s][nmaT _s][nmaT _s][nma ⁵ C _s][nmaA _s][nmaA _s][nmaT _s][nma ⁵ C _s][nmaA _s][nmaT _s][nmaT _s][nma ⁵ C _s][nmaG _s][nmaG _s][nmaG _s][nmaA _s][nmaA _s][nmaT _s][nmaA _s][nmaA][nmaG][nmaA][nmaG][nmaT][nmaT _s][nma ⁵ C]	969
419	TCCATCCATCCTTGAGTTCTTTCCAG	355	[iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iT _s][iT _s][iG _s][iA _s][iG _s][iT _s][iT _s][i ⁵ C _s][iT _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iG]	970
420	TCCATCCATCCTT[n ₆]GAGTTCTTTCCAG	497/249	[iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iT _s][iT _s][n ₆ s][iG _s][iA _s][iG _s][iT _s][iT _s][i ⁵ C _s][iT _s][iT _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iG]	971/972
421	CTCTCCATCCATCTTCTTTCCAGGAA	357	[i ⁵ C _s][iT _s][i ⁵ C _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][iT _s][iT _s][i ⁵ C _s][iT _s][iT _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iG _s][iG _s][iA _s][iA]	973
422	TCCATCCATCCTTGAGTTCTTTCCAG	355	[iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iT _s][iT _s][iG _s][iA _s][iG _s][iT _s][i ⁵ C _s][iT _s][iT _s][i ⁵ C _s][iA _s][iG]	974
423	CATCCATCTATCCGGGATAAGAGTTC	159	[i ⁵ C _s][iA][iT][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iG _s][iG _s][iG _s][iA _s][iT _s][iA _s][iA _s][iG _s][iA _s][iG _s][iT _s][iT _s][i ⁵ C]	975
424	CATCCATCTATCCGGGATAAGAGTTC	159	[i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][iT][i ⁵ C][i ⁵ C _s][iG _s][iG _s][iG _s][iA _s][iT _s][iA _s][iA _s][iG _s][iA _s][iG _s][iT _s][iT _s][i ⁵ C]	976
425	CATCCATCTATCCGGGATAAGAGTTC	159	[i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iG _s][iG _s][iG][iG][iA _s][iT _s][iA _s][iA _s]	977

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[iGs][iAs][iGs][iTs][iTs][i ⁵ C]	
426	CATCCATCTATCCGGGATAAGAGTTC	159	[i5Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iTs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iG][iT][iTs][i ⁵ C]	978
427	CATCCATCTATCCGGGATAAGAGTTC	159	[i5Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iTs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iG][iA][iG][iT][iTs][i ⁵ C]	979
428	ATCCATCTATCCGGGATAAGAGTTC	358	[iAs][iTs][i5Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iGs][iAs][iGs][iTs][iTs][i5C]	980
429	ATCCATCTATCCGGGATAAGAGTT	359	[iAs][iTs][i5Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iTs][i ⁵ Cs][i5Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iAs][iGs][iTs][iT]	981
430	ATCCATCTATCCGGGATAAGAGTTC	358	[iAs][iTs][i5Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	982
431	ATCCATCTATCCGGGATAAGAGTT	359	[iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iAs][iGs][iTs][iT]	983
432	CAACTCATCCATCAGAGTTCTTTCCA	360	[i5Cs][iAs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i5Cs][iAs][iTs][i ⁵ Cs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ Cs][iTs][iTs][iTs][i ⁵ Cs][iA]	984
433	GCCACGAATCTACGGGATAAGAGTTCT	361	[iGs][i ⁵ Cs][i5Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i5Cs][iTs][iAs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iAs][iGs][iTs][iTs][i ⁵ Cs][iT]	985
434	CCACGAATCTACCGGATAAGAGTTC	14	[i ⁵ Cs][i ⁵ C][iA][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i5Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	986
435	CCACGAATCTACCGGATAAGAGTTC	14	[i5Cs][i5Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iA][i ⁵ C][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	987
436	CCACGAATCTACCGGATAAGAGTTC	14	[i5Cs][i5Cs][iAs][i5Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i5Cs][i ⁵ Cs][iGs][i	988

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			G[[iG]][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][i ⁵ C]	
437	CCACGAATCTACCGGATAAGAGTTC	14	[i5Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iG][iT][iTs][i ⁵ C]	989
438	CCACGAATCTACCGGATAAGAGTTC	14	[i5Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i5Cs][i ⁵ Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iA][iG][iA][iG][iT][iTs][i ⁵ C]	990
439	CCACGAATCTACCGGATAAGAGTTCT	8	[i5Cs][i ⁵ C][iA][i5Cs][iGs] [iAs][iAs][iTs][i ⁵ Cs][iTs] [iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs] [iAs][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i ⁵ Cs] [iT]	991
440	CCACGAATCTACCGGATAAGAGTTCT	8	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iA][i ⁵ C][i ⁵ Cs][iGs][iGs] [iAs][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i5Cs] [iT]	992
441	CCACGAATCTACCGGATAAGAGTTCT	8	[i5Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i5Cs][i ⁵ Cs][iGs][i G][iA][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i ⁵ Cs] [iT]	993
442	CCACGAATCTACCGGATAAGAGTTCT	8	[i5Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i5Cs][i ⁵ Cs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iT][iT][i ⁵ Cs] [iT]	994
443	CCACGAATCTACCGGATAAGAGTTCT	8	[i5Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][iGs][i Gs][iAs][iTs][iAs][iAs][iG][iA][iG][iT][iT][i ⁵ Cs][iT]	995
444	CACGAATCTACCGGATAAGAGTTCT	362	[i5Cs][iAs][i ⁵ Cs][iGs][iAs] [iAs][iTs][i ⁵ Cs][iTs][iAs] [i5Cs][i5Cs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs] [iTs][iTs][i ⁵ Cs][iT]	996
445	CCACGAATCTACGATAAGAGTTCT	363	[i5Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs] [iTs][iTs][i ⁵ Cs][iT]	997
446	CACGAATCTACCGGATAAGAGT	364	[i5Cs][iAs][i ⁵ Cs][iGs][iAs] [iAs][iTs][i ⁵ Cs][iTs][iAs] [i ⁵ Cs][i ⁵ Cs][iGs][iGs][iG s][iGs][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iT]	998

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
447	CCACGAATCTACGGGGATAAGAGT	365	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iT]	999
448	CACGAATCTACCGGATAAGAGTTTC	366	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][i ⁵ C]	1000
449	CACGAATCTACCGGGATAAGAGTT	367	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	1001
450	CCACGAATCTACGGATAAGAGTTTC	368	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][i ⁵ C]	1002
451	CCACGAATCTACGGGATAAGAGTT	369	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	1003
452	CACGAATCTACCGATAAGAGTTCT	362	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][i ⁵ Cs][iT]	1004
453	CCACGAATCTACGGGGATAAGAGT	365	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	1005
454	CCACGAATCTACCGGGATAAGAGTT	22	[i ⁵ Cs][i ⁵ C][iA][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iT]	1006
455	CCACGAATCTACCGGGATAAGAGTT	22	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ C][i ⁵ Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iT][iT]	1007
456	CCACGAATCTACCGGGATAAGAGTT	22	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iT][iT]	1008
457	CCACGAATCTACCGGGATAAGAGTT	22	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iT][iT]	1009
458	CCACGAATCTACCGGGATAAGAGTT	22	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iT][iT]	1010

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			Gs][iGs][iGs][iAs][iT][iA][iA][iG][iA][iG][iT][iT]	
459	ACGAATCTACCCAGGGGATAAGAGTT	370	[iAs][i5Cs][iGs][iAs][iAs][iT][i5Cs][iT][iAs][i5Cs][i5Cs][i5Cs][iGs][iGs][iGs][iAs][iT][iAs][iAs][iGs][iAs][iT][iT]	1011
460	CTAGCCACGAATCTAAGAGTTCTTTC	371	[i5Cs][iT][iAs][iGs][i5Cs][i5Cs][iAs][i5Cs][iGs][iAs][iT][i5Cs][iT][iAs][iAs][iGs][iAs][iT][iT][i5Cs][iT][iT][iT][i5C]	1012
461	CCACGAATCTACCATAAGAGTTCTTT	372	[i5Cs][i5Cs][iAs][i5Cs][iGs][iAs][iAs][iT][i5Cs][iT][iAs][iAs][i5Cs][i5Cs][iAs][iT][iAs][iAs][iGs][iAs][iGs][iT][iT][iT][i5Cs][iT][iT][iT][iT][iT]	1013
462	CCATCCATCATCCGGGATAAGAGTCC	373	[i5Cs][i5Cs][iAs][iT][i5Cs][i5Cs][iAs][iT][i5Cs][iAs][iT][i5Cs][iGs][iGs][iGs][iAs][iT][iAs][iAs][iGs][iAs][iGs][iT][i5Cs][i5C]	1014
463	TCCATCCATCATCGGGGATAAGAGTT	120	[iT][i5C][i5C][iAs][iT][i5Cs][i5Cs][iAs][iT][i5Cs][iAs][iGs][iGs][iGs][iAs][iT][iAs][iAs][iGs][iAs][iGs][iT][iT]	1015
464	TCCATCCATCATCGGGGATAAGAGTT	120	[iT][i5Cs][i5Cs][iAs][iT][i5Cs][i5Cs][iAs][iT][i5Cs][iAs][iT][i5Cs][iGs][iGs][iGs][iAs][iT][iAs][iAs][iGs][iAs][iT][iT]	1016
465	TCCATCCATCATCGGGGATAAGAGTT	120	[iT][i5Cs][i5Cs][iAs][iT][i5Cs][i5Cs][iAs][iT][i5Cs][iAs][iT][i5Cs][iGs][iGs][iGs][iAs][iT][iAs][iAs][iGs][iAs][iT][iT]	1017
466	TCCATCCATCATCGGGGATAAGAGTT	120	[iT][i5Cs][i5Cs][iAs][iT][i5Cs][i5Cs][iAs][iT][i5Cs][iAs][iT][i5Cs][iGs][iGs][iGs][iAs][iT][iAs][iAs][iGs][iAs][iT][iT]	1018
467	TCCATCCATCATCGGGGATAAGAGTT	120	[iT][i5Cs][i5Cs][iAs][iT][i5Cs][i5Cs][iAs][iT][i5Cs][iAs][iT][i5Cs][iGs][iGs][iGs][iAs][iT][iAs][iAs][iGs][iAs][iT][iT]	1019
468	TCCATCCATCATCGGGGATAAGAGTTC	121	[iT][i5C][i5C][iAs][iT][i5Cs][i5Cs][iAs][iT][i5Cs][iAs][iT][i5Cs][iGs][iGs][iGs][iAs][iT][iAs][iAs][iGs][iAs][iT][iT][i5C]	1020

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
469	TCCATCCATCATCGGGATAAGAGTTC	121	[iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iA][iT _s][i ⁵ C _s][iGs][iGs][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C]	1021
470	TCCATCCATCATCGGGATAAGAGTTC	121	[iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iGs][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C]	1022
471	TCCATCCATCATCGGGATAAGAGTTC	121	[iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iGs][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C]	1023
472	TCCATCCATCATCGGGATAAGAGTTC	121	[iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iGs][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C]	1024
473	TCCATCCATCATCGGATAAGAGTTCT	122	[iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iGs][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C]	1025
474	CCATCCATCATCCGATAAGAGTTCTT	123	[i ⁵ C _s][i ⁵ C][iA][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C][iT _s][iT _s]	1026
475	CCATCCATCATCCGATAAGAGTTCTT	123	[i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C][iT _s][iT _s]	1027
476	CCATCCATCATCCGATAAGAGTTCTT	123	[i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C][iT _s][iT _s]	1028
477	CCATCCATCATCCGATAAGAGTTCTT	123	[i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C][iT _s][iT _s]	1029
478	CCATCCATCATCGGGATAAGAGT	374	[i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C][iT _s][iT _s]	1030
479	CCATCCATCATCGGGATAAGAGT	374	[i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s][iGs][iAs][iT _s][iAs][iAs][iGs][iAs][iGs][iT _s][iT _s][i ⁵ C][iT _s][iT _s]	1031

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs]	
480	CCATCCATCATCGGATAAGAGTTC	48	[1 ⁵ Cs][1 ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][1 ⁵ Cs][iAs][iTs][1 ⁵ Cs][iGs][iAs][iTs][iAs][iGs][iAs][iGs][iTs][1 ⁵ C]	1032
481	TCATCCATCTAGCGGGATAAGAGTTC	375	[iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	1033
482	CATCCATCTAGCCGGGATAAGAGTTC	376	[i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	1034
483	TCCATCTAGCCACGGGATAAGAGTT	377	[iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][iTs][iTs]	1035
484	ATCCATCTAGCCGGGATAAGAGTT	378	[iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][iTs][iTs]	1036
485	ATCCATCTAGCCGGGATAAGAGT	379	[iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iAs][iGs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][iTs][iTs]	1037
486	CCATCCATCATCCGGGATAAGAGTT	380	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][iTs][iTs]	1038
487	ATATATCCATCCACTAGTTCCTGGGG	381	[iAs][iTs][iAs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iAs][iGs][iTs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iGs][iGs][iGs]	1039
488	ATATATCCATCCA[n6]CTAGTTCCTGGGG	506/259	[iAs][iTs][iAs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][n6s][i ⁵ Cs][iTs][iAs][iGs][iTs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iGs][iGs][iGs]	1040/1041
489	ATATATCCATCCACTAGTTCCTGGGG	381	[iAs][iTs][iAs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iAs][iGs][iTs][iTs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iGs][iGs][iGs]	1042
490	TAGTTCATCATAAGAGTTCTTTCCA	383	[iTs][iAs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs]	1043

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[iAs][iT _s][iAs][iAs][iG _s] [iAs][iG _s][iT _s][iT _s][i ⁵ C _s] [iT _s][iT _s][iT _s][i ⁵ C _s][i ⁵ C _s] [iA]	
491	TCATCTGTTCAACAACTAG	384	[iT _s][i ⁵ C _s][iAs][iT _s][i ⁵ C _s] [iT _s][iG _s][iT _s][iT _s][i ⁵ C _s] [iAs][iAs][i ⁵ C _s][iAs][iAs] [iAs][i ⁵ C _s][iT _s][iAs][iG]	1044
492	ATCTGTTCAATCAGCATTTATTCAAC	385	[iAs][iT _s][i ⁵ C _s][iT _s][iG _s] [iT _s][iT _s][i ⁵ C _s][iAs][iAs] [iT _s][i ⁵ C _s][iAs][iG _s][i ⁵ C _s] [iAs][iT _s][iT _s][iT _s][iAs] [iT _s][iT _s][i ⁵ C _s][iAs][iAs] [i ⁵ C]	1045
493	ATCTACCCACCAGCATTTAT	386	[iAs][iT _s][i ⁵ C _s][iT _s][iAs] [i ⁵ C _s][i ⁵ C _s][i ⁵ C _s][iAs][i ⁵ C _s] [i ⁵ C _s][iAs][iG _s][i ⁵ C _s][iAs] [iT _s][iT _s][iT _s][iT _s][iAs] [iT]	1046
494	CCACGAATCTACCGTTCAATCATTCA	387	[i ⁵ C _s][i ⁵ C _s][iAs][i ⁵ C _s][iG _s] [iAs][iAs][iT _s][i ⁵ C _s][iT _s] [iAs][i ⁵ C _s][i ⁵ C _s][iG _s][iT _s] [iT _s][iT _s][i ⁵ C _s][iAs][iAs][iT _s] [i ⁵ C _s][iAs][iT _s][iT _s][i ⁵ C _s] [iA]	1047
495	CACGAATCTACCCATTACCAGCATT	388	[i ⁵ C _s][iAs][i ⁵ C _s][iG _s][iAs] [iAs][iT _s][i ⁵ C _s][iT _s][iAs] [i ⁵ C _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s] [iT _s][iT _s][i ⁵ C _s][iAs][i ⁵ C _s] [i ⁵ C _s][iAs][iG _s][i ⁵ C _s][iAs] [iT _s][iT]	1048
496	ACGAATCTACCCAGTTCAATCATTCA	389	[iAs][i ⁵ C _s][iG _s][iAs][iAs] [iT _s][i ⁵ C _s][iT _s][iAs][i ⁵ C _s] [i ⁵ C _s][i ⁵ C _s][iAs][iG _s][iT _s] [iT _s][iT _s][i ⁵ C _s][iAs][iAs][iT _s] [i ⁵ C _s][iAs][iAs][iT _s][i ⁵ C _s] [iA]	1049
497	CCACGAATCTACCGTTCAATCATTCA	387	[i ⁵ C _s][i ⁵ C][iA][i ⁵ C][iG _s] [iAs][iAs][iT _s][i ⁵ C _s][iT _s] [iAs][i ⁵ C _s][i ⁵ C _s][iG _s][iT _s] [iT _s][i ⁵ C _s][iAs][iAs][iT _s] [i ⁵ C _s][iAs][iT _s][iT _s][i ⁵ C _s] [iA]	1050
498	CCACGAATCTACCGTTCAATCATTCA	387	[i ⁵ C _s][i ⁵ C _s][iAs][i ⁵ C _s][iG _s] [iAs][iAs][iT _s][iT _s][i ⁵ C _s][iT _s] [iAs][i ⁵ C _s][i ⁵ C _s][iG _s][iT _s] [iT _s][iT _s][i ⁵ C _s][iAs][iAs][iT _s] [iT _s][i ⁵ C _s][iA][iT][iT][i ⁵ C _s] [iA]	1051
499	CACGAATCTACCCATTACCAGCATT	388	[i ⁵ C _s][iA][i ⁵ C][iG][iAs][iAs] [iT _s][i ⁵ C _s][iT _s][iAs][i ⁵ C _s] [i ⁵ C _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s] [iT _s][i ⁵ C _s][iAs][i ⁵ C _s][i ⁵ C _s] [iAs][iG _s][i ⁵ C _s][iAs][iT _s] [iT]	1052
500	CACGAATCTACCCATTACCAGCATT	388	[i ⁵ C _s][iAs][i ⁵ C _s][iG _s][iAs] [iAs][iT _s][i ⁵ C _s][iT _s][iAs] [i ⁵ C _s][i ⁵ C _s][i ⁵ C _s][iAs][iT _s] [iT _s][iT _s][i ⁵ C _s][iAs][i ⁵ C _s] [i ⁵ C _s][iAs][iG][i ⁵ C][iA][iT _s] [iT]	1053

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
501	CCATCCATGTACTCAATCAT	390	[1 ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iT]	1054
502	GAATCTACCCATCTGTTCAATC	106	[zGs][zAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][z ⁵ Cs][zAs][zTs][z ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][zTs][1 ⁵ C]	1055
503	CGAATCTACCCAATCTGTTCAATC	36	[z ⁵ Cs][zGs][zAs][zAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][z ⁵ Cs][zAs][zAs][zTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][zAs][zAs][zTs][1 ⁵ C]	1056
504	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][1A]	1057
505	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	1058
506	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iT][iT][i ⁵ C][iA][iA][iTs][i ⁵ C]	1059
507	CGAATCTACCCACATCTGTTCAAT	111	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iT]	1060
508	ACGAATCTACCCATCTGTTCAATC	113	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1061
509	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ⁵ C][iG][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1062
510	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C][i ⁵ C][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1063
511	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1064

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
512	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ⁵ Cs][iGs][iAs][iAs] [iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs] [i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i As][iTs][i ⁵ Cs][iTs][iGs][i Ts][iTs][i ⁵ Cs][iA][iA][iT s][i ⁵ C]	1065
513	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iG][iA][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs] [i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ C]	1066
514	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C] [i ⁵ C][iAs][iAs][iTs][i ⁵ Cs] [iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ C]	1067
515	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C] s][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C] [iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ C]	1068
516	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C] s][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iA][iA][iTs][i ⁵ C]	1069
517	CGAATCTACCCACCTGTTCAATCA	391	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C] s][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	1070
518	CGAATCTACCCACATCTGTTCAAT	111	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C] s][i ⁵ Cs][iAs][i ⁵ Cs][iAs][i Ts][i ⁵ Cs][iTs][iGs][iTs][i Ts][i ⁵ Cs][iAs][iAs][iT]	1071
519	CGAATCTACCCACTCTGTTCAATC	392	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C] s][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1072
520	CGAATCTACCCAATCTGTTCAATCA	393	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C] s][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][i A]	1073
521	CGAATCTACCCACTCTGTTCAATCA	394	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C] s][i ⁵ Cs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	1074
522	ACGAATCTACCCCATCTGTTCAATC	395	[iAs][i ⁵ Cs][iGs][iAs][iAs] [iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs] [i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i Ts][i ⁵ Cs][iTs][iGs][iTs][i Ts][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1075
523	ACGAATCTACCCAATCTGTTCAATC	396	[iAs][i ⁵ Cs][iGs][iAs][iAs] [iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs]	1076

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[[i ⁵ Cs][i ⁵ Cs][iAs][iAs][iT s][i ⁵ Cs][iTs][iGs][iTs][iT s][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	
524	CACGAATCTACCCGTGTTCAATCATTC	397	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTsCs][i As][iAs][iTs][i ⁵ Cs][iAs][i Ts][iTs][i ⁵ C]	1077
525	ACGAATCTACCCATGTTCAATCATTC	398	[iAs][i ⁵ Cs][iGs][iAs][iAs] [iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iG s][iTs][iTs][i ⁵ Cs][iAs][iA s][iTs][i ⁵ Cs][iAs][iTs][iT s][i ⁵ C]	1078
526	CCACGAATCTACCCGTGTTCAATCATT	399	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][iT]	1079
527	ACGAATCTACCCACTGTTCAATCATT	400	[iAs][i ⁵ Cs][iGs][iAs][iAs] [iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i Ts][iGs][iTs][iTs][i ⁵ Cs][i As][iAs][iTs][i ⁵ Cs][iAs][i Ts][iT]	1080
528	CCACGAATCTACCCATCTGTTCAATC	401	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG s][iAs][iAs][iTs][i ⁵ Cs][iT s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1081
529	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ C][i ⁵ C] [i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1082
530	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iT s][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C][iAs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1083
531	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ C][i ⁵ C] [i ⁵ C][iAs][iAs][iTs][i ⁵ Cs] [iTs][iGs][iTs][iTs][i ⁵ Cs] [iAs][iAs][iTs][i ⁵ C]	1084
532	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C][i ⁵ C][iAs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1085
533	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C][i ⁵ C][iAs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ C]	1086
534	CGAATCTACCCAATCTGTTCAATC	36	[i ⁵ Cs][iGs][iAs][iAs][iTs] [i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ C][i ⁵ C][iAs][iAs][iTs][i ⁵ Cs][i ⁵ C][iAs][iAs][iTs][i ⁵ Cs]	1087

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[[iTs][iGs][iTs][iTs][1 ⁵ Cs][iAs][iAs][1Ts][1 ⁵ C]	
535	CGAATCTACCCAATCTGTTCAATC	36	[1 ⁵ Cs][1Gs][1As][1As][iTs][i ³ Cs][iTs][iAs][i ³ Cs][1 ⁵ C][i ³ C][1As][1As][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ³ Cs][1As][1As][1Ts][1 ⁵ C]	1088
536	CGAATCTACCCAATCTGTTCAATC	36	[1 ⁵ Cs][1Gs][1As][iAs][iTs][i ³ Cs][iTs][iAs][i ³ Cs][1 ⁵ C][1 ⁵ C][1As][1As][1Ts][i ³ Cs][iTs][iGs][iTs][iTs][1 ⁵ Cs][iAs][1As][1Ts][1 ⁵ C]	1089
537	CGAATCTACCCAATCTGTTCAATC	36	[1 ⁵ Cs][1Gs][1As][iAs][iTs][i ³ Cs][iTs][iAs][i ³ Cs][i ³ C][i ³ C][1As][1As][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ³ Cs][iAs][1As][1Ts][1 ⁵ C]	1090
538	CGAATCTACCCAATCTGTTCAATC	36	[1 ⁵ Cs][1Gs][iAs][1As][1Ts][i ³ Cs][iTs][iAs][1 ⁵ Cs][i ³ C][1 ⁵ C][1As][1As][1Ts][i ³ Cs][iTs][iGs][iTs][iTs][1 ⁵ Cs][1As][1As][1Ts][1 ⁵ C]	1091
539	AGCCACGAATCTGTTCAATC	402	[1As][1Gs][1 ⁵ Cs][1 ⁵ Cs][iAs][i ³ Cs][iGs][iAs][iAs][iTs][1 ⁵ Cs][iTs][iGs][iTs][iTs][i ³ Cs][iAs][iAs][1Ts][1 ⁵ C]	1092
540	CCACCAACTCATTCATCTGT	403	[1 ⁵ Cs][1 ⁵ Cs][iAs][i ³ Cs][i ⁵ Cs][iAs][iAs][i ³ Cs][iTs][i ³ Cs][iAs][iTs][iTs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][1Gs][1T]	1093
541	CGAATCTACCCACATCTGTTCAATCA	2	[1 ⁵ Cs][1Gs][iAs][iAs][iTs][i ³ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][iAs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ³ Cs][iA][iA][iT][i ³ Cs][iA]	1094
542	CGAATCTACCCACATCTGTTCAATCA	2	[1 ⁵ Cs][iGs][1As][1As][iTs][i ³ Cs][iTs][iAs][i ³ Cs][i ⁵ Cs][i ³ Cs][iAs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ³ Cs][iA][iA][iT][i ³ Cs][iA]	1095
543	CGAATCTACCCACATCTGTTCAATCA	2	[1 ⁵ Cs][iGs][iAs][iAs][1Ts][1 ⁵ Cs][iTs][iAs][1 ⁵ Cs][i ⁵ Cs][1 ⁵ Cs][iAs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ³ Cs][iA][iA][iT][i ³ Cs][iA]	1096
544	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ³ Cs][1Ts][1As][i ³ Cs][i ³ Cs][1 ⁵ Cs][iAs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ³ Cs][iA][iA][iT][i ³ Cs][iA]	1097
545	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ³ Cs][iTs][iAs][1 ⁵ Cs][1 ⁵ Cs][i ³ Cs][iAs][i ³ Cs][iAs][iTs][i ³ Cs][iTs][iGs][iTs][iTs][i ³ Cs][iA][iA][iT][i ³ Cs][iA]	1098

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
546	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] s][l ⁵ Cs][lAs][i ⁵ Cs][iAs][i T _s][i ⁵ Cs][iT _s][iGs][iT _s][i T _s][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	1099
547	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] s][i ⁵ Cs][iAs][l ⁵ Cs][lAs][i T _s][i ⁵ Cs][iT _s][iGs][iT _s][i T _s][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	1100
548	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] s][i ⁵ Cs][iAs][i ⁵ Cs][iAs][l T _s][l ⁵ Cs][iT _s][iGs][iT _s][i T _s][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	1101
549	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][l ⁵ Cs] s][i ⁵ Cs][iAs][i ⁵ Cs][iAs][i T _s][i ⁵ Cs][lT _s][lGs][iT _s][i T _s][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	1102
550	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] s][i ⁵ Cs][iAs][i ⁵ Cs][iAs][i T _s][i ⁵ Cs][iT _s][iGs][lT _s][l T _s][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	1103
551	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][l ⁵ Cs][i ⁵ Cs] s][i ⁵ Cs][iAs][i ⁵ Cs][iAs][i T _s][i ⁵ Cs][iT _s][iGs][iT _s][i T _s][i ⁵ Cs][iA][lA][iT][i ⁵ Cs][iA]	1104
552	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] s][l ⁵ Cs][iAs][i ⁵ Cs][iAs][i T _s][i ⁵ Cs][iT _s][iGs][iT _s][i T _s][i ⁵ Cs][iA][lA][lT][i ⁵ Cs][iA]	1105
553	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] s][i ⁵ Cs][iAs][i ⁵ Cs][iAs][i T _s][i ⁵ Cs][iT _s][iGs][iT _s][i T _s][i ⁵ Cs][iA][iA][iT][l ⁵ Cs][lA]	1106
554	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][lAs][lAs][iT _s] [i ⁵ Cs][iT _s][iAs][l ⁵ Cs][l ⁵ Cs] s][i ⁵ Cs][iAs][i ⁵ Cs][iAs][i T _s][i ⁵ Cs][iT _s][iGs][iT _s][i T _s][i ⁵ Cs][iA][lA][lT][i ⁵ Cs][iA]	1107
555	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][lT _s]][l ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] s][i ⁵ Cs][iAs][i ⁵ Cs][iAs][i T _s][i ⁵ Cs][iT _s][iGs][iT _s][i T _s][l ⁵ Cs][lA][iA][iT][i ⁵ Cs s][iA]	1108

TABLE 11--continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
556	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iTs][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	1109
557	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C][iA][i ⁵ C][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	1110
558	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iA][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iT][i ⁵ C][iT][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	1111
559	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C][iA][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	1112
560	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iG][iA][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][iTs][i ⁵ C][iTs][iGs][iTs][iTs][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	1113
561	CGAATCTACCCACATCTGTTCAATCA	404	[x][dt][dc][da][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iTs][i ⁵ Cs][iA][iA][iT][i ⁵ Cs][iA]	1114
562	CGAATCTACCCACATCTGTTCAATCA	2	[nma ⁵ Cs][nmaGs][nmaAs][nmaAs][nmaTs][nma ⁵ Cs][nmaTs][nmaAs][nma ⁵ Cs][nma ⁵ Cs][nma ⁵ Cs][nmaAs][nma ⁵ Cs][nmaAs][nmaTs][nma ⁵ Cs][nmaTs][nmaGs][nmaTs][nmaTs][nma ⁵ Cs][nmaAs][nmaAs][nmaTs][nma ⁵ Cs][nmaA]	1115
563	ACGAATCTACCCACATCTGTTCAATC	93	[nmaAs][nma ⁵ Cs][nmaGs][nmaAs][nmaAs][nmaTs][nma ⁵ Cs][nmaTs][nmaAs][nma ⁵ Cs][nma ⁵ Cs][nma ⁵ Cs][nmaAs][nma ⁵ Cs][nmaAs][nmaTs][nma ⁵ Cs][nmaTs][nmaGs][nmaTs][nmaTs][nma ⁵ Cs][nmaAs][nmaAs][nmaTs][nma ⁵ C]	1116
564	CGAATCTACCCACATCTGTTCAATCA	2	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iT][i ⁵ Cs][iAs][iAs][iTs][i ⁵ Cs][iA]	1117
565	ACGAATCTACCCACATCTGTTCAATC	93	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iTs][iGs][iTs][iT][iT][i ⁵ Cs][iAs][iAs][iTs][iT][i ⁵ C]	1118

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
566	CCATCCATCATCTGTTCAAT	405	[1 ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][iT _s][iG _s][iT _s][iT _s][i ⁵ Cs][iAs][iAs][1T]	1119
567	CCATCCATGTACTGTTCAAT	406	[1 ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iG _s][iT _s][iAs][i ⁵ Cs][iT _s][iG _s][iT _s][iT _s][i ⁵ Cs][iAs][iAs][1T]	1120
568	CCATCCATGTACTTTCAATC	407	[1 ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iG _s][iT _s][iAs][1 ⁵ Cs][iT _s][iT _s][iT _s][i ⁵ Cs][iAs][iAs][iT _s][1 ⁵ C]	1121
569	CCATCCATGTACTTCAATCA	408	[1 ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iG _s][iT _s][iAs][1 ⁵ Cs][iT _s][iT _s][i ⁵ Cs][iAs][iAs][iT _s][i ⁵ Cs][1A]	1122
570	ACGAATCTACCCAATCCTTTTATCTA	409	[iAs][i ⁵ Cs][iG _s][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iT _s][iT _s][iT _s][iT _s][iAs][iT _s][i ⁵ Cs][iT _s][iA]	1123
571	CGAATCTACCCACATCCTTTTATCTA	410	[i ⁵ Cs][iG _s][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iT _s][i ⁵ Cs][iT _s][iT _s][iT _s][iT _s][iT _s][iAs][iT _s][i ⁵ Cs][iT _s][iA]	1124
572	CACGAATCTACCCCTTTTATCTACTC	411	[i ⁵ Cs][iAs][i ⁵ Cs][iG _s][iAs][iAs][iT _s][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iT _s][iT _s][iT _s][iT _s][iAs][iT _s][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][iT _s][i ⁵ C]	1125
573	ACGAATCTACCCATCCTTTTATCTAC	412	[iAs][i ⁵ Cs][iG _s][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iT _s][iT _s][iT _s][iT _s][iT _s][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ C]	1126
574	ACGAATCTACCCACTTTTATCTACTC	413	[iAs][i ⁵ Cs][iG _s][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iT _s][iT _s][iT _s][iT _s][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][iT _s][i ⁵ C]	1127
575	CACGAATCTACCCATCCTTTTATCTA	414	[i ⁵ Cs][iAs][i ⁵ Cs][iG _s][iAs][iAs][iT _s][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iT _s][iT _s][iT _s][iT _s][iT _s][iT _s][iT _s][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][iT _s][iA]	1128
576	CCATCCATGTACTTTTATC	415	[1 ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iG _s][iT _s][iAs][i ⁵ Cs][iT _s][iT _s][iT _s][iT _s][iT _s][iT _s][iAs][iT _s][i ⁵ Cs][iT _s][iAs][iT _s][1 ⁵ C]	1129

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
577	CCATCCATGTACTCTTTTAT	416	[1 ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][iT ⁵ s][iT ⁵ s][iAs][1T]	1130
578	CCATCCATGTACTTGTACTC	417	[1 ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][1 ⁵ C]	1131
579	CGAATCTACCCACCATCCATGTACTC	418	[i ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][iT ⁵ s][iAs][i ⁵ Cs][1 ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ C]	1132
580	CGAATCTACCCACATCCATGTACTCA	419	[i ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][iT ⁵ s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ Cs][iA]	1133
581	GCCACGAATCTACTCCATGTACTCAC	420	[iG ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ Cs][iAs][i ⁵ C]	1134
582	ACGAATCTACCCACATCCATGTACTC	421	[iAs][i ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][iT ⁵ s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ C]	1135
583	ACGAATCTACCCAATCCATGTACTCA	422	[iAs][i ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][iT ⁵ s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ Cs][iA]	1136
584	CGAATCTACCCACTCCATGTACTCAC	423	[i ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][iT ⁵ s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ Cs][iAs][i ⁵ C]	1137
585	CACGAATCTACCCCATCCATGTACTC	3	[i ⁵ Cs][iAs][i ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][iT ⁵ s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ C]	1138
586	ACGAATCTACCCATCCATGTACTC	424	[iAs][i ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][iT ⁵ s][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s][iT ⁵ s][iAs][i ⁵ Cs][iT ⁵ s][i ⁵ C]	1139
587	ACGAATCCATCCATGTACTC	425	[1As][1 ⁵ Cs][iG ⁵ s][iAs][iAs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][i ⁵ Cs][i ⁵ Cs][iAs][iT ⁵ s][iG ⁵ s]	1140

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			s][iTs][iAs][i ⁵ Cs][lTs][l ⁵ C]	
588	ACGAATCATCCATGTACTCA	426	[lAs][l ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iAs][iTs][l ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][l ⁵ Cs][lA]	1141
589	CGAATCTCATCCATGTACTC	427	[l ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lTs][l ⁵ C]	1142
590	CGAATCTATCCATGTACTCA	428	[l ⁵ Cs][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][l ⁵ Cs][lA]	1143
591	GAATCTACATCCATGTACTC	429	[lGs][lAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lTs][l ⁵ C]	1144
592	AATCTACCATCCATGTACTC	430	[lAs][lAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lTs][l ⁵ C]	1145
593	ACGAATCCATCCATGTACTC	425	[lAs][l ⁵ C][iG][iAs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lTs][l ⁵ C]	1146
594	ACGAATCCATCCATGTACTC	425	[lAs][l ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iA][i ⁵ C][lTs][l ⁵ C]	1147
595	GAATCTACCCATCTCTCCAT	431	[lGs][lAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][lTs][l ⁵ C]	1148
596	TCCATCATCCATGTACTCAC	432	[lTs][l ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][lAs][l ⁵ C]	1149
597	GAATCTACCCATCCATGTAC	433	[lGs][lAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][lAs][l ⁵ C]	1150
598	CGAATCTACCATCCATGTAC	82	[l ⁵ Cs][iG][iA][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	1151
599	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iA][i ⁵ C][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	1152

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
600	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] [iAs][iT _s][i ⁵ C][i ⁵ Cs][iAs] [iT _s][iGs][iT _s][iAs][i ⁵ C]	1153
601	CGAATCTACCATCCATGTAC	82	[i ⁵ Cs][iGs][iAs][iAs][iT _s] [i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs] [iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][iG][iT _s][iAs][i ⁵ C]	1154
602	CCATCCATGTACTTCACCCA	434	[i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs] [i ⁵ Cs][iAs][iT _s][iGs][iT _s] [iAs][i ⁵ Cs][iT _s][iT _s][i ⁵ Cs] [iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs] [iAs]	1155
603	CAATCATCCATCCATGTACT	435	[i ⁵ Cs][iAs][iAs][iT _s][i ⁵ Cs] [iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s] [iGs][iT _s][iAs][i ⁵ Cs] [iT _s]	1156
604	CAATCATCCATCCATGTACT	435	[i ⁵ Cs][iAs][iT _s][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s] [iGs][iT _s][iAs][i ⁵ Cs][iT _s]	1157
605	CAATCATCCATCCATGTACT	435	[i ⁵ Cs][iAs][iT _s][i ⁵ Cs] [iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s] [iGs][iT _s][iAs][i ⁵ Cs][iT _s]	1158
606	GTTCAATCCATCCATGTACT	436	[iGs][iT _s][iT _s][i ⁵ Cs][iAs] [iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s] [iGs][iT _s][iAs][i ⁵ Cs] [iT _s]	1159
607	TTCATCCATCCATGTACT	437	[iT _s][iT _s][i ⁵ Cs][iAs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s] [iGs][iT _s][iAs][i ⁵ Cs][iT _s]	1160
608	TCAATCACCATCCATGTACT	438	[iT _s][i ⁵ Cs][iAs][iAs][iT _s] [i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s] [iGs][iT _s][iAs][i ⁵ Cs][iT _s]	1161
609	TTTATCCATCCATGTACT	439	[iT _s][iT _s][iT _s][iT _s][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s] [iGs][iT _s][iAs][i ⁵ Cs][iT _s]	1162
610	CTTTATCCATCCATGTACT	440	[i ⁵ Cs][iT _s][iT _s][iT _s][iT _s] [iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s] [iGs][iT _s][iAs][i ⁵ Cs][iT _s]	1163
611	TCACCCACCATCCATGTACT	441	[iT _s][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs] [i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs] [iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs] [iT _s][iGs][iT _s][iAs][i ⁵ Cs] [iT _s]	1164
612	CCATCCATGTACTCCATCCATGTACT	442	[i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs] [i ⁵ Cs][iAs][iT _s][iGs][iT _s] [iAs][i ⁵ Cs][iT _s][i ⁵ Cs][i ⁵ Cs] [i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs]	1165

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	
613	CCATCCATGTACT[n3]CCATCCATGTACT	246/246	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][n3s][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1166/1167
614	CCATCCATCATCC[n6]GGGATAAGAGTTC	198/253	[i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][n6s][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iTs][iTs][i ⁵ C]	1168/1169
615	GCCACGAATCTACTCCATCCATGTAC	445	[iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][iC]	1170
616	GCCACGAATCTACCCATCCATGTACT	446	[iGs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1171
617	CACGAATCTACCCATCCATGTAC	447	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	1172
618	ACGAATCTACCCACCATCCATGTACT	448	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1173
619	CCACGAATCTACCCCATCCATGTACT	449	[i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1174
620	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1175
621	CACGAATCTACCCCATCCATGTACT	1	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1176
622	ACGAATCTACCCCATCCATGTAC	37	[iAs][i ⁵ C][iG][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C]	1177

TABLE 11-continued

[illegible]

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
635	CGAATCTTCCATCCATGTAC	458	[¹⁵ Cs][lGs][iAs][iAs][iT _s][¹⁵ Cs][iT _s][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1190
636	GAATCTATCCATCCATGTAC	459	[lGs][lAs][iAs][iT _s][¹⁵ Cs][iT _s][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1191
637	GAATCTACCATCCATGTACT	455	[lGs][lA][iA][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1192
638	GAATCTACCATCCATGTACT	455	[lGs][lA][iA][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1193
639	GAATCTACCATCCATGTACT	455	[lGs][lAs][iAs][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1194
640	GAATCTACCATCCATGTACT	455	[lGs][lAs][iAs][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1195
641	GAATCTACCATCCATGTACT	455	[lGs][lAs][iAs][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1196
642	GAATCTACCATCCATGTACT	455	[lGs][lAs][iAs][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1197
643	GAATCTACCATCCATGTACT	455	[lGs][lAs][iAs][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1198
644	GAATCTACCATCCATGTACT	455	[lGs][lAs][iAs][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1199
645	CGAATCTACCATCCATGTAC	82	[¹⁵ Cs][lG][iA][iAs][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][lAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1200
646	CGAATCTACCATCCATGTAC	82	[¹⁵ Cs][lG][iA][iA][iT _s][¹⁵ Cs][iT _s][iAs][¹⁵ Cs][¹⁵ Cs][lAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1201
647	ACGAATCTCCATCCATGTAC	457	[lAs][¹⁵ C][iG][iAs][iAs][iT _s][¹⁵ Cs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1202
648	ACGAATCTCCATCCATGTAC	457	[lAs][¹⁵ Cs][iGs][iAs][iAs][iT _s][¹⁵ Cs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][¹⁵ Cs][¹⁵ Cs][iAs][iT _s][iGs][iT _s][lAs][¹⁵ C]	1203

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
649	ACGAATCCCATCCATGTACT	452	[1As][1 ⁵ C][iG][iAs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iAs][1 ⁵ Cs][1T]	1204
650	CGAATCTCCATCCATGTACT	454	[1 ⁵ Cs][G][iA][iAs][iT _s][i ⁵ Cs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][1 ⁵ Cs][1 ⁵ Cs][iAs][iT _s][iGs][iT _s][iAs][1 ⁵ Cs][1T]	1205
651	CGAATCTCCATCCATGTACT	454	[1 ⁵ Cs][1Gs][iAs][iAs][iT _s][i ⁵ Cs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iA][1 ⁵ Cs][1T]	1206
652	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iAs][i ⁵ Cs][iT _s]	1207
653	GAATCTACCATCCATGTACT	455	[1Gs][iAs][iAs][iT _s][i ⁵ Cs][iT _s][1As][1 ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iAs][i ⁵ Cs][1T]	1208
654	GAATCTACCATCCATGTACT	455	[1Gs][iA][iA][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iAs][i ⁵ Cs][1T]	1209
655	GAATCTACCATCCATGTACT	455	[1Gs][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][1T][iA][i ⁵ Cs][1T]	1210
656	GAATCTACCATCCATGTACT	455	[1Gs][iA][iA][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iA][i ⁵ Cs][1T]	1211
657	GAATCTACCATCCATGTACT	455	[1Gs][iA][iA][iT _s][i ⁵ Cs][iT _s][iA][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iAs][i ⁵ Cs][1T]	1212
658	GAATCTACCATCCATGTACT	455	[1Gs][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iA][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iA][i ⁵ Cs][1T]	1213
659	GAATCTACCATCCATGTAC	460	[1Gs][1As][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][1As][1 ⁵ C]	1214
660	GAATCTACCATCCATGTACT	455	[1Gs][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iAs][i ⁵ Cs][1T]	1215
661	GAATCTACCATCCATGTACT	455	[1Gs][iAs][iAs][iT _s][i ⁵ Cs][iT _s][iAs][1 ⁵ Cs][1 ⁵ Cs][iAs][iT _s][i ⁵ Cs][i ⁵ Cs][iAs][iT _s][iGs][iT _s][iAs][iCs][1T]	1216
662	GAATCTACCATCCATGTACT	455	[1Gs][iAs][iAs][iT _s][i ⁵ Cs][iT _s][1As][i ⁵ Cs][i ⁵ Cs][1As]	1217

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][1T]	
663	GAATCTACCCATCCATGTACT	461	[1G _s][iA _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][1 ⁵ C _s][1 ⁵ C _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][1T]	1218
664	CGAATCTACCATCCATGTACT	462	[1 ⁵ C _s][iG _s][iA _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][1 ⁵ C _s][1 ⁵ C _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][1T]	1219
665	ACGAATCTACCATCCATGTACT	463	[1A _s][i ⁵ C _s][iG _s][iA _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][1 ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][1T]	1220
666	CACGAATCTACCATCCATGTACT	464	[1 ⁵ C _s][iA _s][i ⁵ C _s][iG _s][iA _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][1T]	1221
667	CCACGAATCTACCATCCATGTACT	465	[1 ⁵ C _s][i ⁵ C _s][iA _s][i ⁵ C _s][iG _s][iA _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][1 ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1222
668	CCACGAATCTACCATCCATGTACT	465	[i ⁵ C _s][i ⁵ C _s][iA _s][i ⁵ C _s][iG _s][iA _s][iA _s][iT _s][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][1 ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1223
669	GAATCTACCATCCATGTACT	455	[1G _s][1A][iA][iT][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1224
670	GAATCTACCATCCATGTACT	455	[iG _s][iA][1A][1T][1 ⁵ C _s][iT _s][iA _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][1 ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1225
671	GAATCTACCATCCATGTACT	455	[iG _s][iA][iA][1T][1 ⁵ C _s][iT _s][iA _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1226
672	GAATCTACCATCCATGTACT	455	[iG _s][iA][iA][iT][i ⁵ C _s][iT _s][1A _s][1 ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1227
673	GAATCTACCATCCATGTACT	455	[iG _s][iA][iA][iT][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1228
674	GAATCTACCATCCATGTACT	455	[iG _s][iA][iA][iT][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1229
675	GAATCTACCATCCATGTACT	455	[iG _s][iA][iA][iT][i ⁵ C _s][iT _s][iA _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][i ⁵ C _s][i ⁵ C _s][iA _s][iT _s][iG _s][iT _s][iA _s][i ⁵ C _s][iT _s]	1230

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			Ts][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	
676	GAATCTACCATCCATGTACT	455	[iGs][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1231
677	GAATCTACCATCCATGTACT	455	[iGs][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1232
678	GAATCTACCATCCATGTACT	455	[iGs][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1233
679	GAATCTACCATCCATGTACT	455	[iGs][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1234
680	GAATCTACCATCCATGTACT	455	[iGs][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1235
681	GAATCTACCATCCATGTACT	455	[iGs][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1236
682	GAATCTACCATCCATGTACT	455	[iGs][iA][iA][iT][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1237
683	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1238
684	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1239
685	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1240
686	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1241
687	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1242
688	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1243

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			s][iGs][iT][iAs][i ⁵ Cs][iT]	
689	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1244
690	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1245
691	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1246
692	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1247
693	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1248
694	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1249
695	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1250
696	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1251
697	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1252
698	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1253
699	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1254
700	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1255

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
701	GAATCTACCATCCATGTACT	455	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ C][i ⁵ C][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs] [iGs][iTs][iAs][i ⁵ Cs][1T]	1256
702	GAATCTACCATCCATGTACT	455	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][iA] [iT][i ⁵ Cs][i ⁵ Cs][iAs][iTs] [iGs][iTs][iAs][i ⁵ Cs][1T]	1257
703	GAATCTACCATCCATGTACT	455	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs] [iTs][i ⁵ C][i ⁵ C][iAs][iTs] [iGs][iTs][iAs][i ⁵ Cs][1T]	1258
704	GAATCTACCATCCATGTACT	455	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs] [iTs][i ⁵ Cs][i ⁵ Cs][iA][iT] [iGs][iTs][iAs][i ⁵ Cs][1T]	1259
705	GAATCTACCATCCATGTACT	455	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs] [iTs][i ⁵ Cs][i ⁵ Cs][iAs][iT s][iG][iT][iAs][i ⁵ Cs][1T]	1260
706	GAATCTACCATCCATGTACT	461	[LGs][iA][iA][iTs][i ⁵ Cs][i Ts][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs] [iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs] [iTs][iGs][iTs][iAs][i ⁵ Cs] [1T]	1261
707	GAATCTACCATCCATGTACT	461	[LGs][iAs][iAs][iT][i ⁵ C][i Ts][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs] [iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs] [iTs][iGs][iTs][iAs][i ⁵ Cs] [1T]	1262
708	GAATCTACCATCCATGTACT	461	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iT][iA][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs] [iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs] [iTs][iGs][iTs][iAs][i ⁵ Cs] [1T]	1263
709	GAATCTACCATCCATGTACT	461	i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs] [iAs][iTs][iGs][iTs][iAs] [i ⁵ Cs][1T]	1264
710	GAATCTACCATCCATGTACT	461	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C] [iA][iTs][i ⁵ Cs][i ⁵ Cs][iAs] [iTs][iGs][iTs][iAs][i ⁵ Cs] [1T]	1265
711	GAATCTACCATCCATGTACT	461	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C s][iAs][iT][i ⁵ C][i ⁵ Cs][iAs] [iTs][iGs][iTs][iAs][i ⁵ Cs] [1T]	1266
712	GAATCTACCATCCATGTACT	461	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C s][iAs][iTs][i ⁵ Cs][i ⁵ C][iA] [iTs][iGs][iTs][iAs][i ⁵ Cs] [1T]	1267
713	GAATCTACCATCCATGTACT	461	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C s][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i As][iT][iG][iTs][iAs][i ⁵ Cs] [1T]	1268
714	GAATCTACCATCCATGTACT	461	[LGs][iAs][iAs][iTs][i ⁵ Cs] [iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ C]	1269

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			s][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	
715	GAATCTACCCATCCATGTACT	461	[lGs][lAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1270
716	GAATCTACCCATCCATGTACT	461	[iGs][iAs][lAs][lTs][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1271
717	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][l ⁵ Cs][lTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][lTs][iAs][i ⁵ Cs][iT]	1272
718	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][lAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1273
719	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1274
720	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1275
721	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1276
722	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][lTs][iAs][i ⁵ Cs][iT]	1277
723	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][lTs][iAs][i ⁵ Cs][iT]	1278
724	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][lAs][i ⁵ Cs][iT]	1279
725	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1280
726	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs][iT][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs][iT][iGs][iT][iAs][i ⁵ Cs][iT]	1281

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			Cs][iAs][iTs][iGs][iTs][iAs][l ⁵ Cs][iT]	
727	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][lTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][iAs][iTs][l ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][lTs][iAs][l ⁵ Cs][lT]	1282
728	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iTs][i ⁵ Cs][lTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][lTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1283
729	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][l ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][l ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1284
730	GAATCTACCATCCATGTACT	455	[iGs][lAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][l ⁵ Cs][iT]	1285
731	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][lTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][lTs][iAs][i ⁵ Cs][iT]	1286
732	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][lTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][lTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1287
733	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][l ⁵ Cs][l ⁵ Cs][iAs][iTs][i ⁵ Cs][l ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1288
734	GAATCTACCATCCATGTACT	455	[enaGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][enaT]	1289
735	GAATCTACCATCCATGTACT	455	[bGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][bT]	1290
736	GAATCTACCCATCCATGTACT	461	[enaGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][enaT]	1291
737	GAATCTACCCATCCATGTACT	461	[scpGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][scpT]	1292
738	GAATCTACCCATCCATGTACT	461	[bGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][bT]	1293

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			As][iTs][iGs][iTs][iAs][i ⁵ Cs][bT]	
739	GAATCTACCATCCATGTACT	455	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][l ⁵ Cs][iT]	1294
740	GAATCTACCATCCATGTACT	455	[iGs][zAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][z ⁵ Cs][iT]	1295
741	GAATCTACCATCCATGTACT	455	[enaGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][ena ⁵ Cs][iT]	1296
742	GAATCTACCATCCATGTACT	455	[iGs][enaAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][ena ⁵ Cs][iT]	1297
743	GAATCTACCATCCATGTACT	455	[iGs][enaAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][enaT]	1298
744	GAATCTACCATCCATGTACT	461	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1299
745	GAATCTACCATCCATGTACT	461	[iGs][lAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1300
746	GAATCTACCATCCATGTACT	461	[iGs][zAs][iAs][iTs][l ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][z ⁵ Cs][lT]	1301
747	GAATCTACCATCCATGTACT	461	[iGs][scpAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][scpT]	1302
748	GAATCTACCATCCATGTACT	461	[enaGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][ena ⁵ Cs][iT]	1303
749	GAATCTACCATCCATGTACT	461	[iGs][enaAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][ena ⁵ Cs][iT]	1304
750	GAATCTACCATCCATGTACT	461	[iGs][enaAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs]	1305

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			[iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][enaT]	
751	GAATCTACCATCCATGTACT	466	[x][dt][dc][da][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][l ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1306
752	GAATCTACCCATCCATGTACT	467	[x][dt][dc][da][lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1307
753	GAATCTACCATCCATGTACT	455	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1308
754	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1309
755	GAATCTACCCATCCATGTACT	461	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1310
756	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1311
757	CGAATCTACCCATCCATGTACT	468	[l ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1312
758	GAATCTACCCATCCATGTACT	469	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1313
759	ACGAATCTACCCATCCATGTACT	470	[lAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1314
760	CGAATCTACCCATCCATGTACT	471	[l ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1315
761	GAATCTACCCCATCCATGTACT	472	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1316
762	CACGAATCTACCCATCCATGTACT	115	[l ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1317

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			Ts][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][iC][LT]	
763	CGAATCTACCCCATCCATGTACT	473	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C][LT]	1318
764	GAATCTACCCACCATCCATGTACT	474	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C][LT]	1319
765	CGAATCTACATCCATGTACT	475	[i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][LT]	1320
766	ACGAATCTACATCCATGTAC	476	[iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C][LT]	1321
767	GAATCTACCCCATCCATGTA	477	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][LT]	1322
768	CACGAATCTATCCATGTACT	478	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][LT]	1323
769	CACGAATCTAATCCATGTAC	479	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ C][LT]	1324
770	CACGAATCTACATCCATGTA	480	[i ⁵ Cs][iAs][i ⁵ Cs][iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][LT]	1325
771	GAATCTACCCCATCCATGT	481	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][LT]	1326
772	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][LT]	1327
773	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][LT]	1328
774	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][LT]	1329

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			s][iGs][iT][iAs][i ⁵ Cs] [iT]	
775	GAATCTACCATCCATGTACT	455	[lGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][l ³ Cs][l ³ Cs][iAs] [iT][l ³ Cs][i ⁵ Cs][iAs][iT] s][iGs][iT][iAs][i ⁵ Cs] [iT]	1330
776	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][l ³ Cs][l ³ Cs][iAs] [iT][i ⁵ Cs][i ⁵ Cs][iAs][iT] s][iGs][iT][iAs][i ⁵ Cs] [iT]	1331
777	GAATCTACCATCCATGTACT	455	[lGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][i ⁵ Cs][l ³ Cs][iAs] [iT][l ³ Cs][i ⁵ Cs][iAs][iT] s][iGs][iT][iAs][i ⁵ Cs][iT]]	1332
778	GAATCTACCATCCATGTACT	455	[lGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][i ⁵ Cs][l ³ Cs][lAs] [iT][i ⁵ Cs][i ⁵ Cs][iAs][iT] s][iGs][iT][iAs][i ⁵ Cs][iT]]	1333
779	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][i ⁵ Cs][l ³ Cs][lAs] [iT][l ³ Cs][i ⁵ Cs][iAs][iT] s][iGs][iT][iAs][i ⁵ Cs][lT]]	1334
780	GAATCTACCATCCATGTACT	455	[iGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][i ⁵ Cs][l ³ Cs][iAs] [iT][l ³ Cs][i ⁵ Cs][iAs][iT] s][iGs][iT][iAs][i ⁵ Cs][lT]]	1335
781	GAATCTACCATCCATGTACT	461	[lGs][iAs][iAs][iT][i ⁵ Cs] [iT][lAs][i ⁵ Cs][l ³ Cs][i ⁵ Cs] s][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs] [iT][iGs][iT][iAs][i ⁵ Cs] [iT]	1336
782	GAATCTACCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs] [iT][lAs][i ⁵ Cs][l ³ Cs][i ⁵ Cs] Cs][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs] [iT][iGs][iT][iAs][i ⁵ Cs] [iT]	1337
783	GAATCTACCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][l ³ Cs][l ³ Cs][i ⁵ Cs] s][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs] [iT][iGs][iT][iAs][i ⁵ Cs] [iT]	1338
784	GAATCTACCATCCATGTACT	461	[lGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][l ³ Cs][l ³ Cs][i ⁵ Cs] s][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs] [iT][iGs][iT][iAs][i ⁵ Cs] [iT]	1339
785	GAATCTACCATCCATGTACT	461	[lGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][i ⁵ Cs][l ³ Cs][i ⁵ Cs] s][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs] [iT][iGs][iT][iAs][i ⁵ Cs] [iT]	1340
786	GAATCTACCATCCATGTACT	461	[iGs][iAs][iAs][iT][i ⁵ Cs] [iT][iAs][i ⁵ Cs][l ³ Cs][i ⁵ Cs] s][iAs][iT][i ⁵ Cs][i ⁵ Cs][iAs] [iT]	1341

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			As][iTs][iGs][lTs][iAs][i ⁵ Cs][lT]	
787	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1342
788	GAATCTACACATCCATGTACT	482	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][lAs][l ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1343
789	GAATCTACACATCCATGTACT	482	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][lAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1344
790	GAATCTACACATCCATGTACT	482	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][iAs][l ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1345
791	GAATCTACACATCCATGTACT	482	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][lAs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1346
792	GAATCTACCCATCCATGTACT	461	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][iT]	1347
793	GAATCTACCCATCCATGTACT	461	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][lAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][i ⁵ Cs][iT]	1348
794	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][lAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][lTs][iAs][i ⁵ Cs][lT]	1349
795	GAATCTACCCATCCATGTACT	461	[iGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][i ⁵ Cs][l ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1350
796	GAATCTA[n3]CCATCCATGTACT	246	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][n3s][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1351
797	GAATCTA[n6]CCATCCATGTACT	246	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][n6s][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1351
798	GAATCTA[n3]CCCATCCATGTACT	534	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][n3s][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][lT]	1352

Exemplary antisense oligonucleotides

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			Cs][iAs][iTs][iGs][iTs][iAs][i ³ Cs][LT]	
799	GAATCTA[n6]CCCATCCATGTACT	534	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][n6s][i ⁵ Cs][i ⁵ Cs][i ³ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ³ Cs][LT]	1352
800	GAATCTACC[n3]ATCCATGTACT	535	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][n3s][iAs][iTs][i ³ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs]	1353
801	GAATCTACC[n6]ATCCATGTACT	535	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][n6s][iAs][iTs][i ³ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][LT]	1353
802	GAATCTACC[n3]CATCCATGTACT	264	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][n3s][i ³ Cs][iAs][iTs][i ³ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ³ Cs][LT]	1354
803	GAATCTACC[n6]CATCCATGTACT	264	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][n6s][l ³ Cs][iAs][iTs][i ³ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][l ³ Cs][LT]	1354
804	GAATCTAC[n3]CATCCATGTACT	264	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][n3s][i ⁵ Cs][iAs][iTs][i ³ Cs][i ³ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][LT]	1354
805	GAATCTAC[n3]CCATCCATGTACT	246	[l ³ Gs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ³ Cs][n3s][i ⁵ Cs][i ³ Cs][iAs][iTs][i ³ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][LT]	1351
806	GAATCTAC[n6]CCATCCATGTACT	246	[lGs][iAs][iAs][iTs][i ⁵ Cs][iTs][iAs][i ⁵ Cs][n6s][i ⁵ Cs][i ³ Cs][iAs][iTs][i ³ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ³ Cs][LT]	1351
807	GAATCTACCATCCATGTACT	455	[lGs][[nmaAs]][[nmaAs]][[nmaTs][[nma ³ Cs]][[nmaTs]][[nmaAs]][[nma ⁵ Cs]][[nma ³ Cs]][[nmaAs]][[nmaTs][[nma ³ Cs]][[nma ³ Cs]][[nmaAs]][[nmaTs][[nmaGs]][[nmaTs]][[nmaAs][[nma ³ Cs]][LT]	1355
808	GAATCTACCCATCCATGTACT	461	[lGs][[nmaAs]][[nmaAs]][[nmaTs][[nma ³ Cs]][[rmaTs]][[nmaAs]][[nma ⁵ Cs]][[nma ³ Cs]][[nma ³ Cs]][[nmaAs][[nmaTs]][[nma ³ Cs]][[nma ³ Cs]][[nmaAs][[nmaTs]][[nmaGs]][[nmaTs][[nmaAs][[nma ³ Cs]][LT]	1356
809	TCTACCCCCATCCATGTACT	489	[lTs][l ³ Cs][iTs][iAs][i ⁵ Cs][i ³ Cs][i ³ Cs][i ³ Cs][i ³ Cs][iAs][iTs][i ³ Cs][i ³ Cs][iAs][iTs][iGs][iTs][iAs][l ³ Cs][LT]	1357
810	ATCTACCCCATCCATGTACT	490	[lAs][lTs][i ⁵ Cs][iTs][iAs][i ³ Cs][i ³ Cs][i ³ Cs][i ³ Cs][i ³ Cs][lAs][iTs][i ⁵ Cs][i ³ Cs][iAs][	1358

TABLE 11-continued

Exemplary antisense oligonucleotides				
ASO #	Nucleobase Sequence (5' → 3')	SEQ ID NO:	Antisense Oligonucleotide Structures (5' → 3')	SEQ ID NO:
			iTs][iGs][iTs][iAs][1 ⁵ Cs][1T]	
811	GTCGCCCATCCATGTACT	491	[1Gs][1Ts][i ⁵ Cs][iGs][i ⁵ Cs][i ⁵ Cs][iGs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][1 ⁵ Cs][1T]	1359
812	GGGGTCGCCATCCATGTACT	492	[1Gs][1Gs][iGs][iGs][iTs][i ⁵ Cs][iGs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][1 ⁵ Cs][1T]	1360
813	GGGTCGCCCCATCCATGTACT	493	[1Gs][1Gs][iGs][iTs][i ⁵ Cs][iGs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][1T]	1361
814	GGTCGCCCCATCCATGTACT	494	[1Gs][1Gs][iTs][i ⁵ Cs][iGs][i ⁵ Cs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][i ⁵ Cs][i ⁵ Cs][iAs][iTs][iGs][iTs][iAs][i ⁵ Cs][1T]	1362

iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; 1A, 1G, 1⁵C, and 1T are 2'-4'methylene bridge (LNA) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; nmaA, nmaG, nma5C, and nmaT are 2'-O-NMA modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage, and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage; ena is ENA; b is BNA(NC), wherein R is Me; z is AmNA, wherein R is Me; scp is scpENA; [dt][dc][da] is a phosphodiester DNA "d(TCA)" linker; n3 is a C3 spacer as shown in Table 16; n6 is a C6 spacer as shown in Table 16; and x is a C16 lipid, 6-[(1-oxohexadecyl)amino]hexylphosphoryl.
†Each 5'-methyl cytidine (5'C) in any of the sequences provided in Table 5 may independently and optionally be replaced with cytidine.

Secondary Structures

[0288] The term “secondary structures” as used herein conforms to its generally known meaning, i.e. stems or loops. In some embodiments, pharmaceutically secondary structures include, but are not limited to, hairpin, bulge, helix, internal loop (or interior loop), external loop (or exterior loop), multi-branched loop and pseudoknot. In some embodiment, the antisense oligonucleotide may induce a secondary structure between the first target region and the second target region in the UNC13A pre-mRNA. In some embodiment, the antisense oligonucleotide may induce a secondary structure between the first target region and the second target region in the UNC13A pre-mRNA by simultaneously binding to the first target region and the second target region. In some embodiment, a secondary structure induced by the oligonucleotide may inhibit inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA. In some embodiment, a secondary structure induced by the oligonucleotide may restore UNC13A expression.

Salts

[0289] The term “salts” as used herein conforms to its generally known meaning, i.e. an ionic assembly of anions and cations. In some embodiments, the antisense oligonucleotide may be in the form of a pharmaceutically acceptable salt. In some embodiments, pharmaceutically acceptable salts comprise inorganic salts, such as monovalent or divalent inorganic salts. In some embodiments, pharmaceutically

acceptable salts include, but are not limited to, sodium, potassium, calcium, magnesium, and ammonium salts. In some embodiments, the pharmaceutically acceptable salt is a sodium salt. In some embodiments, the pharmaceutically acceptable salt is a potassium salt. In some embodiments, the pharmaceutically acceptable salt is a calcium salt. In some embodiments, the pharmaceutically acceptable salt is a magnesium salt. In some embodiments, the pharmaceutically acceptable salt is an ammonium salt. In some embodiments the ammonium salt is represented by the formula: N(R)₃, wherein each R is H or an alkyl having from 1 to 6 carbon atoms.

Pharmaceutical Compositions

[0290] Pharmaceutical compositions comprising one or more antisense oligonucleotides, either alone or in combination with prophylactic agents, therapeutic agents, and/or pharmaceutically acceptable carriers are provided. The pharmaceutical compositions comprising antisense oligonucleotides provided herein are for use in, but not limited to, diagnosing, detecting, or monitoring a disease, in preventing, treating, managing, or ameliorating a disease or one or more symptoms thereof, and/or in research. In some embodiments, a pharmaceutical composition may further comprise any other suitable therapeutic agent for treatment of a subject, e.g., a human subject having an UNC13A-associated disease (e.g., ALS, FTD, Alzheimer’s disease, or LATE). In some embodiments, the other therapeutic agents may enhance or supplement the effectiveness of the com-

plexes disclosed herein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 100%. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p.Val149Gly, p.Ala141Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Ala90Val, p.Arg116Gly, p.Asn87Ser, p.Asp102Gly, p.Asp125Val, p.Asp91Ala, p.Gln23Lcu, p.Glu101Gly, p.Glu101Lys, p.Glu50Lys, p.Gly13Arg, p.Gly148Ser, p.Gly38Arg, p.Gly42Asp, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.Gly94Asp, p.Gly94Cys, p.Gly94Ser, p.His121Gln, p.His44Arg, p.His47Arg, p.Ile113Thr, p.Ile114Thr, p.Ile150Thr, p.Leu127Ser, p.Leu145Phe, p.Leu145Ser, p.Leu39Val, p.Leu85Phe, p.Phe21Ile, p.Phe65Leu, p.Thr138Ile, p.Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p.Val149Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Asn87Ser, p.Asp102Gly, p.Gly13Arg, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.His121Gln, p.His44Arg, p.Ile150Thr, p.Leu39Val, p.Leu85Phe, p.Thr138Ile, p.Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p.Val149Gly or any combination thereof. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0291] An aspect of the disclosure includes pharmaceutical compositions comprising any one of the antisense oligonucleotides disclosed herein. In some embodiments, the pharmaceutical composition comprises any one of the antisense oligonucleotides disclosed herein and a pharmaceutically acceptable carrier. The pharmaceutical compositions disclosed herein are formulated for administration to a subject. In some embodiments, formulations as disclosed herein comprise an excipient. In some embodiments, an excipient confers to a composition improved stability, improved absorption, improved solubility and/or (e.g., and) therapeutic enhancement of the active ingredient. In some embodiments, an excipient is a pH modifier (e.g., sodium citrate, sodium phosphate, a tris base, or sodium hydroxide), a vehicle (e.g., a buffered solution, petrolatum, dimethyl sulfoxide, or mineral oil), a tonicity agent (e.g., sodium chloride, calcium chloride, magnesium chloride, potassium chloride, dextrose or sucrose or D-mannitol), a solubilizing agent (e.g., polysorbate, cyclodextrin), a binder, a disintegrant or a lubricant. Any one of the disclosed antisense oligonucleotides, when added to pharmaceutically acceptable excipients, can be packaged into kits, containers, packs, or dispensers. The pharmaceutical compositions disclosed herein can be packaged in pre-filled syringes or vials. In

some embodiments, such carriers enable pharmaceutical compositions to be formulated as, for example, tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspension and lozenges for the oral ingestion by a subject. In some embodiments, a pharmaceutically acceptable carrier or diluent is sterile water, sterile saline, sterile buffer solution or sterile artificial cerebrospinal fluid.

[0292] In some embodiments, the delivery vehicle can be used to deliver an antisense oligonucleotide to a cell or tissue. A delivery vehicle is a compound that improves delivery of an antisense oligonucleotide to a cell or tissue. A delivery vehicle can include, or consist of, but is not limited to a polymer, such as an amphipathic polymer, a membrane active polymer, a peptide, a melittin peptide, a melittin-like peptide (MLP), an anti-transferrin peptides, an anti-transferrin antibodies, a lipid, a reversibly modified polymer or peptide, or a reversibly modified membrane active polyamine. In some embodiments, any one of the antisense oligonucleotides or pharmaceutical compositions disclosed herein can be combined with lipids, nanoparticles, polymers, liposomes, micelles, DPCs, peptides, antibodies, virus vectors (e.g., AAV vector), or other delivery systems available in the art.

[0293] In some embodiments, a pharmaceutical composition is formulated to be compatible with its intended route of administration. Non-limiting examples of routes of administration include intravenous, intramuscular, intraperitoneal, intracerebrospinal, subcutaneous, intra-articular, intrasynovial, intrathecal, or intracerebroventricular routes. In some embodiments, the route of administration is intrathecal, or intracerebroventricular routes. In some embodiments, the route of administration is subcutaneous.

Kits

[0294] An aspect of the disclosure includes kits comprising any one of the antisense oligonucleotides disclosed herein or a pharmaceutical composition disclosed herein. In some embodiments, the kit is for treating an UNC13A-associated disease disclosed herein (e.g., ALS, FTD, Alzheimer's disease or LATE). In some embodiments, the kit further comprises an additional agent disclosed herein.

Methods

[0295] Some aspects of the present disclosure provide methods for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell. In some embodiments, a method for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell comprises contacting the cell with any one of the antisense oligonucleotides disclosed herein or any one of the pharmaceutical compositions disclosed herein, thereby inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in the cell.

[0296] Some aspects of the present disclosure provide methods for restoring UNC13A expression (e.g., mRNA level and/or protein level) in a cell. In some embodiments, a method for restoring UNC13A expression comprises contacting the cell with any one of the antisense oligonucleotides disclosed herein or any one of the pharmaceutical compositions disclosed herein, thereby restoring UNC13A expression (e.g., mRNA level and/or protein level) in the cell. In some embodiments, contacting the cell with any one of the antisense oligonucleotides disclosed herein restores

the UNC13A expression level (e.g., mRNA level and/or protein level) by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 95%, relative to UNC13A expression in cells not contacted with an antisense oligonucleotide. In some embodiments, the cell is in vitro (e.g., in a cell culture). In some embodiments, the cell is in vivo (e.g., in a subject).

[0297] The disclosure also provides methods for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell of a subject. In some embodiments, a method for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell comprises administering to the subject any one of the antisense oligonucleotides disclosed herein or any one of the pharmaceutical compositions disclosed herein, thereby inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in the cell.

[0298] The disclosure also provides methods for restoring UNC13A expression (e.g., mRNA level and/or protein level) in a subject compared to baseline pre-treatment levels. In some embodiments, methods comprise administering to the subject any one of the antisense oligonucleotides disclosed herein or any one of the pharmaceutical compositions disclosed herein. The UNC13A expression level (e.g., mRNA level and/or protein level) in the subject is restored in a cell, group of cells, tissue, blood, and/or other fluid of the subject. In some embodiments, the above-described methods further comprise determining UNC13A expression (e.g., mRNA level and/or protein level) in a sample(s) from the subject (e.g., UNC13A level in a blood or serum sample(s)). The level of UNC13A in a sample may be measured by general methods known in the art.

[0299] In some embodiments, in any one of the methods disclosed herein, a subject is non-human primate, or rodent. In some embodiments, a subject is a human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 100%, relative to expression of TDP-43 in a healthy subject or a subject that does not have an UNC13A-associated disease (e.g., ALS, FTD, Alzheimer's disease, or LATE). In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5 Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly, p.Ala141Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Ala90Val, p.Arg116Gly, p.Asn87Ser, p.Asp102Gly, p.Asp125Val, p.Asp91Ala, p.Gln23Leu, p.Glu101Gly, p.Glu101Lys, p.Glu50Lys, p.Gly13Arg, p.Gly148Ser, p.Gly38Arg, p.Gly42Asp, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.Gly94Asp, p.Gly94Cys, p.Gly94Ser, p.His121Gln, p.His44Arg, p.His47Arg, p.Ile113Thr, p.Ile114Thr, p.Ile150Thr, p.Leu127Ser, p.Leu145Phe, p.Leu145Ser, p.Leu39Val, p.Leu85Phe, p.Phe211Ile, p.Phe65Leu, p.Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly, p.Ala146Thr, p.Ala5Ser,

p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Asn87Ser, p.Asp102Gly, p.Gly13Arg, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.His121Gln, p.His44Arg, p.Ile150Thr, p.Leu39Val, p.Leu85Phe, p.Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly or any combination thereof. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof. In some embodiments, a subject is a patient, e.g., a human patient that has or is suspected of having a disease. In some embodiments, the subject is a human patient who has or is suspected of having an UNC13A-associated disease and/or a disease associated with TDP-43 dysfunction in the subject. In some embodiments, the UNC13A-associated disease and/or a disease associated with TDP-43 dysfunction in the subject is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is ALS or FTD. In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is LATE. In some embodiments, the neurodegenerative disease is frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington's disease, multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen's encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression.

[0300] An aspect of the disclosure includes a method of treating a subject having an UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression). In some embodiments, the method comprises administering to the subject any one of the antisense oligonucleotides disclosed herein, or a pharmaceutical composition disclosed herein. In some embodiments the method results in treating the subject having the UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression). In some embodiments, the disease is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is ALS. In some embodiments, the neurodegenerative disease is FTD. In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is LATE. In some embodiments, the neurodegenerative disease is frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington's disease, multiple

sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen's encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression.

[0301] An aspect of the disclosure includes a method of treating a subject having a disease associated with TDP-43 dysfunction in the subject. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 100% relative to expression of TDP-43 in a healthy subject. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p.Val149Gly, p.Ala141Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Ala90Val, p.Arg116Gly, p.Asn87Ser, p.Asp102Gly, p.Asp125Val, p.Asp91Ala, p.Gln23Leu, p.Glu101Gly, p.Glu101Lys, p.Glu50Lys, p.Gly13Arg, p.Gly148Ser, p.Gly38Arg, p.Gly42Asp, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.Gly94Asp, p.Gly94Cys, p.Gly94Ser, p.His121Gln, p.His44Arg, p.His47Arg, p.Ile113Thr, p.Ile114Thr, p.Ile150Thr, p.Leu127Ser, p.Leu145Phe, p.Leu145Ser, p.Leu39Val, p.Leu85Phe, p.Phe21Ile, p.Phe65Leu, p.Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p.Val149Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Asn87Ser, p.Asp102Gly, p.Gly13Arg, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.His121Gln, p.His44Arg, p.Ile150Thr, p.Leu39Val, p.Leu85Phe, p.Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p.Val149Gly or any combination thereof. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene muta-

herein. In some embodiments the method results in treating the subject having a disease associated with TDP-43 dysfunction. In some embodiments, the disease is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is ALS. In some embodiments, the neurodegenerative disease is FTD. In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is LATE. In some embodiments, the neurodegenerative disease is frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington's disease, multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen's encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression.

[0302] In some embodiments, the above-described methods further comprise administering an additional agent for the treatment of an UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) and/or a disease associated with TDP-43 dysfunction in a subject.

[0303] Having now described some embodiments in detail, practice of the invention will be more fully understood from the following examples, which are presented herein for illustration only and should not be construed as limiting the invention in any way.

EXAMPLES

Example 1: Synthesis of Oligonucleotides

[0304] The automated solid-support synthesis of 331 oligonucleotides was performed on nS8-II oligonucleotide synthesizer (GeneDesign). Antisense oligonucleotides correspond to ASOs 1-331 and 350-814 in Tables 3, 5, 9, and 11. All the oligonucleotides were synthesized using low-loaded Unylinker™ universal solid support (1 μmol scale, 42.6 μmol/g, ChemGenes, N-4000-10). The protocol for solid-support synthesis is shown in Table 6 below.

TABLE 6

Step	Reaction	Reagent	Reaction time
1	Detritylation	Dichloroacetic Acid-Toluene (3:97)	90 seconds
2	Coupling	Activator solution:0.1M Amidite solution = 40:60 (12 eq.)	300 seconds
3	Oxidation or Thiolation	Iodine Solution (abt. 0.05 mol/L)[Pyridine:Water(9:1) (for oxidation) or 0.1M ((dimethylamino-methylidene)amino)-3H-1,2,4-dithiazoline-3-thione in pyridine (for thiolation)	30 seconds (for oxidation) or 400 seconds (for thiolation)
4	Capping	1-Methylimidazole-Acetonitrile (2:8) + Acetic Anhydride-2,6-Lutidine-Acetonitrile (2:3:5)	90 seconds

tion comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof. In some embodiments, the method comprises administering to the subject any one of the antisense oligonucleotides disclosed herein or a pharmaceutical composition disclosed

[0305] At the end of the solid phase synthesis, cyanoethyl protecting groups were removed by a 10 minute treatment with 10% diethylamine in acetonitrile. The solid support was dried with an argon gas flush and placed into a 2 mL microtube. 500 μL of concentrated aqueous ammonia was

added to the tube and the support was treated for 8 hours at 60° C. As for AmNA (R=Me)-containing oligonucleotides, 500 μ L of tert-butylamine-MeOH-water (1/1/2) solution was added to the tube and the support was treated for 7 hours at 65° C. (Masaki Yamagami et. al., 1P-40 Synthesis and Properties of Antisense Oligonucleotides Containing AmNA and GuNA, The 46th International Symposium on Nucleic Acids Chemistry & The 3rd Annual Meeting of Japan Society of Nucleic Acids Chemistry on October 29th). The antisense oligonucleotides were isolated with the cartridge purification system (Glen-Pak™ DNA Purification Cartridge, #60-5200, according to the manual provided) to afford an aqueous solution containing the purified oligonucleotide. For distillation of volatile components, a Thermo Scientific SAVANT SPD2010 Speed Vac Concentrator was used. Quantification of the antisense oligonucleotides was conducted based on UV absorbance measurement on Unchained Labs LUNATIC. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μ m, 2.1 $\times$ 50 mm) column. The eluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100% MeOH). The sample was lyophilized to afford a white powder. Measured molecular weights by mass spectrometry confirmed the molecular weight of the antisense oligonucleotides. A subset of the antisense oligonucleotides was selected for testing in Example 2.

Alternative Method for Synthesis of Oligonucleotides

[0306] The automated solid-support synthesis of oligonucleotides was performed on nS8-II oligonucleotide synthesizer (GeneDesign). All the oligonucleotides were synthesized using low-loaded Unylinker™ universal solid support (1.5 μ mol scale, 42.6 μ mol/g, ChemGenes, N-4000-10). The detailed protocol for nS8-II is shown in Table 7 below.

TABLE 7

Step	Reaction	Reagent	Reaction time
1	Detritylation	Dichloroacetic Acid-Toluene (3:97)	90 sec
2	Coupling	Activator solution:0.1M Amidite solution = 40:60 (12 eq.)	300 sec
3	Oxidation or Thiolation	Iodine Solution (abt. 0.05 mol/L) Pyridine:Water(9:1) (for oxidation) or 0.1M ((dimethylamino-methylidene)amino)-3H-1,2,4-dithiazoline-3-thione in pyridine (for thiolation)	30 seconds (for oxidation) or 400 seconds (for thiolation)
4	Capping	1-Methylimidazole-Acetonitrile (2:8) + Acetic Anhydride-2,6-Lutidine-Acetonitrile (2:3:5)	90 sec

[0307] At the end of the solid phase synthesis, cyanoethyl protecting groups were removed by a 10 minute treatment with 10% diethylamine in acetonitrile. The synthesis was

repeated three times on a 1.5 μ mol scale, and those were combined into one, resulting in a total of 4.5 μ mol scale. The subsequent experiment was then carried out using this combined amount. The solid support was dried with an argon gas flush and placed into a 2 mL microtube. 500 μ L of conc. Aq. ammonia was added to the tube and the support was treated for 8 hours at 60° C. Oligonucleotides were isolated with the cartridge purification system (Glen-Pak™ DNA Purification Cartridge, #60-5200, according to the manual provided) to afford aqueous solution containing the purified oligonucleotide. For distillation of volatile components, a Thermo Scientific SAVANT SPD2010 SpeedVac Concentrator was used. The resulting solution was filtered through a 0.20 μ m filter (ADVANTEC, DISMIC: 13CP020AS). UV absorbance of the solution was measured and then the sample was lyophilized to afford a white powder. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μ m, 2.1 $\times$ 50 mm) column. The eluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100% MeOH). The data represents the calculated values obtained through deconvolution processing using Waters MassLynx, such as MaxEntl because the mass spectrum was observed not as molecular ion peaks, but as multivalent ions. Quantification of oligonucleotides was conducted based on UV absorbance measurement on Unchained Labs LUNATIC. Measured molecular weights by mass spectrometry confirmed the molecular weight of the antisense oligonucleotides.

Synthesis of 2'-O-NMA-Modified Oligonucleotides

[0308] The automated solid-support synthesis of 2'-O-NMA-modified oligonucleotides was performed on nS8-II oligonucleotide synthesizer (GeneDesign). All the oligonucleotides were synthesized using low-loaded Unylinker™

universal solid support (1 μ mol scale, 42.6 μ mol/g, ChemGenes, N-4000-10). The protocol for solid-support synthesis is shown in Table 17.

TABLE 17

Step	Reaction	Reagent	Reaction time
1	Detritylation	Dichloroacetic Acid-Toluene (3:97)	90 seconds
2	Coupling	Activator solution:0.1M Amidite solution = 40:60 (12 eq.)	300 seconds
3	Oxidation or Thiolation	Iodine Solution (abt. 0.05 mol/L) Pyridine:Water(9:1) (for oxidation) or 0.1M ((dimethylamino-methylidene)amino)-3H-1,2,4-dithiazoline-3-thione in pyridine (for thiolation)	30 seconds (for oxidation) or 400 seconds (for thiolation)

TABLE 17-continued

Step	Reaction	Reagent	Reaction time
4	Capping	1-Methylimidazole-Acetonitrile (2:8) + Acetic Anhydride-2,6-Lutidine-Acetonitrile (2:3:5)	90 seconds

[0309] At the end of the solid-phase synthesis, cyanoethyl protecting groups were removed by treating the solid support with 10% diethylamine in acetonitrile for 10 minutes. The solid support was then dried using an argon gas flush and transferred into a 2 mL microtube. Next, 500 μ L of concentrated aqueous ammonia was added to the tube, and the mixture was incubated at 65° C. for 15 minutes. Subsequently, 500 μ L of 40% methylamine was added, and the solution was further incubated for 15 minutes with an aqueous ammonia-to-methylamine ratio of 1:1. Volatile components were removed using a Thermo Scientific SAVANT SPD2010 SpeedVac Concentrator. The resulting solution was filtered through a 0.20 μ m filter (ADVANTEC, DISMIC: 13CP020AS) and purified using a C18 reverse-phase column (Xbridge @Oligonucleotide BEH C18, 10 mm $\times$ 50 mm, 130A, 2.5 μ m) via HPLC, employing a gradient of 10%-70% of mobile phase B over 30 minutes. Mobile phase A consisted of 240 mM hexafluoroisopropanol and 7

such as MaxEnt1 because the mass spectrum was observed not as molecular ion peaks, but as multivalent ionfragment ions. Quantification of oligonucleotides was conducted based on UV absorbance measurement on Unchaind Labs LUNATIC. Measured molecular weights by mass spectrometry confirmed the molecular weight of the antisense oligonucleotides.

Synthesis of C16-Conjugated Oligonucleotides

[0311] Oligonucleotide was chemically synthesized using the deoxynucleoside phosphoramidite approach by AKTA oligopilot plus 10 System. (solid support: UnyLinker™ NittoPhase® 341 μ mol/g). 5'-TFA-Amino Modifier C6 CE-Phosphoramidite (Glen research, 10-1916-02) was subjected at the end of coupling cycle. The protocol for solid-support synthesis is shown in Table 18.

TABLE 18

Step	Reaction	Reagent	Volume	Reaction time
1	Detritylation	Dichloroacetic Acid-Toluene (3:97)	Until trityl UV disappears	
2	Coupling	Activator solution:0.1M Amidite solution = 40:60	4 eq.	10 min (for DNA)/20 min (for MOE)
3	Oxidation or Thiolation	Iodine Solution (abt. 0.05 mol/L)[Pyridine:Water(9:1) (for oxidation) or 0.1M ((dimethylamino-methylidene)amino)-3H-1,2,4-dithiazolone-3-thione in pyridine (for thiolation)	3.5 CV (for oxidation) or 3 CV (for thiolation)	10 min
4	Capping	1-Methylimidazole-Acetonitrile (2:8) + Acetic Anhydride-2,6-Lutidine-Acetonitrile (2:3:5)	2CV	10 min

mM triethylamine in an aqueous solution containing 5% methanol, while mobile phase B was 100% methanol. The purified DMT-ON fraction was collected and freeze-dried. The resulting freeze-dried powder was dissolved in 1 mL of distilled water and subjected to detritylation using a cartridge purification system (Glen-Pak™ DNA Purification Cartridge, #60-5200) according to the manufacturer's instructions, yielding an aqueous solution containing the purified oligonucleotide. Volatile components were again removed using the Thermo Scientific SAVANT SPD2010 SpeedVac Concentrator. The final solution was filtered through a 0.20 μ m filter (ADVANTEC, DISMIC: 13CP020AS), and its UV absorbance was measured. The purified sample was then lyophilized to afford a white powder. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μ m, 2.1 $\times$ 50 mm) column. The eluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100% MeOH). [0310] The data represents the calculated values obtained through deconvolution processing using Waters MassLynx,

[0312] At the end of the solid phase synthesis, cyanoethyl protecting groups were removed by a 10 min treatment with 10% diethylamine in acetonitrile. After transferring the solid phase support from the column reactor to 15 mL Falcon tube, aq. ammonia (4 mL) was added to the tube and the support was treated for 8 hours at 60° C. After filtering the reaction solution, nitrogen gas was blown against the filtrate to remove ammonia. The solution was then lyophilized. The residue was dissolved in water (2 mL), and 300 μ L of 3 M aq. Sodium acetate solution (pH=7.0) was added to the solution. EtOH was added to the solution, and the resulting white suspension was centrifuged. The supernatant was removed by decantation, and the residue was dissolved in water. Quantification of the antisense oligonucleotides was conducted based on UV absorbance measurement on Unchaind Labs LUNATIC. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μ m, 2.1 $\times$ 50 mm) column. The eluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution

(100% MeOH). The sample was lyophilized to afford a C6-amino conjugated oligonucleotide without further purification.

[0313] The crude C6 amino conjugated oligonucleotide was dissolved with water. N,N-diisopropylethylamine (100 eq.) and palmitic acid N-hydroxysuccinimide ester (10 eq.) dissolved in DMSO was added to the substrate solution. The reaction mixture was incubated at 45 degree for 10 h. 3 M aq. Sodium acetate solution (pH=7.0) was added to the solution. EtOH was added to the solution, and the resulting white suspension was centrifuged. The supernatant was removed by decantation, and the residue was dissolved in water. The mixture was purified by HPLC (XBridge, OST C18 column, 2.5 μ m, 10 $\times$ 50 mm0.1M TEAA in DW/0.1M TEAA in MeCN, 10%-60% for 30 min). The fraction containing the target product was evaporated, and the residue was dissolved in water, and 3M NaOAc aq. was added to the solution for anion exchange. EtOH was added to the solution, and the resulting white suspension was centrifuged. The supernatant was removed by decantation, and the residue was dissolved with water. The solution was then lyophilized to give C16-conjugated oligonucleotide. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μ m, 2.1 $\times$ 50 mm) column. The eluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100% MeOH).

[0314] The data represents the calculated values obtained through deconvolution processing using Waters MassLynx, such as MaxEntl because the mass spectrum was observed not as molecular ion peaks, but as multivalent ionfragment ions. Quantification of oligonucleotides was conducted based on UV absorbance measurement on Unchaind Labs LUNATIC. Measured molecular weights by mass spectrometry confirmed the molecular weight of the antisense oligonucleotides.

Example 2: Activities of Antisense Oligonucleotides In Vivo

[0315] This example demonstrates that antisense oligonucleotides disclosed herein inhibited inclusion of the cryptic exon (CE) into UNC13A mRNA.

[0316] Single stranded Adeno-associated viral vector 9 (ssAAV9) encoding the UNC13A exon20-intron 20-exon 21 sequence and part of the intron 21 sequence with alternative primer binding site upstream and downstream of the UNC13A sequences driven by a human synapsin I promoter (ssAAV9-SYN1-hUNC13A (SEQ ID NO: 328)) was produced by VectorBuilder. ssAAV9-SYN1-hUNC13A (2 μ L of 1.0 $\times$ 10¹³ genome copies/mL) was injected into the right dorsal hippocampus of 8-week-old mice. One week later, to knock down the endogenous mouse Tardbp (TDP-43), the mice received an antisense oligonucleotide (2 μ L of 2.5 mM) targeting mouse Tardbp (TDP-43) (5'-AAGGCTcatatgtACTTT-3' (SEQ ID NO: 317) (hereinafter referred to as "Tardbp ASO") with a phosphorothioate backbone, lowercase letters indicating deoxyribonucleosides and uppercase letters indicating 2'-O-methoxyethyl-modified nucleosides, and bases of "C" or "c" indicating 5'-methyl cytosine, or phosphate-buffered saline (PBS) (2 μ L) (control; Tardbp ASO (-)) by intrahippocampal injection. The antisense oligonucleotides targeting mouse Tardbp were synthesized by Gene Design Co., Ltd. (Osaka, Japan). One-week post-

injection, test antisense oligonucleotides (10 μ L of 1.0 mM) or PBS (10 μ L) (control; test antisense oligonucleotide (-)) or were intracerebroventricularly injected. Test antisense oligonucleotides correspond to ASOs 1, 2, 8, 9, 14, 16, 17, 80, 144, 145, 149, 164, 193, 194, 214, 220, 222, 235, 238, 276, and 328 in Tables 3 and 5. Mice were euthanized at 14 days post-test antisense oligonucleotides injection and the right dorsal hippocampus was isolated. Total RNA was extracted using QIAzol Lysis Reagent (Qiagen, 79306) and RNeasy Mini Kit (Qiagen, 74106) and was reverse-transcribed with SuperScript IV VILO Master Mix (Invitrogen, 11756050) following the manufacturers' protocol. RT-qPCR was performed with TaqPath qPCR Master Mix, CG (Applied Biosystems, A15297) on the QuantStudio 7 Flex Real-Time PCR System. Expression levels of mouse Tardbp ((TDP-43) and mouse Actb (Actb) were evaluated using Thermofisher TaqMan Gene Expression Assays: Tardbp (TDP-43) (ID: Mm01257504_g1), Actb (ID: Mm00607939_s1). For detecting the splicing of the UNC13A cryptic exon (CE) and normal splicing, the following sequences of primers and probes were synthesized at Integrated DNA technologies (IDT) and were used: UNC13A_CE FWD 5'-3' CCCCCGTAC-CATGTCCAGTACA (SEQ ID NO: 320), UNC13A_CE REV 5'-3' CATTACCAGCATTATTCAACAAA (SEQ ID NO: 321), UNC13A_CE Probe 5'-3'/56-FAM/CTGCAT-GAG/ZEN/CTGCCT/3IABKFQ; UNC13A minigene_Normal FWD 5'-3' CTCACTAAAGGGAACAAGCGA (SEQ ID NO: 322), UNC13A minigene_Normal REV 5'-3' GTG-GAACAGGTTCTCATGC (SEQ ID NO: 323), UNC13A minigene_Normal Probe 5'-3'/56-FAM/CAGTGTGGA/ZEN/GATCAAAGGCGAGGA/3IABKFQ/(SEQ ID NO: 324).

[0317] Each target expression level was calculated using the absolute quantification methods. The expression level of mouse Tardbp was normalized by mouse Actb and the expression level of UNC13A cryptic exon (CE) was normalized by minigene-derived normal UNC13A (detected by using UNC13A minigene_Normal primers and probe) expression level. To calculate TDP-43 knock-down level in Tardbp ASO treated group, the average expression value of the control group (Tardbp ASO non-treated and test antisense oligonucleotide non-treated group) were used for the normalization (100%: Tardbp ASO (-), test antisense oligonucleotide (-) group). For the value of UNC13A CE normalized by minigene-derived normal UNC13A, the average expression value of the control group (Tardbp ASO treated and test antisense oligonucleotide non-treated group) were used for the normalization (100%: Tardbp ASO (+) test antisense oligonucleotide (-)). If multiple tests were performed, the average of those values was calculated.

[0318] Results are shown in Table 8 below.

TABLE 8

ASO #	% TDP-43 level in mice that received the Tardbp ASO relative to mice that received PBS	% inclusion of UNC13A CE in mice treated with the indicated ASO relative to mice treated with PBS
1	38.3	60.8
2	31.0	45.5
8	35.4	55.2
9	33.7	21.5
14	38.1	57.7
16	34.9	63.0

TABLE 8-continued

ASO #	% TDP-43 level in mice that received the Tardbp ASO relative to mice that received PBS	% inclusion of UNC13A CE in mice treated with the indicated ASO relative to mice treated with PBS
17	26.3	64.3
80	30.2	47.0
144	34.6	54.6
145	33.2	62.9
149	34.5	47.0
164	31.7	70.0
193	30.7	68.5
194	34.1	42.2
214	29.0	54.4
220	30.5	19.3
222	32.8	54.6
235	35.1	65.7
238	37.4	34.7
276	31.8	53.2
328	30.7	50.5

[0319] The results show that all antisense oligonucleotides tested in Table 8 showed inhibition of the inclusion of the CE into UNC13A mRNA.

[0320] The same experiment was run with additional test antisense oligonucleotides corresponding to ASOs 81, 107, 324, 327, 369, 373, 396, 486, 494, 495, 514, 521, 532, 561, 603, 629, 631, 632, 633, 650, 660, 663, 692, 726, and 328 in Tables 3, 9, 5, 11. Results are shown in Table 12 below.

TABLE 12

ASO #	% TDP-43 level in mice that received the Tardbp ASO relative to mice that received PBS	% inclusion of UNC13A CE in mice treated with the indicated ASO relative to mice treated with PBS	Amount
81	28.5	37.3	10 nmol
107	26.4	45.1	10 nmol
324	27.4	40.8	10 nmol
327	33.1	35.3	10 nmol
369	36.7	46.8	3 nmol
373	30.3	21.2	10 nmol
396	33.1	48.8	3 nmol
486	33.1	51.5	10 nmol
494	27.3	63.8	10 nmol
495	28.6	47.2	10 nmol
514	33.8	48.6	10 nmol
521	33.2	49.4	10 nmol
532	46.7	35.3	10 nmol
561	47.7	21.1	3 nmol
603	25.7	49.4	10 nmol
629	26.1	51.4	10 nmol
631	34.7	56.9	10 nmol
632	42.0	39.0	10 nmol
633	24.5	63.1	3 nmol
650	27.0	62.5	3 nmol
660	23.9	51.9	3 nmol
663	26.0	42.6	3 nmol
692	34.4	60.2	10 nmol
726	43.9	57.8	10 nmol

[0321] The results show that all antisense oligonucleotides tested in Table 12 showed inhibition of the inclusion of the CE into UNC13A mRNA.

Example 3: Activities of Antisense Oligonucleotides In Vivo (hUNC13A BAC Tg Mice)

[0322] This example demonstrates that antisense oligonucleotides (ASO) disclosed herein inhibited inclusion of the cryptic exon (CE) into UNC13A mRNA in human UNC13A BAC transgenic (Tg) mouse model. This mouse model was generated as follows by Institute of Immunology Co., Ltd. Following a modified version of the method by Abe et al. (Exp Anim. 2004 53(4):311-320. Establishment of an efficient BAC transgenesis protocol and its application to functional characterization of the mouse Brachyury locus), human CH17-170L19 BAC clone (BACPAC GENOMICS, INC.) including human UNC13A gene was purified using a plasmid extraction kit, and linearized by reacting with *PI-SceI*. The linearized DNA was separated in an agarose gel and extracted using electroelution and dialyzed in TE buffer prepared for microinjection. The purified DNA fragments were assessed for high purification without fragmentation by applying them to pulse-field gel and determine the DNA concentration using a NanoDrop spectrophotometer (Asahi Techno Glass Corporation). Dilute the DNA solution to 0.5 ng/ μ l and store it at 4° C. until used for injection. For the verification of the Human UNC13A Locus contained in BAC clone, PCR using following primers were conducted: PCR-A FWD 5'-3' GGTCTGCACAGGAGGAAACC (SEQ ID NO: 536), PCR-A REV 5'-3' CCACACGACCCCTCCAGCAG (SEQ ID NO: 537). For the verification of purified BAC clone by PCR with following primers: PCR-B FWD 5'-3' CTACGAGCATGTCATGAAGTTGC (SEQ ID NO: 538), PCR-B REV 5'-3' AAGCCCATTCATTCTAGTGCTG (SEQ ID NO: 539), PCR-C FWD 5'-3' ACATCCCTCAGAGTGTGTGTCA (SEQ ID NO: 540), PCR-C REV 5'-3' CCTGGGGAGGAGAATGAGGTAAA (SEQ ID NO: 541). The DNA were injected into the fertilized egg pronucleus of C57BL/6J mice. After injection, the embryos were injected into the one-sided fallopian tubes of female mice. Genotyping and southern blotting analysis for the F0 mouse were performed by using primers and probes as follows: PCR-1 FWD 5'-3' ACATCCCTCAGAGTGTGTGTCA (SEQ ID NO: 540), PCR-1 REV 5'-3' CCTGGGGAGGAGAATGAGGTAAA (SEQ ID NO: 541), PCR-2 FWD 5'-3' TCCATTGAAGTGCATGTGTGTAG (SEQ ID NO: 542), PCR-2 REV 5'-3' GGGACCATAAGAGCTAAAGTTGG (SEQ ID NO: 543), PCR-3 FWD 5'-3' GCATTTGAAAGGTCTCAACTGTATT (SEQ ID NO: 544), PCR-3 REV 5'-3' TTGTTGCTGATGAATCTTGTGTG (SEQ ID NO: 545), PCR-4 FWD 5'-3' GGGAATTTATCTGAAGACCAAGC (SEQ ID NO: 546), PCR-4 REV 5'-3' TCAGCTTTTGAATGGAGAATGAA (SEQ ID NO: 547), PCR-5 FWD 5'-3' TACTCTCGCTGTCTCCTTGTGTC (SEQ ID NO: 548), PCR-5 REV 5'-3' ACTGAAAGAGGCTCAGGAACACT (SEQ ID NO: 549), PCR-6 FWD 5'-3' GACAGTGACTACCGCAGTGAAAC (SEQ ID NO: 550), PCR-6 REV

5'-3' CCACAAGAAAATGTGAATTCTGC (SEQ ID NO: 551), PCR-7 FWD 5'-3' AGTGGGCTTAATTTGCATT-CATT (SEQ ID NO: 552), PCR-7 REV 5'-3' GCACAGGTATAATCAGACCCACA (SEQ ID NO: 553), PCR-8 FWD 5'-3' GGTGGAAGAAGCACTGAGAGG (SEQ ID NO: 554), PCR-8 REV 5'-3' AGAAGAAGGG-GAAAGTCAGAGGT (SEQ ID NO: 555), PCR-9 FWD 5'-3' AGTTACACCGTCGTCCTATCAT (SEQ ID NO: 556), PCR-9 REV 5'-3' ATGACCTT-GAAGAAGTGGCTTTC (SEQ ID NO: 557), PCR-10 FWD 5'-3' CCTCACAGTTGTAGACCATGGAA (SEQ ID NO: 558), PCR-10 REV 5'-3' CACACCTCCTTCCA-GATGTTGTA (SEQ ID NO: 559), PCR-11 FWD 5'-3' CTACGAGCATGTATGAAGTTGC (SEQ ID NO: 558), PCR-11 REV 5'-3' AAGCCCATTCATTCTAGTGCTG (SEQ ID NO: 559), PCR-12 FWD 5'-3' CCCTGTCTCGTGTCCCTCAGA (SEQ ID NO: 560), PCR-12 REV 5'-3' CATGGGCAAGCAGTGTGTTCTA (SEQ ID NO: 561), PCR-Internal control (mouse Actb) FWD 5'-3' AGAGAGCTCACCATTACCATCT (SEQ ID NO: 562), REV 5'-3' TCCTAGGCTCTCAAAACAAACC (SEQ ID NO: 563), Southern blotting Probe 5'-3' AGCCACGGGTATCCCATGGCCCGTC-CACACCAGCCTGGCTATGCTAACTCCCAA-CCATGCACCAAGTCTGGCCCCACA-CATCCCCAGCCAGCCTCCTTCCCAATGTCAGC-CTGGGGTTTCTAGATCTAGAGGGAG-GAGAGAAAAGTGGGCTGAAAGTGGGCTGA AGT-CATGGTTAGGGATCAAGGTCAGGGTTAGTTCAGC-CATCAAAGATCAGAGGTCA GGGTTAGAATTGGTGT-TAGGCAAAGGGTCACTGCTTAGGGCCAAATTG-GAGGTCA TTGTCAGGGTTAGG-GAAAGGTCCTGGGGCCGGCTCAGTAGGGGAGTAGG AATCAAG GGAAGCAATCAG-CATCAGGGGCAAGCTGACTGTTTA-GATGGGCAGCAGGGTTG (SEQ ID NO: 564). The F0 mice and subsequent generations were backcrossed with C57BL/6J.

[0323] To knock down the endogenous mouse Tardbp (TDP-43), eight to eleven week old mice received a heteroduplex oligonucleotide (10 μ L of 2.5 mM) targeting mouse Tardbp (TDP-43) (antisense strand: 5'-ATATGcacgctgat-tCCTTT-3' (SEQ ID NO: 1363), with phosphorothioate backbone, lowercase letters indicating deoxyribonucleosides, uppercase letters with underlines indicating ENA-modified nucleosides and bases of "C" or "c" indicating 5'-methyl cytosine, complementary strand: 5'-aaaggaatcacgctgcatat-3' (SEQ ID NO: 1364) with phosphate backbone, lowercase letters indicating deoxyribonucleosides and bases of "c" indicating 5'-methyl cytosine (hereinafter referred to as "Tardbp HDO") or 5% dextrose (5% Dex) (10 μ L) as vehicle control (Tardbp HDO (-)) by intracerebroventricular (ICV) injection on day 0 and day 7. The antisense strand of Tardbp HDO was synthesized by KNC Laboratories Co., Ltd. (Kobe, Japan). The complementary strand of Tardbp HDO was synthesized by Gene Design Co., Ltd. (Osaka, Japan). On day 1 and day 21, test

ASOs (10 μ L of 0.3 mM) or DPBS (10 μ L) as vehicle control (test ASO (-)) were injected by ICV. Test ASOs correspond to ASOs 211, 372, 391, 392, 393, 394, 395, 511, 564, 565, 658 in Tables 3, 5, 9, and 11. Mice were euthanized at 21 days after the last test ASO injection, and the cerebral cortex and the spinal cord were isolated. Total RNA was extracted using QIAzol Lysis Reagent (Qiagen, 79306) and Quick-Gene RNA Tissue Kit (Kurabo, RT-S2) with DNase (QIAGEN, 79254) treatment, and was reverse-transcribed with SuperPrep II Cell Lysis & RT Kit for qPCR (Toyobo, SCQ-401) following the manufacturers' protocol.

[0324] RT-qPCR was performed with THUNDERBIRD Probe qPCR Mix (Toyobo, QPS-101) on the Viia7, QuantStudio 7 Flex or QuantStudio 12K Flex Real-Time PCR System (Applied Biosystems). Expression levels of mouse Tardbp (TDP-43) was evaluated using ThermoFisher TaqMan Gene Expression Assays: Tardbp (TDP-43) (ID: Mm01257504_g1). For detecting the splicing of the UNC13A cryptic exon (CE) and mouse Rbfox3, the following sequences of primers and probes were synthesized at Integrated DNA technologies (IDT) and were used: UNC13A_CE FWD 5'-3' CCCCCGTAC-CATGTCCAGTACA (SEQ ID NO: 320), UNC13A_CE REV 5'-3' CATTACACCAGCATTTATTCAACAAA (SEQ ID NO: 321), UNC13A_CE Probe 5'-3'/56-FAM/CTGCAT-GAG/ZEN/CTGCCT/3IABKFQ/. Rbfox3 FWD 5'-3' CCGCTCGTTAAAAATGATCTCC (SEQ ID NO: 565), Rbfox3 REV 5'-3' CGACTACATGTCTCCAACATCC (SEQ ID NO: 566), Rbfox3 Probe 5'-3'/56-FAM/TTCCCGAAT/ZEN/TGCCCGAACATTTGC/3IABKFQ/ (SEQ ID NO: 567). RT-qPCR protocol was as below; 95° C. for 60 sec, 45 cycles of 95° C. for 15 sec and 60° C. for 60 sec.

[0325] Each target expression level was calculated using the absolute quantification methods. The expression level of mouse Tardbp (TDP-43), UNC13A cryptic exon (CE) were normalized by mouse Rbfox3 expression level. To calculate TDP-43 knock-down level in Tardbp HDO treated group, the average expression value of the control group (Tardbp HDO (-) and test ASO (-) group (PBS only)) were used for the normalization (100%: Tardbp HDO (-) and test ASO (-)). For the value of UNC13A CE normalized by Rbfox3, the average expression value of the control group (Tardbp HDO (+) and test ASO (-) group) were used for the normalization (100%: Tardbp HDO (+) and test ASO (-)).

[0326] For statistical analysis, post-hoc analysis was performed for CE expression data of cerebral cortex and spinal cord. Aspin-Welch's test for Tardbp HDO (-) and test ASO (-) group vs. Tardbp HDO (+) and test ASO (-) was performed to confirm the success of the TDP-43 knock-down. Then, Dunnett's multiple comparison test for Tardbp HDO (+) and test ASO (-) group vs. Tardbp HDO (+) and test ASO (+) groups was performed. Test ASOs that showed significant decrease ($p < 0.05$) in CE inclusion are shown in Table 13 below.

TABLE 13

ASO #	Concentration (nmol)	Cerebral Cortex		Spinal cord	
		% of TDP-43 level in mice that received the Tardbp HDO and PBS relative to mice that received PBS only (normalized by Rbfox3 level)	% of inclusion of UNC13A CE in mice treated with the Tardbp HDO and indicated ASO relative to mice treated with Tardbp HDO and PBS (normalized by Rbfox3 level)	% of TDP-43 level in mice that received the Tardbp HDO and PBS relative to mice that received PBS only (normalized by Rbfox3 level)	% of inclusion of UNC13A CE in mice treated with Tardbp HDO and indicated ASO relative to mice treated with Tardbp HDO and PBS (normalized by Rbfox3 level)
211	3 + 3	26.6	28.1	23.1	40.4
372	3 + 3	24.1	6.7	24.9	5.7
391	3 + 3	37.5	24.6	30.5	18.5
392	3 + 3	18.0	53.1	20.1	16.8
393	3 + 3	19.3	31.1	20.9	15.7
394	3 + 3	25.3	10.5	23.1	9.0
395	3 + 3	19.8	30.7	21.3	22.6
511	3 + 3	33.0	47.3	29.4	43.3
564	3 + 3	30.1	29.7	26.4	34.5
565	3 + 3	36.9	23.2	41.4	36.5
658	3 + 3	24.3	44.3	23.3	41.4

Example 4: Activities of Antisense Oligonucleotides In Vitro (shTDP—SH-SY5Y Cells with LNP TF)

[0327] SH-SY5Y cell line (Parental SH-SY5Y Cell line from ATCC; CRL-2266, Lot: 58462094) with doxycycline (dox) inducible expression (Tet-on system) of TDP-43 specific shRNA for knock down was generated (hereinafter referred to as shTDP—SH-SY5Y). TurboRFP and microRNA30-adapted short hairpin RNA is expressed under the Tet operator and minimal CMV promoter and rtTA3 is expressed under the UbC promoter. microRNA30-adapted short hairpin RNA sequence is as below: ttgaat-gaggcttcagtactttacagaatcgttgctgcacatcttggaacacttgctgg-gaatacttctcagggttaacccaacagaaggctc gagaaggtatat-tgctgttgacagtgagcgacacactacaattgatatacaaatagtggaagccacagatg-attgatatacaattgtagtgtgttg cctactgcctcgggaattcaaggggctactttaggagcaattatctgtttactaaaact-gaataaccttgctatctctttgatacatattttacaaagct gaataaaatggtataaaat-taaatcacttt (SEQ ID NO: 568). shTDP-SH-SY5Y cells were grown in 43.5% DMEM, high glucose, HEPES (Gibco, 12430054), 43.5% Ham's F-12 Nutrient Mix (Gibco, 11765047), 10% Fetal Bovine Serum (Gibco, A31606-01 or Clontech, 631105), 1xMEM NEAA Solution (Gibco, 11140050), 2 mM Sodium Pyruvate (Gibco, 11360070), and 2 µg/mL Blastidin S HCl (Gibco, A1113903) at 37° C., 5% CO2. To evaluate UNC13A splicing abnormality and its correction by ASOs, around 60% TDP-43 knock-down was induced by treating cells with 4 µg/mL of doxycycline (Clontech, 631311) for 14 days.

Manufacturing Methods

[0328] (1) A lipid mixture (cationic lipid:DPPC:cholesterol:GM-020=60:10.6:28:1.4, in mole ratio) was dissolved in 90% EtOH/10% RNase-free water to afford an approximately 14.4 mg/mL lipid solution. The cationic lipid was 3-{[4-(dimethylamino) butanoyl]oxy}-2,2-bis {[(9Z)-tetradec-9-enoyloxy]methyl}propyl (9Z)-tetradec-9-enoate, described in WO2019131839,

the contents of which, related to cationic lipids, are incorporated herein by reference. The lipid solution and 10 mM Citrate buffer, pH3 were mixed by using the apparatus NanoAssemblr (Precision Nanosystems) at room temperature with a flow rate ratio of 3 mL/min:9 mL/min to afford a dispersion containing a lipid particle. By using a Slyde-A-Lyzer (molecular weight cutoff: 20k, Thermo Fisher Scientific), the dispersion obtained was dialyzed with water at 4° C. for 1 hour and with 10 mM MES/0.9% NaCl buffer, pH5 for 2 days. Subsequently, filtration was performed with a 0.2 µm syringe filter (IWAKI CO., LTD.) to prepare a composition for nucleic acid introduction, and thereafter the composition was stored at 4° C. The lipid concentration of the composition was analyzed using HPLC and the size was measured using Zetasizer Nano ZS (Malvern Instruments). The results are shown in Table 14.

[0329] (2) At culturing day 11 with doxycycline treatment, shTDP-SH-SY5Y cells were seeded at 60,000 viable cells/well in 96-well plate (Corning, 3598). 0.853 uM of the composition of (1) diluted by saline were gently mixed with 2 uM of test compound in equal amount to form a mixed composition, diluted by 1 mM MES, pH 5.5 in saline and/or DPBS (–), and incubated for 10 min at room temperature. The mixed composition was added to the culture media at the final concentration of 50 nM test compound within 30 min after mixing lipid-based nanoparticles and test compound. The cells were harvested on day 14 after doxycycline treatment by using FastLane Cell Multiplex Kit (QIAGEN, 216513) according to the manufacturer's instructions.

TABLE 14

	Size (nm)	Polydispersity index (PDI)	Cationic lipid concentration
Nanoparticle 1	69.1	0.065	3.73 mM

[0330] RT-qPCRs were run on a QuantStudio™ 7 Flex Real-Time PCR System (Applied Biosystems) or ViiA™ 7 Real-Time PCR System (Applied Biosystems). The experiments were run in duplicate. The following PCR probe assay for UNC13A CE synthesized at Integrated DNA technologies (IDT) were used: UNC13A_CE FWD 5'-3' CCCC GTACCATGTCCAGTACA (SEQ ID NO: 320), UNC13A_CE REV 5'-3' CATTCAACCAGCATTTATTCAACAAA (SEQ ID NO: 321), UNC13A_CE Probe 5'-3'/56-FAM/CTGCATGAG/ZEN/CTGCCT/3IABKFQ/. For housekeeping gene GAPDH expression analysis was performed by using ThermoFisher® TaqMan Gene Expression Assay Hs99999905_m1. PCR protocol was as below; 50° C. for 30 min, 95° C. for 15 min, 45 cycles of 94° C. for 15 sec and 60° C. for 60 sec.

[0331] Relative expression of UNC13A CE was calculated by normalizing to GAPDH expression (deltaCt), then the average expression value of the control group (saline) without ASO treatment were used for the normalization (100%: dox (+) ASO (-)). Test ASOs correspond to ASOs 1-331 and 350-814 in Tables 3, 5, 9, and 11. Results are shown in Table 15 below.

TABLE 15

ASO #	Relative expression of UNC13A CE
1	25
2	30
3	25
4	96
5	80
6	54
7	46
8	43
9	8
10	52
11	51
12	33
13	62
14	20
15	43
16	14
17	27
18	63
19	52
20	51
21	56
22	28
23	27
24	64
25	57
26	90
27	46
28	72
29	44
30	62
31	41
32	75
33	22
34	5
35	18
36	15
37	31
38	42
39	42
40	22
41	5
42	5
43	2
44	4
45	4

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
46	3
47	2
48	3
49	5
50	7
51	7
52	7
53	48
54	47
55	65
56	37
57	43
58	36
59	9
60	13
61	76
62	63
63	33
64	27
65	65
66	62
67	42
68	28
69	56
70	49
71	59
72	47
73	71
74	74
75	38
76	26
77	31
78	34
79	33
80	25
81	16
82	25
83	23
84	23
85	31
86	18
87	19
88	26
89	30
90	21
91	30
92	36
93	57
94	11
95	16
96	12
97	25
98	8
99	8
100	8
101	5
102	5
103	3
104	7
105	2
106	12
107	24
108	18
109	21
110	38
111	27
112	44
113	5
114	68
115	27
116	32
117	33
118	26
119	16

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
120	6
121	13
122	31
123	6
124	10
125	38
126	40
127	51
128	48
129	81
130	83
131	34
132	50
133	40
134	36
135	70
136	56
137	25
138	43
139	41
140	69
141	18
142	11
143	47
144	18
145	36
146	7
147	11
148	23
149	11
150	27
151	11
152	15
153	14
154	74
155	41
156	65
157	77
158	31
159	24
160	32
161	31
162	28
163	34
164	46
165	41
166	30
167	38
168	96
169	55
170	37
171	19
172	21
173	95
174	95
175	89
176	85
177	91
178	85
179	85
180	95
181	40
182	54
183	54
184	78
185	60
186	20
187	51
188	54
189	61
190	47
191	37
192	35
193	32

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
194	20
195	48
196	43
197	34
198	47
199	47
200	95
201	97
202	78
203	61
204	74
205	42
206	61
207	58
208	75
209	15
210	43
211	33
212	35
213	32
214	30
215	39
216	24
217	46
218	27
219	26
220	17
221	24
222	27
223	25
224	35
225	27
226	29
227	25
228	17
229	27
230	15
231	19
232	33
233	27
234	26
235	13
236	51
237	20
238	17
239	24
240	18
241	22
242	15
243	76
244	64
245	67
246	61
247	58
248	81
249	65
250	75
251	48
252	72
253	43
254	40
255	73
256	87
257	28
258	65
259	96
260	78
261	81
262	45
263	85
264	43
265	82
266	56
267	67

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
268	98
269	86
270	23
271	11
272	61
273	67
274	88
275	100
276	20
277	20
278	40
279	42
280	36
281	86
282	78
283	106
284	105
285	62
286	27
287	31
288	46
289	73
290	89
291	68
292	18
293	70
294	34
295	16
296	18
297	16
298	77
299	21
300	57
301	42
302	41
303	19
304	6
305	29
306	19
307	31
308	3
309	93
310	97
311	99
312	99
313	99
314	59
315	99
316	77
317	12
318	13
319	12
320	17
321	31
322	45
323	28
324	14
325	22
326	100
327	18
328	12
329	21
330	6
331	26
350	12
351	26
352	12
353	18
354	24
355	12
356	16
357	8
358	24
359	32

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
360	32
361	46
362	41
363	32
364	24
365	13
366	11
367	12
368	10
369	8
370	29
371	23
372	3
373	2
374	3
375	6
376	4
377	3
378	5
379	4
380	4
381	12
382	14
383	10
384	5
385	3
386	4
387	8
388	3
389	23
390	9
391	11
392	12
393	8
394	7
395	5
396	14
397	8
398	11
399	15
400	11
401	12
402	13
403	13
404	17
405	16
406	14
407	10
408	12
409	45
410	40
411	12
412	28
413	40
414	20
415	23
416	30
417	7
418	2
419	31
420	14
421	38
422	26
423	16
424	21
425	28
426	12
427	15
428	57
429	51
430	68
431	48
432	56
433	30

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
434	25
435	25
436	22
437	17
438	16
439	26
440	33
441	36
442	33
443	47
444	62
445	58
446	14
447	24
448	51
449	33
450	37
451	28
452	31
453	38
454	29
455	36
456	40
457	23
458	31
459	63
460	61
461	54
462	68
463	14
464	8
465	18
466	8
467	4
468	18
469	8
470	11
471	7
472	4
473	3
474	14
475	13
476	9
477	10
478	38
479	15
480	47
481	20
482	29
483	37
484	67
485	56
486	27
487	27
488	24
489	45
490	49
491	2
492	36
493	55
494	24
495	37
496	38
497	15
498	16
499	16
500	17
501	25
502	32
503	46
504	15
505	31
506	32
507	62

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
508	56
509	37
510	34
511	31
512	28
513	19
514	17
515	15
516	16
517	68
518	42
519	28
520	46
521	44
522	61
523	67
524	27
525	43
526	48
527	48
528	49
529	27
530	22
531	36
532	20
533	31
534	16
535	40
536	16
537	30
538	26
539	43
540	42
541	15
542	15
543	18
544	14
545	22
546	15
547	18
548	26
549	15
550	18
551	11
552	6
553	8
554	20
555	19
556	29
557	14
558	15
559	16
560	20
561	23
562	19
563	20
564	27
565	42
566	36
567	55
568	47
569	46
570	59
571	69
572	51
573	54
574	65
575	69
576	49
577	36
578	62
579	28
580	35
581	69

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
582	36
583	45
584	58
585	21
586	20
587	10
588	38
589	40
590	31
591	29
592	12
593	6
594	5
595	38
596	11
597	11
598	42
599	36
600	35
601	27
602	27
603	16
604	13
605	11
606	19
607	26
608	22
609	28
610	22
611	36
612	35
613	27
614	15
615	18
616	30
617	19
618	31
619	28
620	23
621	23
622	20
623	26
624	22
625	26
626	19
627	41
628	41
629	23
630	37
631	27
632	12
633	17
634	21
635	37
636	20
637	20
638	25
639	23
640	21
641	22
642	26
643	26
644	31
645	17
646	19
647	7
648	9
649	18
650	16
651	32
652	19
653	10
654	11
655	13

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
656	7
657	8
658	8
659	10
660	21
661	15
662	9
663	24
664	9
665	7
666	5
667	6
668	7
669	14
670	14
671	20
672	27
673	15
674	27
675	27
676	20
677	14
678	16
679	22
680	19
681	13
682	28
683	13
684	18
685	21
686	26
687	21
688	40
689	34
690	21
691	23
692	19
693	29
694	14
695	18
696	32
697	14
698	16
699	18
700	15
701	13
702	18
703	23
704	18
705	20
706	23
707	23
708	25
709	22
710	20
711	18
712	24
713	20
714	23
715	26
716	31
717	34
718	38
719	29
720	40
721	31
722	40
723	33
724	32
725	38
726	16
727	16
728	24
729	26

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
730	16
731	24
732	23
733	34
734	16
735	32
736	13
737	14
738	13
739	14
740	13
741	12
742	14
743	13
744	16
745	13
746	18
747	17
748	22
749	19
750	14
751	16
752	18
753	19
754	25
755	22
756	28
757	24
758	34
759	25
760	23
761	27
762	17
763	27
764	21
765	37
766	59
767	48
768	49
769	62
770	65
771	69
772	23
773	35
774	42
775	27
776	38
777	22
778	21
779	30
780	32
781	24
782	34
783	45
784	30
785	24
786	27
787	36
788	64
789	68
790	70
791	66
792	24
793	22
794	36
795	38
796	53
797	60
798	54
799	58
800	50
801	47
802	33
803	45

TABLE 15-continued

ASO #	Relative expression of UNC13A CE
804	58
805	43
806	66
807	10
808	8
809	36
810	28
811	57
812	69
813	67
814	69

Additional Embodiments

- [0332] 1. An antisense oligonucleotide comprising a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region, wherein the first target region and/or the second target region each comprises an UNC13A sequence, wherein the antisense oligonucleotide modulates splicing of an UNC13A cryptic exon.
- [0333] 2. The antisense oligonucleotide of embodiment 1, wherein the UNC13A cryptic exon is in intron 20 of an UNC13A pre-mRNA.
- [0334] 3. The antisense oligonucleotide of embodiment 1 or embodiment 2, wherein the first target region and the second target region are in intron 20 of an UNC13A pre-mRNA.
- [0335] 4. The antisense oligonucleotide of any one of embodiments 1-3, wherein the first target region is adjacent to the second target region in an UNC13A pre-mRNA.
- [0336] 5. The antisense oligonucleotide of any one of embodiments 1-4, wherein the first target region is downstream of the second target region in an UNC13A pre-mRNA.
- [0337] 6. The antisense oligonucleotide of any one of embodiments 1-5, wherein the first target region is more than 1 nucleoside downstream of the second target region.
- [0338] 7. The antisense oligonucleotide of any one of embodiments 1-6, wherein the first target region and the second target region are both in the cryptic exon.
- [0339] 8. The antisense oligonucleotide of any one of embodiments 1-6, wherein the first target region is downstream of the cryptic exon and the second target region is in the cryptic exon.
- [0340] 9. The antisense oligonucleotide of any one of embodiments 1-6, wherein the first target region and the second target region are both downstream of the cryptic exon.
- [0341] 10. The antisense oligonucleotide of any one of embodiments 1-8, where in the first target region and/or the second target region comprises a nucleotide sequence of any one of SEQ ID NOs: 313-316.
- [0342] 11. The antisense oligonucleotide of any one of embodiments 1-10, wherein the first target region comprises the nucleotide sequence of SEQ ID NO: 315, and the second target region comprises the nucleotide sequence of SEQ ID NO: 315.

- [0343] 12. The antisense oligonucleotide of any one of embodiments 1-10, wherein the first target region comprises the nucleotide sequence of SEQ ID NO: 315, and the second target region comprises the nucleotide sequence of SEQ ID NO: 314.
- [0344] 13. The antisense oligonucleotide of any one of embodiments 1-10, wherein the first target region comprises the nucleotide sequence of SEQ ID NO: 315, and the second target region comprises the nucleotide sequence of SEQ ID NO: 313.
- [0345] 14. The antisense oligonucleotide of any one of embodiments 1-10, wherein the first target region comprises the nucleotide sequence of SEQ ID NO: 313, and the second target region comprises the nucleotide sequence of SEQ ID NO: 313.
- [0346] 15. The antisense oligonucleotide of any one of embodiments 1-14, wherein the first antisense sequence and/or the second antisense sequence is 6-15 nucleosides in length, optionally wherein the first antisense sequence and/or the second antisense sequence is 8-15 nucleosides in length.
- [0347] 16. The antisense oligonucleotide of any one of embodiments 1-15, wherein the first antisense sequence comprises at least 8 contiguous nucleobases of any one of SEQ ID NOS: 191-245, and/or wherein the second antisense sequence comprises at least 8 contiguous nucleobases of any one of SEQ ID NOS: 194, 199, 212, 226, 227, and 246-306.
- [0348] 17. The antisense oligonucleotide of any one of embodiments 1-16, wherein the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOS: 191-245, and/or wherein the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOS: 194, 199, 212, 226, 227, and 246-306.
- [0349] 18. The antisense oligonucleotide of any one of embodiments 1-17, wherein the first antisense sequence and the second antisense sequence are immediately adjacent to each other.
- [0350] 19. The antisense oligonucleotide of any one of embodiments 1-18, wherein one or more nucleobases are between the first antisense sequence and the second antisense sequence.
- [0351] 20. The antisense oligonucleotide of any one of embodiments 1-17, wherein one or more nucleobases at the 3' end of the first antisense sequence is complementary to one or more nucleobases at the 3' end of the second target region, and/or one or more nucleobases at the 5' end of the second antisense sequence is complementary to one or more nucleobases at the 5' end of the first target region.
- [0352] 21. The antisense oligonucleotide of any one of embodiments 1-20, comprising the nucleobase sequence of any one of SEQ ID NOS: 1-190.
- [0353] 22. The antisense oligonucleotide of any one of embodiments 1-21, comprising one or more modified nucleosides.
- [0354] 23. The antisense oligonucleotide of embodiment 22, wherein the one or more modified nucleosides comprise a non-bicyclic 2' modified nucleoside, a 2'-4' bridged nucleoside, or combinations thereof.
- [0355] 24. The antisense oligonucleotide of embodiment 23, wherein the non-bicyclic 2'-modified nucleoside is a 2'-O-methoxyethyl (2'-MOE) modified nucleoside, a 2'-O-Methyl (2'-O-Me) nucleoside, or a 2'-fluoro (2'-F) modified nucleoside, and/or the 2'-4' bridged nucleoside is a locked nucleic acid (LNA, 2'-4' methylene bridge), an ethylene-bridged nucleic acid (ENA, 2'-4' ethylene bridge), or a constrained ethyl nucleic acid (cEt, 2'-4' ethylene bridge).
- [0356] 25. The antisense oligonucleotide of any one of embodiments 1-24, wherein the antisense oligonucleotide comprises one or more 2'-MOE modified nucleosides and one or more LNAs.
- [0357] 26. The antisense oligonucleotide of any one of embodiments 1-24, wherein each nucleoside of the antisense oligonucleotide is a 2'-MOE modified nucleoside.
- [0358] 27. The antisense oligonucleotide of any one of embodiments 1-26, comprising one or more modified internucleoside linkages.
- [0359] 28. The antisense oligonucleotide of any one of embodiments 1-27, wherein each internucleoside linkage of the oligonucleotide is a modified internucleoside linkage.
- [0360] 29. The antisense oligonucleotide of embodiment 27 or embodiment 28, wherein the modified internucleoside linkage is a phosphorothioate linkage.
- [0361] 30. The antisense oligonucleotide of any one of embodiments 1-29, wherein each internucleoside linkage of the oligonucleotide is a phosphorothioate linkage.
- [0362] 31. The antisense oligonucleotide of any one of embodiments 1-29, wherein the antisense oligonucleotide comprises a mix of phosphodiester linkages and phosphorothioate linkages.
- [0363] 32. The antisense oligonucleotide of any one of embodiments 1-22, wherein the antisense oligonucleotide is a phosphorodiamidate oligomer (PMO).
- [0364] 33. The antisense oligonucleotide of any one of embodiments 1-32, wherein each cytosine of the oligonucleotide is not a 5-methyl-cytosine.
- [0365] 34. The antisense oligonucleotide of any one of embodiments 1-32, wherein each cytosine of the oligonucleotide is a 5-methyl-cytosine.
- [0366] 35. The antisense oligonucleotide of any one of embodiments 1-32, wherein one or more of the cytosines of the oligonucleotide is a 5-methyl-cytosine.
- [0367] 36. The antisense oligonucleotide of any one of embodiments 1-35, comprising a structure as provided in Table 5.
- [0368] 37. The antisense oligonucleotide of any one of embodiments 1-36, wherein the antisense oligonucleotide inhibits inclusion of the UNC13A cryptic exon into an UNC13A mature mRNA.
- [0369] 38. The antisense oligonucleotide of any one of embodiments 1-37, wherein the antisense oligonucleotide is in the form of a pharmaceutically acceptable salt.
- [0370] 39. A composition comprising the antisense oligonucleotide of any one of embodiments 1-38.
- [0371] 40. The composition of embodiment 39, further comprising a pharmaceutically acceptable carrier.
- [0372] 41. A method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell, comprising contacting the cell with the antisense oligo-

- nucleotide of any one of embodiments 1-38, or the composition of embodiment 39 or embodiment 40.
- [0373] 42. A method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject, comprising administering to the subject the antisense oligonucleotide of any one of embodiments 1-38, or the composition of embodiment 39 or embodiment 40.
- [0374] 43. A method of treating a disease associated with abnormal UNC13A expression in a subject, comprising administering to the subject the antisense oligonucleotide of any one of claims 1-38, or the composition of embodiment 39 or embodiment 40.
- [0375] 44. A method of treating a disease associated with TDP-43 dysfunction in a subject, comprising administering to the subject the antisense oligonucleotide of any one of claims 1-38, or the composition of embodiment 39 or embodiment 40.
- [0376] 45. The method of embodiment 43 or embodiment 44, wherein the disease is a neurodegenerative disease.
- [0377] 46. The method of any one of embodiments 43-45, wherein the neurodegenerative disease is amyotrophic lateral sclerosis (ALS).
- [0378] 47. The method of any one of embodiments 43-45, wherein the neurodegenerative disease is frontotemporal dementia (FTD).
- [0379] 48. The method of any one of embodiments 42-47, wherein the subject is human.
- [0380] 49. The method of any one of embodiments 42-48, wherein the subject has decreased expression of TDP-43 gene or protein.
- [0381] 50. The method of any one of embodiments 42-49, wherein the subject does not possess an SOD-1 gene mutation.
- [0382] 51. The method of any one of embodiments 42-50, wherein the subject possesses an UNC13A gene mutation in intron 20.
- [0383] 52. The method of embodiment 51, wherein the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.
- [0384] 53. A pharmaceutical composition for use in a method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject, comprising the antisense oligonucleotide of any one of embodiments 1-38.
- [0385] 54. A pharmaceutical composition for use in a method of treating a disease associated with abnormal UNC13A expression in a subject comprising the antisense oligonucleotide of any one of embodiments 1-38.
- [0386] 55. A pharmaceutical composition for use in a method of treating a disease associated with TDP-43 dysfunction in a subject comprising the antisense oligonucleotide of any one of embodiments 1-38.
- [0387] 56. The pharmaceutical composition of any one of embodiments 53-55, further comprising a pharmaceutically acceptable carrier.
- [0388] 57. The pharmaceutical composition of any one of embodiments 53-56, wherein the subject is human.
- [0389] 58. The pharmaceutical composition of any one of embodiments 55-57, wherein the subject has decreased expression of TDP-43 gene or protein.
- [0390] 59. The pharmaceutical composition of any one of embodiments 55-58, wherein the subject does not possess an SOD-1 gene mutation.
- [0391] 60. The pharmaceutical composition of any one of embodiments 55-59, wherein the subject possesses an UNC13A gene mutation in intron 20.
- [0392] 61. The pharmaceutical composition of embodiment 60, wherein the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.
- [0393] 62. An antisense oligonucleotide of any one of embodiments 1-38 for use as a medicament.
- [0394] 63. The antisense oligonucleotide of embodiment 62, wherein the medicament is administered to a subject.
- [0395] 64. An antisense oligonucleotide of any one of embodiments 1-38 for use in a method of treating a disease associated with abnormal UNC13A expression in a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein.
- [0396] 65. An antisense oligonucleotide of any one of embodiments 1-38 for use in a method of treating a disease associated with TDP-43 dysfunction in a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein.
- [0397] 66. The antisense oligonucleotide of any one of embodiments 63-65, wherein the subject is human.
- [0398] 67. The antisense oligonucleotide of any one of embodiments 63-66, wherein the subject has decreased expression of TDP-43 gene or protein.
- [0399] 68. The antisense oligonucleotide of any one of embodiments 63-67, wherein the subject does not possess an SOD-1 gene mutation.
- [0400] 69. The antisense oligonucleotide of any one of embodiments 63-68, wherein the subject possesses an UNC13A gene mutation in intron 20.
- [0401] 70. The antisense oligonucleotide of embodiment 69, wherein the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

EQUIVALENTS

[0402] The disclosure may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting of the disclosure. Scope of the disclosure is thus indicated by the appended claims rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are therefore intended to be embraced herein.

SEQUENCE LISTING

The patent application contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US20250283083A1>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

1-75. (canceled)

76. An antisense oligonucleotide comprising the following structure:

$[i^5Cs][iGs][iAs][iAs][iTs][i^5Cs][iTs][iAs][i^5Cs][i^5Cs]$
 $[i^5Cs][iAs][i^5Cs][iAs][iTs][i^5Cs][iTs][iGs][iTs][iTs]$
 $[i^5Cs][iAs][iAs][iTs][i^5Cs][iA]$ (SEQ ID NO: 571),
 wherein iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage, or a pharmaceutically acceptable salt thereof.

77. A method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell of a human subject, comprising administering to the subject the antisense oligonucleotide of claim 76.

78. The method of claim 77, wherein the method restores UNC13A expression in the cell.

79. A method of treating a neurodegenerative disease in a human subject, comprising administering to the subject the antisense oligonucleotide of claim 76.

80. The method of claim 79, wherein the neurodegenerative disease is amyotrophic lateral sclerosis (ALS).

81. The method of claim 79, wherein the neurodegenerative disease is frontotemporal dementia (FTD).

82.-87. (canceled)

88. An antisense oligonucleotide consisting of the following structure:

$[i^5Cs][iGs][iAs][iAs][iTs][i^5Cs][iTs][iAs][i^5Cs][i^5Cs]$
 $[i^5Cs][iAs][i^5Cs][iAs][iTs][i^5Cs][iTs][iGs][iTs][iTs]$
 $[i^5Cs][iAs][iAs][iTs][i^5Cs][iA]$ (SEQ ID NO: 571),
 wherein iA, iG, i⁵C, and iT are 2'-O-methoxyethyl (2'-MOE) modified adenosine, guanosine, 5'-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage, or a pharmaceutically acceptable salt thereof.

89. A method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell of a human subject, comprising administering to the subject the antisense oligonucleotide of claim 88.

90. The method of claim 89, wherein the method restores UNC13A expression in the cell.

91. A method of treating a neurodegenerative disease in a human subject, comprising administering to the subject the antisense oligonucleotide of claim 88.

92. The method of claim 91, wherein the neurodegenerative disease is amyotrophic lateral sclerosis (ALS).

93. The method of claim 91, wherein the neurodegenerative disease is frontotemporal dementia (FTD).

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